Introduction to Basic Probability

Introduction to Basic Probability

A Flipped Classroom Approach

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Introduction

Welcome

In this part of the book, we discuss some of the logistics. Specifically, we talk about who the book is for, how it can be used, and the resources it provides for educators and students in their probability and statistics endeavors. First, however, we provide a brief history of how this book came to be.

In 2020, as the educational world was rocked by COVID-19, many of us started experimenting with different ways to engage our students. We experimented with new and old ways to engage students: this book is largely a product of that time and of the "flipped classroom" approach. The paradigm of flipped classroom that I followed in my probability and statistics class is simple:

1. Provide students with a set of *notes*. Each chapter in this book roughly corresponds to a set of notes provided to students.
2. Provide students with short, "to-the-point" videos. These *pre-lectures* help students engage with the material alongside the notes.
3. In class, students are given a set of *worksheets*. Again, each chapter of our book provides indicative worksheets that you can adapt for your classes.

I hope you have as much fun reading and using the book as I had writing it.

Intended audience

The book is primarily intended for undergraduate engineering students who attempt their first ventures in probability and statistical modeling. What sets the book apart from some of the other (excellent) literature on the topic is its focus on hands-on learning through worksheets and worked examples. If you are an educator, my goal was to provide you with as many exercises as possible. These exercises and activities have been shown to work well in practice.

At a second level, the book is also designed from the perspective of industrial engineering (IE). Many of the applications and motivation in the book are drawn directly from problems faced by modern industrial engineers. If you are in industrial engineering, I hope you appreciate this perspective; if you are in a different area, it may offer you a glimpse of the *IE world*.

Logistics and other information

The book consists of 12 theory and activities chapters, followed by a concluding chapter. At the end of each of the first 12 chapters, I provide a series of in-class activities for students to engage in. Some chapters (e.g., Chapters 1–3) require a single lecture to go through; others, especially in the space of estimation (Chapters 11 and 12) could take approximately 2 weeks of class time to go through.

1. Random experiments, sample spaces, and other notation

Theory

Most of the chapters in the book are divided into two main sections: a **theory** section that will include all of the material, and an **activities** section that will include a series of exercises and activities for students to undertake.

Chapter objectives

In this chapter, our goal is to **build the necessary vocabulary**. Throughout the book, we will introduce notions and definitions, although some of the most basic definitions are provided in this first chapter. The learning objectives follow.

Learning outcomes: Random experiments, sample spaces, and other notation

After this chapter, you will be able to:

- Give examples of experiments, sample spaces, and events.

- Explain sets and describe how they are used.
- Use Venn diagrams to represent events.
- Understand and apply basic set operations.
- Give examples of and recognize mutually exclusive events.
- Calculate the cardinality of a set or event.

Chapter questions

- What is a **random experiment**? What is a **sample space**? What is an **event**?
- What is the **cardinality** of an event?
- **Venn diagrams and set notation**: can we use them for representing events and sample spaces?
  - $A \cup B$
  - $A \cap B$
  - $\overline{A}$ (complement of A)
  - $A \setminus B$
- What does it mean that two events are **mutually exclusive**?
- What are some **useful identities** for set operations?

Motivating examples

A game of Monopoly

Is Monopoly a game of luck or strategy? Perhaps both? Imagine that it is your turn and your friends have built hotels *everywhere* on the board. As it is your turn, you are about to roll two dice and hope you get the sum of the two dice to be a 6 or a 7 to avoid paying your friends and declaring bankruptcy. Everyone expects you to lose! That said, what are the true "chances" you will roll a 6 or a 7, after considering all the scenarios?

At the end of this first chapter, we should have all the necessary definitions in order to be able to quantify risks and chances, not only for Monopoly, but for all engineering applications.

Let's play some cards

Continuing with games, assume that you are playing a card game on a deck with 52 cards of 4 different suits: ♥, ♣, ♦, ♠. There are 13 cards of each suit, being ordered as: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 followed by three face cards noted as J, Q, K. The game is simple: *pick a card, any card*. If that card is red, you win; otherwise, you lose.

Should we play? How should we play? Are there ways to again mathematically quantify our risks and our gains?

Definitions

Random experiments

A **random experiment** is defined as an activity or situation where the outcome obtained may be different, even when executed the same way. This randomness in the outcome could be due to the inherent nature of the experiment, due to different levels of skill required to get a better result, or due to differences in instrumentation.

Example 1.1. Random experiments

Random experiments are all around us.

- Flipping a coin is a random experiment, as its outcome can be either heads or tails. One could even go further and quantify the "chances" of getting one outcome versus another (as in, "50-50" using a fair coin).
- Rolling a six-sided die is a random experiment, with the outcome being integers from 1 to 6.

The difficulty in understanding whether a process truly constitutes a random experiment is that there is no objective measure. For example, measuring the air quality inside a home at a specific time sounds like it should be *deterministic* in nature: that is, there is a *true* value and we are measuring to find it. However, from a mathematical standpoint, we could say that this process is indeed a random experiment, as it can depend on the instruments used to measure it, the person measuring it, among others. In simple terms, if an outcome cannot be adequately guaranteed, then it can be viewed as a random experiment.

Exercise 1.1. Are those random experiments?

What are some other random experiments you can think of?

- Is the time at which the next bus arrives at the bus stop a random experiment?
- How about baking? Is the quality of a cake a random experiment?
- Is a football game a random experiment?
- Finally, is measuring the width of a coffee table at a furniture store using a ruler a random experiment?

Before we move to the next section, let us return to our card game from the Motivating Examples. What do you think? Is our card game a random experiment? In essence, if you always follow the same strategy on different decks (e.g., pick the first card at the top, or pick the first card at the bottom of the deck), are you guaranteed the same result?

Sample spaces

With the term **sample space**, we refer to the set of all possible outcomes that can be obtained from a random experiment. If an outcome is not in the sample space of a random experiment, then that outcome cannot happen as a result of that random experiment. We typically reserve the notation S as the sample space and we write $S = \{outcomes\}$. We can then say that $s \in S$ to denote an outcome s belonging to the sample space.

Example 1.2. Sample spaces

Sample spaces of random experiments include:

- In a game of tic-tac-toe (random experiment), we can *win*, *lose*, or *tie*. Hence, $S = \{\text{win, lose, tie}\}$.
    - At the end of a game of tic-tac-toe, these are the only allowed outcomes.
- In a graded course (random experiment), we can get an A, A-, B+, B, B-, …
    - The sample space above does not include A+: this means that you cannot receive an A+ as your grade.
- The duration of a random tv show episode (random experiment), is any positive real number.
    - You cannot watch a *negative amount of* TV.

A sample space can have a finite or countably infinite number of possible outcomes (e.g., "1, 2, 3, 4, 5, or 6" or any integer number) or it can be an interval of real numbers (e.g., any number between 0 and 1, that is $[0, 1]$). We call the first type of sample space **discrete**. We will focus on that type of sample spaces for now. The second type of sample space (where the outcome is a real number belonging to some interval) is called **continuous**. The rest of this chapter is devoted to discrete sample spaces.

Exercise 1.2. Discussing sample spaces

Sample spaces should include all possible outcomes for a random experiment.

- Is… food poisoning a possible outcome of cooking? What are the other outcomes in that random experiment of you cooking?
- Is… snow a possible outcome for tomorrow's weather? What are the other outcomes in the random experiment of the weather tomorrow?

Example 1.3. Monopoly

Assume again it is your turn to roll the dice for a round of Monopoly. What is the sample space for rolling two dice?

Let (i, j) represent the tuple of the first die (i) and the second die (j). Then, the sample space is:

- $(1,1),(1,2),(1,3),(1,4),(1,5),(1,6)$
- $(2,1),(2,2),(2,3),(2,4),(2,5),(2,6)$
- $(3,1),(3,2),(3,3),(3,4),(3,5),(3,6)$
- $(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)$
- $(5,1),(5,2),(5,3),(5,4),(5,5),(5,6)$

- $(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)$

We can even write $S = \{(i,j) : i,j \in \mathbb{Z}, 1 \leq i,j \leq 6\}$, which basically states that the sample space is defined as all possible tuples of integer numbers between 1 and 6.

This seems like it can get overly complicated fast. We will need a more efficient way to *count* outcomes soon.

Example 1.4. Longest distance to a hospital

Even though we will focus on discrete sample spaces for now, let us first consider an example of a continuous sample space. What is the sample space of the longest distance we are located from a hospital at a given time?

Let d be the distance (as the crow flies) from a hospital at any time. Then we must have $d > 0$. That said, we can perhaps refine this further. If we are in the United States, then the longest distance between any two points (not necessarily from or to a hospital) is approximately 5,859 miles. Therefore, we can say that d can be defined as the set of real numbers between 0 and 5,859. Equivalently, the sample space of d is the set of real numbers between 0 (at the hospital already) to 5859. Furthermore, assuming that we are only interested in the conterminous United States ("Lower 48" or the lower 48 states including Washington, D.C.) then that same sample space can now be written as a real number between 0 and 2892 (the corresponding maximum distance in the Lower 48).

Exercise 1.3. More practice with sample spaces

Give two examples of sample spaces.

- Provide the sample space of a random experiment with a finite number of possible outcomes.
- Provide the sample space of a random experiment with outcomes defined over an interval of real numbers.

Once again, before we move to more definitions, let us think back to our card game. The number of outcomes is finite, that is for sure, so our sample space is discrete. But what is our sample space? As is the case with many random experiments, there are multiple ways to describe the sample space: $S = \{1\heartsuit, 2\heartsuit, \ldots, K\heartsuit, 1\clubsuit, 2\clubsuit, \ldots, K\spadesuit\}$ or $S = \{red, black\}$ or even $S = \{\heartsuit, \clubsuit, \diamondsuit, \spadesuit\}$. The selection of the proper sample space is guided by what we are trying to achieve. In our motivation, we spoke about the color of the suit, so a sample space of $S = \{red, black\}$ seems like the simpler and better choice.

Events

The term **event** is used to define a subset (collection) of outcomes from the sample space. It can include only one such outcome or it can contain many of the outcomes. An event can be a combination of outcomes, the negation of an outcome, or other logical relationship between outcomes. An event is called *simple* if only one outcome can satisfy it or *compound* if it requires multiple outcomes. Some examples to clarify these definitions follow.

Example 1.5. Events

Events are essentially collections of outcomes

- In a graded class (random experiment) with a sample space of $S = \{A, A-, B+, B, B-, C+, C, C-, D+, D, F\}$, then an event can be to **pass the class** or to **get a B or higher**.
- When picking a single card from a deck of cards, then an event can be to **pick a Queen (Q)** or **to pick a Queen of clubs (Q♣)**. The first event is compound (there are four outcomes that can satisfy it as in Q♣, Q♠, Q♥, Q♦) whereas the second one is simple (only one outcome can satisfy it as in Q♣).
- In tomorrow's weather (random experiment), an event can be **no rain**. This is the negation of the event/outcome rain, and it means any outcome in the sample space **except for rain.**

Let us practice with a couple of exercises:

Exercise 1.4. Events

Answer the following questions.

- Let T be the time until the next bus arrives. Is $T \leq 10 \text{ minutes}$ a valid event?
- Define an event for the temperature tomorrow.
- In a board game where players roll two six-sided dice, is "getting a 10" a simple or a compound event? You may assume that the sample space is defined as $S = \{(i, j) : i, j \in \mathbb{Z}, 1 \leq i, j \leq 6\}$ (like we did earlier).

We are almost there with our definitions! Before we move to the next section, we revisit our card game. Let E be our event of "picking a red card." Assume that our sample space is defined as $S = \{red, black\}$. Then, event E is a **simple event** as there is only one outcome that satisfies it.
That said, let's say that the sample space was $S = \{1\heartsuit, 2\heartsuit, \ldots, K\heartsuit, 1\clubsuit, 2\clubsuit, \ldots, K\spadesuit\}$. Then, the event E is now a **compound event** as there are 26 outcomes that satisfy it. We can write that picking a red card corresponds to event $E = \{1\heartsuit, 2\heartsuit, \ldots, K\heartsuit, 1\diamondsuit, \ldots, K\diamondsuit\}$.

Sets and set operations

With the term **set** we mean a collection of *items*: these items are named **elements** in this context and are mathematical in nature. For example, the set of prime numbers includes integer numbers $2, 3, 5, 7, 11, \ldots$ or the set of grades higher than a B- are $A, A-, B+, B$. A set with no elements is **empty**; a set with one element is a **singleton**. As you can probably tell, our sample space definition can be viewed in this light as the definition of a set.

Set operations are a very, very useful way to describe events based on relationships between the outcomes of the random experiment. The most common set operations (and the ones I will predominantly use in this book) are:

1. The **union of two events** E_1, E_2 as $E_1 \cup E_2$. The union can be viewed as an "either/or, or both relationship," when discussing about the relationship between two events.
2. The **intersection of two events** E_1, E_2 as $E_1 \cap E_2$. The intersection can be viewed as "both events E_1 **and** E_2 should happen," when discussing about only two events.
3. The **complement** of an event E as $\overline{E}$ (sometimes it is also written as E^c or E'). The complement can be viewed as "*any* other event but E can happen."
4. The **relative complement** (sometimes termed as the difference) of an event E_2 from event E_1 as $E_1 \setminus E_2$. The relative complement can be viewed as "all outcomes in E_1 that are not included in E_2."

Let us check some examples before proceeding.

Example 1.6. Prime factorization

In mathematics, we define the prime factorization of an integer number as the set of all prime numbers that can perfectly divide it. Hence, let us define our set E_1 as all the factors of integer number 2025. That is, $E_1 = \{1, 3, 5, 9, 15, 25, 27, 45, 75, 81, 135, 225, 405, 675, 2025\}$. Now, define E_2 as the set of all prime numbers: $E_2 = \{2, 3, 5, 7, 11, 13, 17, 19, 23, \ldots\}$.

Then, the prime factorization of 2025 can be written as $E_1 \cap E_2$. In other words, the set of prime factors of integer number 2025 is the intersection between the set of its factors (E_1) and the set of prime numbers (E_2).

Example 1.7. Good day for a barbecue

Assume that tomorrow's weather can be represented as a random experiment with sample space $S = \{\text{sun}, \text{clouds}, \text{rain}\}$. Then, we could say that a good day for a barbecue is a day without rain. If $E = \{\text{rain}\}$, then a good day for a barbecue is $\overline{E}$.

Sets and Venn diagrams

Set operations can be nicely described through the use of Venn diagrams. Consider the following examples, where the whole sample space consists of A, B, and C. In all of the following cases, the shaded area shows the set operation visually.

1. A and B should both happen ⟐ $A \cap B$:

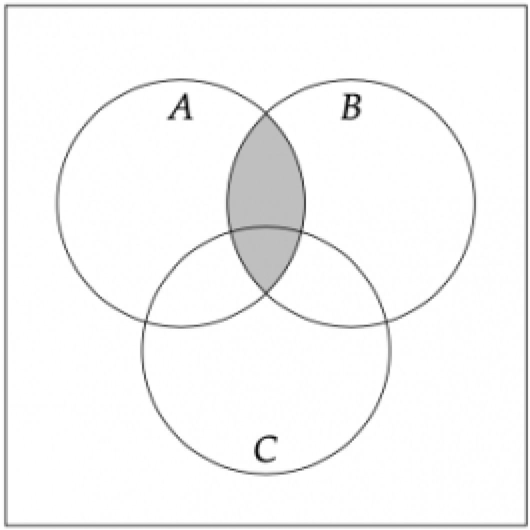

Figure 1.1. The visual representation in Venn diagram for [latex]A \cap B[/latex].

2. B or C should happen ⟐ $B \cup C$:

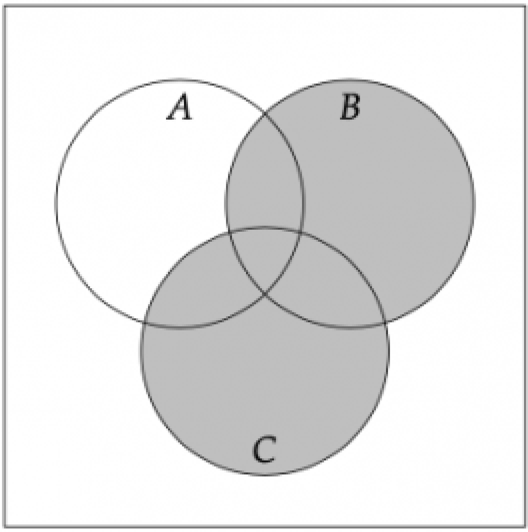

Figure 1.2. The visual representation in Venn diagram for [latex]B \cup C[/latex].

3. Neither A nor B should happen � $\overline{A} \cap \overline{B}$.

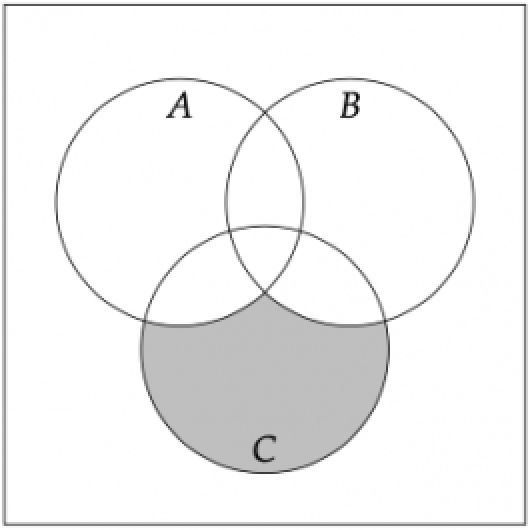

Figure 1.3. The visual representation in Venn diagram for [latex]\overline{A} \cap \overline{B}[/latex].

Since we assumed that $S = A \cup B \cup C$ (that is, $A, B,$ and C make up the whole sample space), then $\overline{A} \cap \overline{B}$ can also be expressed as:

1. Only C should happen but not A nor B� $C \setminus (A \cup B)$.
2. A or B should not happen � $\overline{(A \cup B)}$.

4. A, B, and C should all happen � $A \cap B \cap C$:

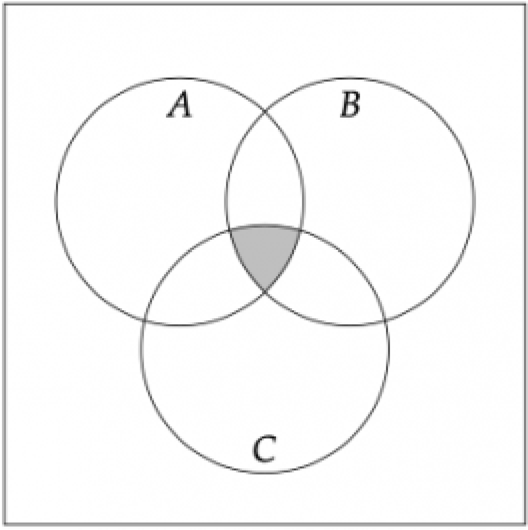

Figure 1.4. The visual representation in Venn diagram for [latex]A\cap B\cap C[/latex],

5. A, but not B, should happen ◇ $A \setminus B$:

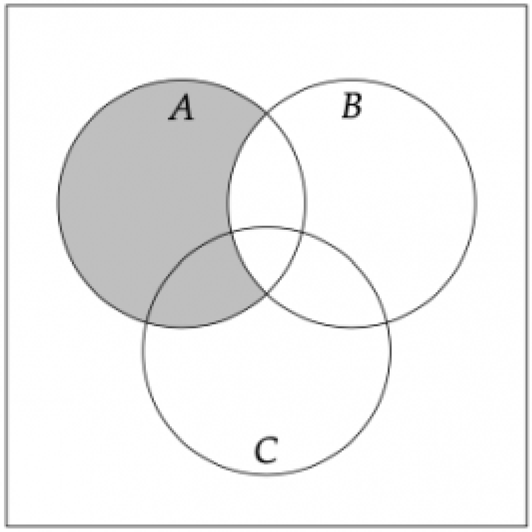

*Figure 1.5. The visual representation in Venn diagram for [latex]*A*setminus B[/latex],*

Let’s practice these together.

Example 1.8. Practice with Venn diagrams

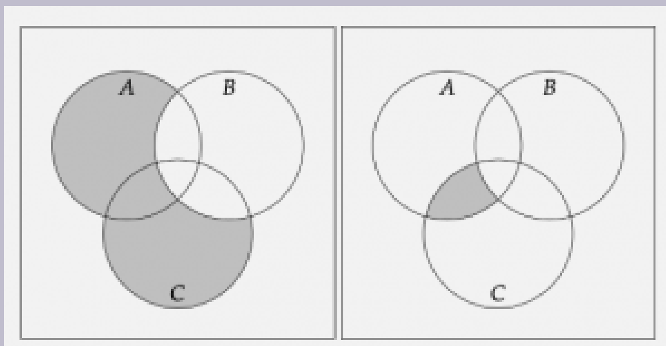

Figure 1.6. The two Venn diagrams for the exercise.

Explain these two Venn diagrams in words and mathematically.

For the first one (left Venn diagram), we want A or C, but not B. In mathematical terms, we can write this as $(A \cup C) \setminus B$. In the second one (right Venn diagram), we are looking at A and C, but not B. Again, mathematically, we have $(A \cap C) \setminus B$.

More set definitions

We have already established some vocabulary for sets, but it is useful to provide more details before proceeding. Specifically, we define the following:

- **Empty set**: a set containing no elements is called an empty or a null set, and is denoted by $\emptyset$.

- **Subsets and supersets**: assume that all elements of set S_2 are contained in another set S_1. Then, we say that set S_1 is a superset of S_2 and that set S_2 is a superset of S_1, and we write that $S_2 \subseteq S_1$.
    - Assume again that all elements of set S_2 are contained in another set S_1. Then, if there is at least one element in S_1 that is not included in S_2, we say that set S_1 is a proper superset of S_2 and we write that $S_2 \subset S_1$ (similarly, we say that set S_2 is a proper subset of S_1).
    - By definition, the empty set is a subset of any other set.
- **Mutually exclusive events**: we say that two events are **mutually exclusive** if they contain no common outcomes. Mathematically, two events E_1, E_2 are mutually exclusive if $E_1 \cap E_2 = \emptyset.$.

Visually, using Venn diagrams, we can see that a subset would all belong inside a superset. For example, if $E_1 \subseteq E_2$, we could expect the following Venn diagram.

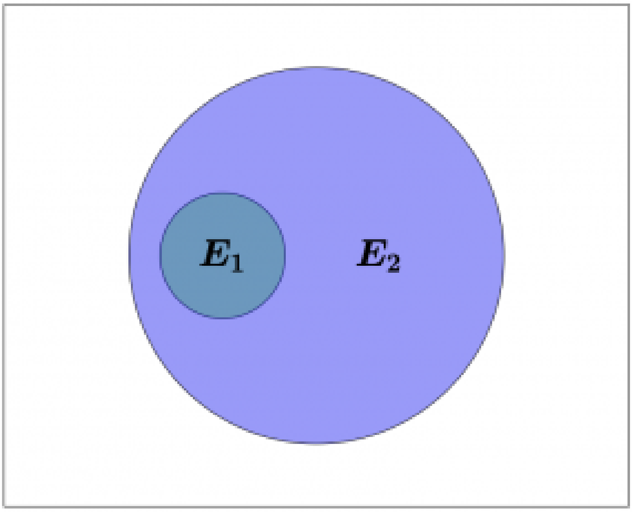

Figure 1.7. The Venn diagram representation of a subset. Here, [latex]E_1[/latex] is a subset of set [latex]E_2[/latex] and hence is wholly contained in [latex]E_2[/latex].

In the case of two mutually exclusive events E_1, E_2, we could expect a Venn diagram like the following:

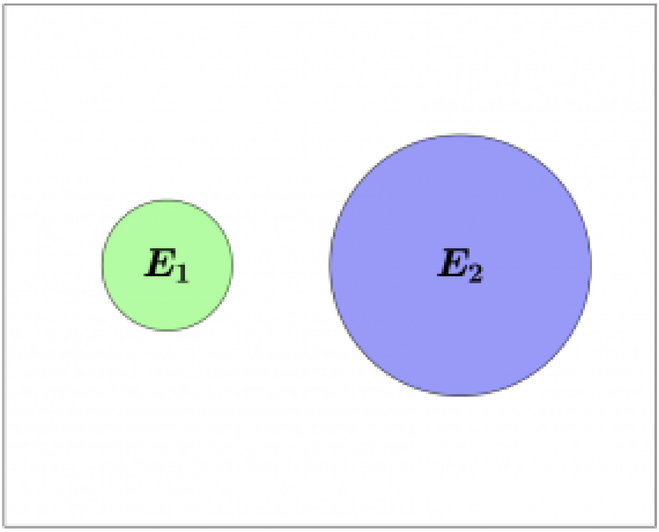

Figure 1.8. The Venn diagram representation of two mutually exclusive events. Here, [latex]E_1[/latex] and [latex]E_2[/latex] do not intersect at all.

Example 1.9. Subsets and supersets

Consider a sample space $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and four events/sets:

1. $A = \{1, 2, 3, 4, 5\}$: the number picked from S is smaller than or equal to 5.
2. $B = \{9, 10\}$: the number picked from S is

greater than 8.

3. $C = \{2, 3, 5, 7\}$: the number picked from S is prime.
4. $D = \{2, 4, 6, 8\}$: the number picked from S is prime.

Then, we have:

- $A \subseteq \overline{B}$. That is, set $A = \{1, 2, 3, 4, 5\}$ is a subset of $\overline{B} = \{1, 2, 3, 4, 5, 6, 7, 8\}$.
  - Even more, we can see that $A \subset \overline{B}$, as there are elements in $\overline{B}$ that are not included in A.
- $B \cap C = \emptyset$. Notice how B and C contain no common elements and hence the intersection between B and C is empty.
  - This means that B and C are mutually exclusive events.
- $A \cup B \cup C \cup D = S$. The union of all four sets makes up the whole sample space: there is no element in the sample space S that does not appear in at least one of the four sets A, B, C, D.

Exercise 1.5. Real life subsets and mutually exclusive events

Can you provide an example of a real-life random experiment and come up with two events such as one of them is a subset of the other? Can you identify two events that are mutually exclusive?

- How about passing a class and getting an A?
- Or not raining tomorrow, but it being cloudy?

Before proceeding to see some set operation laws (such as the famous De Morgan laws) and the definition of cardinality, we go once again back to our motivational card game. Assume that the sample space $S=\{1\heartsuit, 2\heartsuit, \ldots, K\heartsuit, 1\clubsuit, 2\clubsuit, \ldots, K\spadesuit\}$, and consider three events:

- $A=$"get a card with the value 3 or less."
- $B=$"get a red card."
- $C=$"get a face card (J, Q, K)."

What is:

1. the union of A and B?
2. the intersection of B and C?
3. the intersection of A and C?
4. the complement of C?

Let us start with the union of A and B:

1. $A \cup B$: "get a card with the value of 3 or less or a red card."
    - The event happens if we get $7\heartsuit$.
    - The event happens if we get $2\diamondsuit$.
    - The event happens if we get $1\spadesuit$.

- The event does not happen if we get 10♣.

2. $B \cap C$: "get a red <u>and</u> face card."

    - The event happens if we get Q♥.
    - The event happens if we get J♦.
    - The event does not happen if we get 1♠.
    - The event does not happen if we get K♣.

3. $A \cap C$: "get a face card that is <u>also</u> less than or equal to 3 in value."

    - The event never happens.
    - In set notation, we have $B \cap C = \emptyset$.
    - B and C are mutually exclusive events.

4. $\overline{C}$: "<u>not</u> get a face card."

    - The event does not happen if we get Q♥.
    - The event does not happen if we get J♦.
    - The event happens if we get 1♠.
    - The event happens if we get 6♣.

Laws for set operations

Assume S is the sample space of our random experiment, and A, B, C are some events. Then, we have the following laws:

1. $A \cup \overline{A} = S$: all the elements in A and all the elements in "not A" ($\overline{A}$) make up the whole sample space.
2. $A \cap \overline{A} = \emptyset$: there are no common elements in A and in "not A" ($\overline{A}$).
3. $\overline{\overline{A}} = A$: the complement of the complement of any event, is the event itself!These three properties are shown in the

following figure.

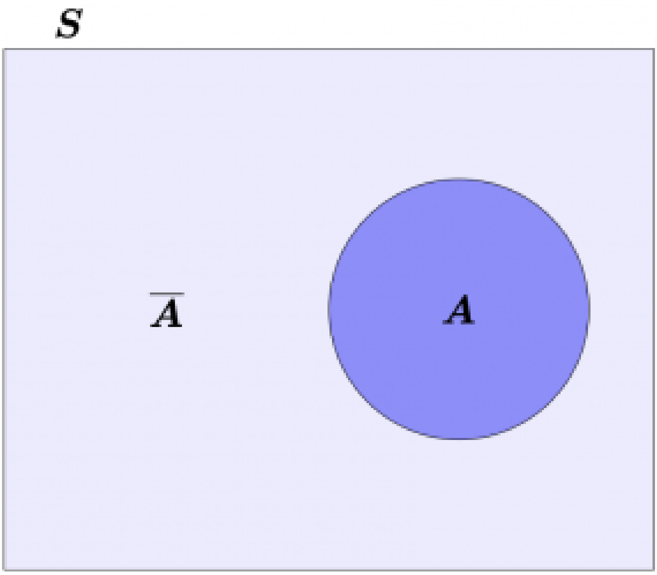

Figure 1.9. The Venn diagram representation of a set and its complement.

We also define the transitive property for the union and intersection. Namely:

1. $A \cup B = B \cup A$ and
2. $A \cap B = B \cap A$.

Finally, we are ready to reveal De Morgan's laws:

1. $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$.
2. $\overline{(A \cap B)} = \overline{A} \cup \overline{B}$.
3. $A \cup (B \cup C) = (A \cup B) \cup C, \;\; A \cap (B \cap C) = (A \cap B) \cap C$.
4. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \;\; A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Cardinality

The **cardinality** of an event E is a measure of its size and it corresponds to the number of elements/outcomes that it contains. It is denoted by $|E|$.

Example 1.10. Cardinality

Consider a graded course, with a sample space (possible grades) of $S = \{A, A-, B+, B, B-, \ldots F\}$.

- Let event E_1 be that you pass the class (i.e., get a score greater than or equal to a C-): we have that $E_1 = \{A, A-, B+, B, B-, C+, C, C-\}$ and its cardinality is $|E_1| = 8$.
- Let event E_2 be that you "score in the B zone" (i.e., get a score greater than or equal to a B-, but lower than or equal to a B+): we have that $E_2 = \{B+, B, B-\}$ and its cardinality is $|E_2| = 3$.

Note that the cardinality is properly defined for events that can be *counted*. More on counting in our next chapter! We now proceed to some important cardinality properties.

1. $E = \emptyset \implies |E| = 0$. This states that an empty set includes no elements.
2. $E_1 \subseteq E_2 \implies |E_1| \leq |E_2|$. Again, this implies that a set that is wholly included in another set has at most the same cardinality.

- A proper subset has a strictly smaller cardinality: $E_1 \subset E_2 \implies |E_1| < |E_2|$

3. Two mutually exclusive events E_1, E_2 have that:
   - $|E_1 \cap E_2| = 0;$
   - $|E_1 \cup E_2| = |E_1| + |E_2|.$

4. Finally, we provide the following statement without proof (but check the activity sheet for the chapter for a way to prove it): for *any* two events E_1, E_2, we have that $|E_1 \cup E_2| = |E_1| + |E_2| - |E_1 \cap E_2|.$

Example 1.11. Back to our motivation card game

Let us finally discuss the actual card problem of our motivation. Assume, once again, that our sample space is defined as:
$S = \{1\heartsuit, 2\heartsuit, \ldots, K\heartsuit, 1\clubsuit, 2\clubsuit, \ldots, K\spadesuit\}.$
Our game states that we win if we pick a red card. There are 13 $\heartsuit$ and 13 $\diamondsuit$ cards in the game. This gives us a cardinality of 26 outcomes. Recall that in total, our sample space consists of 52 outcomes, that is $|S| = 52$. **One might want to reason then that we have 26 favorable outcomes in a total of 52 outcomes…**

This is a good time to check an example from industrial engineering and operations research!

Example 1.12. Quality control

A circuit board manufacturer inspects 200 boards at the end of a production run and flags each board for three possible defect types:

- A = solder defect.
- B = component misaligned.
- C= surface contaminated.

The inspection from quality control resulted in the following counts:

Table 1.1. Counts for different conditions of a circuit board after inspection.

Condition	Count
Solder defect only (A)	40
Misalignment only (B)	30
Contamination only (C)	25
Both A and B	10
Both A and C	8
Both B and C	5
All three	3

1. **How many boards had at least one defect?**
2. **How many boards had no defects?**
3. **Are events A and C mutually exclusive?**

We can answer all questions using set notation:

1. We can apply the cardinality formula and extend it

to three sets:

$$\begin{aligned} & |A \cup B \cup C| \\ &= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \\ &= 40 + 30 + 25 - 10 - 8 - 5 + 3 \\ &= 75 \text{ boards.} \end{aligned}$$

2. Now that we know that 75 boards have at least one defect, this means the remaining set is $S \setminus (A \cup B \cup C)$ with cardinality $|S \setminus (A \cup B \cup C)| = 200 - 75 = 125$ boards.
3. No, A and C cannot be mutually exclusive as $A \cap C$ is not empty: it has cardinality equal to 8 (see "Both A and C" in the table).

Let's practice this in the next two exercises.

Exercise 1.6. Shift scheduling

Another popular industrial engineering problem is in shift scheduling. A fulfillment center operates three shifts, referred to as **Day** (D), **Evening** (E), and **Night** (N), and employee three types of employees in **Receiving** (R), **Picking** (P), and **Shipping** (S). The current schedule looks like the following:

Table 1.2. The number of employees at each shift.

	Day	Evening	Night	Total
Receiving	12	8	6	**26**
Picking	30	25	15	**70**
Shipping	18	12	10	**40**
Total	**60**	**45**	**31**	**136**

One employee is selected at random to participate at an anonymous survey. Let:
– D = employee works the day shift
– P = employee works in the Picking department

Answer the following questions:

1. What is $|D|$? What is $|P|$?
2. What is $|D \cap P|$?
3. What is $|D \cup P|$?
4. What is the cardinality of a Picking department employee who is *not* on the day shift? That is, what is $|P \setminus D|$?

Exercise 1.7. Practicing with cardinalities

A class at the university is taught by three different professors. The number of students that took the class and the grades they received are shown in the following table.

Table 1.3. The count of different letter grades for the same class under different professors.

Letter Grade	Professor 1	Professor 2	Professor 3	Total
A	108	20	30	158
B	44	49	46	139
C	11	15	15	41
D	0	1	8	9
Total	163	85	99	347

Answer the following questions.

- What is the cardinality of the event "receiving an A in Professor 1's class?"
- What is the cardinality of the event "receiving an A or being in Professor 1's class?"
- Are the events of "getting an A in Professor 1's class" and "getting a B in Professor 1's class" mutually exclusive events?
- What is the cardinality of the intersection of the events "getting an A" and "not being in Professor 1's class?"

Example 1.13. Back to Monopoly

In the beginning of this chapter, we only needed to roll a 6 or a 7 to “survive” another round (so to speak) of Monopoly. Let us finish this chapter with the following thought process:

- The sample space of rolling two dice is: $S = \{(1,1),(1,2),(1,3),\ldots,(1,6),(2,1),\ldots,(6,6)\}$.
- Counting all possible outcomes, we have that $S = |36|$.
- The event “roll a 6” contains 5 outcomes: $\{(1,5),(2,4),(3,3),(4,2),(5,1)\}$.
- The event “roll a 7” contains 6 outcomes: $\{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$.
- “Roll a 6” and “roll a 7” are mutually exclusive events. They cannot happen at the same time (they share no common outcomes).
- Hence, we have a total of $5+6=11$ favorable outcomes.

While we have yet to quantify probabilities and risks, it sounds like we have 11 favorable outcomes out of a total of 36. This doesn’t sound as small as we originally thought after all, no?

Activities

Activity 1: Playing with coins

Problem 1

A coin is tossed 4 times in a row and we mark whether it has come up Heads or Tails each time. Explain (in a sentence) why this is a random experiment. [1]

Problem 2

In the experiment from Problem 1, what is the sample space S? [2] What is the cardinality of the sample space $|S|$?

Problem 3

Consider again the experiment from Problem 1. What is the cardinality of the following events:

1. Consider what happens if we repeat this process. Do we always get the same sequence?
2. In general, there are multiple ways to define a sample space. Here, we may want to focus on each individual coin toss as it happens. For example, $\{Heads, Heads, Tails, Heads\}$ could be a potential outcome, whereas $\{Tails, Tails, Tails, Heads\}$ would be another.

- Get the sequence Heads, Tails, Tails, Heads.
- The third coin toss comes up Heads.
- Get at least three Heads.

Food for Thought 1.1.

Hmm.. You know what's interesting? When all outcomes in the sample space are equally likely, then the "likelihood" of an event (*probability* perhaps?) is equal to the cardinality of the event over the cardinality of the sample space! For example, "Get at least three Heads" has cardinality 5, and the sample space S has cardinality $|S| = 16$ for a "likelihood" of

$$5/16 = 31.25\%.$$

Problem 4

Consider the events in Problem 3. Are the first and the second events mutually exclusive? How about the first and the third events? Finally, what can you say about the second and the third events? Justify (in a sentence) your answer.

Activity 2: Random number generators

Problem 5

Let us stay with **discrete sample spaces**: that is, sample spaces where the set of outcomes is discrete. Consider a six-sided die with sides equal to the numbers $S = \{1, 2, 3, 4, 5, 6\}$. Now, define the following:

- A: the number on the die is even ($2, 4, 6$).
- B: the number on the die is smaller than or equal to 3 ($1, 2, 3$).
- C: the number on the die is 2.

What is $A \cap \overline{B}$, $A \cap B$, and $B \cap C$? Enumerate the outcomes that fall in each of them.

Problem 6

Assume that we have a random number generator that produces **real** numbers between 0 and 100. That is, the generator can produce numbers such as 33.33 or 0.08, but no negative numbers or numbers above 100. In set notation/range, we have sample space $S = [0, 100]$. Now, let us define the following events:

- $A = [80, 100]$: the number produced is greater than or equal to 80.
- $B = [20, 50]$: the number produced is greater than or equal to 20, but smaller than or equal to 50.
- $C = [35, 85]$: the number produced is greater than or equal to 35, but smaller than or equal to 85.

Calculate the range of values for each of the following events. For example, if asked to find what $A \cap C$, you would answer $[80, 85]$, as $A \cap C$ includes values that are both in A and in C at the same time. Be careful with the inclusion and exclusion notation $[\ ,\ ]$ and $(\ ,\ )$, respectively!
What are the range of values of:

- $\overline{C} =$
- $C \setminus A =$
- $B \cap C =$
- $(\overline{A} \cap B) \cup (\overline{A} \cap C) =$

Activity 3: Set and cardinality properties

We saw many interesting properties in the first chapter. Now, it is time to study and follow their derivations. To begin with, consider the first of the two De Morgan's law we saw in the first chapter: $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$.

Problem 7

In the two Venn diagrams below, we mark in gray the event $(A \cup B)$ and $\overline{(A \cup B)}$ (corresponding to the left hand side). We have solved this for you, so you can move to Problem 8.

Answer to Problem 7.

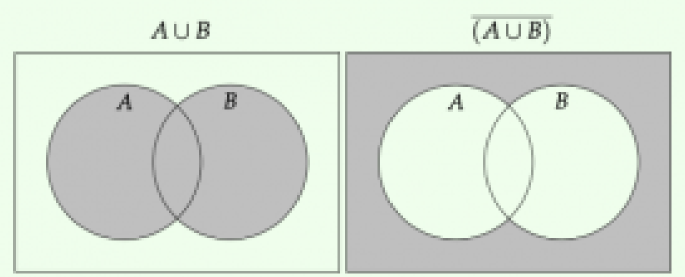

Figure 1.10. The two Venn diagram representations for Problem 7.

Problem 8

In the three Venn diagrams provided, mark the events $\overline{A}$, $\overline{B}$, and $\overline{A} \cap \overline{B}$ (corresponding to the right hand side of the DeMorgan's law).

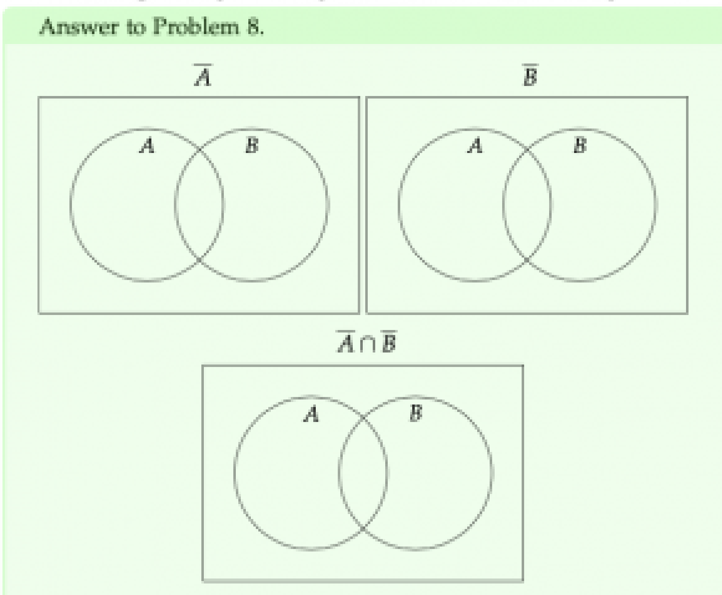

Figure 1.11. Three empty Venn diagrams for you to complete for Problem 8.

Food for Thought 1.2.

Note how the last Venn diagram in Problem 8 (the one for $\overline{A} \cap \overline{B}$) is the same as the last Venn diagram in

Problem 7 (the one corresponding to $\overline{(A \cup B)}$, "proving" De Morgan's law.

Problem 9

Based on the previous constructive derivation you did, what can you say about the extension of this De Morgan's law to more than 2 events? Specifically, what is $\overline{(A \cup B \cup C)}$? Use the diagrams below to do something similar to what you did in Problems 7 and 8.

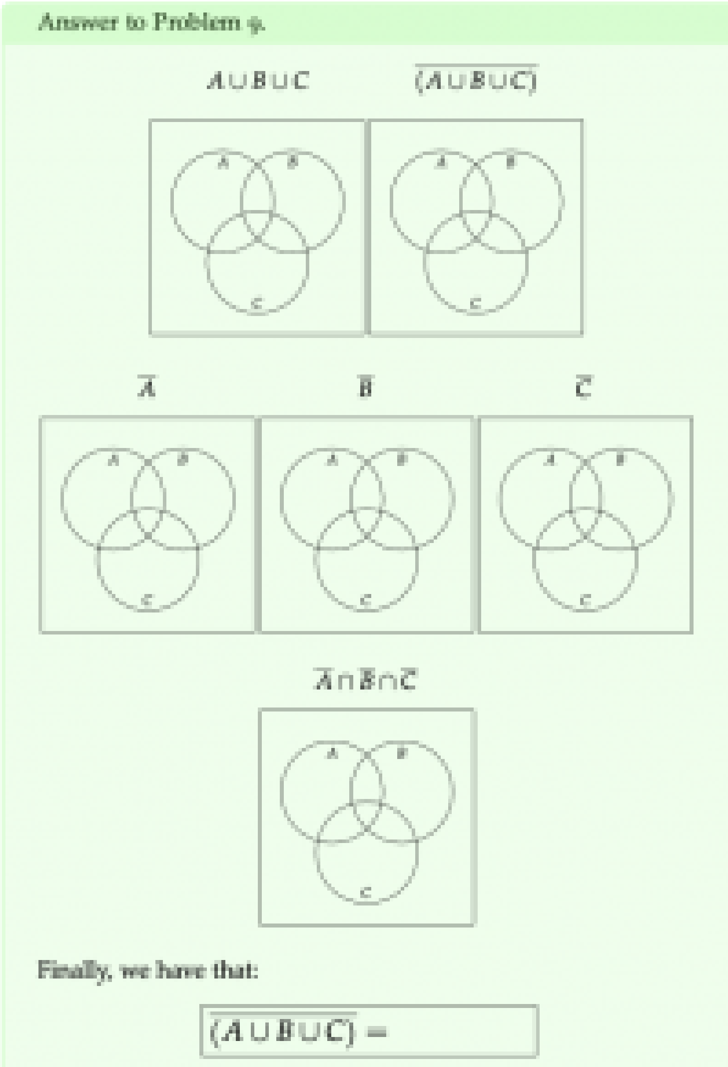

Figure 1.12. The empty Venn diagrams and the missing equality for Problem 9.

Problem 10

Consider two **mutually exclusive** events A, B. What can you say about the cardinality of $A \cap B$ and of $A \cup B$? You may express your answer as a function of the individual cardinalities $|A|, |B|$.

Problem 11

It is true that for two general sets A, B, we have that $|A \cup B| = |A| + |B| - |A \cap B|$. Let us construct a proof for the statement using your answers in Problem 10.
Consider an event X that is comprised of **three mutually exclusive events** X_1, X_2, X_3. What can you say about the cardinality of X and the cardinalities of X_1, X_2, X_3? Choose the (one) correct answer.

- $|X| < |X_1| + |X_2| + |X_3|$
- $|X| \leq |X_1| + |X_2| + |X_3|$
- $|X| = |X_1| + |X_2| + |X_3|$
- $|X| \geq |X_1| + |X_2| + |X_3|$
- $|X| > |X_1| + |X_2| + |X_3|$

Food for Thought 1.3.

This is true, in general, no matter how many mutually exclusive events we can break an event down to. Say that event X consisted of n mutually exclusive events $X_1, X_2, \ldots, X_n$, we could write that its cardinality

$|X|$ is equal to the sum of the individual event cardinalities: $|X| = \sum_{i=1}^{n} |X_i|$.

Problem 12

Based on your observation in Problem 11, can we think of $A \cup B$ as three mutually exclusive events? Check the following Venn diagram and mark the **three mutually exclusive events** that describe the greyed area. How are they described mathematically?

Answer to Problem 12.

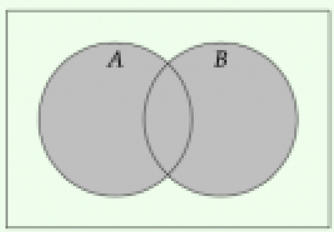

Figure 1.13. The Venn diagram representation of [latex]A\cup B[/latex] to help you in Problem 12.

Problem 13

Combine your observations from Problems 11 and 12 to derive that $|A \cup B| = |A| + |B| - |A \cap B|$.

2. Counting and a first definition of probability

Theory

Chapter objectives

In this chapter, our goal is to **learn to count and use counting to estimate probability**.

Learning outcomes: Counting

After this chapter, you will be able to:

- interpret probabilities and recall fundamental probability properties.
- count how many outcomes satisfy an event.
- recall the multiplication rule to count.
- use the multiplication rule to count.
- differentiate between permutations and combinations.
- use permutations and combinations to count.
- differentiate between different types of permutations.

Chapter questions

1. How do we "count"? What do we mean by "counting" in the context of the class?
2. What is the **multiplication** rule for counting?
3. How do we use the multiplication rule to obtain:
    - **permutations**, i.e., arrangement of n elements in groups of r (where <u>order matters</u>)?
        - Or even permutations of all n elements?
    - **combinations**, i.e., arrangement of n elements in groups of r (where <u>order does not matter</u>)?
    - different types of permutations?
4. And, finally, how does it all relate to probability?

Motivation

A first attempt at defining probability

When we discuss **probability** there are two (often competing) worldviews:

1. the <u>frequentist view</u>**,** which states that the probability of an event happening represents a relative frequency ("long-run frequency") of the times that the event happens versus the number of times the random experiment is conducted;
2. the <u>Bayesian view</u>, which states that the probability of an event is a subjective measure quantifying the perceived likelihood of an event ("degree of belief" in an event happening).

Here, we provide a first definition of probability:

Definitions and theorems 2.1. Probability

With every event E of a sample space S of a random experiment, we associate a **real number** called probability and denoted as $P(E)$ to represent the likelihood of that event happening. Probabilities satisfy three main rules (the *Kolmogorov axioms* of probability):

1. $P(E) \geq 0$, for any event E.
2. $P(S) = 1$, that is if an event comprises the whole sample space then its probability is 1.
3. If $E_1, E_2, \ldots, E_m$ are m **mutually exclusive events**, then $P\left(E_1 \cup E_2 \cup \ldots \cup E_m\right) = P\left(E_1\right) + P\left(E_2\right) + \ldots + P\left(E_m\right)$, or even more concisely:
$$P\left(\bigcup_{i=1}^{m} E_i\right) = \sum_{i=1}^{m} P(E_i).$$

From the definition, we can deduce that all probabilities are in $[0, 1]$, where a probability of 0 implies that an event can never happen, and a probability of 1 implies that an event is certain (will always happen).

Counting and equally likely outcomes

When the outcomes in a discrete random experiment with sample space S are **equally probable**/equally likely, we assume that the

probability of each outcome happening is $\frac{1}{|S|}$. Hence, our question becomes:
“how can we count all favorable outcomes and contrast them to all possible outcomes to derive a measure of probability?”
Why would that be useful?

Counting

The multiplication rule

In the previous chapter, we would always fully enumerate all possible outcomes. For example, rolling two dice results in a total of 36 outcomes:
$S = \{(1,1),(1,2),(1,3),\ldots,(1,6),(2,1),\ldots,(6,6)\}$.
What happens if I need to find the cardinality of the sample space of rolling 10 dice, though? Resorting to full enumeration to count sounds like unnecessary work.

Example 2.1. A new burrito restaurant

In a new burrito place, you are allowed to choose *only one* of two types of tortillas (flour and wheat), *only one* of four types of “meat options” (chicken, pork, steak, no meat), and *only one* of two types of beans (refried and black beans). A food critic needs to try one burrito every day until they have tried all possible burritos. How many days will they be visiting the restaurant to do that?

Our intuition tells us to multiply each number of options in each step; and our intuition would be right! When our outcomes arise from a sequence of k steps (here $k = 3$), each of them with $n_i, i = 1, \ldots k$ options, i.e., n_1 options in step 1 ($n_1 = 2$ for the tortilla), n_2 options in step 2 ($n_2 = 4$ for the meat options), and so on, then:

Definitions and theorems 2.2. Multiplication rule

the number of possible outcomes is
$n_1 \cdot n_2 \cdot \ldots \cdot n_k$.

Two key observations:

1. at each step i, we can have to pick exactly one of the n_i available options.
2. the order does not matter.

Example 2.2. A new burrito restaurant

Hence, in our burrito place example, we have 3 steps (tortilla type, meat type, bean type), leading to a total of $2 \cdot 4 \cdot 2 = 16$ combinations (in the figure below, we show the 8 possible outcomes for a wheat tortilla; a similar figure would show the other 8 possible outcomes for a flour tortilla).

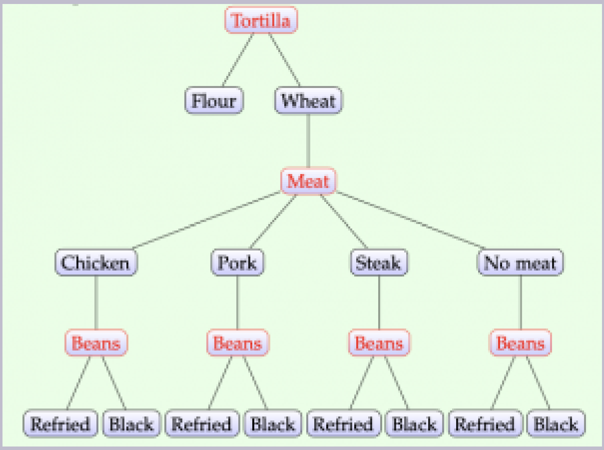

Figure 2.1. A full enumeration of all the options available in Example 2.2. At each red intersection we need to make a decision.

Example 2.3. Greek license plates

In Greece, a vehicle is required to have a license plate with 3 letters (from the Greek alphabet!) and 4 numbers (integer numbers between 0 and 9). How many plates can there be, seeing as the Greek alphabet has 24 letters?

Figure 2.2. An example of a greek license plate, of the form [latex]XXX \#\#\# \#[/latex], where [latex]X[/latex] is one of 24 alphabet letters, and [latex]X[/latex] is one of 10 integer numbers.

A vehicle will have a license plate in the form of $XXX\#\#\#\#$, where X is one of 24 alphabet letters, and X is one of 10 integer numbers (0 through 9). Hence, in total, we have:

$\#\text{license plates} = 24 \cdot 24 \cdot 24 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 138240000.$

Permutations

A **permutation** is an **ordered sequence of elements** selected from some set. For example, consider the sample space

$S = \{1, 2, 3\}$. All permutations are: $(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)$. Specifically, for a sample space S with n possible outcomes (**cardinality** is $|S| = n$), then:

Definitions and theorems 2.3. Permutations of n elements

the number of permutations is $P_n = n!$

$n!$ is the **factorial** of integer number n and is calculated as $n! = n \cdot (n-1) \cdot (n-2) \ldots 1$. By definition, we assume that $0! = 1$.

We can see how this formula is *immediately* derived from the multiplication rule. Assume that we have n elements being placed in order. There are n positions: the first position, the second position, ..., the n-th position. For the first position, we have n options (any of our elements); for the second position, though, we only have $n - 1$ options (we have to exclude the element that was placed in the first position). This logic leads to the following number of options for each of the positions:

Table 2.1. The number of possible options per position for a permutation of all n items.

position	1	2	...	n
# options	n	$n-1$	...	1

Using the multiplication rule immediately leads to $n \cdot (n-1) \cdot \ldots 1 = n!$.

Example 2.4. A random draw

5 people have been named the winners of a competition. There are 5 different books that will be given to them. How many different outcomes (assignments of winners to books) do we have?

There are $P_5 = 5! = 120$ possible outcomes (assignments of winners to books).

We can also opt to select only $r < n$ from the available elements in the set. For example, consider the sample space $S = \{1, 2, 3, 4\}$. The permutations of $r = 2$ elements from that set are:
$(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)$
.
When selecting r outcomes from a total of n outcomes, then:

Definitions and theorems 2.4. Permutations of r < n elements

the number of permutations is $P_{n,r} = \dfrac{n!}{(n-r)!}$

We can again show this using the multiplication rule for counting. We again want to place elements in order; since we only want to place $r < n$ elements, this means that we will not be placing all of them in some position, but instead only up to position r. See the updated table of positions, where we stop placing elements at the r-th position now:

Table 2.2. The number of possible options per position for a permutation of $r < n$ items.

position	1	2	...	r
# options	n	$n-1$	...	$n-r-1$

Using the multiplication rule now leads to

$$n \cdot (n-1) \cdot \ldots (n-r-1) = \frac{n!}{(n-r)!}.$$

Example 2.5. A random draw (continued)

A total of 100 people are participating in a draw. 5 of the participants will be named winners and get one of 5 different books. How many different outcomes (assignments of winners to books) do we have now?

There are $P_{100,5} = \dfrac{100!}{(95)!} = 9034502400$

possible outcomes (assignments of winners to books).

We note here that $\dfrac{100!}{95!}$ can be difficult to compute in a calculator due to the magnitude of the numbers involved. On the other hand, simplifying this to the equivalent $100 \cdot 99 \cdot 98 \cdot 97 \cdot 96$ can prove a lot easier.

Another type of permutation arises when some of the outcomes are the same (for example, two entries in a competition belonging to the same person). In that case, there are fewer **distinguishable permutations**. Formally, we assume that:

1. we have k different types of outcomes;
2. the number of options for each type of outcome is known to be n_i (n_1 outcomes of type 1, n_2 outcomes of type 2, $\cdots$, n_k ourcomes of type k);
3. summing all options gives us the total number of outcomes available n, or equivalently $n_1 + n_2 + \ldots + n_k = n$.

Why is that different?

Example 2.6. Words with friends

As an example, assume we are given the following 5 letters in Scrabble: $2 \times E, 2 \times S, 1 \times T$. The possible possible 5-letter “words” we can create are: EESST, SEEST, ESEST, SESET, ESSET, SSEET, TEESS, ETESS, EETSS, EESTS, SEETS, ESETS, STEES, TSEES, ESTES, SETES, TESES, ETSES, SSTEE, TSSEE, STSEE, TSESE, STESE, ETSSE, TESSE, SETSE, ESTSE, SSETE, ESSTE, SESTE.

You may have noticed how that number is much smaller than $5!$: this is attributed to the fact that some of the options are the same. For example, name one of the letters “E” as if it is the first (E1), while the other is now the second (E2). Do the same with the two “S” letters we have. We can see how the following two permutations are indistinguishable!

E1 E2 S1 S2 T vs. E2 E1 S1 S2 T

This is why we need a new type of counting. If we have k types of elements with n_i objects of each type I ($i = 1, \ldots, k$) such that $\sum_{i=1}^{k} n_i = n$, then:

Definitions and theorems 2.5. Indistinguishable permutations of n elements

the number of permutations is calculated as $P_n^{n_1, n_2, \ldots, n_k} = \dfrac{n!}{n_1! \cdot n_2! \cdot \ldots n_k!}$.

Example 2.7. A full game of Scrabble

How many different 7-letter words (maybe nonsensical) can we create in a game of Scrabble, where we have $2 \times A, 1 \times B, 2 \times S, 1 \times T, 1 \times X$?

There are $k = 5$ different letters with $n_1 = 2, n_2 = 1, n_3 = 2, n_4 = 1, n_5 = 1.$ Hence, the total number of distinguishable, 7-letter words we can create is:

$P_7^{2,1,2,1,1} \dfrac{7!}{2! \cdot 1! \cdot 2! \cdot 1! \cdot 1!} = 1260 \text{ words.}$

Combinations

In all of the discussion so far, **order matters**. Often, though, we do not care about it. For example, consider the following applications of interest:

1. creating a group of 4 people out of a full class for a project.
2. checking the numbers on ten dice.
3. picking the winning numbers in a lottery such as PowerBall.

Then, we define a **combination** as an unordered subset of $r < n$ outcomes selected from a sample space with $n$outcomes. To motivate this a little better, we revisit an earlier example.

Example 2.8. A random draw (with a twist)

A total of 10 people are participating in a draw. 5 of the participants will be named winners and get a book: it will be the same book for any winner! How many different outcomes do we have in this setup?

Let's assume that we number the people participating in the draw using numbers 1 through 10. Some of the permutations of choosing $r = 5$ of them would be: (1, 2, 3, 4, 5), (1, 2, 3, 5, 4), (1, 2, 4, 3, 5), (1, 2, 4, 5, 3), (1, 2, 5, 3, 4), (1, 2, 5, 4, 3), …

We note here that all of these configurations… are the same, if they are all getting the same book! How many of the permutations are the same, when the order doesn't matter?

Let us see if we can derive how many of the permutations are actually the same when order no longer matters. We may have some insight already: if we pick r elements, then the order they are placed in no longer matters. That is, with $r = 5$ elements like in our example, all permutations including the same 5 elements are the same. These are $P_5 = 5! = 120$ in total. Or, in general:

Definitions and theorems 2.6. Combinations of $r < n$ elements

the number of combinations is calculated as

$$C_{n,r} = \frac{P_{n,r}}{r!} = \frac{n!}{(n-r)! \cdot r!}.$$

The number of all possible combinations is also commonly written as $C_{n,r} = \binom{n}{r}$. The notation $\binom{n}{r}$ is also read as "n choose r."

A common difficult when first learning about permutations and combinations is to recognize which one is the proper selection for a given application. Let us practice with this!

Exercise 2.1. Permutation or combination?

Are the following better described as a permutation or as a combination? Before answering, ask yourselves whether "order matters" or not!

- Choosing 5 students out of 80 candidates to participate in a group?
- Locating 10 different facilities in 10 cities in the USA?
- Creating a board of 6 students out of 250 students, such as one serves as a president, another as a vice president, one as a treasurer, one as a secretary, one as a media coordinator, and one as a director of outreach?

Example 2.9. Distinguishing between permutations and combinations

You have to select between 10 students for 3 positions. Y

- Assume that you are allowed pick the same student for all three positions, if you'd like. How many possible outcomes are there if the 3 positions are different?
  We need to use the **multiplication rule** to calculate $10 \cdot 10 \cdot 10 = 1000$ possible outcomes.
- What if you are not allowed to select the same student more than once? Assume again the 3 positions are different.

We need to use a **permutation** (10 students for 3 different positions): $P_{10,3} = 720$ potential outcomes.

- What if all positions are actually for the same type of work?
  We now have a **combination** (10 students for 3 positions): $C_{10,3} = 120$ outcomes.

From counting to probability

As mentioned in the Motivation section, **probability** is a real number between 0 and 1 that quantifies how likely an outcome (event) is. Adopting the frequentist view of probabilities (stating the probability of an outcome is the relative frequency with which that outcome appears over all possible outcomes), we could possibly count the number of outcomes that are favorable and divide by the total number of possible outcomes and thus estimate probability.

Exercise 2.2. Quality control

A package is set to leave a factory and be sent to a retailer. The package contains 100 items. We already know that exactly 3 of the 100 items are defective. The quality control team over at the retailer works as follows: they select a sample of 6 items from the 100, and check them. If there are 0 defective items in the selected sample of 6, they accept the package and sell its contents; otherwise, they send the package back. What is the probability that the quality control rejects the package and sends it back?

To answer this question, we decompose the problem into its components. We need to know:

1. how many ways are there to select 6 items from the 100?
2. how many ways are there to have 1, 2, or 3 defective items in the 6 selected?

Let us begin by addressing the first question.

Example 2.10. Quality control: How many ways are there to select 6 items from the 100 in the package?

Step 1: How many ways are there to select 6 items from the 100 in the package? This is a **combination** and we get:

$$C_{100,6} = \binom{100}{6} = \frac{100!}{6! \cdot 94!} = 1192052400 \text{ ways.}$$

For the second question, we need to think slightly differently. Let x be the number of defective items and $6 - x$ the number of non-defective items in the sample of 6. Then:

Example 2.11. Quality control: How many ways are there to have 1, 2, or 3 items in the sample of 6 selected?

Step 2: How many ways are there to have 1, 2, or 3 defective items from the 3 available in the selected sample of 6? This is another **combination**, albeit requiring more calculations.

- **Step 2a**: How to select $x = 1$ defective items in the sample? We would need to pick 1 out of the 3 defective and 5 out of the 97 non-defective! Mathematically, we have $C_{3,1}$ $C_{3,1} = 3$ and $C_{97,5} = \binom{97}{5} = 64446024.$We should now use the multiplication rule between the two, as we have to pick one option from the 3 possible options from the first selection and one option from the 64446024 possible ones in the second selection, for a total of $3 \cdot 64446024 = 193338072 \text{ ways.}$
- **Step 2b**: How to select $x = 2$ defective items in the sample? Again, using the same logic as earlier, we have $C_{3,2} = \binom{3}{2} = 3$ and $C_{97,4} = \binom{97}{4} = 3464840,$ with the total being now $3 \cdot 3464840 = 10394520 \text{ ways.}$
- **Step 2c**: How to select $x = 3$ defective items in the sample? Finally, $C_{3,3} = \binom{3}{3} = 1$ and

$C_{97,3} = \binom{97}{3} = 147440,$ and a total of $1 \cdot 147440 = 147440 \text{ ways.}$

So, we have the three favorable outcomes ($x = 1, 2, 3$) and the total number of outcomes. We combine all in this last part:

Example 2.12. Quality control: Calculating the probability of failing the inspection

Step 3: Let E be the event that we fail inspection.

First, we make the argument that the three events (selecting $x = 1$, $x = 2$, or $x = 3$ defective in the sample) are mutually exclusive, and hence the cardinality of the union of the three events is equal to the summation of the individual cardinalities. So, the total number of favorable outcomes is $193338072 + 10394520 + 147440 = 203880032$.

We divide this number by the total number of outcomes to get a probability of

$$P(E) = \frac{203880032}{1192052400} = 0.171.$$

Exercise 2.3. Picking 3 cards from a deck

You pick 3 cards at random from a deck with 52 cards. What is the probability that all 3 are face cards? What is the probability that 2 are face cards?

Summary

Counting can be quite hard the first times we work with it. It is very important that we discuss and establish our assumptions first. Does the “order matter?” Do I have same elements appearing? In lieu of a summary, let us practice with the “type” of counting we can use (multiplication rule, permutation, combination, indistinguishable permutation) depending on the circumstances/context.

Example 2.13. Ranking

Assume we are **ranking** $n = 6$ engineering majors in terms of percentage of graduates who go on to pursue graduate degrees. **How many different rankings can we produce?**

This is a **permutation of** n **items,** and hence we can produce $6! = 720$ different rankings. Each position in the ranking can be filled by one major (that cannot be repeated again down the line). Say that the majors are A, B, C, D, E, F: then one possible ranking could be ABCDEF whereas another one would be BADCEF, and so on. The first position can be filled by 6 majors (any of the six majors can be ranked first!); the second place can be filled by 5 majors (any of the five *remaining* majors, excluding the one that is ranked first); and so on.

Example 2.14. Choosing when order doesn't matter

Assume we are **selecting** $r = 3$ out of $n = 6$ engineering majors to provide three awards to for their service. **How many different selections can we produce?**

This is a little unclear. Are the $r = 3$ majors receiving the same award? Or are they receiving different awards? To see why this makes a difference, assume that the winning majors are A, B, and C. If they are all receiving the same recognition/award, then this is one possible selection. If they are receiving different awards, then this is six selections: ABC, ACB, BAC, BCA, CAB, CBA. So, let's try this again:

Example 2.14. Choosing when order doesn't matter (corrected)

Assume we are **selecting** $r = 3$ out of $n = 6$ engineering majors to provide threeawards to for their service. **The awards are the same for all selected majors. How many different selections can we produce?**

This is now a **combination** of r items out of a total of n, and then the answer would be:

$$C_{n,r} = \frac{n!}{r! \cdot (n-r)!} = \frac{6!}{3! \cdot 3!} = 20.$$

The selection has to be one of the following: ABC, ABD, ABE, ABF, ACD, ACE, ACF, ADE, ADF, AEF, BCD, BCE, BCF, BDE, BDF, BEF, CDE, CDF, CEF, DEF.

Example 2.15. Choosing when order does matter

Assume we are **selecting** $r = 3$ out of $n = 6$ engineering majors to provide three **different awards to for their service. How many different selections can we produce?**

This is now a **permutation** of r items out of a total of n, and then the answer would be:

$P_{n,r} = \frac{n!}{(n-r)!} = \frac{6!}{3!} = 120.$ To see how this is derived, consider our answer earlier. There were 20 possible selections of three majors to award. Now, since we are "ranking" each major in the selection, each of the possible selections (say, e.g., ABC) is actually corresponding to $r! = 6$ possible rankings! Those would be ABC, ACB, BAC, BCA, CAB, CBA. In total, then we have $P_{n,r} = P_r \cdot C_{n,r} = 3! \cdot 20 = 120.$

Example 2.16. Choosing when <u>some</u> order does matter

Assume we are **selecting** $r = 3$ out of $n = 6$ engineering majors to provide two **different** awards to for their service: one first place award, and two honorable mentions. **How many different selections can we produce?**

This is now a permutation with **indistinguishable outcomes**. To see why this is the case, assume the selection of ABC as the awarded selection. If A is ranked first, then the order of the other two does not matter: that is, we have three possible rankings now. A first with BC honorable mentions, B first with AC honorable mentions, and finally C awarded first place and AB being the honorable mentions. In total, we would get $3 \cdot C_{n,r} = 3 \cdot 20 = 60$ possible selections!

The formula for indistinguishable permutations would be $\frac{n!}{n_1!n_2!n_3!}$, where $n_1 = 1$ is the number of first place awards, $n_2 = 2$ is the number of honorable mentions, and $n_3 = 3$ is the number of non-awarded majors. Our final answer then is $\frac{6!}{1! \cdot 2! \cdot 3!} = \frac{720}{1 \cdot 2 \cdot 6} = 60.$

Activities

Activity 1: Texas hold'em

Texas hold'em is a variant of poker (a card game) where each player is given 2 starting cards out of the 52 cards in a deck. There are 13 cards of each suit; and there are four suits: ♥, ♦, ♠, ♣. The 13 cards are from 1 to 10, along with three special cards named J, Q, K.

Problem 1

How many combinations of starting cards are there? For the sake of this problem consider that order does not matter (and hence a starting card setup of ♥3, ♠7 and ♠7, ♥3 are one and the same). Also assume that the same numbers of different suits are different setups (and hence ♦5, ♠K and ♣5, ♦K are two different starting setups). As a hint, consider that you have 52 cards to pick from, and you need to see how many ways you have of picking 2 of them — without any care for the order.

Problem 2

We define a pair as two same cards (of different suit). For example, ♥K, ♣K and ♥4, ♦4 are pairs. Considering again that order does not matter, and that the same numbers of different suits are different setups (and, for example, ♥4, ♦4 and ♠4, ♥4 are two different setups), how many starting setups that are pairs are there? If it helps, consider how many pairs of a single card (say, "7"s) you can create. Then, whatever that number is, you can multiply it by all 13 possible cards.

Problem 3

Considering your answers in Problems 1 and 2, what is the probability that we are dealt a pair in our 2 starting cards?

Activity 2: Elections and turnout

Assume that a small town has two candidates running for mayor: Candidate A and Candidate B. A small town has 45 residents that are allowed to vote for their next mayor. Now, typically in an election, you have to count all votes, and the candidate with the most votes wins. That said, the town does not want to invest the resources to count all 45 votes and are contemplating a new system. In this new system, \textbf{only three voters} are selected at random and asked to vote. Then, the candidate that gets at least two votes is proclaimed the winner. You may assume that all voters will pick exactly one of the two candidates. That is, they are not allowed to vote for another person.

Problem 4

Assume we have independently polled all voters and we are aware that Candidate A is set to win with 30 votes compared to 15 votes for Candidate B. If the town implements this new system, what is the probability that Candidate B is actually the one that wins the election? For a hint, look at the Quality Control example in our notes.

Problem 5

How does the calculation change if we were to pick 5 voters among all registered ones (instead of 3)?

Problem 6

What do you observe when comparing your answers to Problem 4 and Problem 5? Which of the two candidates would prefer a bigger number of voters go to the polls?

Activity 3: Deriving the permutations and combinations formulae

Problem 7: From multiplications to permutations

In the notes, we saw that the formula for a permutation of n items is $P_n = n!$. Use the multiplication rule to show that this is indeed the case.

Problem 8: From multiplications to permutations of $r < n$ items

Similarly, in the notes earlier, we derived the formula for a permutation of $r < n$ items from a total of n items. Again, use the multiplication rule to show that $P_{n,r} = \dfrac{n!}{(n-r)!}$.

Problem 9: From permutations to combinations

Recall that a permutation $P_{n,r}$ is an ordered sequence of r items out of n possible items, whereas a combination $C_{n,r}$ is an unordered sequence of the same r items. Use the multiplication rule to show that $P_{n,r} = C_{n,r} \cdot r!$. Then, use that fact to derive the combinations formula.

Activity 4: Summary of results

Problem 10: Committing to memory

We have seen several different formulae for counting in this chapter. In your answer, provide one example and write the formula associated with each setup.

Committing to memory

Write the formula and an example in the space provided.

- **Multiplication rule:**

Formula	Example

- **Permutations of n items:**

Formula	Example

- **Permutations of $r < n$ items:**

Formula	Example

- **Combinations of $r < n$ items:**

Formula	Example

- **Distinguishable permutations of groups of n items:**

Formula	Example

3. Basic probability theory

Theory

Chapter objectives

In this chapter, our goal is to **introduce the basic theory behind probability.**

Learning outcomes: Random experiments, sample spaces, and other notation

After this chapter, you will be able to:

- recall and explain the basic properties that probability has.
- calculate the probability of an event.
- apply set operations in probability calculations.
- define and provide examples of conditional probabilities.
- apply the conditional probability formula.
- recognize independence.

Chapter questions

1. What are the basic rules/axioms of probability?
2. What is:

    - the probability of an event *not happening* (complement)?
    - the probability of *two or more events all happening*? (intersection)?
    - the probability of *at least one of two or more events happening* (union)?

3. What is a conditional probability? Can we give examples of conditional probabilities?
4. How are conditional probabilities calculated?

    Conditional probabilities will be very important in the next class!
5. What does it mean that two events are **independent**?

Motivating examples

Will I miss my flight?

Flying from a small airport almost always requires a layover in another airport. For example, flying from Urbana-Champaign to New York City usually is done through Chicago with two legs: Urbana-Champaign to Chicago, and Chicago to New York City. My layover is only 45 minutes in Chicago, so I am naturally worried about making my connection. I would feel much better if I knew whether my first flight leaves on time or not. **What is the probability that I make my second flight given that my first flight is delayed by 15 minutes or more?**

Data collection

A company has undertaken the large effort of contact tracing and testing for a communicative disease in a metropolitan area. It is expected that from the people that live in the area, 1% has been in close contact with an already known case of the disease, 15% has been working in close contact with multiple people as an essential worker, and 6% has traveled to a location (in and outside the USA) with high risk of contagion for that particular disease. A random person is selected for a test, but will only be administered the test if they fall within one of the three categories above. **What is the probability that the person will be administered the test?**

Probabilities

We repeat here the definitions of probability from the previous chapter. As a reminder, there are two interpretations of **probability**:

1. a more objective definition relating probability to the *relative frequency* of favorable outcomes versus all outcomes.
2. a more subjective definition relating probability to "degree of belief" or "risk."

In both cases, we can use **basic probability theory**, the collection of rules and axioms about probability, to calculate the likelihood of an event happening or not. Once again, we repeat here the definition of probability:

Definitions and Theorems 3.1. Probability

With every event E of sample space S of a random experiment, we associate a **real number** called probability and denoted as $P(E)$ to represent the likelihood of that event happening. Probabilities satisfy three main rules (the *Kolmogorov axioms* of probability):

1. $P(E) \geq 0$, for any event E.
2. $P(S) = 1$, that is if an event comprises the whole sample space then its probability is 1.
3. If $E_1, E_2, \ldots, E_m$ are m **mutually exclusive events**, then $P(E_1 \cup E_2 \cup \ldots \cup E_m) = P(E_1) + P(E_2) + \ldots + P(E_m)$, or even more concisely:
$$P\left(\bigcup_{i=1}^{m} E_i\right) = \sum_{i=1}^{m} P(E_i).$$

From the definition, we can deduce that all probabilities are in $[0, 1]$. An impossible event is one that can never happen (has a probability of 0); a certain even is one that always happens (has a probability of 1). Every sample space has a probability of 1, as we will see later. As an example, consider a sample space S that consists of three events: A, B, C. Then the event that neither A nor B nor C happen is *impossible*; the event that either A or B or C happens is *certain*.
We sometimes present probability as a percentage (%): for example, a probability of 0.4 can be also written as a probability of 40%.

Example 3.1. Probabilities of mutually exclusive events

Assume that two events E_1, E_2 are mutually exclusive: for example, let E_1 be the event that you get an A in a class, while E_2 is the probability that you get an A-. Your personal believe is that you have a 30% chance at an A and a 20% chance at an A-. Then, the probability that you get *at least* an A- in the class is
$P(\text{“at least an } A- \text{ in IE 300”}) = P(E_1) + P(E_2) = 50\%.$

An immediate consequence of the definition of probability and the *Kolmogorov axioms* of probability is the following:

- $P(\overline{E}) = 1 - P(E)$. In plain terms, the probability of an event “not happening” is 1 minus the probability of it happening.

Example 3.2. Probabilities of complements of events

A flight is only canceled in the case of extreme weather. During that time of year, extreme weather affects an airport 1% of the time. What is the probability that a flight tomorrow proceeds as usual?

Let E be the probability of an extreme weather event happening. Then, the flight will proceed as usual with

probability
$P(\overline{E}) = 1 - P(E) = 1 - 0.01 = 0.99.$

- $P(\emptyset) = 0, P(S) = 1$. As we discussed earlier, an event with no outcomes is empty and can never happen, whereas an event that encompasses the whole sample space is certain to happen.
- If **[latex]E_1 \subseteq E_2[/latex]**, then $P(E_1) \leq P(E_2)$. Recall that we say that one event E_1 is contained in another event E_2 and write that $E_1 \subseteq E_2$ if all outcomes that satisfy E_1 are included in E_2. It should come as no surprise that when $E_1 \subseteq E_2$, we have that $P(E_1) \leq P(E_2)$. For example, recall the weather events: "sunny" implies "no rain", which in turn implies that "sunny" has a smaller probability than "no rain".

Example 3.3. When $E_1 \subseteq E_2$

A pizza restaurant advertises delivery in 30 minutes or less. Assume that the probability of an order being delivered in 30 minutes or less is 0.9. Then, the following two can immediately be derived from this statement:

- the probability of an order being delivered in 15 minutes or less is **at most** 0.9.
- the probability of an order being delivered in 1 hour or less is **at least** 0.9.

Exercise 3.1 Emails and spam

An email is categorized as one of the following 2 (mutually exclusive) categories: spam or not-spam. Emails that are not-spam are also categorized as one of 3 (mutually exclusive again) categories: urgent, normal priority, and advertisements. Answer the following questions.

- **True or False?** If the probability of a message being spam is 0.45, then the probability of a message being not-spam is 0.55.
- **Brief answer**. What is the probability that an urgent email is spam?
- **Multiple choice**. An email is urgent with probability 0.1 and an email is of normal priority with probability 0.2. Pick one:
    1. $P(not\text{-}spam) < 0.3$.
    2. $P(not\text{-}spam) = 0.3$.
    3. $P(not\text{-}spam) \geq 0.3$.

Unions and intersections of events

For any two events E_1, E_2, define their union $E = E_1 \cup E_2$. We then have that $E_1 \subseteq E$, as event E includes all outcomes in E_1, and $E_2 \subseteq E$, because similarly event E includes all outcomes in E_2. This leads to the following: $P(E_1) \leq P(E)$, $P(E_2) \leq P(E)$, $P(E) \leq P(E_1) + P(E_2)$. It is

interesting to take a minute and recall when the equality holds (we can check back to the axioms of probability for that).
We can use a similar deduction of the intersection of two events. Specifically, let $E = E_1 \cap E_2$. Then, we must have that $E \subseteq E_1$ and $E \subseteq E_2$, as all events in the intersection belong to both E_1 and E_2. This leads to: $P(E) \leq P(E_1)$, $P(E) \leq P(E_2)$. **With these in mind, how can we calculate $P(E_1 \cup E_2)$ exactly?** We have already seen how this calculation works for two mutually exclusive events: the third Kolmogorov axiom of probability gives us that $P(E_1 \cup E_2) = P(E_1) + P(E_2)$, when E_1 and E_2 are mutually exclusive.
To derive the calculation, we turn to basics.

Example 3.4. Deriving that $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

We have already seen how $E_1 \cup E_2$ can be written as the union of three mutually exclusive events as in the image:

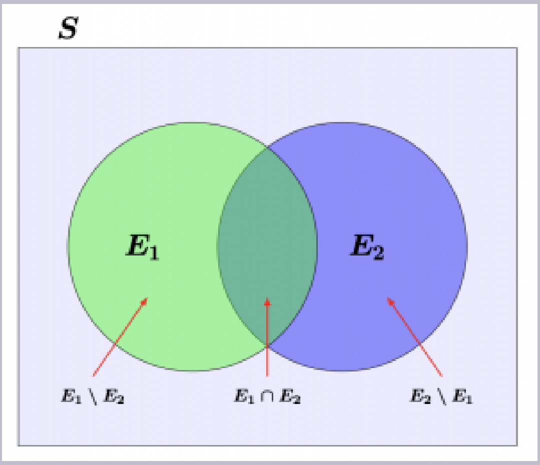

Figure 3.1. The Venn diagram representation of [latex]E_1 \cup E_2[/latex].

Those are:

1. $E_1 \setminus E_2$ (the left side of the image, including all outcomes in E_1 except for outcomes in E_2);
2. $E_1 \cap E_2$ (the middle side of the image, including all outcomes that are both in E_1 and E_2); and
3. $E_2 \setminus E_1$ (the right side of the image, including all outcomes in E_2 excluding outcomes in E_1).

From the third Kolmogorov axiom, the one discussing about mutually exclusive events, we have that:

$$P(E_1 \cup E_2) = P(E_1 \setminus E_2) + P(E_1 \cap E_2) + P(E_2 \setminus E_1).$$

Now, we note that E_1 (and E_2, respectively) can also be written as the union of two mutually exclusive events: $E_1 = (E_1 \setminus E_2) \cup (E_1 \cap E_2)$ (and $E_2 = (E_2 \setminus E_1) \cup (E_1 \cap E_2)$, respectively). This gives that:

$$P(E_1) = P(E_1 \setminus E_2) + P(E_1 \cap E_2) \implies P(E_1 \setminus E_2) = P(E_1) - P(E_1 \cap E_2).$$

Combining everything, we get:

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2).$$

Let us check this with an example involving **equally probable events**.

Example 3.5. Deriving that $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$ *for equally probable events*

Let us assume that we have a sample space S consisting of some number of **equally probable** events. Then, we can provide another, more intuitive, reasoning behind the $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$ formula.

- First, recall that for any two sets $E_1, E_2 \subseteq S$, we have shown that $E_1 \cup E_2| = |E_1| + |E_2| - |E_1 \cap E_2|.$ (see Chapter 1 and the **cardinality** discussion).
- Second, recall that probability can be viewed as the fraction of favorable outcomes to all possible outcomes (equal to $|S|$ here). Dividing both sides of by $|S|$, we get:

$$\frac{|E_1 \cup E_2|}{|S|} = \frac{|E_1|}{|S|} + \frac{|E_2|}{|S|} - \frac{|E_1 \cap E_2|}{|S|} \implies$$
$$\implies P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2).$$

Example 3.6. Practice with $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

Two flights are delayed individually with probabilities 0.25 and 0.15, respectively. However, they are also affecting each other, so they may both be delayed at the same time with probability 0.1. **What is the probability that at least one of the flights is delayed on a given day?**

Let E_1 be the event that flight 1 is delayed and E_2 the event that flight 2 is delayed. Hence, "at least one flight being delayed" can be expressed as $E_1 \cup E_2$. Using the formula, we get

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) = 0.25 + 0.15 - 0.1 = 0.3.$$

Let us turn to an example from industrial engineering and operations research next.

Example 3.7. Machine downtime and the union of events

Two machines on a production line (let them be Machine A and Machine B) fail during a shift. Historical data has shows the following probability values:

- $P(\text{Machine A fails}) = 0.12$.
- $P(\text{Machine B fails}) = 0.08$.
- $P(\text{both fail simultaneously}) = 0.02$.

Answer the following questions:

1. What is the probability that **at least one** machine fails during a shift?
2. What is the probability that the shift runs with **no machine failures**?

Solution

For the first question, define events A="Machine A fails", B = "Machine B fails". We then have: $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.12 + 0.08 - 0.02 = 0.18.$

For the second one, we can rewrite "no failures" as the complement of "at least one failure". Hence: $P(\text{no failures}) = 1 - P(A \cup B) = 1 - 0.18 = 0.82.$

Exercise 3.2. From data to probabilities

Recall the grades from 3 different professors for the same class, as shown during a previous chapter:

Table 3.1. The number of letter grades assigned to students per different professor in a class.

Letter Grade	Professo r 1	Professo r 2	Professo r 3	Total
A	108	20	30	158
B	44	49	46	139
C	11	15	15	41
D	0	1	8	9
Total	163	85	99	347

Assuming you call on one student out of the total 347 students, answer the following questions:

1. What is $P(E_1)$? Define event E_1 that you pick a student from Professor 1's class?
   $P(E_1) = 163/347 = 0.4697$.
2. What is $P(E_2)$? Define event E_2 that you pick a student who received an A in the class?
   $P(E_2) = 158/347 = 0.4553$.
3. What is $P(E_1 \cap E_2)$? That is, event $E_1 \cap E_2$ is one where you pick a student who was both in Professor 1's class and received an A in the class?
   $P(E_1 \cap E_2) = 108/347 = 0.3112$.
4. How about the probability that you pick *either* a student from Professor 1's class *or* a student who received an A in the class?

Exercise 3.3. Back to the motivating example

Before moving to the next section, we think back to the motivating example of collecting data for a communicative disease through selective testing. Recall that we had the following probabilities:

- a 1% probability for a person to have been in close contact with a known case;
- a 15\% probability for a person to work as an essential worker; and
- a 6\% probability that a person has traveled to a location with high contagion risk.

However, when estimating the probability that a person qualifies for the test, we have found that only 17% of the population would indeed qualify, rather than 22% that the summation of the three events would give. Based on your knowledge so far, **does that make sense? How could that happen?**

This last question should get us thinking about union of more than 2 events. For the next part see also the Activities section later in the chapter.

Deriving the probability of the union of more than 2 events

Assume that in the following picture, the sample space S is the whole rectangle and A, B, C are some events.

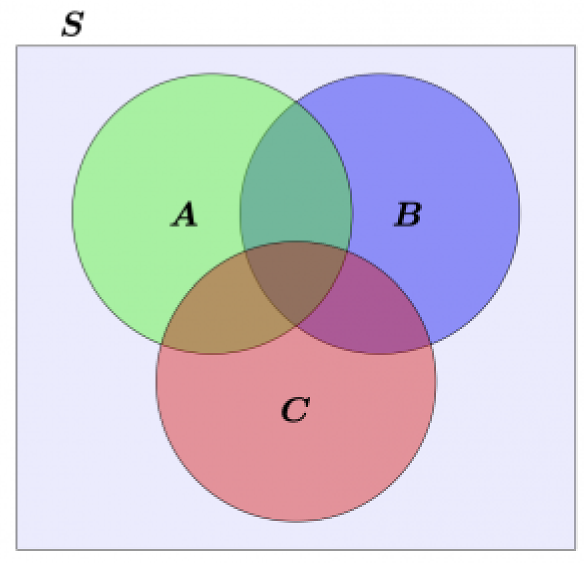

Figure 3.2. The Venn diagram representation of three events [latex]A, B, C[/latex] *in some sample space* [latex]S[/latex]*.*

Like we did earlier, we will separate these three events into **seven mutually exclusive parts** of the Venn diagram. Specifically, we will separate the three events into:

- $E_1 = (A \setminus B) \cap (A \setminus C)$.
- $E_2 = (B \setminus A) \cap (B \setminus C)$.
- $E_3 = (C \setminus A) \cap (C \setminus B)$.
- $E_4 = (A \cap B) \setminus C$.
- $E_5 = (A \cap C) \setminus B$.
- $E_6 = (B \cap C) \setminus A$.
- $E_7 = A \cap B \cap C$.

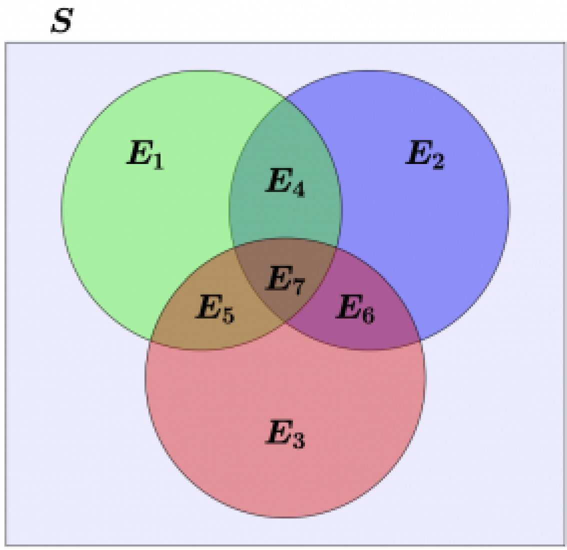

Figure 3.3. The Venn diagram representation of multiple events [latex]E_1, E_2, E_3, E_4, E_5, E_6, E_7[/latex] in some sample space [latex]S[/latex].

Using this numbering, we can deduce that:

$P(A \cup B \cup C) = P(E_1) + P(E_2) + P(E_3) + P(E_4) + P(E_5) + P(E_6) + P(E_7)$.

Let us focus on event A:

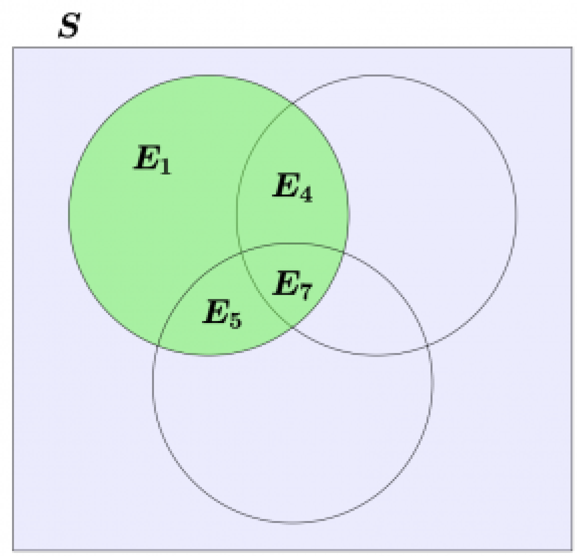

Figure 3.4. The Venn diagram representation of multiple events [latex]E_1, E_4, E_5, E_7[/latex] in some sample space [latex]S[/latex]. These specific events compose event [latex]A[/latex].

First, we observe that $P(E_7) = P(A \cap B \cap C)$. Second, we note that $P(E_1) = P(A) - P(E_4) - P(E_5) - P(E_7)$.. Last, we also note that $P(E_4) = P(A \cap B) - P(E_7), P(E_5) = P(A \cap C) - P(E_7)$. Similarly, we obtain for event B:

- $P(E_7) = P(A \cap B \cap C)$.
- $P(E_2) = P(B) - P(E_4) - P(E_6) - P(E_7)$.
- $P(E_4) = P(A \cap B) - P(E_7)$.
- $P(E_6) = P(B \cap C) - P(E_7)$.

Finally, focusing on event C:

- $P(E_7) = P(A \cap B \cap C)$.
- $P(E_3) = P(C) - P(E_5) - P(E_6) - P(E_7)$.
- $P(E_5) = P(A \cap C) - P(E_7)$.

- $P(E_6) = P(B \cap C) - P(E_7)$.

Replacing all of those in
$P(A \cup B \cup C) = P(E_1) + P(E_2) + P(E_3) + P(E_4) + P(E_5) + P(E_6) + P(E_7)$
, leads to:
$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$.
In general, for m events $E_1, E_2, \ldots, E_m$, whose union probability we are looking for, we do the following:

1. Add the probabilities of the individual events.
2. Subtract the probabilities of the intersections of any two events.
3. Add the probabilities of the intersections of any three events.
4. Continue subtracting the probabilities of the intersections of any $4, 6, \ldots$ events and adding the probabilities of the intersections of any $5, 7, \ldots$, events.

Conditional probabilities

Motivation

It is common and expected to want to recalculate our probabilities ("rethink our chances/risks") as more information becomes available, or under certain updating conditions. For example, the probability that I miss the second leg of my flights is immediately affected by any delays I might experience in the first leg of my flight. In such cases, we turn to **conditional probability**, or the calculation of a probability given a condition.

Definitions and theorems 3.2 Conditional probability

Conditional probability is defined as the probability that an event E_1 happens given that event E_2 has already happened: this is written as $P(E_1|E_2)$.

We note here that this is commonly read as "the probability of E_1 given E_2" or "the probability of E_1 such that E_2 has happened."

Example 3.8. Conditional probabilities

Conditional probabilities are fundamental in estimating risk and in quantifying uncertainty, especially in a dynamic world as more information becomes available. Examples include:

- What is the probability that I miss my second flight, given that my first flight was delayed?
- What is the probability that an order is successfully delivered on time, given that it is processed by our warehouse in Tennessee?
- After calibrating a machine, the quality of the first three processed parts seems to have dropped. What is the probability that calibration was done incorrectly?

Before proceeding to show the formula for calculating a conditional probability, let us think about what we might need. Assume two events E_1, E_2 with already calculated probabilities:

- $P(E_1) = 0.5.$
- $P(E_2) = 0.25.$
- $P(E_1 \cap E_2) = 0.2.$

Note the last probability (of the intersection): since it is positive, then this means that these two events are not mutually exclusive. As an aside, with these probabilities, we can also calculate the probability of the union ($P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) = 0.55$), even though we do not need this result.

With these probabilities at hand, we ask you to think about the conditional probability of E_1 given E_2. In other words, if the original probability of E_1 happening is 50%, does your perception/probability of E_1 change if you were told that E_2 has already happened?

Example 3.9. Changing our perception

The probability that E_1 happens is 0.5 (50%). Does this perception change if we are told that E_2 has already happened?

Let us try to come up with a visual parallel to the provided probabilities. In the figure here, we have 20 dots, out of which 10 (50%) are blue and 5 (25%) are red. 20% of them (4 in number) are both red and blue for a "purplish" color. Note that the four purple dots are both red and blue and hence they represent the intersection of both events.

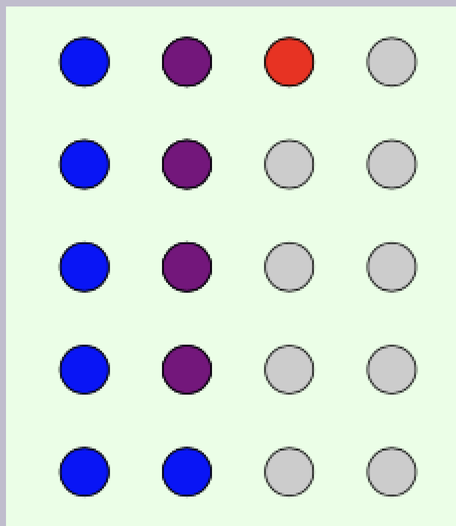

Figure 3.5. 20 dots colored blue, red, and purple (both red and blue). Specifically, 6 of them are blue, 1 of them is red, while another 4 of them are purple.

Now, we assume that we know E_2 has happened. This leaves us with only 5 cases:

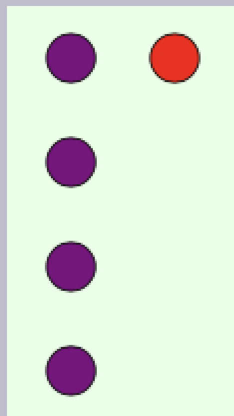

Figure 3.6. A subset of the original dots; here we only have five dots leftover, out of which one is red and four are purple.

It is easy to tell that knowing that E_2 has happened leaves us with only 5 cases, out of which 4 correspond to E_1. Does that mean that our perception should *change*? Is $P(E_1|E_2)$ different than we would have expected without prior information about E_2?

Definition

We are now ready to provide the mathematical definition of a **conditional probability** and supplement the previous definition.

Definitions and theorems 3.3. Conditional probability and its calculation

The **conditional probability** of one event E_1 conditional to event E_2, defined only for events that can happen (that is, $P(E_2) > 0[latex]), is calculated by [latex]P(E_1|E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)}$.

Example 3.10. Changing our perception (continued)

In the previous example, we were given that:

- $P(E_1) = 0.5.$
- $P(E_2) = 0.25.$
- $P(E_1 \cap E_2) = 0.2.$

Hence, we can calculate:

- $P(E_1|E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{0.2}{0.25} = 0.8.$
  Note that this coincides with our visual that had 4 purple dots out of a total of 5 red dots, for a probability of 80%!
- $P(E_2|E_1) = \frac{P(E_2 \cap E_1)}{P(E_1)} = \frac{0.2}{0.5} = 0.4.$

We mentioned mutually exclusive events earlier: as a reminder, two mutually exclusive events must have no common outcomes and hence $P(E_1 \cap E_2) = 0$. This immediately means that knowing that one mutually exclusive event has happened *excludes* the other from happening, or in mathematical terms: $P(E_2|E_1) = 0 \text{ and } P(E_1|E_2) = 0$.The multiplication rule for probabilities
A very straightforward rewriting of the conditional probability formula gives us a very important result. Solving for the numerator of the right hand side in the conditional probability formula gives us that for any two events A, B, we must have that $P(A \cap B) = P(A|B) \cdot P(B)$. This will come quite in handy in the next chapter.

Example 3.11. Using the multiplication rule for probabilities

Two power stations work together to provide electricity to a county in the United States. When both of them fail, the county loses power. During a winter storm, the first power station (S_1) loses power with probability $P(\overline{S}_1) = 0.1$; the second loses power with probability $P(\overline{S}_2 = 0.05$. That said, they affect one another (seeing as one losing power implies a surge in demand for the other) and we have calculated that $P(\overline{S}_1|\overline{S}_2) = 0.3$. What is the probability that both power stations lose power during an upcoming winter storm?

This is the definition of the multiplication rule for probabilities! We want $P(\overline{S}_1 \cap \overline{S}_2)$, that is the probability that both power stations lose power at the same time. Based on the rule, we have: $P(\overline{S}_1 \cap \overline{S}_2) = P(\overline{S}_1|\overline{S}_2) \cdot P(\overline{S}_2) = 0.3 \cdot 0.05 = 0.015.$

This is the best time to review another critical industrial engineering application: that of preventive maintenance. Predictive maintenance is one of the most important industrial engineering applications involving conditional probability.

Example 3.12 Predictive maintenance

A factory monitors a critical for its operations machine using a sensor. Based on maintenance records:

- $P(\text{machine fails within 30 days}) = 0.10$.
- $P(\text{sensor gives an alert}) = 0.18$.
- $P(\text{machine fails within 30 days AND sensor gives an alert}) = 0.09$.

Let F = "machine fails within 30 days" and H = "sensor gives an alert."

Answer the following questions:

1. What is $P(F|H)$?
2. What is $P(H|F)$?
3. If no alert is sent out by the sensor, what is the probability that the machine still fails within 30 days?

For the first part, we immediately apply the conditional probability formula: $P(F|H) = \frac{P(F \cap H)}{P(H)} = \frac{0.09}{0.18} = 0.5.$. Hence, there is a 50% chance. Notice how without the sensor information, that probability is at 10%! This means that the sensor is highly informative for preventive maintenance.

Moving on to the second question, we flip our conditional probability calculations. The numerator will still be the same, but the denominator changes: $P(H|F) = \frac{P(F \cap H)}{P(F)} = \frac{0.09}{0.10} = 0.9.$ The interpretation here is also useful: when the machine fails, we did get a sensor alert prior to it 90% of the time. Again, this comes to show the informative nature of the sensor.

For this last pat, now, we want to calculate $P(F|\overline{H})$. Recall that this can be computed (using again the conditional probability formula) as: $P(F|\overline{H}) = \frac{P(F \cap \overline{H})}{P(\overline{H})}$. We will calculate each part one by one.

First, let us use set notation to represent $F \cap \overline{H}$. Note that $F = (F \cap H) \cup (F \cap \overline{H})$, where $F \cap H$ and $F \cap \overline{H}$ are mutually exclusive events. This means that: $P(F \cap \overline{H}) = P(F) - P(F \cap H) = 0.10 - 0.09 = 0.01$. Next, we calculate $P(\overline{H}) = 1 - P(H) = 1 - 0.18 = 0.82.$ Finally, we can combine to get

$$P(F|\overline{H}) = \frac{P(F \cap \overline{H})}{P(\overline{H})} = \frac{0.01}{0.82} = 0.012.$$

This means that when we receive no alert from the sensor, the probability of a machine failure goes does from 10% to 1.2%, again confirming the usefulness of the sensor. The absence of an alert is strong evidence that the machine will not be failing in the next 30 days.

Independence

This last discussion must have us thinking about the impact that events have on one another. Specifically, we want to quantify how impactful one event is to another; for example, knowing that a train is running late probably implies delays in the whole transportation system. On the other hand, me having pizza tonight probably will not affect the stock market in France tomorrow.

We typically say that two entities are **independent** if actions of one are completely unaffected (and do not themselves affect) the actions of the other. In probability theory, we say that two events are **independent events** if knowledge that one has happened (or not) does not affect our perception for the probability of the other. In mathematical terms, we say the following.

Definitions and theorems 3.4. Independence of two events

Two events E_1, E_2 are independent if we have that: $P(E_2|E_1) = P(E_2)$ and $P(E_1|E_2) = P(E_1)$. Equivalently, we may write that two events E_1, E_2 are independent if $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$.

Exercise 3.4. Thinking about independence

Can you think of two events from real life that are independent? How about of two events that are clearly dependent?

Example 3.13. Independence and mutual exclusivity

Two events are mutually exclusive. Are they independent?

Most definitely not! The opposite is true: two mutually exclusive events A, B will have that $P(A \cap B) = 0$ (as they share no outcomes), meaning that the conditional probabilities $P(A|B) = 0$ and $P(B|A) = 0$. For two events to be independent, we should have had that

$P(A) = P(A|B)$ ($P(B) = P(B|A)$, respectively).

Example 3.14. Machine downtime and independence

We now revisit Example 3.7. As a reminder, two machines (Machine A and Machine B) fail during a shift and we had obtained the following probabilities:

- $P(\text{Machine A fails}) = 0.12$.
- $P(\text{Machine B fails}) = 0.08$.
- $P(\text{both fail simultaneously}) = 0.02$.

Do Machines A and B fail independently of one another?

Again, the answer is no! If they were failing independently, then we'd have $P(A \cap B) = P(A) \cdot P(B)$, whereas in our case we have $P(A \cap B) = 0.02$ and $P(A) \cdot P(B) = 0.12 * 0.08 = 0.0096.$ Hence, the two machines fail together more often than we would have anticipated if they were independent.

Why can that be? Perhaps the two machines share a common power component? Or maybe one of them failing leads the other one to work more, failing more easily in turn?

Activities

Activity 1: A simple game with dice

Problem 1

Consider a game where you roll two fair dice (i.e., where each number on the side of the dice from 1 to 6 has an equal probability of appearing). What is the probability that the sum of the numbers on the two dice is 7?

Problem 2

What is the probability the sum of the numbers on the two dice is 7 given that the first of the two dice rolled on a 3?

Problem 3

Based on your answers on Problems 1 and 2, what can you claim about the independence of the events "the sum of the two dice is equal to 7" and "the first dice is a 3"? Is that true for all pairs of events "the sum of the two dice is 7" and "the first dice is a i" where $i = 1, 2, 3, 4, 5, 6$?

Problem 4

Prove or disprove[1] the following statement:
When throwing two dies, the two events "the sum of the two dies is

$j = 2, 3, \ldots 12$" and "the first die is a $i = 1, 2, \ldots, 6$" for $i < j$ are independent events.

Activity 2: Quality control revisited

Problem 5

A manufacturing facility is making 2 different products. Every product can be classified as defective (D) or non-defective (ND). In addition to that, some products appear to have cosmetic damage (C) or not (NC). The company has collected data for both products over the last 400 items for each. You may treat these numbers as "probabilities": for example a product 1 is defective and has cosmetic damage with probability $5/400$, whereas a product 2 that is known to be defective has cosmetic damage with probability $2/20$.

1. To disprove a statement, you may do so by counterexample. That is, you can find an example where the statement is not true.

Table 3.2. Product 1 defective and cosmetic damage information (counts).

	Cosmetic damage?		
Defective?	Yes (C)	No (NC)	Total
Yes (D)	5	23	**28**
No (ND)	24	348	**372**
Total	**29**	**371**	**400**

Table 3.3. Product 2 defective and cosmetic damage information (counts).

	Cosmetic damage?		
Defective?	Yes (C)	No (NC)	Total
Yes (D)	2	18	**20**
No (ND)	38	342	**380**
Total	**40**	**360**	**400**

You pick up an item at random from the recent production of Product 1. If you see it has cosmetic damage, does this alter your perception that the product is defective? To answer that, you could calculate the probability that an item is defective (let it be $P(D)$) and contrast it with the probability that it is defective *given* that it has cosmetic damage (let it be $P(D|C)$)? What if these probabilities are equal to one another? What if they are not?

Problem 6

Using the data provided from the previous problem, what can you deduce about Product 2? Does knowing that it has cosmetic damage alter your perception that the product is defective?

Activity 3: Independent vs. mutually exclusive events

Problem 7

Assume that we have two events A, B that are **mutually exclusive**. We already know that $P(A) = 0.3$ and $P(B) = 0.4$. What is $P(A|B)$? What is $P(A \cap B)$? What is $P(A \cup B)$?

- $P(A|B) =$
- $P(A \cap B) =$
- $P(A \cup B) =$

Problem 8

Assume that we have two events A, B that are **independent**. We already know that $P(A) = 0.3$ and $P(B) = 0.4.$ What is $P(A|B)$? What is $P(A \cap B)$? What is $P(A \cup B)$?

- $P(A|B) =$
- $P(A \cap B) =$
- $P(A \cup B) =$

Activity 4: The birthday problem

A class has 90 students. If a class had 366 students (assume for now with me that February 29th does not exist and, hence, every year has exactly 365 days), then we would be guaranteed that at least two of the students in the class share the same birthday.
The question though becomes: in a class of 90, like the one in this exercise, what is the probability that two of the students have the same birthday?

Problem 9

Let's start simple. If there are only two students in the class, what is the probability that they share the same birthday?

Problem 10

With your answer in Problem 9 in mind, add a third person in the mix. What is the probability that the third person has the same birthday with either the first or the second person? As a hint, consider the event that you do not have the same birthday as E and then calculate $P(\overline{E}) = 1 - P(E)$. You may also consider the following: how many possible triplets of birthday dates can you create such that no two birthdays are the same? And how many possible triplets of birthday dates can you create in total? Probability can de calculated as the first number over the second on, if only we knew how to count the number of events...

Food for thought 3.1.

In many instances when interested in finding the probability of an event E, it is easier to calculate $P(\overline{E})$ and use that to find $P(E) = 1 - P(\overline{E})$, rather than trying to calculate $P(E)$ immediately!

Problem 11

Based on your reasoning in Problems 9 and 10, you must have reached a probability of $\frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdot \ldots \cdot \frac{365-n+1}{365}$ for the event that no two people share a birthday in a group of n people. Using the fact that $P(E) = 1 - P(\overline{E})$, we can deduce that the probability that two people share a birthday is:
$P(\text{share birthday}) = 1 - \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdot \ldots \cdot \frac{365-n+1}{365}$.
What does that expression evaluate to in a class of 90 students? What does it evaluate to in a class of 25 students?

Food for thought 3.2.

Next time a person in any class of a significant size shares the same birthday with you, remember that this is decidedly **not** the biggest coincidence in the world, but instead a rather common observance!

Say we had done this for many values of n, starting from $n = 1$ (one person alone has a 0% chance of sharing the birthday with someone else), $n = 2$ (two people sharing a birthday with a $1/365$ chance), $n = 3$ (triplet of people where at least two are sharing a birthday), and so on, until $n = 160$ people. We would have then obtained a figure like the one below. What do you observe?

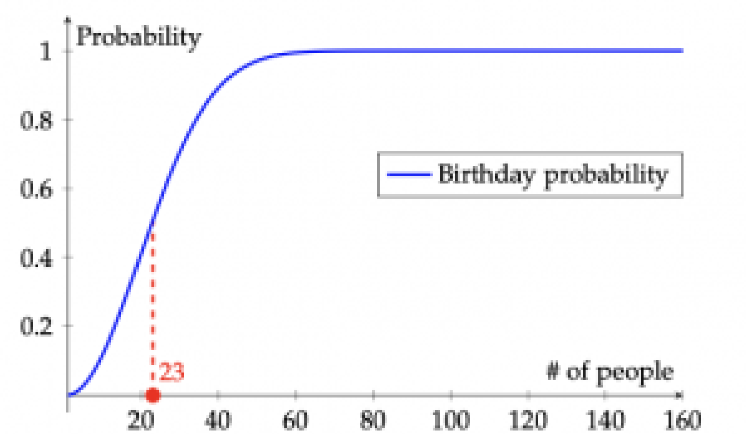

Figure 3.7. A visualization of the birthday paradox. Here we show the probability of two people sharing the same birthday in a group. At a group of about 23 people, the probability of two people sharing a birthday is 50%.

4. Bayes' theorem

Theory

Chapter objectives

In this chapter, our goal is to **introduce, derive, and use one of the most fundamental results in probability theory: Bayes’ theorem.**

Learning outcomes: Bayes’ theorem

After this chapter, you will be able to:

- recall and explain the law of total probability.
- use the law of total probability to calculate probabilities.
- formulate Bayes’ theorem.
- describe Bayes’ theorem.
- explain what Bayes’ theorem implies for probabilities.
- apply Bayes’ theorem in calculating probabilities.

Chapter questions

1. What is the **multiplication** rule for probabilities?
2. What does the **law of total probability** state?
3. What do we need to use Bayes' theorem?

    - **states** S_i, or descriptive events that are hard to tell;
    - **state probabilities**, $P(S_i)$, also called "priors";
    - **outcomes** O�, or observable events;
    - **probabilities** of outcomes given the states P(O�|S_i), also called **likelihoods**.

4. What does the **Bayes' theorem** state?

    - Final output: probabilities of states given the outcomes P(S_i|O�), also called **posteriors**.

P(S_i|O�) = [P(S_i) · P(O�|S_i)] / [Σ P(S_i) · P(O�|S_i)]

- It can also be viewed as: P(hypothesis|evidence) = [P(evidence|hypothesis) · P(hypothesis)] / P(evidence)

Motivating examples

The Mantoux test

The Mantoux (sometimes called the Mendel–Mantoux) test is a diagnostic tool for tuberculosis (TB). In the test, a does of tuberculin is injected: after some time, the reaction on the skin is measured and a positive or negative reaction is given. It is assumed that about 0.05% of the children in the world have TB. The test is pretty accurate, with 99% success rate — that is a person with TB receives a positive result 99% of the time, and a person without TB receives a negative result 99% of the time.

The Mantoux test is mandatory in schools in most European countries. A randomly selected child did the test, which came up positive. **Are you 99% certain that the child has TB?**

Pilot season

Studios typically make decisions about shows based on a single episode made early on, called a "pilot." This pilot episode is viewed by a carefully selected audience who then reports either favorable or unfavorable reviews. A show is considered highly successful, moderately successful, or unsuccessful depending on its performance while on air.

Historically, 95% of highly successful shows received favorable reviews, 50% of moderately successful shows received favorable reviews, and 10% of unsuccessful shows received favorable reviews.

You are one of the producers of a new TV show, and are showing the pilot episode to a major studio. The audience loved it and gave generally favorable reviews. **Will your show definitely be a huge success?**

The law of total probability

Motivation

The Spring 2020 semester saw a rapid change of plans for most university courses due to the COVID-19 pandemic. Specifically, some universities allowed students to change their registrations to accommodate a so-called “credit/no credit” grade for their classes. This means that students would no longer receive a letter grade or score, but instead a sign denoting whether they received credit for the class or not. In the next motivation example, we focus on one such case.

Example 4.1 Credit/No credit

A University course requires students to end up with an average of 70 or above to qualify for credit, whereas an average of 60 is enough to qualify for a passing grade. That is, if a student opts for credit/no credit, then they would need to average at least a 70; if they opt out of credit/no credit, then a 60 would suffice to pass the class. A student believes they will end up with a score between 65 and 80 — so they are definitely passing the class — but they are thinking of opting for the credit/no credit option. Unfortunately, the class is missing two important grades: the final project and a final (non-cumulative) exam.

We are commonly facing problems like this in every day life. Decision-making under uncertainty revolves around us making decisions where the outcomes are not guaranteed. In such cases, the decision-maker *weighs the different futures* and aims to quantify the probability of a favorable outcome. Let’s revisit the student from the example.

Example 4.2 Credit/no credit or fully graded?

The student has been quite enjoying the material of the final exam and they are optimistic that they will score very highly. They believe that they will end up with a score of 90/100 with probability 50% or a score of 80/100 with probability 50%. They are not as confident for the final project, where they believe they received either a 50/100, a 60/100, or a 70/100 (with probabilities 30%, 60%, and 10%, respectively). So, say, they go ahead and put all these eventualities in a table.

A table can be quite handy to keep track of all of the events whose outcomes are uncertain. In the case of the student, they would have to be enumerate a total of 6 cases (why?), which are presented next.

Example 4.3 Credit/no credit or fully graded?

Table 4.1 Student created table with six scenarios

Scenario	Final project	Final exam	Final grade
#1	50	80	65
#2	50	90	68
#3	60	80	69
#4	60	90	72
#5	70	80	73
#6	70	90	76

Given these scenarios, if the student picks credit/no credit, **what is the probability they do not receive credit?**

Before proceeding to the derivation of the **law of total probability**, we note that enumerating all scenarios can be quite daunting (remember the counting rules, and how big that number can get when enumerating all cases).

Derivation

We begin our derivation considering only two events. Specifically, consider two events $A, B,$ marked in the figure below as the blue area of the rectangle and the circle in the middle, respectively. We also mark the complement of A as $\overline{A}$ in the figure.

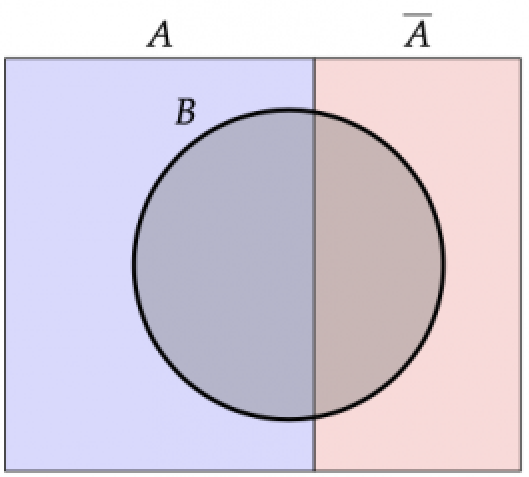

Figure 4.1. A Venn diagram representation of two events [latex]A, B[/latex].

Say, we are interested in the probability of B happening. We can present this as a function of A as follows:

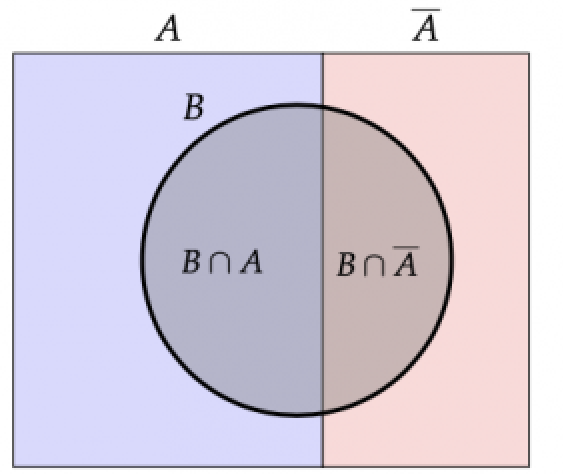

Figure 4.2. A Venn diagram representation of two events [latex]A, B[/latex]. Here we also show the intersection of [latex]B[/latex] with [latex]A[/latex] and the complement of [latex]A, \overline{A}[/latex].

From this second figure, we observe that B can be written as a union of two mutually exclusive events as in $B = (B \cap A) \cup \left(B \cap \overline{A}\right) \implies P(B) = P\left(B \cap A\right) + P\left(B \cap \overline{A}\right).$ Finally, we recall here that $P(A \cap B) = P(A) \cdot P(B|A),$ using the multiplication rule we saw during our previous chapter., which we can replace in the earlier equation to get: $P(B) = \underbrace{P\left(B \cap A\right)}_{P(A) \cdot P(B|A)} + \underbrace{P\left(B \cap \overline{A}\right)}_{P(\overline{A}) \cdot P(B|\overline{A})} \implies P(B) = P(A) \cdot P(B|A) + P(\overline{A}) \cdot P(B|\overline{A}).$

This is the **law of total probability for two events.**

Definitions and Theorems 4.1. The law of total probability for two events

For two events A, B, we can write the law of total probability as:

$$P(B) = P(A) \cdot P(B|A) + P(\overline{A}) \cdot P(B|\overline{A}).$$

Example 4.4 The law of total probability for two events

Two urns contain red and blue balls. The first urn contains 3 red and 3 blue balls, while the second urn contains 5 red and 2 blue balls. We pick one ball at random from the first urn and (without seeing its color) place it in the second urn. **What is the probability we pick a blue ball from the second urn now?**

We distinguish two cases:

1. They pick a red ball from the first urn; now the second urn has 6 red and 2 blue balls.
2. They pick a blue ball from the first urn; now the second urn has 5 red and 3 blue balls.

Both scenarios have a $\frac{3}{6} = 50\%$ probability of happening. Hence, using the law of total probability, we have:

$$\begin{aligned} P(\text{pick blue ball}) =& P(\text{pick blue ball}|\text{Scenario 1}) \cdot P(\text{Scenario 1}) + \\ &+ P(\text{pick blue ball}|\text{Scenario 2}) \cdot P(\text{Scenario 2}) = \\ &= \frac{2}{8} \cdot 0.5 + \frac{3}{8} \cdot 0.5 = 0.3125. \end{aligned}$$

It appears as if we are reducing our probability calculation to an "average" of sorts of all possible future. More on that interpretation shortly.

Exercise 4.1. Practicing with the law of total probability for two events

A recreational runner will do the 100 meter dash in less than 18 seconds with probability 15% when the weather conditions are poor and with probability 90% when the weather conditions are good. During spring, the weather conditions are good with probability 80% (and bad the remaining time). **What is the probability that the runner finishes the dash in less than 18 seconds on a random spring day?**

Use and interpretation

First of all, the law of total probability can be generalized to more than 2 states. Assume we have m mutually exclusive and

collectively exhaustive events. The definition follows:

Definitions and Theorems 4.2. Collectively exhaustive events

With the term **collectively exhaustive** events, we mean events whose union is the whole sample space. Formally, let S be the sample space, and let $E_i, i = 1, \ldots, m$ be some events. Then, events E_i are collectively exhaustive if $E_1 \cup E_2 \cup \ldots \cup E_m = \bigcup_{i=1}^{m} E_i = S.$

Recall now the definition of **collectively exhaustive events**. Again, S to be the sample space, and let $E_i, i = 1, \ldots, m$ be some events. If these events are *both* collectively exhaustive and mutually exclusive, then we have *both* that $\bigcup_{i=1}^{m} E_i = S$, and $E_i \cap E_j = \emptyset$ for any two events $E_i, E_j, i \neq j.$ In other words, these events both make up the whole sample space and share no common outcomes with one another! Dividing a sample space in such events that are both collectively exhaustive and mutually exclusive can be particularly interesting. An example of a series of mutually exclusive and collectively exhaustive events is given in the figure below, where $S = A_1 \cup A_2 \cup A_3 \cup A_4$ and, as is shown,
$A_1 \cap A_2 = A_1 \cap A_3 = A_1 \cap A_4 = A_2 \cap A_3 = A_2 \cap A_4 = A_3 \cap A_4 = \emptyset.$
Four collectively exhaustive and mutually exclusive events. For example, they could represent numbers of unique website visitors in a given day. A_1 could then be up to 1000 visitors, A_2 could

represent between 1001 and 3000 visitors, A_3 between 3001 and 5000 visitors, and A_4 5001 or more visitors.

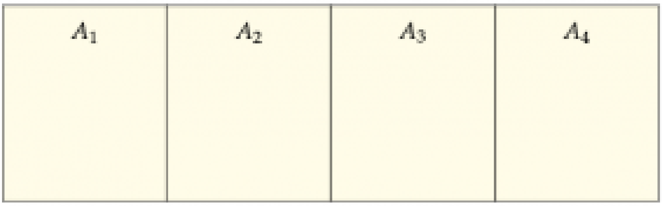

Figure 4.3. Four collectively exhaustive and mutually exclusive events. For example, they could represent numbers of unique website visitors in a given day. [latex]A_1[/latex] could then be up to 1000 visitors, [latex]A_2[/latex] could represent between 1001 and 3000 visitors, [latex]A_3[/latex] between 3001 and 5000 visitors, and [latex]A_4[/latex] 5001 or more visitors.

Example 4.5 Collectively exhaustive events

In the student example, the final project can be viewed as three collectively exhaustive and mutually exclusive events, since the student can only have received a score of 50, 60, or 70. On the other hand, the final exam score has two collectively exhaustive and mutually exclusive events (a score of 80 or 90).

When dealing with $m > 2$ mutually exclusive and collectively exhaustive events $A_i, i = 1, \ldots, m,$ the law of total probability becomes:

$$\begin{aligned}P(B) &= P(A_1 \cap B) + P(A_2 \cap B) + \ldots + P(A_m \cap B) = \\ &= P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2) + \ldots + P(A_m) \cdot P(B|A_m) = \\ &= \sum_{i=1}^{m} P(A_i) \cdot P(B|A_i).\end{aligned}$$

Definitions and Theorems 4.3. The law of total probability

Formally, the **law of total probability** is a theorem that allows us to calculate the probability of any event B given the conditional probabilities of that event B given collectively exhaustive and mutually exclusive events $A_i, i = 1, \ldots, m$ as

$$P(B) = \sum_{i=1}^{m} \underbrace{P(A_i \cap B)}_{P(A_i) \cdot P(B|A_i)} = \sum_{i=1}^{m} P(A_i) \cdot P(B|A_i).$$

The summation can be viewed as a "weighted average;" specifically, we weigh each possible path that leads to event B happening given another event A_i based on the individual probability of A_i happening. This is why we may sometimes informally refer to the value as "overall" or "average" probability. Let's see what that means with an example.

Example 4.6 Credit or no credit?

Let's go back to the table of scenarios the student had prepared. Let us rewrite the eventualities:

- If they receive a 50 in the final project, then they definitely do not get credit, as both scenarios 1 and 2 have them score below the 70 needed.
- If they receive a 60 in the final project, then they have a 50% of getting credit, as scenario 4 is the one that has them score above the 70 needed.
- If they receive a 70 in the final project, then they definitely get credit, as both scenarios 5 and 6 have them score above the 70 needed.

Let A_1, A_2, A_3 be the events of getting a 50, 60, or 70 in the final project and let C be the event of receiving credit in the class. Then, in probability terms, we have:

$$P(C) = P(A_1) \cdot P(C|A_1) + P(A_2) \cdot P(C|A_2) + \\ + P(A_3) \cdot P(C|A_3) = 0.3 \cdot 0 + 0.6 \cdot 0.5 + 0.1 \cdot 1 = 0.4.$$

Here is one more to help you practice:

Exercise 4.2. Practicing with the law of total probability

You have just booked a two-leg (two-flight) trip. The flights are very close to one another and you are worried you will miss your second flight in the case of a delay. If the first flight is not delayed (which happens 60% of the time), you will be certainly fine (won't miss the flight); if the first flight is delayed up to 30 minutes (which happens 20% of the time), you might miss the second flight with probability 50%; finally, if the first flight is delayed by more than 30 minutes (which happens 20% of the time), you will definitely miss the second flight. **What is the probability you miss the second flight?**

The law of total probability as a tree

Decision trees are fundamental decision-making tools; they help visualize different eventualities and different decisions we can make along the way. The law of total probability plays quite an important role in the theory of decision trees. That said, in this setup, we only focus on visualizing the law of total probability through the use of a "tree of possible futures": think of it as a multiverse of sorts, where each future is presented. We do so using the following motivating example.

Example 4.7. The law of total probability as a tree

Assume that we are interested in whether a student passes a class with two exams or not. The first exam has two questions, and hence a student can answer $0, 1,$ or 2 questions correctly. On the other hand, the second exam only has 1 question, so the student can answer 0 or 1 questions correctly. For the student to pass, they'd need to answer 2 questions or more in all of the exams in the class.

Right off the bat, we see how the first exam can significantly hurt the chances of the student. If they get a 0 in that, then they have no path to passing the class. On the other hand, if they answer both questions, then they have already passed and can skip the final! Let's assume that the student answers 0 questions 10% of the time, 1 question 70% of the time, and 2 questions 20% of the time in the first exam, and are equally probable (50% of the time they answer 0 and 50% of the time they answer 1) in the final exam. Finally, assume that the **two exams are independent.**

Now, here is where we can make it visually interesting. We can depict these two exams/events as a *tree* depicting all possible futures:

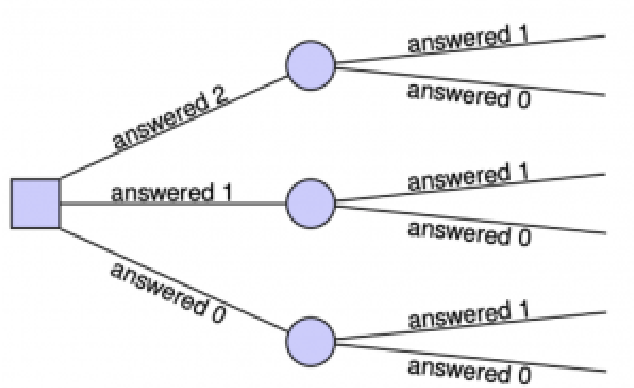

Figure 4.4. A tree showing six possible futures, where the student answers 0, 1, or 2 questions in the first exam and 0 or 1 questions in the second exam.

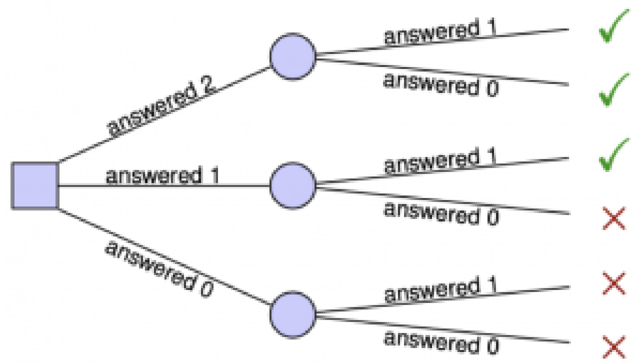

Figure 4.5. The same tree as in Figure 4.4, where the "desired futures" are marked with a green checkmark.

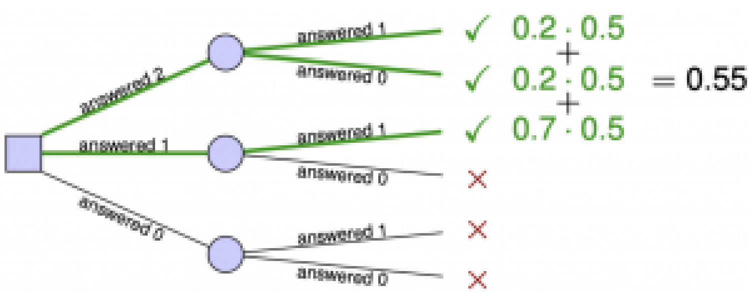

Figure 4.6. The same tree as in Figures 4.4 and 4.5, where the probability of each "desired future", as well as the total probability, are calculated.

Among those, our **favorable outcomes** are the three top branches (2 questions in the first exam and 1 in the second exam, 2 questions in the first exam and 0 in the second exam, or 1 question in the first exam and 1 in the second exam):

Since the two exams are independent, then we can **multiply the probabilities** that lead to each favorable outcome/branch. Moreover, we are now ready to **add the unique probabilities** that lead to the student passing:This finally leads to our calculation of:

$$
\begin{aligned}
P(\text{pass}) = {} & P(\text{pass}|\text{answered 2 in the first exam}) \cdot P(\text{answered 2 in the first exam}) + \\
& + P(\text{pass}|\text{answered 1 in the first exam}) \cdot P(\text{answered 1 in the first exam}) + \\
& + P(\text{pass}|\text{answered 0 in the first exam}) \cdot P(\text{answered 0 in the first exam}) = \\
& = 1 \cdot 0.2 + 0.5 \cdot 0.7 + 0 \cdot 0.1 = 0.55.
\end{aligned}
$$

We note here that $P(\text{pass}|\text{answered 2 in the first exam}) = 1$ (answering two questions in the first exam immediately has you passing the class), $P(\text{pass}|\text{answered 1 in the first exam}) = 0.5$ (answering one question in the first exam has you depend on the second exam, where you are equally probable of answering zero or one questions), and $P(\text{pass}|\text{answered 0 in the first exam}) = 0$ (answering zero questions in the first exam leaves you without a possible way to pass the class).

Brief summary and review of definitions

This is a good time to stop briefly and remind ourselves of a series of definitions we have seen so far. Before we move on to the very important Bayes' theorem derivation and use, let us remind ourselves what each of these mean:

1. **Independent events**: events that are unaffecting one another. Put differently, one event happening changes nothing for the other event. Mathematically, two (or more) events are independent if $P(A|B)$ is the same as $P(A)$ (and vice-versa): $P(A|B) = P(A)$.
2. **Mutually exclusive events**: events that cannot both happen at the same time. In other terms, one excludes the other. Mathematically, two (or more) events are mutually exclusive if $A|B = \emptyset$ and hence $P(A|B) = 0$.
3. **Collectively exhaustive events**: events whose union describes anything and everything that can happen. In other words, it has to be at least one of these events that happens.

Mathematically, two (or more) events are collectively exhaustive if $A \cup B$ is the same as S and hence $P(A \cup B) = 1$.

What do you think about the following?

Exercise 4.3. Review of definitions

- Can two events be… both mutually exclusive and collectively exhaustive?
- How about both mutually exclusive and independent?
- Finally, can two events be both collectively exhaustive end independent?

Finally, we remind the theory behind conditional probabilities and the law of total probability, so that it is as fresh in our minds as possible, prior to discussing Bayes' theorem.

- **Conditional probability**: $$\underset{\substack{\uparrow \\ \text{"B given A"}}}{P(B|A)} = \frac{\overset{\substack{\text{"A and B"} \\ \downarrow}}{P(A \cap B)}}{P(A)}.$$
- **Multiplication rule of probabilities**: based on the definition of conditional probabilities, we have $$P(A \cap B) = P(A) \cdot P(B|A) \quad \Big(= P(B) \cdot P(A|B) \Big).$$
- **Law of total probability** (which we just saw):

$$P(A) = \sum_{i=1}^{n} P(A|B_i) \cdot P(B_i), \text{ if } \underbrace{B_1 \cup B_2 \cup \ldots \cup B_n = S}_{\text{collectively exhaustive}} \text{ and } \underbrace{B_i \cap B_j = \emptyset}_{\text{mutually exclusive}} \text{ for all } i, j.$$

Bayes' theorem

States of the world

Consider the following paradigm. You wake up in the morning, and unbeknownst to you, the world is at a *certain* ***state***. Let's call this state "good" or "bad." In a "good" world, 90% of the people you interact with is happy and smiling and welcoming. In a "bad" world, only 5% of the people you talk to are happy and smiling and welcoming. Unfortunately, you have no idea which state the world is in today. What could you do to find out? What if the first person you see on the street smiles at you and waves good morning? Does that make you feel more or less confident that you have woken up in a "good" world today?

The above paradigm, as far-fetched as it sounds, applies in multiple aspects of our life. A student could have studied for hours for an exam and then went on to answer a multiple choice question correctly, or could have gotten lucky and could give the correct answer by picking one of the multiple choices at random. A diagnostic test could come back positive, and this could mean that the patient is indeed positive to a disease, or it could be a mistake (referred to as a false positive) and the patient is fine. Even worse, a diagnostic test could come back negative, when the patient is unfortunately positive (this is called a false negative) to the disease. In all the above cases, there is a "state of the world" that we are querying through tests, observations, experiences, whose outcomes we read and try to relate to.

States vs. outcomes

We contrast states to outcomes as follows. We consider that a state is fleeting and unknowable; hence, we perform a test and make an observation of its ***outcome***. However, the outcome of the test does not necessarily reveal the state, as no test is perfect. In other words, we query for an unknown state by observing the outcomes we get. This should become clearer with the following example.

Example 4.8. The Mantoux test: states and outcomes

In the Motivation section, we saw the Mantoux test.

- The states of the world (unknowable for certain) are whether a kid has TB or not.
- The test here is the Mantoux test. Its outcomes are either a positive or a negative test.

We now provide two key observations:

1. **The test outcome is not equivalent to the state.** A positive Mantoux test does not always mean a person with TB. A low score in a test does not always mean a student who did not study. A good review does not necessarily mean you will like a movie.
2. **Looking for something rate, we will encounter many false**

positives. Think of what happens when searching in a vast desert for an oasis. Most time, the oases you see on the way are simply mirages.

A two-state example

We present a two-state example, adapted by Daniel Kahneman's "Thinking, Fast and Slow" book.

Example 4.9. Farmer or librarian?

You sit next to someone in a flight and you start talking. The person tells you that they are from the USA, they discuss with you how much they enjoy reading books in their free time, and that they enjoy learning about other cultures. They then ask you: "We've been talking for a while. Guess what my occupation is. **Do you think I work as a librarian or as a farmer?**"

Our mind can create some connections based on what we know and what we *think we know* (our biases, our prior experiences, our instincts). We know that there are more farmers than librarians in the USA (roughly 3 million farmers compared to 300,000 librarians). We also *think we know* that librarians probably enjoy books and learning about other cultures. We may jump to a conclusion, but if we do the math, we will see that our mind can rely too much on prior beliefs rather than context.
In summary: we formulate a hypothesis (for the state of the

world) and, then, given evidence (outcomes of a test) that we leverage, we check to see if we are right or wrong. If this has gotten your attention, let us proceed with formalizing that mathematically.

Definitions

Let's collect here some definitions and notations that will be useful throughout the derivation of Bayes' theorem:

- $S_i, i = 1, \ldots, n$: n states of the world, which have to be **mutually exclusive** and **collectively exhaustive**.
- $O_j, j = 1, \ldots, m$: m outcomes of a test we administer trying to understand the true state of the world.

We also need to define our beliefs for what the state of the world is. These are called **prior probabilities**, as they reflect prior beliefs and biases. Examples include the probability that a random student has studied or not, or the probability that a random person is a librarian or a farmer. These probabilities are provided/estimated before we see the outcomes of the test:

- $P(S_i)$: prior probability of state S_i.

Example 4.10. Prior probabilities

Back in the motivation for the chapter, we discussed the Mantoux test is a diagnostic tool for tuberculosis (TB). In the motivating text, we wrote “It is assumed that about 0.05% of the children in the world have TB.”

As discussed earlier, we have two states: S_1 (kid has TB) and S_2 (kid does not have TB). Note that $S_1 = \overline{S}_2$, and hence the two states are mutually exclusive and collectively exhaustive. Then, the prior probabilities are $P(S_1) = 0.05\%$ and $P(S_2) = 99.95\%$.

Additionally, we define **likelihood probabilities** that represent the probability we see a certain outcome of the test when the world is in a certain state. Examples include the probability that a student does well in an exam given that they have studied or that a person works as a librarian given that they enjoy reading books.

- $P(O_j|S_i)$: likelihood probability of seeing outcome O_j given that we are in state S_i.

Moreover, we have already established that $P(O_j \cap S_i)$ is the probability that we both experience outcome O_j and we are in state S_i. This is called a **joint probability**:

- $P(O_j \cap S_i)$: joint probability of both seeing outcome O_j and we are in state S_i.

Finally, we may calculate the probability that a random test returns a certain outcome O_j. This is referred to as a **marginal probability**. An example would be the probability of an “A” in an exam, or the probability of a positive Mantoux test, regardless of any other information.

- $P(O_j)$: marginal probability of seeing outcome O_j.

Finally, don't lose sight of what we are searching for! This would be $P(S_i|O_j)$: also referred to as the **posterior probability** of being in state S_i given that we observed outcome O_j. We may now proceed to the derivation of Bayes' theorem.

Derivation and use

For a change, let us provide **Bayes' theorem** *before* deriving it.

Definitions and Theorems 4.4. Bayes' theorem

Bayes' theorem is a fundamental result in probability theory that allows us to invert a conditional probability. Specifically:

- Given n mutually exclusive and collectively exhaustive events $S_1, S_2, \ldots, S_n$ (called states), with known prior probabilities $P(S_i)$;
- and m outcomes $O_1, O_2, \ldots, O_m$ with known likelihood probabilities $P(O_j|S_i)$;
- then the posterior probability $P(S_i|O_j)$ can be calculated as

$$P(S_i|O_j) = \frac{P(S_i) \cdot P(O_j|S_i)}{\sum_{i=1}^{n} P(S_i) \cdot P(O_j|S_i)}.$$

A quick mnemonic for the Bayes' theorem can be that *posterior = likelihood* x *prior / evidence.*

We now break down the derivation in steps.

- Step 1: **conditional probabilities**. What can you say about $P(S_i|O_j)$? We could use the **conditional probability** formula to get: $P(S_i|O_j) = \dfrac{P(S_i \cap O_j)}{P(O_j)}$.
- Step 2: **joint probability**. Which other conditional probability employs $P(S_i \cap O_j)$? Recall that (from the definition of conditional probabilities) both $P(A|B)$ and $P(B|A)$ have $P(A \cap B)$ in the numerator. Hence, $P(O_j|S_i) = \dfrac{P(S_i \cap O_j)}{P(S_i)} \implies P(S_i \cap O_j) = P(S_i) \cdot P(O_j|S_i)$. Replacing this in the original conditional probability formula, we get something that looks *close enough* to the Bayes' theorem formula: $P(S_i|O_j) = \dfrac{P(S_i \cap O_j)}{P(O_j)} = \dfrac{P(S_i) \cdot P(O_j|S_i)}{P(O_j)}$.
- Step 3: **marginal probability**. Finally, what can we say about $P(O_j)$ in the denominator? Let's go back to the multiplication rule for probabilities: $P(O_j) = \displaystyle\sum_{i=1}^{n} P(S_i) \cdot P(O_j|S_i)$. Replacing this in the denominator gives us Bayes' theorem:

$$P(S_i|O_j) = \frac{P(S_i) \cdot P(O_j|S_i)}{\sum_{i=1}^{n} P(S_i) \cdot P(O_j|S_i)}.$$

Bayes' theorem essentially relates the posterior probability $P(S_i|O_j)$ to our prior probabilities $P(S_i)$ and our likelihood probabilities $P(O_j|S_i)$. We also present this visually in the next figure.

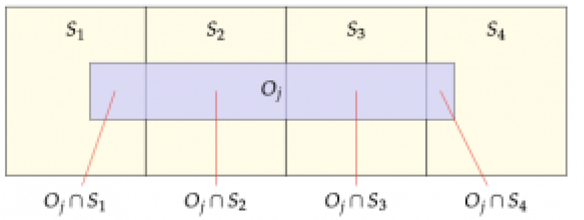

Figure 4.7. Consider 4 mutually exclusive and collectively exhaustive states [latex]S_1, S_2, S_3, S_4.[/latex] When outcome [latex]O_j[/latex] happens, note how the probabilities of each state change. For example, given [latex]O_j[/latex], we have [latex]P(S_1|O_j)=\frac{P(O_j\ cap S_1)}{P(O_j)}.[/latex] Replacing [latex]P(O_j\cap S_1)[/latex] by [latex]P(S_1)\cdot P(O_j|S_1)[/latex] and [latex]P(O_j)[/latex] by [latex]P(S_1)\cdot P(O_j|S_1)+P(S_2)\cdot P(O_j|S_2)+P(S_3)\cdot P(O_j|S_3)+P(S_4)\cdot P(O_j|S_4)[/latex] gives the result.

Example 4.11. Back to the Mantoux test

Going back to the Mantoux test, let's fill in the information we need to answer the question: "**what is the probability a kid has TB given that the test came back positive?**"

- First, we recall the states we established earlier. Namely, S_1: kid has TB; S_2: kid does not have TB.
- Second, we note again the outcomes: O_1: positive Mantoux test; O_2: negative Mantoux test.
- Next, we define our prior probabilities.
  $$P(S_1) = 0.0005,\ P(S_2) = 0.9995.$$

- Finally, we establish our likelihood probabilities: $P(O_1|S_1)=0.99, P(O_1|S_2)=0.01$; $P(O_2|S_1)=0.01, P(O_2|S_2)=0.99.$

Using the Bayes' theorem, we have:

$$P(S_1|O_1)=\frac{P(S_1)\cdot P(O_1|S_1)}{P(S_1)\cdot P(O_1|S_1)+P(S_2)\cdot P(O_1|S_2)}=$$
$$=\frac{0.0005\cdot 0.99}{0.0005\cdot 0.99+0.9995\cdot 0.01}=0.0472.$$

We deduce that a positive Mantoux test implies a 4.72% **chance** of actually having TB.

Exercise 4.4. Pilot season

Answer the probability question in the pilot episode of the motivation. Here it is again:

Studios typically make decisions on shows based on a single episode made early on, called a “pilot.” This pilot episode is viewed by a carefully selected audience who then reports either favorable or unfavorable reviews. A show is considered highly successful, moderately successful, or unsuccessful depending on its performance while on air.

Historically, 95% of highly successful shows received favorable reviews, 50% of moderately successful shows received favorable reviews, and 10% of unsuccessful shows received favorable reviews.

You are one of the producers of a new TV show, and are showing the pilot episode to a major studio. The audience loved it and gave generally favorable reviews. **What is the probability your show is indeed a huge success?**

Bayes' theorem and reliability

Bayes' theorem can be particularly useful in reliability analysis and engineering. When a failure happens, we often cannot tell which part needs replacing without invasive observations. As one example, consider the following example:

Example 4.12. Reliability analysis and Bayes' theorem

A machine consists of **5 independent components** (marked as A, B, C, D, and E in the figure below). For the machine to work either A and B (top pathway) or C, D, and E (bottom pathway) need to work. The probability that each individual component works is given in the figure.

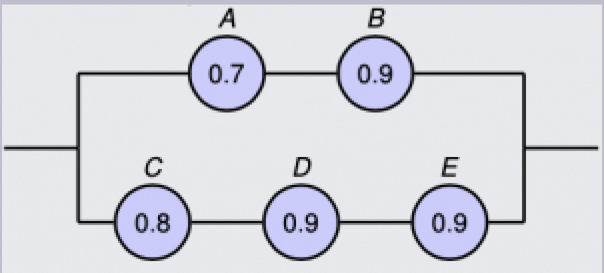

Figure 4.8. The circuit with the 5 individual components. Components A (with probability 70%) and B (90%) make up the top pathway; components C (with probability 80%), D (with probability 90%), and E (with probability 90%) make up the bottom pathway. At least one pathway needs to work for the circuit to operate.

We observe the machine and it is working: **what is the probability that component A is not working?** Mathematically, what is

$$P(\text{A is not working}|\text{machine is working})?$$

- First, we study what is the probability that the circuit is indeed working. This can be done in a series of different ways. Here we show two of them!

  1. Let W be the event that the circuit is working. W can be written as a union of two (independent) events: $\text{top} \cup \text{bottom}$. We have:

     - For the top pathway: $P(\text{top}) = P(A \cap B) = P(A) \cdot P(B) = 0.7 \cdot 0.9 = 0.63.$ We are allowed to multiply the individual probabilities as each component works independently from the others.

- For the bottom pathway:

$$\begin{aligned} P(\text{bottom}) = P(C \cap D \cap E) = \\ = P(C) \cdot P(D) \cdot P(E) = \\ = 0.8 \cdot 0.9 \cdot 0.9 = 0.648. \end{aligned}$$

- Finally, we can calculate

$$\begin{aligned} P(W) = P(\text{top} \cup \text{bottom}) = \\ = P(\text{top}) + P(\text{bottom}) - P(\text{top} \cap \text{bottom}) = \\ = 0.63 + 0.648 - 0.63 \cdot 0.648 = 0.86976. \end{aligned}$$

2. We could instead create a "truth table" with all 2^5 possible scenarios for each component. Let 1 mean that the event is true and 0 that the event is false (i.e., the component/circuit is not working). Then, we would have:

Table 4.2. The truth table, corresponding to all possible events leading to the component working or not.

A	B	C	D	E	W	Probability
1	1	1	1	1	1	0.40824
1	1	1	1	0	1	0.04536
…	…	…	…	…	…	
0	1	1	0	1	0	0.01944
…	…	…	…	…	…	

Summing up all the favorable (working) outcomes, we would get the same probability calculation 0.86976.

- Second, we need $P(W|\overline{A})$. That is, the probability that the machine is working given that component A has failed. Visually:

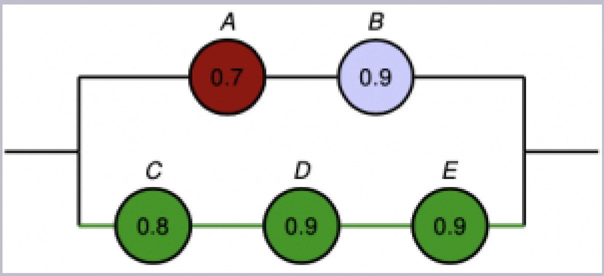

Figure 4.9. A visual representation of the circuit, where A has failed and appears red. C, D, and E are all green (operating) as the machine is known to be working.

We then have:

$$P(W|\overline{A}) = P(\text{bottom}) = 0.648.$$

- Third, we combine everything into Bayes' theorem:

$$P(\overline{A}|W) = \frac{P(W|\overline{A}) \cdot P(\overline{A})}{P(W)} = \frac{0.648 \cdot 0.3}{0.86976} = \frac{0.1944}{0.86976} = 0.2235.$$

We will have you practice with one more "reliability" question during the activities of the chapter.

Another visual representation of Bayes' theorem

Assume that we get a representative population from two states: Kansas and New York. Assuming that Kansas is about 9 times smaller than New York, we pick 90 people from New York and 10 from Kansas, represented pictorially in the next figure, where each red dot is a New York resident, and each blue dot is a Kansas resident.

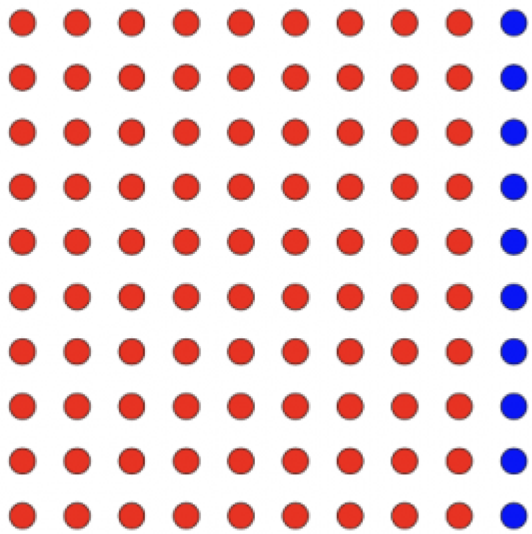

Figure 4.10. A 10×10 grid (100 dots), representing a population consisting of 90 residents of New York (in red) and 10 residents of Kansas (in blue).

Let us assume that only 20% of the population of New York works in an agriculture-related job. The same percentage is 80% for Kansas. Without loss of generality, we show that with the following grid where agriculture-related jobs are shaded in green (light green for New York agriculture-related employees, dark green for Kansas).

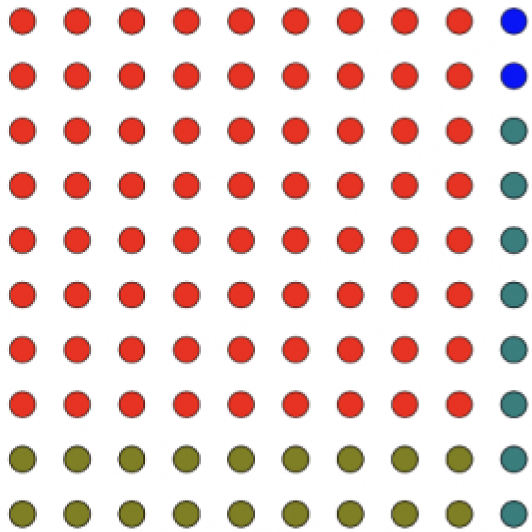

Figure 4.11. A grid showing 100 dots, each representing a resident. In red, we show 90 dots for residents of New York; in blue, we show 10 dots for residents of Kansas. We also show agriculture-related jobs in green. Specifically, a New York resident working an agriculture-related job is shown in light green, whereas dark green is reserved for Kansas residents working agriculture-related jobs.

Finally, assume the person next to you is flying from Kansas to New York (and has made it clear that they are either from Kansas or New York) and works in a farm. While your original bias may be that the person has to be in Kansas (look at the percentage of agriculture-related jobs for the Kansas population!), Bayes' theorem states that the true probability they are from Kansas is only $8/26 = 0.31$. Formally:

$$P(\text{Kansas}|\text{farmer}) = \frac{P(\text{Kansas} \cap \text{farmer})}{P(\text{farmer})} = \frac{0.1 \cdot 0.8}{0.1 \cdot 0.8 + 0.9 \cdot 0.2} = 0.31.$$

We let you *experiment* with such biases in the next question as an exercise.

Exercise 4.5. Bayes' theorem and political beliefs

Assume that in a country there are two predominant political parties. We have collected 100 registered voters, 50 from each of the parties. In the next figure, assume that the first five columns are voters of Party A (in red) and the other five columns are voters of Party B (in blue).

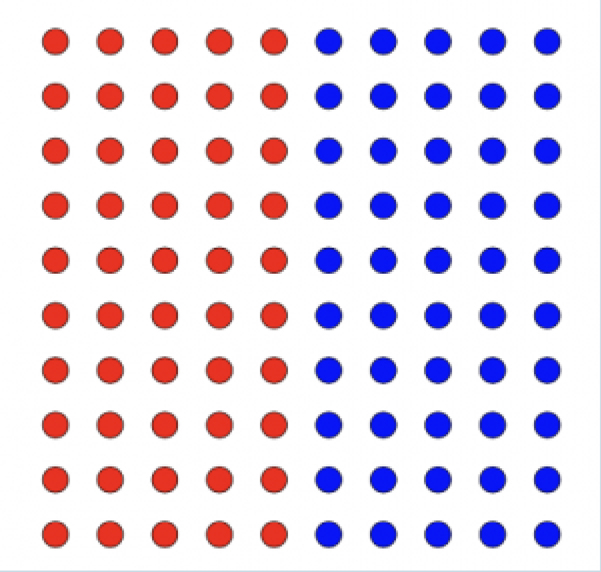

Figure 4.12. 100 indicative voters, with 50 supporting Party A (in red), and 50 supporting Party B (in blue).

In a poll, we ask them how they feel about a certain issue and count how many agree and how many disagree. The voters that overwhelmingly agree appear in shades of green in depending on the party they are registered with.

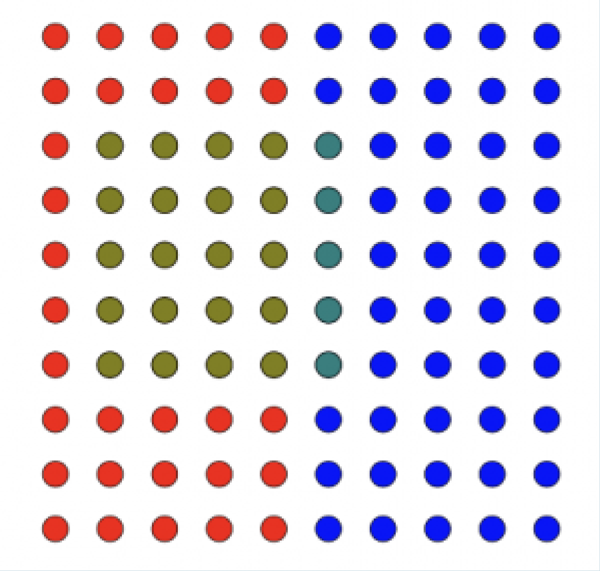

Figure 4.13. The same 100 voters, but now there are 20 from Party A who support a statement and another 5 from Party B who support a statement.

Bayes' theorem states that selecting a random person is a registered voter of Party A (hypothesis) is 50%. However, if the person picked agrees overwhelmingly with a given statement (evidence), then our perception changes. Now, the probability that the person is a registered voter of Party A becomes $\frac{20}{25} = 80\%$.

What can you say for the probability of a person being a

registered voter of Party B given that they do not overwhelmingly agree with the statement?

Summary of Bayes' theorem

Bayes' theorem is very useful for "flipping" a conditional probability question. When we need $P(A|B)$, but it proves elusive or difficult, whereas $P(B|A)$ perhaps is easier or readily available, then Bayes' theorem can be the tool we need. To work with Bayes' theorem, we need the following:

1. Define your states S_i and what are your outcomes O_j.
2. Typically, Bayes' theorem is phrased in terms of "what is the probability of this **state** <u>given</u> that **outcome**?"
3. Don't forget to check whether your state probabilities are mutually exclusive and collectively exhaustive!
   - This means that $\sum P(S_i) = 1$ for all possible states S_i.
4. You could have many states; you could have many outcomes; you do not necessarily have as many states as outcomes.

Activities

Activity 1: The law of total probability

Problem 1: A game of tennis

In a game of tennis [1], two players compete for the next point. The person serving has (historically) an advantage and wins the next point with probability $70\%.$ This probability changes to 50% (so both players are equally probable to win the next point), if the person serving loses the point (you may assume that this is because tennis is also a sport of momentum and morale). When the person serving wins the point again, the probability returns to $70\%.$

What is the probability the player serving wins two points in a row? What is the probability the player serving wins the second point (no matter what happens in the first point)? What is the probability the other player wins the second point (no matter what happens in the first point)?

Problem 2: Decision-making under uncertainty

A company needs to make a decision on whether to open a new facility or not. Their chief economic strategist has recommended

1. The Wimbledon Open (one of the four major tournaments) is happening while writing this chapter, hence the motivation for the question!

one of three options:

1. Invest in a new facility.
2. Play it safe and invest in renovations in existing facilities.
3. Cut down on all expenses.

The market next year can be bullish, bearish, or stagnant[2]. The economic strategist has identified the following probabilities of success for the company depending on the state of the economy:

Table 4.3. The different options and their probabilities of success, depending on whether the market was bullish, bearish, or stagnant.

Option	Bullish	Bearish	Stagnant
#1	80%	5%	30%
#2	50%	50%	80%
#3	10%	95%	40%

What should the company choose to do assuming that next year's market is expected to be bullish with probability 30%, bearish with probability 30%, and stagnant with probability 40%? To answer the question, get the probability of success for each of three options and pick the one with the highest.

2. A bullish market is recognized by an increase in prices, due to the market participants having optimistic views of the economy; a bearish market on the other hand sees a decline in prices, as market participants become pessimistic of the economy. Finally, a stagnant economy sees neither increases or declines in prices and prices stay in a constant level.

Activity 2: Using Bayes' theorem

In this activity, we will bring smaller and bigger applications of the Bayes' theorem. We already saw an interesting one that has to do with the Mantoux test (any diagnostic tool in general). Here, we will see examples from fair grading to detecting spam messages and blood supply logistics.

Problem 3: Fair grader

A quiz in a class is scheduled to have just one multiple choice question with 5 different answers. A student decides to only study half the material for the quiz: hence with probability 50% they will know the answer to the question and will answer the question correctly for sure, and with probability 50% they will guess an answer at random (recall: this implies that all 5 answers are **equally probable** events). A student receives a perfect score in the quiz! **What is the probability they guessed the answer?**

Problem 4: SPAM?

Gmail has observed that the word "inheritance" appears in 20% of all spam emails. It appears at only 0.1% of all known non-spam email communications. Roughly, the estimate right now is that 45% of all email communications are spam. You received an email with the subject Inheritance: **what is the probability it is spam?**

Problem 5: A need for blood

In Greece, blood type distribution is as follows: 44.4% type O, 37.9% type A, 13% type B, and 4.7% type AB. You may assume that blood types are mutually exclusive and collectively exhaustive; that is no one can have two or more blood types, and everyone must have exactly one blood type. However for people born in the 1960s or earlier, typing was not correctly done. We now know that:

- for a person with blood type A, they would be (correctly) found as A with probability 85%;
- for a person with blood type O, they would be (incorrectly) found as A with probability 5%;
- for a person with blood type B, they would be (incorrectly) found as A with probability 15%;
- for a person with blood type AB, they would be (incorrectly) found as A with probability 25%;

A person born in the period before the 1960s is brought to a hospital and needs blood. They are listed as having type A blood: **what is the probability they actually need type A blood?**

Activity 3: Reliability and Bayes' theorem

Winter storms are common in the Midwest and the North. During a winter storm, smaller communities may lose power as electricity substations may fail. Specifically, a smaller county in Illinois is relying for all of its electricity on the following three substations, as shown in the next figure:

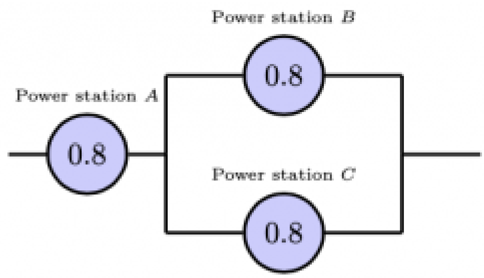

Figure 4.14. Three power stations located as follows. Power station A is on the left and is required for all other pathways. Power Stations B and C are located on top and on bottom. Both power stations A and B or both power stations A and C are necessary for power to reach the county.

During a particularly bad winter storm, each of the three power stations may stop working with probability 20%; that is, they work well with probability 80%. For the county to have power, we need either Power Stations A and B to work, Power Stations A and C to work, or all Power Stations A, B, and C to work. You may assume that all Power Stations function **independently**.
You may define the following events if they help in your calculations:

- A, B, C: power station $A/B/C$ is working.
- E: county has power.
- $\overline{E}$: county does not have power.

Problem 7

What is the probability that the county does not have power during a winter storm?

Problem 8

During a winter storm, people reported a large explosion in Power Station B, which immediately stopped working. What is the probability the county has power given that Power Station B is not working? Based on your answers in Problems 7 and 8, would you say that events B and E are **independent**[3] or not?

Problem 9

For this question, disregard the setup in Problem 8. That is, please assume that all three power stations are working properly initially. Now, assume that the county has just lost power. There is only one crew available to fix the issue, so they have to visit one of the power stations and see if it has any issues. What is the probability that A has failed given that the county does not have power? Which power station should you instruct the crew to visit first?

Activity 4: Flipping a flipped classroom

Problem 7

Create one exercise using the Bayes' theorem. To help you, you can follow the next steps:

1. Find an interesting application where you need to know something and you may run a test to figure it out.

3. You may contrast $P(E)$ to $P(E|\overline{B})$.

2. Look online to see if you can find probabilities that appear realistic.
3. Pose the question of searching for a probability.
4. Solve the question. Bonus ⯑ if the result is counterintuitive!

5. Discrete random variables

This chapter should be spread across two or more lectures, due to its larger size and the importance of the fundamental notions presented. The activity set found at the end of this chapter is also longer in length to have you practice with more of the notions. In a first lecture, you can focus first on Activities 1-4, leaving by the remaining activities for the next lecture(s).

Theory

Chapter objectives

In this chapter, our goal is to **introduce, define, and use discrete random variables.**

Learning outcomes: Discrete random variables

After the next chapter, you will be able to:

- define discrete and continuous random variables.
- differentiate between discrete and continuous random variables.
- differentiate between when to use cumulative distribution functions, probability mass functions,

and probability density functions.

After the next chapter alone, you will be able to

- give examples of at least four common different discrete distributions.
- recall when to and how to use:
    - binomially distributed random variables.
    - hypergeometric distributed random variables.
    - geometric distributed random variables.
    - Poisson distributed random variables.

Chapter questions

1. What is the difference between **discrete** and **continuous random variables**?
2. What are the **axioms** of discrete probability distributions?
3. What is the **probability mass function**? What is the **cumulative distribution function**?
4. How do we distinguish between:
    - the **binomial** distribution?

      n tries, need x successes, each success has probability p.
    - the **hypergeometric** distribution?

      n tries, need x successes, have a population of N items among which there are K successes.
    - the **geometric** distribution?

need first success in the x-th try, each success has the same probability p.

5. How does the uniform distribution work?
6. How does the Poisson distribution work?
7. When provided with a *rate of events* in a Poisson distribution, how do I change the time scale (e.g., from events per minute to events per hour, etc.)? Does a simple transformation work?
    - Does the same transformation work for probabilities? Is it true that double the rate implies double the probability?
8. What is the memorylessness property?
9. What is the relationship between Poisson random variables (which are <u>discrete</u>) and time (which is a <u>continuous</u> random variable)?

Motivating examples

2-engine vs. 4-engine planes

Suppose that for a flight to be completed successfully (which we would certainly prefer, in any flight) we need at least half of the engines to be operational at the end of the trip. In the case of a 2-engine aircraft, this means at least one has to be operational; in a 4-engine aircraft, at least two must be operational. We are considering purchasing a new aircraft for recreational purposes, and we are contemplating a 2-engine or a 4-engine plane. **Which one would be safer?**

Big in Japan

Since 1900, there have been several earthquakes of seismic intensity over 7.0 on the Richter scale either in or affecting the Kanto region of Japan. One of the most notable earthquakes is the 1923 Great Kanto Earthquake that devastated Tokyo and Yokohama. In general, especially for those of us living in seismically active regions/zones (i.e., areas with high seismic/earthquake activity), we probably grew up hearing statements such as "the probability of an earthquake affecting location X within the next Y years is $Z\%.$" Hey, this is what we do in this book!
We make the following assumptions for big earthquakes:

1. Big earthquakes are independent events — that is the fact that a big earthquake happened does not increase or decrease the probability of another big earthquake soon.
2. The probability of an earthquake occurring is the same throughout the year[1].

What is the probability that there will be a large earthquake in the Kanto region in the next year? What is the probability that there will be a large earthquake in the Kanto region in the next decade?

1. This is a property also called homogeneity. More on that later.

Random variables

While this chapter is devoted to *discrete* random variables, we provide here a general definition of both discrete and continuous random variables for completeness. That said, this chapter on discrete random variables and the next chapter on continuous random variables share nomenclature and notions so we provide a list of definitions here, for convenience.

The world around us is the result of a series of random processes, whose outcomes affect the way we perceive things. When the next bus arrives depends on traffic (random), demand for bus stops along the way (random), the weather (random); the number of attempts it takes for a student to pass a class is also a random process. Mathematically, we need to somehow define these stochastic outcomes and processes, using numerical representations. This next section does exactly that. We begin with the definition of **random variables**.

Definitions and Theorems 5.1. Random variables

With the term **random variable**, which are also sometimes termed random quantities or stochastic variables, we mean a real-valued[2] function defined over the sample space.

This next definition makes this a little more formal:

Definitions and Theorems 5.2. Random variables

A random variable X maps each outcome in the sample space S to a set of measurable outcomes (usually real-valued). That is, $X : S \to \mathbb{R}$. Using the "belongs to" symbol ($\in$), we say that the probability that random variable X takes on some value in set A is written as $P(X \in A)$.

For example, let X be the side that a dice falls on. Then, the probability that it takes on the value of 2 can be written as $P(X = 2)$. On the other hand, define T as the random variable that represents the time until the next bus arrives. Then, the probability that the time until the next bus arrives is smaller than or equal to 10 minutes can be written as $P(T \leq 10 \text{ minutes})$.

2. While there are instances where the function does not take real values, this is often the case in engineering applications, and will always be the case in this textbook.

Classification

As mentioned earlier, in this chapter we focus on *discrete random variables* (as opposed to *continuous random variables* in the next chapter). However, we first provide definitions of what discrete and continuous random variables entail, before talking about the different functions that are used to describe them.

Definitions and Theorems 5.3. Discrete random variables

Discrete random variables are **random variables** that are allowed to take countable, discrete values.

Discrete random variables typically take whole (integer) numbers or a value from a list of possible numbers. Counts or categories are commonly modeled as discrete random variables.

Example 5.1. Discrete random variables

- The side a dice falls is a discrete random variable. Let X be that number. Then X can take on value $1, 2, 3, 4, 5, 6.$
- The number of customers that are currently seated at a restaurant is a discrete random variable. Let X be the number of customers; then, X can take on

values $0, 1, 2, \ldots$.

- The result of a coin toss is a discrete random variable. Let X be Heads or Tails (or, numerically, we could assign numbers 0 and 1).

On the other hand, the definition of continuous random variables allows for real values. Specifically:

Definitions and Theorems 5.4. Continuous random variables

Continuous random variables are random variables that are allowed to take an infinite set of any real value within a range.

Compared to discrete random variables, continuous random variables can represent any measurement or observation that takes real values (such as the temperature or time elapsed).

Example 5.2. Continuous random variables

- The time until the next bus arrives at a bus stop is a continuous random variable. Let X be that number.

Then X can take on any nonnegative value $[0, +\infty)$.

- The remaining lifetime of a lightbulb is a continuous random variable. Let X be that number. Then X can also take on any nonnegative value $[0, +\infty)$.
- The weight of a "1 pound" ground meat package is a continuous random variable; while we would like the weight to be exactly 1 pound, due to errors in measurement that can be 10% smaller or bigger. Let X be the true weight. Then X can take any value $[0.9, 1.1)$ pounds.

Exercise 5.1. Discrete or continuous?

Classify these random variables as discrete or continuous:

- the time it takes for a biker to go from one side of campus to the other.
- the number of red lights the biker has to stop at when going from one side of the campus to the other.
- the distance a biker traverses to go from one side of campus to the other.
- the number of times the biker changed speed gear while going from one side of campus to the other.

Probability distributions and functions

When a random variable behaves a specific way, we say that it follows a probability **distribution**. A probability distribution is described by two distribution functions:

1. The **probability mass function** for discrete random variables or **probability density function** for continuous random variables.
2. The **cumulative distribution function**.

We formally define those where needed for discrete (next section) and continuous (next chapter) random variables.

Discrete random variables

The probability mass function

Let X be a discrete random variable.

Definitions and Theorems 5.5. Probability mass function

We define the **probability mass function** (*pmf*) $p(x)$ of a discrete random variable X as the probability that the random variable takes a specific value x. Mathematically, we have $p(x) = P\left(X = x\right)$.

Some textbooks may use $f(x)$ for the probability mass function. In the next three chapters, we will use $p(x)$ for discrete random variables and their probability mass functions, and reserve $f(x)$ for continuous random variables and their probability *density* functions.

Example 5.3. Probability mass function

Let $p(x) = \frac{x}{6}$ for a random variable X taking on integer values $1, 2, 3$. Then, the probability that X is equal to 2 can be calculated as $p(2) = \frac{2}{6} = \frac{1}{3}$..

From the definition, we need to be very careful with distinguishing between X (upper case) and x (lower case)! X refers to the random variable; x is a given value. Typically, we use the upper case for the random variable ($X, Y, T, \ldots$) and the lower cases for the potential values ($x, y, t, \ldots$). As an example: we can write $P(X = 7)$ and read it as "what is the probability that *random variable* X is equal to the *value* 7?"

Example 5.4. Probability mass function

What is the probability that a store has **exactly 20** customers enter in the next hour?

This is a question that should be answered using the probability mass function of a discrete random variable! We do the following steps:

- First, let X be the number of customers entering the store in the next hour.
  - X is a random variable; furthermore, it is a discrete random variable as the number of customers has to be a nonnegative integer value.
- Second, we note that we are interested in seeing $x = 20$ customers showing up. That is, we want to calculate $P(X = 20)$.
- Finally, we derive that probability using the ***probability mass function*** and calculate it as $p(20)$.

The cumulative distribution function

Once again, let X be a discrete random variable.

Definitions and Theorems 5.6. Cumulative distribution function

We define the **cumulative distribution function** (*cdf*) $F(x)$ of a discrete random variable X as the probability that the random variable takes a value *up to* a specific value x. Mathematically, we have $F(x) = P(X \leq x)$.

In other words, the probability that random variable X is smaller than or equal to a given value x. Equivalently, we may sum up all of the individual probabilities for all outcomes in the sample space that are smaller than or equal to the given value x:

$$F(x) = P(X \leq x) = \sum_{y:y \leq x} P(X = y) = \sum_{y:y \leq x} p(y).$$

Example 5.5. Cumulative distribution function

Let $p(x) = \frac{x}{6}$ for a random variable X taking on integer values $1, 2, 3$. Then, the probability that X is less than or equal to 2 can be calculated as

$$F(2) = p(1) + p(2) = \frac{1}{6} + \frac{2}{6} = 0.5.$$

Let us check one more example for the cdf.

Example 5.6. Cumulative distribution function

What is the probability that a store has **fewer than or 20** customers enter in the next hour?

We follow the approach from our earlier solution where we defined the discrete random variable (X as the number of customers entering the store in the next hour. That said, we now use the cumulative distribution function and get $P(X \leq 20) = F(20)$.

An immediate result from the definition of the cdf is that if we are interested in the probability of seeing more than a certain value x we may write that $P(X > x) = 1 - P(X \leq x) = 1 - F(x)$. Now, consider $P(a < X \leq b)$, that is the probability the random variable is greater than a (but not equal to it) and smaller than or equal to b. Combining the two definitions (of $P(X \leq x)$ and $P(X > x)$) we get that $P(a < X \leq b) = F(b) - F(a)$.

Example 5.7. 2-engine vs. 4-engine aircraft

We saw that a plane performs a trip safely if at least half of its engines are operational. Let X be the number of engines that have failed during a trip with a 2-engine aircraft, and Y the number of engines that have failed during a trip with a 4-engine aircraft. Then, we should be looking for:

- $P(X \leq 1)$: for the probability of a successful trip with a 2-engine aircraft.
- $P(Y \leq 2)$: for the probability of a successful trip with a 4-engine aircraft.

We should be using the **cumulative distribution function** to answer these questions.

Example 5.8. Earthquakes

In a similar fashion, let X be the number of earthquakes in the Kanto region in the next year and Y be the same number in the next decade. Then, we should be looking for:

- $P(X = 1)$: for the probability of one big earthquake in the next year.
- $P(Y = 1)$: for the probability of one big earthquake in the next decade.

Here, we would be using the **probability mass functions** for the two random variables.

Exercise 5.2. Practicing with the functions

Should you use the pmf or the cdf?

- To avoid paying your friends in a game of Monopoly you need to get a 6 or less when throwing two dies.
- An exam has 10 multiple choice questions. What is the probability you answer all of them correctly?
- An exam has 10 multiple choice questions. What is the probability you answer more than or equal to 8 questions correctly?
- Two people are playing a game that is best out of three. What is the probability the first player wins with a score of 2 to 1?

Properties of discrete random variables

Assume a discrete random variable X with n outcomes $x_i, i = 1, \ldots, n$. Then, the probability mass function $p(x)$ of random variable X has to satisfy the following three rules, a direct consequence of **Kolmogorov's axioms of probability**:

1. $p(x_i) = P(X = x_i)$, for every outcome $x_i, i = 1, \ldots, n$.
2. $p(x_i) \geq 0$.
3. $\sum_{i=1}^{n} p(x_i) = 1.$

Exercise 5.3. Practicing with the basic properties

Earlier we defined $p(x) = \frac{x}{6}$ for a random variable X taking on integer values $1, 2, 3$. Is this a valid probability mass function?

Example 5.9. Urns and balls

Two balls are drawn from an urn containing 5 red and 4 black balls. Define a random variable X as the number of red balls drawn. What is its probability mass function?

- X has three outcomes: $0, 1, 2$.
  - To select 0 red balls, we need to pick two of the four black balls and none of the red balls: $$p(0) = P(X = 0) = \frac{C_{4,2}}{C_{9,2}} = \frac{6}{36}.$$
  - To select 1 red ball, we need to pick one of the four black balls and one of the red balls: $$p(1) = P(X = 1) = \frac{C_{4,1} \cdot C_{5,1}}{C_{9,2}} = \frac{20}{36}.$$
  - To select 2 red balls, we need to pick none of the four black balls and two of the red balls: $$p(2) = P(X = 2) = \frac{C_{5,2}}{C_{9,2}} = \frac{10}{36}.$$
- We note that this probability mass function satisfies all three properties:

1. Every outcome is associated with one probability mass function value.
2. All probability mass function values are nonnegative.
3. The summation of the three probability mass function values is equal to 1:
$$\sum_{x=1}^{3} p(x) = \frac{6}{36} + \frac{20}{36} + \frac{10}{36} = \frac{36}{36} = 1.$$

Example 5.10. Calculating a probability mass function based on observations

A sample space is described by two mutually exclusive outcomes A and B. We have observed that for some real number x, the pmf is $P(A) = 3 \cdot x$ and $P(B) = 10 \cdot x^2$. **What is the probability mass function?**

In essence, we are asked to find x. We will need to use the valid pmf rules. We need to verify what x should be in order to satisfy $P(A), P(B) \geq 0$ and $P(A) + P(B) = 1$.

- First of all, it is easy to see how $x \geq 0$, as otherwise it would turn $P(A)$ negative.
- Then, replacing the pmf in $P(A) + P(B) = 1$ we have:

$$P(A) + P(B) = 1 \Longrightarrow \; 3 \cdot x + 10 \cdot x^2 = 1$$
$$\Longrightarrow \; 10 \cdot x^2 + 3 \cdot x - 1 = 0 \; \Longrightarrow$$
$$\Longrightarrow \; x = \begin{cases} -0.5 \\ 0.2 \end{cases}$$

- Recall that from the two solutions, we can discard the negative one (see the first point), hence $x = 0.2$. Replacing, we get: $P(A) = 0.6$ and $P(B) = 0.4$.

For the cumulative distribution function of a discrete random variable X, as a consequence of the earlier axioms, we have:

1. $0 \leq F(x) \leq 1$.
2. If $x \leq y$, then $F(x) \leq F(y)$.

Let us compute the cumulative distribution function for a few discrete random variables.

Example 5.11. A fair dice

Consider a fair, six-sided dice and let X be the (discrete) random variable that the dice takes on values $S = \{1, 2, 3, 4, 5, 6\}$. The probability mass function is $p(x) = \dfrac{1}{6}$, for $x = 1, 2, 3, 4, 5, 6$, and 0 otherwise. What is the cumulative distribution function $F(x)$?

By definition, $F(x) = \sum_{y \le x} p(y)$. Since the probability mass function is equal to $\frac{1}{6}$ for each outcome, then we have

$$F(x) = \begin{cases} 0, & x < 1, \\ \frac{1}{6} \cdot x, & x = 1, 2, 3, 4, 5, 6, \\ 1, & x > 6. \end{cases}$$

We can also visualize the cumulative distribution function. In this next example, we do just that.

Example 5.12. Urns and balls (continued)

Consider the previous sample space $S = \{0, 1, 2\}$ with pmf $p(0) = \frac{6}{36}, p(1) = \frac{20}{36}, p(2) = \frac{10}{36}$ from the earlier example with the urn containing red and black balls. Then, the cdf can be calculated as:

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{6}{36}, & 0 \le x < 1 \\ \frac{26}{36}, & 1 \le x < 2 \\ 1, & x \ge 2. \end{cases}$$

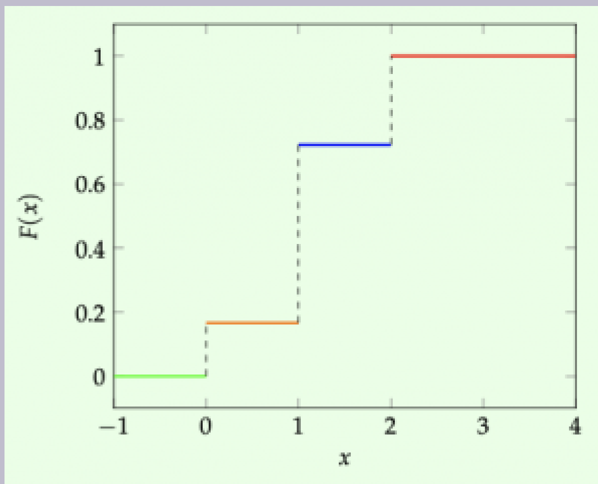

Figure 5.1. The cumulative distribution function [latex]F(x)[/latex] for Example 5.12.

Note the "jumps" in the visualization of $F(x)$. Those happen because (for a discrete random variable) the value of the cdf only changes when we encounter one of the discrete outcomes of the sample space.

Exercise 5.4. Practicing with probability mass functions and cumulative distribution functions

Let the sample space be $S = \{1, 2, 3\}$ with a probability mass function of

$$p(1) = \frac{1}{2}, p(2) = \frac{1}{3}, p(3) = \frac{1}{6}.$$

1. Verify this is a valid pmf.
2. Compute the cdf $F(x)$.
3. Plot the cdf (like we showed in the previous example).

Commonly used discrete distributions

In this part of the chapter, we will study some very commonly used and observed discrete distributions. We begin our discussion from the simplest and most basic of distributions (*Bernoulli* random variables), and then we will add more special cases and details as we move on, before finishing this chapter with another ubiquitous distribution, in the form of the *Poisson distribution.*

The Bernoulli distribution

Consider a single experiment, performed only once, with nothing fancy but two probable outcomes: 1) success, and 2) failure. Now, define a random variable X, based on that single experiment: $X = \begin{cases} 0, & \text{if the experiment fails} \\ 1, & \text{if the experiment succeeds} \end{cases}$. Assuming that the random variable satisfies $P(X = 1) = p$ and $P(X = 0) = q = 1 - p$, then this random variable is said to follow a **Bernoulli distribution**.

Definitions and Theorems 5.7. The Bernoulli distribution

The Bernoulli distribution is the probability distribution of a **discrete** random variable which is allowed to take the value of 1 with some positive probability p and the value of 0 with probability $1 - p$.

The Bernoulli distribution is common when modeling the outcome of a single experiment that can have a binary success-failure, yes-no, true-false outcome. The key is that we are only interested in the outcome of a single experiment, and not the repetition of many (more on that distinction, shortly).

Example 5.13. Examples of Bernoulli distributed random variables

- Will it rain tomorrow? If it rains with probability $p = 0.2$, then tomorrow's rain (yes-no) follows a Bernoulli distribution.
- Is the next coin toss a Heads or a Tails? If each outcome happens with probability p and $1 - p$, respectively, then the next coin toss follows a Bernoulli distribution.
- Does the experimental drug work on Patient A? If it does with probability $p = 0.77$, then the outcome of the drug on the patient is a Bernoulli distributed

random variable.

In all of the above, we have a "success" (rain tomorrow; Heads; drug works) and a "failure" (no rain tomorrow; Tails; drug does not work), and the probability of success p (and consequently the probability of failure $q = 1 - p$) is known a priori.

Based on the definition of Bernoulli random variables, the probability mass function and the cumulative distribution function:

- **Probability mass function**: $$p(x) = \begin{cases} P(X = 0) = q = 1 - p, \\ P(X = 1) = p \end{cases}$$
- **Cumulative distribution function**: $$F(x) = \begin{cases} 0, & \text{if } x < 0 \\ 1 - p, & \text{if } 0 \leq x < 1 \\ 1, & \text{if } x \geq 1 \end{cases}$$

Example 5.14. Urns and balls

An urn contains 40 black and 10 red balls. You pick at random one ball from the urn. Let X be the number of black balls you pick from the urn. What is the probability mass function of X?

- We make the argument that X is a Bernoulli distributed random variable. The outcome of the experiment is 0 black balls, or 1 black ball. The probability of picking 1 black ball from the urn is $p = \frac{40}{50} = 0.8$ (and hence, we get $q = 1 - p = 0.2$).

- Since it is a Bernoulli random variable, then the probability mass function can be written as: $p(x) = \begin{cases} 0.2, & \text{if } x = 0, \\ 0.8, & \text{if } x = 1. \end{cases}$

What if we consider more than one experiments? What if, say, we picked 5 balls and wanted to get 3 black ones?

The binomial distribution

Assume that we have a Bernoulli distributed random variable, and we repeat that experiment *multiple times*. For example, we have a fair coin (with 50% chance of each outcome) that we toss $n = 5$ times, or we observe the performance of a new experimental drug on $n = 110$ patients. This leads to a new distribution, **the binomial distribution**:

Definitions and Theorems 5.8. The binomial distribution

The binomial distribution with parameters an integer n and $p \in (0,1)$ is the probability distribution of a discrete random variable which is allowed to take the values $\{0,1,\ldots,n\}$, depending on the number of successes observed when performing n independent experiments, each with an outcome modeled as a Bernoulli random variable with probability of success p.

The setup is quite simple and can be summarized as follows.

1. n independent experiments: this means that the outcome of one experiment does not affect the others.
2. each experiment has the same probability p (respectively, $q = 1 - p$) of resulting in a success (respectively, failure): this means that each experiment can be modeled as a simple Bernoulli random variable.

Letting X be the number of successes obtained, we say that X follows a binomial distribution, or that it is binomially distributed. We sometimes may also write that $X \sim \mathrm{binom}(n,p)$, as n and p are the only parameters necessary to fully define a binomial distribution.

Example 5.15. The binomial distribution

Multiple experiments could mean multiple coin tosses, or a control group of 100 patients, or a best-of-five game series!

- A coin is tossed multiple times. The number of Heads that come up in $n = 10$ coin tosses is a binomially distributed quantity.
    - Note that in the case of a fair coin, we would further have the probability of Heads ("success") be equal to $p = 0.5$.
    - We may have as few as 0 Heads in $n = 10$ coin tosses, or even as many as 10 Heads in $n = 10$ coin tosses.
- A control group of $n = 110$ patients is given an experimental drug that successfully works with probability $p = 0.77$. The number of patients that are successfully treated is a binomially distributed random variable.
- You play monopoly with your friends, and you win with probability $p = 0.25$. The number of wins you experience in the next week of games ($n = 7$ games, assuming you play once a day) follows a binomial distribution.

Example 5.16. Coin tosses

Probably the most common example (and possibly the least “inspired”) to explain binomial random variables comes from coin tosses. Assume we possess a “fair” coin with probability of Heads $p = 0.5$, and probability of Tails $q = 1 - p = 0.5$. What is the probability that there will be **exactly 2** Heads in $n = 3$ tosses of the coin?

This would be a binomial distribution with $n = 3$, $p = 0.5$, and where we'd use the probability mass function $p(x)$ (if we had it) for $x = 2$ Heads. We would then report the probability of 2 Heads as $P(X = 2) = p(2) = \ldots$

We *omit* the derivation (it is actually part of your activities for the chapter!), and we jump to the formula for the probability mass function. Generalizing on the Bernoulli distribution, we get:

- **Probability mass function**: $p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \; x = 0, 1, \ldots, n$
- **Cumulative distribution function**: $F(x) = \sum_{y=0}^{\lfloor x \rfloor} \binom{n}{y} p^y (1-p)^{n-y}$, where $\lfloor x \rfloor$ is the greatest integer value that is smaller than or equal to x.
  - For example,

$$\begin{aligned} F(2.2) &= \sum_{y=0}^{\lfloor 2.2 \rfloor} \binom{n}{y} p^y (1-p)^{n-y} = \\ &= \sum_{y=0}^{2} \binom{n}{y} p^y (1-p)^{n-y} = \\ &= \binom{n}{0} p^0 (1-p)^n + \binom{n}{1} p^1 (1-p)^{n-1} + \binom{n}{2} p^2 (1-p)^{n-2} = \\ &= P(X=0) + P(X=1) + P(X=2). \end{aligned}$$

- As a reminder, $\binom{n}{x} = \dfrac{n!}{x! \cdot (n-x)!}$ is the number of **combinations** of choosing x items out of n items.

It seems we now have all we need to answer the first motivating question: should we choose a 2-engine or a 4-engine plane?

Example 5.17. 2-engine versus 4-engine

From our motivating example, let p be the probability that an engine fails during a trip, and hence $q = 1 - p$ is the probability it does not fail. Let X be the number of engines that have failed during a trip with a 2-engine aircraft, and Y the number of engines that have failed during a trip with a 4-engine aircraft. For the success of each of the two planes, we are looking for $P(X \leq 1)$ and $P(Y \leq 2)$.

Both are **binomially distributed** with different parameters (number of experiments/engines is different).

- $X \sim \text{binom}(2, p)$
- $Y \sim \text{binom}(4, p)$

Let us compare these two quantities:

$$
\begin{aligned}
P(X \leq 1) = F(1) = \sum_{y=0}^{1} p^y (1-p)^{2-y} = \quad & (P(X=0) + P(X=1) =) \\
= \binom{2}{0} \cdot p^0 \cdot (1-p)^2 + \binom{2}{1} \cdot p^1 \cdot (1-p)^1 = & \\
= 1 - 2p + p^2 + 2 \cdot p - 2 \cdot p^2 = & \\
= 1 - p^2 &
\end{aligned}
$$

$$
\begin{aligned}
P(Y \leq 2) = F(2) = \sum_{y=0}^{2} p^{y}(1-p)^{4-y} = {} & (P(Y=0) + P(Y=1) + P(Y=2) =) \\
&= \binom{4}{0} \cdot p^{0} \cdot (1-p)^{4} + \binom{4}{1} \cdot p^{1} \cdot (1-p)^{3} + \binom{4}{2} \cdot p^{2} \cdot (1-p)^{2} = \\
&= 1 - 4 \cdot p + 6 \cdot p^{2} - 4 \cdot p^{3} + p^{4} + 4 \cdot p - 12 \cdot p^{2} + 12 \cdot p^{3} - 4 \cdot p^{4} \\
&+ 6 \cdot p^{2} - 12 \cdot p^{3} + 6 \cdot p^{4} = \\
&= 1 + 3 \cdot p^{4} - 4 \cdot p^{3}
\end{aligned}
$$

Can we compare the two? To prefer a 2-engine plane we need its probability of success to be higher, that is we need:
$1 - p^{2} \geq 1 + 3 \cdot p^{4} - 4 \cdot p^{3} \implies -3 \cdot p^{4} + 4 \cdot p^{3} - p^{2} \geq 0 \implies$
$\implies -3 \cdot p^{2} + 4 \cdot p - 1 \geq 0.$

Let's plot $z = -3 \cdot p^{2} + 4 \cdot p - 1$ and see what we get! Since for $z \geq 0$ we prefer a 2-engine aircraft, it suffices to see when z is nonnegative in the plot!

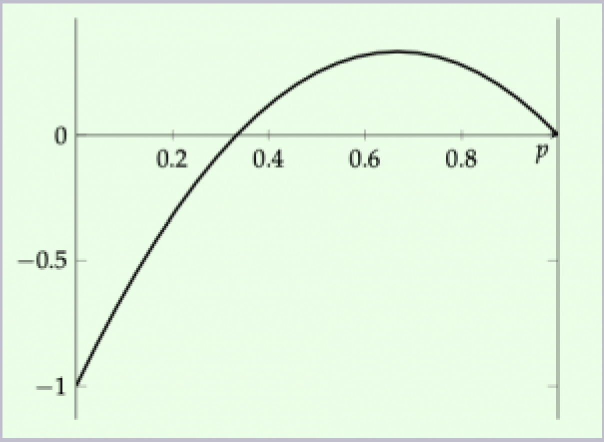

Figure 5.2. A plot visualizing the difference in probability between the reliabilities of 2-engine and 4-engine planes. When the function plotted is positive, we prefer the 2-engine plane's reliability.

We observe that for probability of engine failure $p \geq \frac{1}{3}$, then a 2-engine plane is favored! We would have gotten the same result by solving the inequality at $-3 \cdot p^2 + 4 \cdot p - 1 \geq 0$ for p.

Exercise 5.5. Practicing with the binomial distribution

An urn contains 40 black and 10 red balls. You pick at random one ball from the urn, check its color, and after checking its color, you put it back in the urn. Let X be the number of black balls you pick from the urn in $n = 10$ tries. What is the probability that $X = 6$? What is the probability that $X \geq 9$?

Food for thought: why is it important in the previous exercise to put the ball back in the urn? What changes if we remove it from the urn before picking a new ball at random?

The hypergeometric distribution

This last question must have us thinking: what if the probability p of a success can change depending on the outcome of the previous experiment? More specifically, what if:

- we had a “population” of N items;
- with $K \leq N$ of them being successes (and hence the remaining $N - K$ are failures);
- and we picked a “sample” of n of them, all at once, without replacing them as we take them out?

While it sounds like a binomial distribution (we repeat an experiment of “picking items” n times and each time we observe a of success/failure with some probability), the fact that the probability does not seem known and depends on a larger population makes this slightly different.

We are officially defining the **hypergeometric distribution** as follows:

Definitions and Theorems 5.9. The hypergeometric distribution

The hypergeometric distribution with parameters integers N, K, n is the probability distribution of a discrete random variable which is allowed to take the values $\{0, 1, \ldots, n\}$, depending on the number of successes observed when performing n experiments, whose success/failure outcomes each correspond to a random choice of a success or a failure from a larger population containing initially K successes and $N - K$ failures.

The functions of interest for the hypergeometric distribution are:

- **Probability mass function**:

$$p(x) = \frac{\binom{K}{x} \cdot \binom{N-K}{n-x}}{\binom{N}{n}} \quad x = 0, 1, \ldots, n$$

- **Cumulative distribution function**:

$$F(x) = \sum_{y=0}^{\lfloor x \rfloor} \frac{\binom{K}{y} \cdot \binom{N-K}{n-y}}{\binom{N}{n}}.$$

And while it sounds like a totally new problem/distribution, we have indeed dealt with this before! Specifically, recall the example from Chapter 2:

Example 5.18. ~~Quality control~~ Example of a hypergeometric distribution!

A package is set to leave a factory and be sent to a retailer. The package contains 100 items. We already know that exactly 3 of the 100 items are defective. The quality control team over at the retailer works as follows: they select a sample of 6 items from the 100, and check them. If there are 0 defective items in the selected sample of 6, they accept the package and sell its contents; otherwise, they send the package back. What is the probability that the quality control rejects the package and sends it back?

When we first encountered this question, we used our counting rules and probability insights, to calculate the answer as $0.171 = 17.1\%.$ We now though change the setup to fit our distribution.

Check how this problem matches the setup of the hypergeometric distribution: $N = 100$ as the total population size; $K = 3$

defective ones in the larger population; $n = 6$ as the selected sample size. If X is the number of defective items picked in the sample, then the X follows a hypergeometric distribution and hence, we expectedly calculate

$$P(X > 0) = 1 - P(X = 0) = 1 - p(0) = 1 - \frac{\binom{3}{0} \cdot \binom{97}{6}}{\binom{100}{3}} = 1 - 0.829 = 0.171.$$

Exercise 5.6. Practicing with the hypergeometric distribution

An urn contains 40 black and 10 red balls. You pick at random a sample of five balls from the urn. Let X be the number of black balls in the sample. What is the probability that $X = 3$?

Note that in the earlier practice, we replaced the ball each time we took it out of the urn, resulting in a binomial distribution; on the other hand, picking a sample *without replacement* leads to a hypergeometric distribution. This is the big difference between the binomial and the hypergeometric distribution: **sampling with/without replacement**.

Example 5.19. Binomial or hypergeometric?

Let us revisit the urn and balls example again. We still 40 black and 10 red balls and pick at random a sample of five balls from the urn. Let X be the number of black balls in the sample.

We need one more detail:

- Are we taking 5 balls from the urn one-by-one, looking at each one's color, and putting it back in, before picking the next?
- Are we selecting 5 items from the urn and look at their colors in the end (i.e., we never put them back in)?

In the first case, we would have ourselves a binomial distribution, as the probability of success (picking a black ball) is always $p = \frac{40}{50} = 0.8.$ In the second case, we should use a hypergeometric distribution.

The main difference is that with replacement, the probability of picking an item stays the same throughout the experiment, no matter how many times it is repeated; without replacement, the probability changes with every selection. More formally, assume we have a large population of N items, K of which are successes. We pick n items from the population. The number of successes X we observe is:

1. distributed as a hypergeometric distribution, if we do not replace the items as we pick them (without replacement), or
2. distributed as a binomial distribution, if we replace the items as we pick them (with replacement).

The geometric distribution

Let us look at another extension of simple **Bernoulli** random variables. Earlier, during our discussion for binomially distributed random variables, we cared about the number of successes in a series of trials. How about the first success though? When[3] did it occur? Since we are talking about a series of experiments, this first success can occur at the first attempt, the second attempt, the third one, and so on. To model this, we introduce the **geometric distribution**:

Definitions and Theorems 5.10. The geometric distribution

The geometric distribution is the probability distribution of a discrete random variable consisting of the number of trials required until we observe the first success, when each success outcome appears independently with probability p.

Example 5.20. The geometric distribution

3. When here refers to the discrete number corresponding to the trial that led to that success. When dealing with the continuous nature of time itself, we will need a different (continuous) distribution..

- Assume again we are in possession of a fair coin. The number of trials needed to get the first Heads is a geometric distributed quantity.
    - On the other hand, if we knew we were only tossing the coin 4 times, and wanted to see how many Heads we had in the end, that would be a binomial distribution.
- A kid learning basketball is shooting free throws with a probability of scoring equal to $p = 25\%$. The number of free throws until the first one is scored is distributed following a geometric distribution.
    - Again, like earlier, if the kid had decided to attempt 5 free throws and was counting how many they would score, that would be a binomial distribution.

It is important that the experiments are all independent trials — that is, the outcome of one does not affect the other. Additionally, the probability of success p is the same throughout and known a priori. The functions of interest for the geometric distribution are:

- **Probability mass function**:
  $p(x) = (1-p)^{x-1} \cdot p \; x = 1, \ldots$
- **Cumulative distribution function**:
  $$F(x) = \sum_{y=0}^{\lfloor x \rfloor} (1-p)^{y-1} \cdot p.$$

Again, the derivation itself is left as a problem in this chapter's set of activities. In plain terms, the probability that we see our first success at the x-th trial, defined for $x = 1, \ldots$ (i.e., $x = 1$ means that the first success is observed at the first trial; $x = 2$ means that the first success is observed at the second trial; and so on) is equal to the product of $x - 1$ failure probabilities ($q^{x-1} = (1-p)^{x-1}$) followed by a single success (with probability p).

Example 5.21. Another geometric distribution

Let us go back to the kid learning to play basketball. As a reminder, they succeed in scoring a free throw with probability $p = 0.25$. What is the probability they score their first free throw in their fourth attempt?

Let X be the number of free throws until scoring the first one. We are looking for $P(X = 4) = (1-p)^3 \cdot p = 0.75^3 \cdot 0.25 = 0.1055.$

Exercise 5.7. Practicing with the geometric distribution

Assume we have a fair coin (that is, Heads and Tails both have 50% chance each). What is the probability:

- the first Heads appears in the 2nd coin toss?
- the first Heads appears in the 5th coin toss?

- the first Heads appears in the 10th coin toss?

How about if we are looking for the first success to happen in the first 3 coin tosses? Then, we would add the individual chances for all possible trials up to 3. Let us do an example with that:

Example 5.22. Learning to play basketball

Again, we go back to the child playing basketball and scoring a free throw with probability $p = 0.25$. What is the probability they score their first free throw in **one of their first four attempts**?

Let X be the number of free throws until scoring the first one. We are now looking for $P(X \leq 4) = \sum_{x=1}^{4} p(x) = P(X=1) + P(X=2) + P(X=3) + P(X=4)$.

We calculate them individually, to see how the geometric distribution behaves here:

- score in their first attempt: $x = 1$ with probability $P(X=1) = p(1) = 0.75^0 \cdot 0.25 = 0.25.$
  - As expected, this is the same as a simple Bernoulli random variable.
- score in their second attempt: $x = 2$ with probability

$$P(X=2)=p(2)=0.75^1\cdot 0.25=0.1875.$$

- score in their third attempt: $x=3$ with probability $$P(X=3)=p(3)=0.75^2\cdot 0.25=0.140625.$$
- score in their fourth attempt: $x=4$ with probability $$P(X=4)=p(4)=0.75^3\cdot 0.25=0.10546875.$$

Summing up, we get $P(X\leq 4)=0.68359375.$

We have now seen the similarities and differences between the binomial distribution and the geometric distribution. However, when looking at a single success in n tries/attempts, we may still be getting confused as to which one is the correct distribution. To discuss that, we focus on one more example from industrial engineering. In some instances, we may buy/order more material than we need, just in case. For example, when baking and needing 2 eggs, we may as well buy 3 in case we make a mistake. Or, when using Narcan on a person that has overdosed on an opioid, then we may have multiple doses available that we try until the person recovers. How should we model such random variables?

Example 5.23. Binomial or geometric? Or… both?

A professor is working on a research project alongside four research assistant students. The next step of the research project will take exactly 1 day and it can be difficult: it is successful with probability $p = 0.25$. The professor is considering two options:

1. Ask one student to do the next step. If the student is successful, then we are done; otherwise, the professor is going to ask the next student. Note that if all four students try the next step and all fail, then the step is failed and the professor is going to have to change the project progress.
2. Ask all students to work on the next step. Then, the next day, the professor will see if at least one student has been successful. Again, if all students fail the step, then the project will have to change to avoid that step.

What is the probability the step is successful in each of the two options?

This example establishes the differences between the binomial and the geometric distribution. Specifically, for each option we define the following random variables:

1. Let X be the number of students/days it takes to complete the step. Then we would be needing $P(X \leq 4)$.
2. Let Y be the number of students that have successfully completed the step out of the $n = 4$ we ask. Then we would be needing $P(Y \geq 1)$.

We make the argument here that X follows a geometric distribution with parameter $p = 0.25$. This is the case as we are interested in the number of tries until the first success. We also establish that Y follows a binomial distribution with parameters $n = 4, p = 0.25$. Again, this can be seen as we will try $n = 4$ times and see the number of successes at the end. Let us calculate these probabilities:

$$\begin{aligned} P(X \leq 4) &= P(X=1) + P(X=2) + P(X=3) + P(X=4) = \\ &= p + (1-p)p + (1-p)^2 p + (1-p)^3 p = \\ &= 0.25 + 0.75 \cdot 0.25 + 0.75^2 \cdot 0.25 + 0.75^3 \cdot 0.25 = \\ &= 0.68359375. \\ P(Y \geq 1) &= P(Y=1) + P(Y=2) + P(Y=3) + P(Y=4) = \\ &= 1 - P(Y=0) = 1 - \binom{4}{0} p^0 (1-p)^4 = \\ &= 1 - 0.25^0 \cdot 0.75^4 = 0.68359375. \end{aligned}$$

We see that the two probabilities (despite the differences in the approach the professor follows) are the same! Successfully finishing the next step has the same probability despite using the resources available one-by-one, or altogether. However, there seems to be some difference in the *time it takes* to complete the step. The first approach will take at least one day and up to four; the second approach will take exactly one day! There also seems to be some difference in the *number of resources* we use. The first approach may only use a single resource at the best case scenario, or up to all four in the worst case, whereas the second approach will definitely use up all of our resources.

We may need to revisit this example later to do a full study on the pros and cons of each approach...

The Poisson distribution

Usually, we motivate the distribution with an example and then proceed to derive what we need. Here, we do the opposite. Despite the ubiquity of Poisson random variables around us (number of goals scored by a soccer player in a season! number of shark attacks of the coast of California during the summer months! number of earthquakes in Greece per decade!), we will motivate the Poisson distribution from a mathematical perspective. Recall the binomial distribution and its probability mass function: $p(x) = \binom{n}{x} \cdot p^x \cdot (1-p)^{n-x}$. Assume that we try way too many experiments (in essence let $n \to \infty$) and define λ as the number of successes we get: that is, $p = \frac{\lambda}{n}$. Let's replace this in the probability mass function itself:

$$p(x) = \binom{n}{x} \cdot p^x \cdot (1-p)^{n-x}$$
$$= \frac{n!}{x! \cdot (n-x)!} \cdot \left(\frac{\lambda}{n}\right)^x \cdot \left(1 - \frac{\lambda}{n}\right)^{n-x}.$$

Let us now employ some of the cool limit properties that we know as $n \to \infty$:

$$\begin{aligned}\lim_{n\to\infty} P(X=x) &= \lim_{n\to\infty} \binom{n}{x} \cdot p^x \cdot (1-p)^{n-x} = \\ &= \lim_{n\to\infty} \frac{n!}{x! \cdot (n-x)!} \cdot \left(\frac{\lambda}{n}\right)^x \cdot \left(1-\frac{\lambda}{n}\right)^{n-x} = \\ &= \lim_{n\to\infty} \frac{\lambda^x}{x!} \cdot \underbrace{\frac{n!}{(n-x)! \cdot n^x}}_{1} \cdot \underbrace{\left(1-\frac{\lambda}{n}\right)^n}_{e^{-\lambda}} \cdot \underbrace{\left(1-\frac{\lambda}{n}\right)^{-x}}_{1} = \\ &= e^{-\lambda}\frac{\lambda^x}{x!}\end{aligned}$$

Definitions and Theorems 5.11. The Poisson distribution

A discrete random variable X taking nonnegative integer values $0, 1, 2, ...$ is distributed as a **Poisson random variable** with parameter (rate) $\lambda > 0$ if it follows a probability mass function

$$p(x) = P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}.$$

Poisson random variables are ubiquitous and have been used to model a wide, *wide* array of diverse applications:

Example 5.24. The Poisson distribution

- The number of phone calls a customer call center receives every day.
  - If we assume that on average they receive $\lambda = 3$ calls per minute, then the number of phone calls received per hour, per day, in the next 20 minutes... are all Poisson distributed random variables!
- The number of shark attacks off the California coast every year is a Poisson distributed quantity.
- The number of home runs in a baseball series.
- The number of patients arriving in an emergency department during the nigh hours.
- The number of website requests in the next day.
- The number of typos in a book.

As can be seen from the examples, Poisson distributed random variables are commonly used to model the number of events that happen in a given interval. Poisson distributed random variables need to satisfy three main conditions:

1. **independence**: an event happening should not affect the rate with which more events happen.
2. **homogeneity**: the rate with which events happen is constant.
3. **no two events can occur at exactly the same time**: Instead there is a small interval of time that separates two consecutive events.

If these conditions are valid, and we are provided with a rate $\lambda > 0$, then the number of events/occurrences/observations in a given time period/space follows a Poisson distribution.

Exercise 5.8. Testing the conditions

Is it fair to assume that typos appearing in a book/notes follow a Poisson distribution? Why/Why not?

The functions of interest for the Poisson distribution are:

- **Probability mass function**:
$$p(x) = P\left(X = x\right) = e^{-\lambda}\frac{\lambda^x}{x!},\ x = 0, 1, \ldots.$$
- **Cumulative distribution function**:
$$F(x) = \sum_{y=0}^{\lfloor x \rfloor} p(y) = \sum_{y=0}^{\lfloor x \rfloor} e^{-\lambda}\frac{\lambda^y}{y!}.$$

Example 5.25. Earthquake probabilities

We will model our motivating example with predicting the probability of an earthquake in the Kanto region of Japan using the Poisson distribution. We need to estimate λ, the rate of events. From the data, we are told that there have been 5 big earthquakes over the last 135 years, and hence we "estimate"

$\lambda = \dfrac{5}{135} = 0.037$ earthquakes per year. We are interested in $P(X = 1)$ for the probability of one big earthquake in the next year.

Hence, we use the pmf for a Poisson random variable with rate $\lambda = 0.037$ and get:

$$P(X = 1) = e^{-0.037} \cdot \frac{0.037^1}{1!} = 0.0357 = 3.57\%.$$

What if we were interested in the probability of observing one large earthquake in the next decade?

Example 5.26. Earthquake probabilities: adapting the rate

We already have that the rate is

$$\lambda = \frac{5}{135} = 0.037 \text{ earthquakes per year.}$$

Now, though, let Y be the random variable denoting the number of earthquakes in the next decade. We need to

adapt the rate to fit the new time period. We do so by multiplying λ by 10 so as to much the new time period. Our new rate is now $\lambda = 0.37 \text{ earthquakes per decade}$. Finally, we get

$$P(Y = 1) = e^{-0.37} \cdot \frac{0.37^1}{1!} = 0.2556 = 25.56\%.$$

In many cases, we are more interested in **at least one event happening**, rather than exactly one event happening. In these instances, we need to calculate $P(X \geq 1) = \sum_{x=1}^{\infty} e^{-\lambda} \frac{\lambda^x}{x!}$, since the number of events can be as large as we want to. Instead, we typically calculate $P(X \geq x)$ as $1 - P(X < x)$.

Let us use that to calculate the probability of *at least one earthquake* in the next year/decade:

Example 5.27. Earthquake probabilities: at least one earthquake

As a reminder we have two Poisson random variables:

1. X denoting the number of big earthquakes in the next year with rate $\lambda = 0.037$.
2. Y denoting the number of big earthquakes in the next decade with rate $\lambda = 0.37$.

We now calculate the two probabilities of at least one event (big earthquake) in some period (next year/decade).

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-0.037} \cdot \frac{0.037^0}{0!} =$$
$$= 1 - 0.96367613534 = 0.0363 = 3.63\%.$$
$$P(Y \geq 1) = 1 - P(Y = 0) = 1 - e^{-0.37} \cdot \frac{0.37^0}{0!} =$$
$$= 1 - 0.6907 = 0.3093 = 30.93\%.$$

Expectedly, the probability of at least one earthquake is bigger during the next decade than during the next year (as the next year is included in the next decade).

Exercise 5.9. Practicing with the Poisson distribution

Assume typos appear in my notes following a Poisson distribution with a rate of $\lambda = 0.5/page.$ What is the probability that no typos exist in the first page? What is the probability that there exist more than 1 typo in the first 10 pages?

Plotting the Poisson distribution also proves an interesting endeavor. Let's remember that all distributions we are discussing are discrete: hence, we will simply plot each point and then connect the points with a line. For example, in the next plot, we show the case for $\lambda = 1$ and how we would connect the different data points.

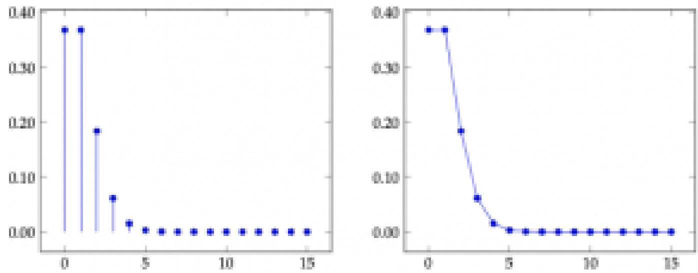

Figure 5.3. A plot that shows the probability values for a Poisson *random variable with rate [latex]\lambda=1[/latex].*

We can do the same for different rates (not necessarily only integer!) as in the next three figures ($\lambda = 2, 5, 10$):

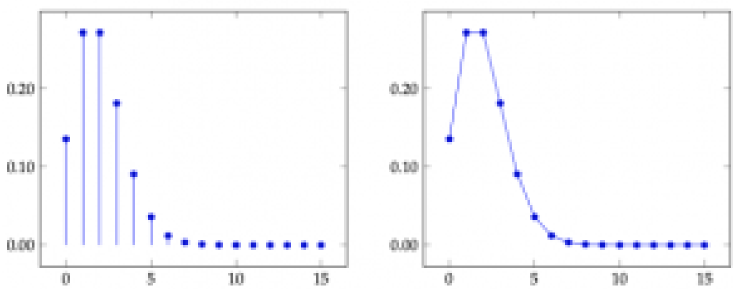

Figure 5.4. A plot that shows the probability values for a Poisson *random variable with rate [latex]\lambda=2[/latex].*

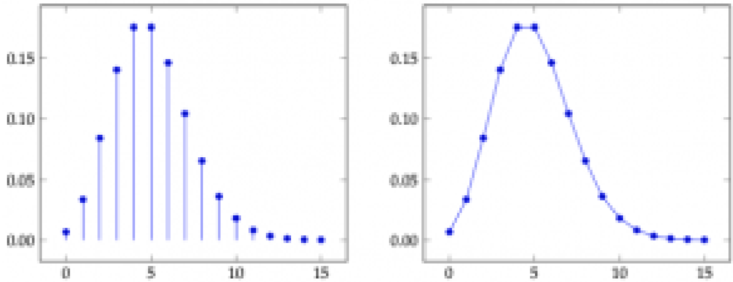

Figure 5.5. A plot that shows the probability values for a Poisson random variable with rate [latex]\lambda=5[/latex].

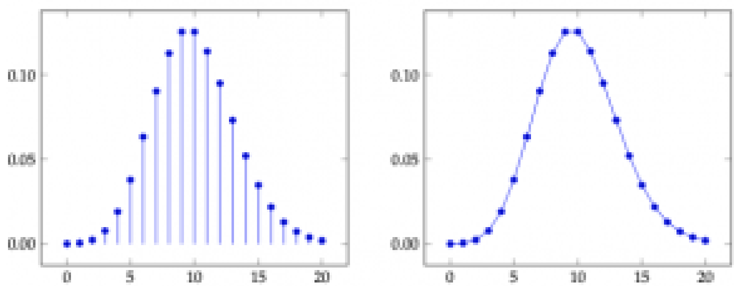

Figure 5.6. A plot that shows the probability values for a Poisson random variable with rate [latex]\lambda=10[/latex].

Finally, take a look at what happens as the rate becomes bigger and bigger… In the next figure, we show $\lambda = 100$:

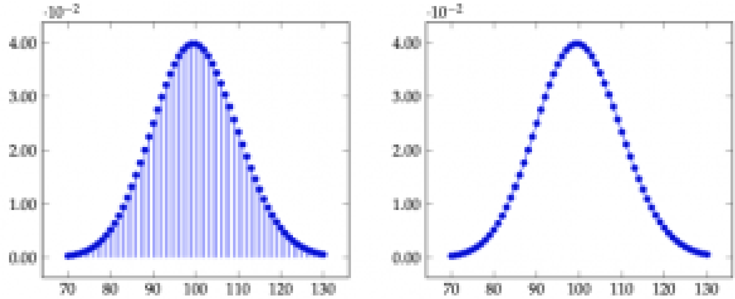

Figure 5.7. The Poisson distribution for [latex]\lambda=100.[/latex]

This unique, bell-curve shape that is symmetrical around the rate is quite interesting…

The uniform distribution

We leave the *simplest* for last. The uniform distribution is the plainest form of a discrete random variable: a random variable with n outcomes, each of which has the same probability of happening. The setup is the following: all n outcomes are **equally probable** (equivalently, each of the outcomes happens with probability $\frac{1}{n}$).

Definitions and Theorems 5.12. The uniform distribution

The **uniform distribution** is the probability distribution of a discrete random variable with n different outcomes

$x_i, i = 1, \ldots, n$ when each outcome is equally probable and has probability of happening $\frac{1}{n}$.

Note that the probability can be zero when $n \longrightarrow \infty$. Also, in a special case, a uniformly distributed random variable takes on integer values in $[a, b]$. In that special case, the possible outcomes of the random variable would be $\{a, a+1, \ldots, b\}$. The functions of interest are:

- **Probability mass function**: $p(x) = \frac{1}{b-a+1}, \ x = a, a+1, \ldots, b$
- **Cumulative distribution function**: $F(x) = \frac{\lfloor x \rfloor - a + 1}{b-a+1}$

Example 5.28. A diamond cutting facility

A demanding customer has shown up in a diamond cutting facility and has asked for **2 custom-made fine-cut diamond castings**. They are willing to buy exactly 2 of those, as long as they are of high quality... Diamond cutting is an expensive process, but you can make a lot of money out of it, and hence decide to take on the order. You plan to buy enough material for $Q = 4$ castings, just to be safe. Assuming that diamond cutting is a purely random process and all outcomes (producing $x = 0, 1, \ldots, Q$ high quality diamonds) are equally probable. **What is the probability you satisfy your customer?**

Let X be the number of high-quality castings obtained out of $Q = 4$ produced. This is a uniformly distributed quantity with equally probable outcomes $0, 1, 2, 3, 4$; that is, each of these outcomes has a $\dfrac{1}{Q+1} = \dfrac{1}{5} = 0.2$ chance of happening. Hence, the probability we satisfy this demanding customer is:

$$P(X \geq 2) = P(X = 2) + P(X = 3) + P(X = 4) = \frac{3}{5}.$$

Note that we could have also answered this question using the cumulative distribution function (replacing $a = 0, Q = b$):

$$P(X \geq 2) = 1 - P(X \leq 1) = 1 - F(1) = 1 - \frac{1 - 0 + 1}{4 - 0 + 1} = 1 - \frac{2}{5} = \frac{3}{5}.$$

An interesting question would be how big should Q be in order for us to satisfy the customer with a certain, given probability…

Activities

Activity 1: Basic discrete random variable questions

A movie magazine gives each movie they review a score from 0 to 4 stars — no movie is allowed to have a fractional number of stars though. Over the years, the magazine has observed that the movies get a score that is distributed as a discrete random variable X with probability mass function

$$p(x) = \begin{cases} \frac{2x+2}{c}, & x = 0, 1, 2, 3, 4 \\ 0, & \text{otherwise.} \end{cases}$$

Problem 1: Valid probability mass functions

What is the value of c for which $p(x)$ is a valid probability mass function?

Food for thought 5.1.

This property can prove very useful. When we have an idea for the probability mass function of a *discrete* random variable, but we are missing part of it (e.g., missing a coefficient's value), then we can sum over all possible values of the probability mass function and it should equate to 1. This summation only holds for discrete random variables. We will discuss more about what we anticipate for continuous random variables that are allowed to take any real value within a range in the next chapter.

Problem 2: Constructing cumulative distribution functions

Using the same probability mass function $p(x)$ that you were given in Problem 1 (replace the value for c that you calculated), what is the cumulative distribution function (cdf) of discrete random variable X? We could write it as a summation! That is, $F(x) = \sum_{y \leq x} p(y)$.

\begin{answerbox}{green!80!white}{Answer to Problem 2.}
\begin{itemize}
\item[] $P(X\leq 2)=$
\end{itemize}
\vspace{70pt}
\end{answerbox}

Problem 3: Calculating probabilities 1

Use the cdf from Problem 2 to calculate the probability that a movie gets up to 2 stars (including 2 stars) out of 4: what is $P(X \leq 2)$?

Problem 4: Calculating probabilities 2

For the same distribution, calculate the following probabilities:[4]

- $P(1 \leq X \leq 3) =$
- $P(1 < X < 3) =$
- $P(1 < X \leq 3) =$

4. For discrete random variables, we can calculate the probability of X being between a and b in many different ways: 1) $P(a \leq X \leq b) = \sum_{x=a}^{b} p(x)$. 2. $P(a \leq X \leq b) = P(a - 1 < X \leq b) = F(b) - F(a - 1)$. Just be careful with $\leq$ vs. $<$.

Activity 2: Binomial, geometric, or hypergeometric?

Problem 5

A foundry has received an order for **5 castings**, made from precious metals. Each casting produced is of high quality (and hence can be sold to the customer) with probability 0.97. All castings are produced independently. You decide to schedule 6 castings for production, knowing full well that the customer only wants 5. **What is the probability you get more than or equal to 5 high quality castings?** To help you get started, let X be the number of high quality castings produced out of the 6 you tried to produce. What is X distributed as? What is $P(X \geq 5)$?

Problem 6

The foundry from earlier has received the same order for **5 castings**. However, instead of producing new ones, they use a batch of older castings already produced. The batch contains 100 castings, 97 of whom are of high quality. They decide to pick 6 of them at random and give them to the customer. **What is the probability that the customer gets more than or equal to 5 high quality castings in the sample of 6 they receive?** Once again, to get you started, let X be the number of high quality castings you get in the group of 6 castings you pick from the batch. What is X distributed as in this case?

Problem 7

A foundry has now received an order for **1 casting**, made from precious metals. Like earlier, each casting produced is of high quality with probability 0.97, independently of any other castings produced. We decide to produce castings one-by-one until the first one of high quality. **What is the probability you get the first high quality casting within the first two castings produced?** Again, as a hint, let X be the number of castings produced until the first one of high quality is observed. What is X distributed as? What is $P(X \leq 2)$?

Problem 8

Let us generalize! A foundry has now received an order for **5 castings**, made from precious metals: again, each casting is of high quality with probability 0.97, independently of others. We decide to produce castings one-by-one until we finally get 5 castings to give to the customer. Derive the probability mass function for this new distribution, which seems to combine the binomial and the geometric!

Activity 3: Overbooking

An airline has found that 10% of people buying a first class ticket do not show up to travel on the day of their flight (and independent of one another). This is why the company has decided to **sell 32 first class tickets for a flight that contains 30 first class seats**. If more than 30 first class ticket holders show up, then the company has to pay a penalty.

Problem 9

What is the probability the company does not pay a penalty for a flight? Equivalently, what is the probability that at most 30 first class passengers show up, even though 32 tickets were booked?

Problem 10

A flight just departed without having to pay a penalty. What is the probability the flight departed with at least one empty first class seat?

Activity 4: Deriving the binomial distribution probability mass function

In this activity, we will use a small example to derive the formula for the probability mass function of the binomial distribution. Hint: as a reminder, we should finally obtain that

$$P(X = x) = \binom{n}{x} p^x \cdot (1-p)^{n-x}.$$

A group of students is participating in a seminar class that requires each student to take a one-question exam in the end of the semester. The class has been historically difficult, and 70% of the students answer the question correctly and pass, while the remaining 30% fails. This semester, the class has 3 students.

Problem 11

This seminar class could end up with:

- all students passing.
- 2 students passing.
- 1 student passing.
- no students passing.

We would like to calculate the probability of each of them. Before that, though, enumerate all the possible scenarios we may have. As a hint, you may assume that the students are given a grade of P or F, then you could have PPP (everyone passes) or PFP (the first and third students pass, the second one fails), among others. **How many scenarios are there in total?** Assume that "order matters", that is we want to be able to distinguish between the first, the second, and the third student.

Problem 12

Assuming that the performance of one student does not affect the performance of any other student, what is the probability that every student passes and what is the probability that every student fails? That is, what is $P(PPP)$ and what is $P(FFF)$?

Problem 13

Under the same independence assumption from Problem 12, what is the probability that the first student is the only one that passes? What is the probability that the second student is the only one that passes? And what is the probability that the third student is the only one that passes? How would that calculation change if instead of 3 students, we had n students?

Problem 14

Based on your answers in Problem 13, what is the probability that **exactly one** student passes? Could we… simply add the three probabilities from Problem 13? Why can we do that?

Problem 15

Can we generalize the result in Problem 14? If we had n students, and we wanted to have exactly $x \leq n$ of them pass, how many probabilities would we have to sum up? Recall that in Problem 14, you had to add 3 of the probabilities, as there were 3 possible scenarios. In general, though, you would want to enumerate all possible *combinations* of x elements from a total of n elements…

Activity 5: Selling online

You have decided to sell an item online. You have given no price in your ad: instead, you ask the interested people reading the ad to submit an offer. Assume that their offers will be a whole dollar amount (no fractional offers), such as $37 or $73, uniformly distributed between 0 and 99 dollars. You would like to make at least $75 from the item, so any offer that is greater than or equal to $75 dollars would be acceptable. In addition, you receive *exactly* one offer every day since posting the ad.

Problem 16

What is the probability a random offer you receive is higher than or equal to $75?

Problem 17

What is the probability you sell the item on exactly the 3rd day? What distribution does this follow?

Problem 18

What is the probability you sell the item in one of the first 3 days?

Problem 19

You decide not to look at your email for 10 days. When opening your email again to check on the (10) offers you have received, what is the probability that at least one of them offers you $75 or more? What distribution does the number of "successful" offers follow? \sidenote{Hint: consider as if you get 10 "tries" and need at least one of them be a "success"...}

Problem 20

You decide not to look at your email for 3 days. What is the probability that you sell the item on the 6th day, given that in the first 3 days you receive no satisfying offers? Contrast this answer to your answer in Problem 17. As a hint, consider using the formula for conditional probabilities and the geometric distribution probability mass function.

Food for thought 5.2.

Will you look at this. Knowing that the first three days are failed offers does not change (either increase or decrease) the probability of getting a success in the next three days! This is observed because our answers in Problems 17 and 20 are the same. We say that the geometric distribution is **memoryless**.

What is more, the **geometric distribution is the only discrete distribution that is memoryless**.

Activity 6: The Poisson distribution

Problem 21

A transportation engineer is collecting data during rush hour on an intersection. They have noticed that the number of vehicles during rush hour follows a Poisson distribution with a rate of 7.5 per minute. **What is the probability that exactly 10 vehicles pass through the intersection in the next minute?**

Problem 22

What is the probability that exactly 10 vehicles pass through the intersection **in the next 2 minutes**?

> *Food for thought 5.3.*
>
> Note how our first order of business with a Poisson distributed random variable is to convert the rate λ in the necessary time units. In Problem 21, we needed $\lambda = 7.5$ per minute, whereas in Problem 22, we had to change this to $\lambda = 15$ every 2 minutes.

Problem 23

During a pandemic, the number of patients in need of a special treatment is Poisson distributed with a rate λ of 1 patient every 4 hours. Assuming a hospital can offer 4 of those treatments per day (24 hours), what is the probability the hospital runs out of available treatments during a day (24 hours) of operations? Assume this means that at least 5 patients show up in a day.

Problem 24

Scientists in the midwestern states have observed an increase in the frequency of floods and river overflows. The most recent estimate is that a devastating flood in some location in the midwest may happen with a rate of 1 every 50 years, whereas earlier estimates had placed that number in a rate of 1 every 200 years. Quantify the change in probability of a devastating flood in the midwest between the two estimates. What is the probability of seeing at least one devastating flood in the next year with the earlier and the more recent estimates?

Problem 25

We saw in class the probability mass function for a Poisson distributed random variable with rate λ: it is $p(x) = P(X = x) = e^{-\lambda} \cdot \frac{\lambda^x}{x!}$. Assume that $\lambda = 3$ per year. What is the probability that there will be no events in the next year? Can you say that this means that the next event will happen more than a year from now? Let T be the time of the next event: what is $P(T > 1 \text{ year})$?

> *Food for thought 5.4.*
>
> So, the **time to the next event** is related to the number of events.. Specifically, if we have $X = 0$ events during time t, this means that the time to the

next event T has to be greater than t. Mathematically:
$X = 0 \implies T > t$

But, <u>time is of a continuous nature</u> whereas the number of events is discrete (has to be integer). Does that mean there is a relationship between a Poisson distributed random variable and a continuous (real-valued) random variable? *Interesting*: let's keep that in mind for our next chapter on continuous random variables.

Activity 7: "Truncated" Poisson distributed random variables

Problem 26

Assume that X is a Poisson-distributed discrete random variable with rate $\lambda = 2$ per 30 minutes. However, if X becomes bigger than or equal to 3, it stays equal to 3. Practically, X follows a Poisson distribution for $x \leq 3$, but is equal to 3 whenever $x > 3$. **What is the probability mass function for this random variable?** In essence, you are asked to compute the probability mass function of random variable $Y = \min\{X, 3\}$. Note that the values that Y is allowed to take are 0, 1, 2, and 3.

Problem 27

Similarly, consider the opposite problem in which discrete random variable X is Poisson distributed with rate $\lambda = 2$, but it now is equal to 3 whenever $x \leq 3$. **What is the probability mass function of this random variable?** In other words, you are now asked to compute the probability mass function of $Y = \max\{X, 3\}$.

6. Continuous random variables

Theory

Chapter objectives

In this chapter, our goal is to **introduce, define, and use continuous random variables.**

Learning outcomes: Continuous random variables

After this chapter, you will be able to:

- model real-life applications using continuous random variables.
- give examples of continuous random variables and differentiate between discrete and continuous random variables.
- use and interpret probability density functions and cumulative distribution functions for continuous random variable distributions.
- give examples of at least four common different continuous distributions.
- recall when to and how to use:

- exponentially distributed random variables.
- Erlang and Gamma distributed random variables.
- normally distributed random variables.
- uniformly distributed random variables.

- define the memorylessness property and apply it to exponentially distributed random variables.
- use Poisson random variables, exponential random variables, and Erlang/Gamma random variables, and provide examples of their relationship.
- Use the standard normal distribution table to calculate probabilities of normally distributed random variables.

Chapter questions

1. What is the **probability density function** (pdf) f(x) of a continuous random variable?

   What is P(X=x) for a continuous random variable?
2. What is the **cumulative density function** (cdf)

   $$F(x) = \int_{-\infty}^{x} f(y)dy?$$

   - Remember! f(x) does **not** reveal probability!

     The probability that a continuous random variable is exactly equal to a value is zero!

3. What is the difference between a discrete uniform distribution and a **continuous uniform distribution**?
4. When provided with a *rate of events* in an **exponential distribution**, what is the probability the next event happens within some time t?
   - pdf for the exponential: $f(t) = \lambda e^{-\lambda t}$.
   - cdf for the exponential: $F(t) = P(T \le t) = 1 - e^{-\lambda t}$.
5. What is the memorylessness property?
   - How does it apply to the exponential distribution?
6. What are the differences between **exponential**, **Erlang**, and **Poisson**?Given a rate of events λ:
   - time until the next event: **exponentially distributed** with rate λ.
   - time until the k-th event: **Erlang distributed** with parameters k, λ.
   - number of events within time t: **Poisson distributed** with rate λt.
7. What parameters do I need to set up a **normally distributed** random variable?
8. What does it mean to "convert to a z value" or check the "z-table"?
9. How do I use the z-table to answer probability questions for normally distributed random variables?

Motivating examples

Time to perform a task

When studying the operations at a manufacturing facility, we observed that a specific task is completed in random time; that said, when asking the employee about it, they nonchalantly report their time to be “almost always 25-30 seconds.”

Even though they may think so, when actually timing their operations, we may see some faster/slower iterations of the same task. For example, on Monday, we timed them at 27.4 seconds; on Tuesday, they were done in 23.31 seconds; on Wednesday, it took them 31.59 seconds!

While the sentiment of “25-30 seconds” is there, this is clearly a random variable. Moreover, it is a **continuous random variable**, as time is a continuous quantity!

The parent of “time studies” is widely considered to be Frederick W. Taylor, the author of “The Principles of Scientific Management.” The book is viewed as the foundation of scientific management and, hence, industrial engineering. In its very basic form, a time study includes three steps:

1. Task observation.
2. Time recording.
3. Data analysis.

We will return to this motivating example shortly.

Big in Japan

In our earlier example, we discussed the probability of seeing a certain number of earthquakes in the Kanto region of Tokyo in

Japan. **What if, though, we are interested in the timing of the next earthquake?** How would we go about modeling this using continuous random variables?

Congratulations, you are our 100,000th customer!

Last time, we discussed about the probability of the next customer arriving in the next hour, next day, next year. What about the probability of the 10th customer arriving at a certain time? Or, consider a printer that starts to fail and needs maintenance after the 1000th job: what is the probability these failures start happening a month from now?

Food poisoning and how to avoid it

A chef is using a new thermometer to tell whether certain foods have been adequately cooked. For example, chicken has to be at an internal temperature of 165 Fahrenheit or more to be adequately cooked; otherwise, we run the risk of salmonella. The restaurant wants to take *no chances*! The chef, then, takes a look at the temperature reading at the thermometer and sees 166. What is the probability that the chicken is adequately cooked, if we assume that the thermometer is right within a margin of error?

Continuous random variables

A continuous random variable is fundamentally different than a discrete in one major aspect: it is allowed to take any value within a range, and hence there are infinite possible outcomes. We remind here the definition of a continuous random variable, that we first saw during the previous chapter:

Definitions and Theorems 6.1. Continuous random variables

Continuous random variables are random variables that are allowed to take an infinite set of any real value within a range.

While we have discussed the differences between discrete and continuous random variables in the previous chapter, we offer one more (indicative!) example of why these two are so different.

Example 6.1. Continuous vs. discrete random variables

Guessing which number I am thinking between 0 and 10 is a tricky proposition.

- If asked to do so in integer numbers (that is, 0 or 1 or 2…) then it is difficult to guess correctly, but not nearly impossible: we'd get a probability of 1 over 11 or a little more than 9%.
- On the other hand, when asked to do so with *any* number… that probability drops to zero!

Continuous random variables, like their discrete random variable counterparts, are described by two distribution functions: the

probability density function (often stylized as pdf) and the **cumulative distribution function** (cdf). We discuss those next.

The probability density function

Contrary to what we did for discrete random variables, we motivate the existence and need for a probability density function using the time study motivation example from the beginning of the chapter. As a reminder:

Example 6.2. Time study

A specific task in a manufacturing facility is completed in time that is random; while the employee performing it claims the task takes "almost always 25-30 seconds," our own small time study showed numbers that are outside the $[25, 30]$ range: we timed them at 27.4 seconds, 23.31 seconds, 31.59 seconds, …

Say we collected $n = 100$ such times and we decide to plot them in a histogram: a histogram is a visual device that collects "similar" times together in a bar. The tallest the bar, the more often that range of times appears. Do not worry about this definition right now; it belongs to a different topic, that of presenting and summarizing data.
In our case, the histogram looks as follows, where we collected together all the times between 0-5 seconds; 5-10 seconds; 10-15 seconds; and so on:

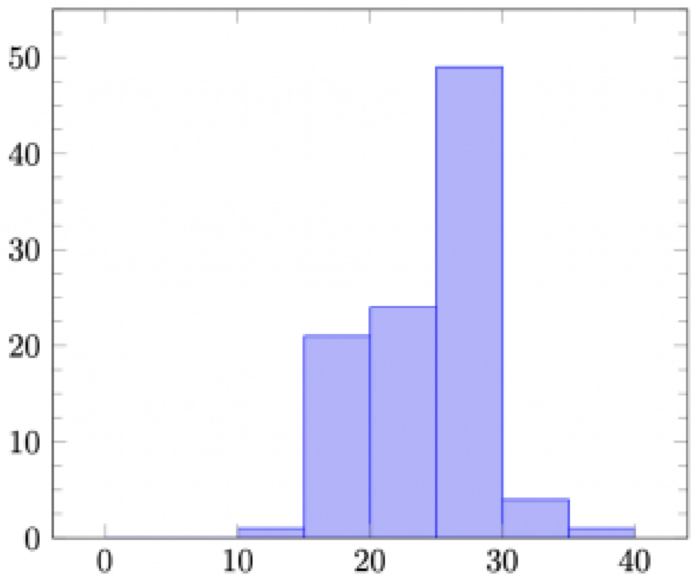

Figure 6.1. A histogram of the times observed. The intervals are of 5 seconds each.

Now, let us try to make the length of each interval smaller: that is, let us plot 0-1 seconds, 1-2 seconds, and so on:

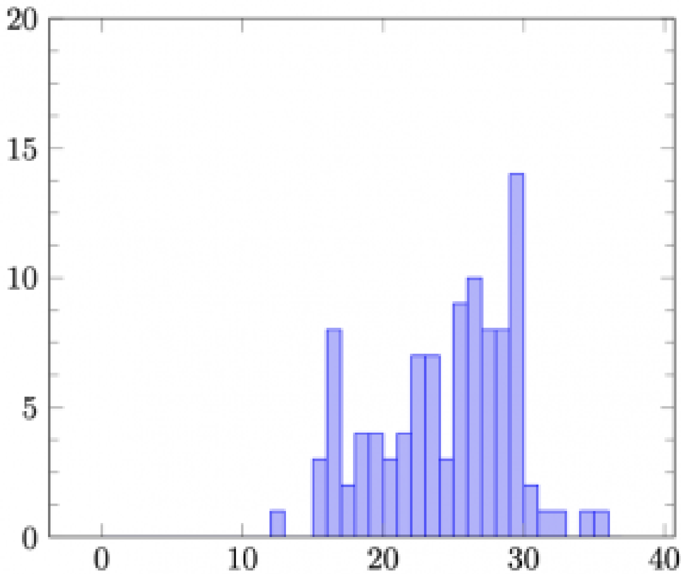

Figure 6.2. Another histogram of the times observed. The intervals are now 1 second each.

If we were to continue this process, and only connect the “tops” of the bars, we would get a continuous curve, similar to the following:

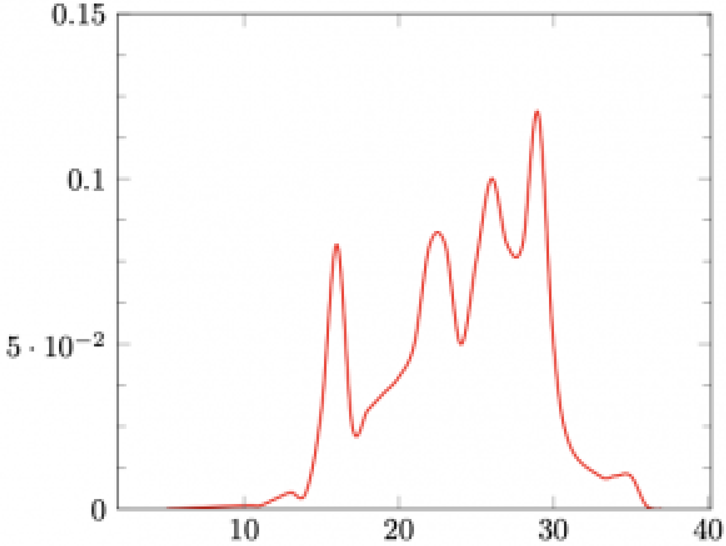

Figure 6.3. The representation of observed times, if we were able to create a continuous mapping.

This curve is the **probability density function**: we can think of the probability that random variable X falls within some range, as the area under this curve in that range. For example, the probability that X is within the range between 25 and 30 seconds (as the worker was anticipating) in the above example can be visually computed as:

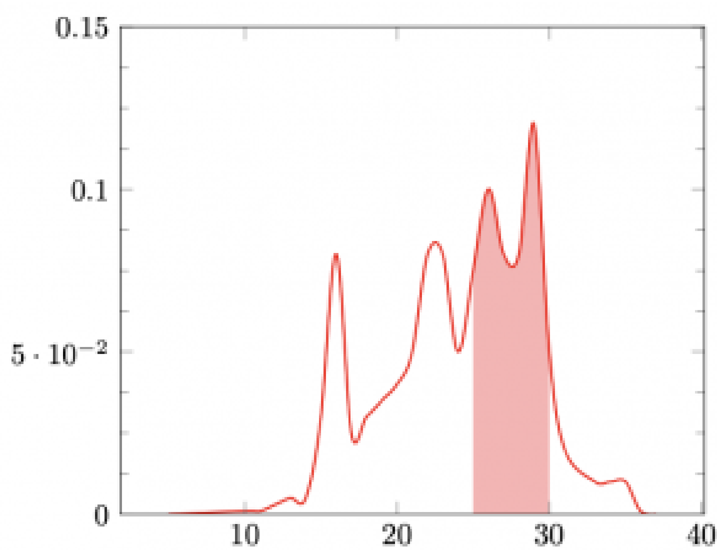

Figure 6.4. A visual representation of the probability that the random time is between 25 and 30 seconds, based solely on our continuous curve from the observations.

Based on that, we may interpret that the area under the *whole curve* has to equal 1, as it is the probability that random variable X takes any value in the sample space! We formalize everything now. Let X be a continuous random variable. Then:

Definitions and Theorems 6.2. Probability density function

We define the **probability density function** (*pdf*) $f(x)$ of a continuous random variable X as the function whose curve presents the relative likelihood that random variable

X takes a value around that location. Mathematically, the area under the curve of the function reveals the probability that random variable X falls within a range of values. A probability density function over sample space S satisfies three rules:

1. $f(x) \geq 0$, for all $x \in S$.
2. $\int\limits_S f(x)dx = 1$.
3. $P(X \in A) = \int\limits_A f(x)dx$.

Due to the fact that every continuous random variable X can be described from its probability density function $f(x)$, it may be useful to redefine continuous random variables as follows:

Definitions and Theorems 6.3. Continuous random variables

A random variable X is a **continuous random variable** if it can take uncountably many values such that there exists some function $f(x)$ called a **probability density function** defined over real values $(-\infty, +\infty)$ such that:

1. $f(x) \geq 0$;

2. $\displaystyle\int_{-\infty}^{+\infty} f(x)dx = 1$;

3. $\displaystyle P(X \in A) = \int_A f(x)dx$.

An immediate consequence of these two definitions is the following. For any continuous random variable X with probability density function $f(x)$, we must have that the probability it takes *exactly* some value x is: $P(X = x) = \int_0^0 f(x)dx = 0$. The probability is indeed zero! This is because any integral with the same limits corresponds to a zero area under the curve of $f(x)$. Practically: the probability that a random variable that is allowed to take uncountably many possible values within a continuous range of values takes on a specific value is indeed zero!

This last note probably changes the way we need to discuss continuous probabilities. What if, instead of asking for the probability that continuous random variable X is exactly equal to some value x, we focus on the probability that continuous random variable X *belongs to some interval of values*?

Example 6.3. Changing the way we think about probabilities

So, for a continuous random variable, instead of the probability that:

- the average temperature tomorrow is exactly 78.3 Fahrenheit ($P(X = 78.3 \text{ F}) = 0$);
- the next bus passes in exactly 3 minutes and 25 seconds ($P(T = 205 \text{ seconds}) = 0$);
- the error of an ammeter (used to measure the current in a circuit) is exactly 0.1 A ($P(E = 0.1 \text{ A}) = 0$);

we may ask for the probability that:

- the average temperature tomorrow is between 78 and 79 Fahrenheit (that is, $P(78 \text{ F} \leq X \leq 79 \text{ F})$);
- the next bus passes between 3 and 4 minutes from now (that is, $P(180 \text{ seconds} \leq T \leq 240 \text{ seconds})$);
- the error of an ammeter is within 0.1 A (that is, $P(0 \text{ A} \leq E \leq 0.1 \text{ A})$).

Hence, we integrate the pdf to calculate probability. Let us check this with an example:

Example 6.4. Using the probability density function

What is the probability that random variable X, which can take any real value with pdf $f(x)$ is between 0 and 10?

That would be: $P\left(0 \leq X \leq 10\right) = \int_{0}^{10} f(x)dx.$

Due to the continuous nature of random variable X, and due to the fact that $P(X = x) = 0$, for any value x, we also get that: $P\left(0 \leq X \leq 10\right) = P\left(0 < X < 10\right) = P\left(0 < X \leq 10\right) = P\left(0 \leq X < 10\right).$

Exercise 6.1. Practicing with the probability density function

Assume that continuous random variable X is distributed with probability density function $f(x)$ in $[0, \infty)$ (that is, it is a nonnegative real number). What is:

- the probability that X lies between 2 and 5?
- the probability that X is less than or equal to 5?
- the probability that X is strictly less than 5? Is this different than the previous calculation?
- the probability that X is exactly equal to 5?

The cumulative distribution function

Due to the nature of continuous random variables, the **cumulative distribution function** takes on an even more important role. As a reminder, here is the definition for a discrete random variable:

Definitions and Theorems 6.4. Cumulative distribution function (discrete random variables)

We define the **cumulative distribution function** (*cdf*) $F(x)$ of a discrete random variable X as the probability that the random variable takes a value *up to* a specific value x. Mathematically, we have $F(x) = P(X \leq x)$.

In other words, the probability that random variable X is smaller than or equal to a given value x. Equivalently, we may sum up all of the individual probabilities for all outcomes in the sample space that are smaller than or equal to the given value x:

$$F(x) = P(X \leq x) = \sum_{y: y \leq x} P(X = y) = \sum_{y: y \leq x} p(y).$$

The idea behind the cumulative distribution function remains the same: "the probability that random variable X is smaller than or equal to some specific value x." However, we need to update the mathematical calculation to reflect the fact that continuous random variables can take uncountably many possible values within a range, and hence there is no "summing" over all values anymore. Instead, we need to integrate:

Definitions and Theorems 6.5. Cumulative distribution function (continuous random variables)

We define the **cumulative distribution function** (*cdf*) $F(x)$ of a continuous random variable X with probability density function $f(x)$ as the probability that the random variable takes a value *up to* a specific value x. Mathematically, we have $F(x) = P\left(X \leq x\right)$.

In other words, the probability that random variable X is smaller than or equal to a given value x. Equivalently, we may integrate the probability density function over all values that are smaller than x:

$$F(x) = P\left(X \leq x\right) = \int\limits_{-\infty}^{x} f(y)dy.$$

Immediately, we see that $F'(x) = f(x)$: the derivative of the cdf gives us the pdf. Moreover, we can make a similar observation about $P(a \leq X \leq b)$ like we made for discrete random variables. Specifically:

$$P(a \leq X \leq b) = F(b) - F(a) = \int\limits_{a}^{b} f(x)dx.$$

Example 6.5. Timing chemical reactions

The time until a chemical reaction is over is measured in milliseconds (ms). The probability that the reaction is over by time x is given by the following cumulative distribution function:

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-0.01x}, & x \geq 0. \end{cases}$$

Answer the following questions.

1. Derive the probability density function.
2. What proportion of chemical reactions are performed in less than or equal to 200 ms?
3. What proportion of chemical reactions take more than or equal to 100 ms and less than or equal to 200 ms?

First, we derive the probability density function. By definition:

$$f(x) = F'(x) = \begin{cases} 0, & x < 0 \\ \left(1 - e^{-0.01x}\right)', & x \geq 0 \end{cases} = \begin{cases} 0, & x < 0 \\ 0.01 \cdot e^{-0.01x}, & x \geq 0 \end{cases}$$

For the second question, we need $F(200)$:

$$P(x \leq 200) = F(200) = 1 - e^{-2} = 0.8647.$$

Finally, we need to compute $P(100 \leq x \leq 200)$:

$$P(100 \leq x \leq 200) = \int_{100}^{200} f(x)dx = \int_{100}^{200} 0.01 \cdot e^{-0.01x} dx =$$

$$= e^{-1} - e^{-2} = 0.2325.$$

To answer this last part, we could have also used the fact that $P(a \leq X \leq b) = F(b) - F(a)$:

$$P(100 \leq x \leq 200) = F(200) - F(100) = 0.2325.$$

Exercise 6.2. Practicing with the probability mass function and the cumulative distribution function

X is a continuous random variable which is allowed to take values in $[0, 1]$. Its probability density function is $f(x) = 2x$. Answer the following questions:

1. What is the cumulative distribution function?
2. What is $P(X \leq 0.5)$?
3. What is $P(0.25 \leq X \leq 0.75)$?

Summary

In summary, we have important functions that describe the behavior of random variables.

- the probability mass/density function (pmf/pdf), denoted by $p(x)$, $f(x)$.
- the cumulative distribution function (cdf), denoted by $F(x)$.

For **discrete random variables**:

$$P(X \leq x) = F(x) = \sum_{y \leq x} p(y)$$

$$P(a \leq X \leq b) = \sum_{x=a}^{b} p(x) = F(b) - F(a-1)$$

$$P(X = x) = p(x).$$

For **continuous random variables**:

$$P(X \leq x) = F(x) = \int_{-infty}^{x} f(y)dy$$

$$P(a \leq X \leq b) = \int_{a}^{b} f(x)dx = F(b) - F(a)$$

$$P(X = x) = 0.$$

\begin{align*}
f(x) &= \begin{cases} \lambda\cdot e^{-\lambda \left(x-\gamma\right)}, & \text{if } x\geq \gamma \\ 0, & \text{if } x< \gamma. \end{cases}
\end{align*}
A nice visual for helping us address probability-related questions is the following:

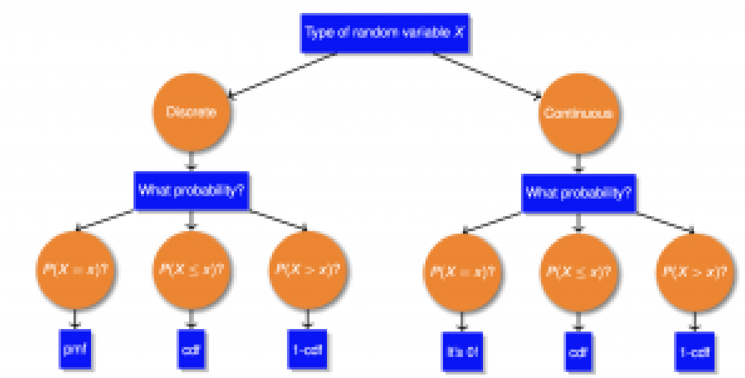

Figure 6.5. A visual for helping us address probability-related questions.

Example 6.6. Discrete and continuous random variable probability functions

1. Assume a discrete random variable X taking integer values between 0 and 10 with a probability mass function of $p(x) = c \cdot x$. What is c, in order for $p(x)$ to be a valid probability mass function?
2. Assume a continuous random variable X taking values between 0 and 10 with a probability density function of $f(x) = c \cdot x$. What is c, in order for $p(x)$ to be a valid probability density function?

We answer those two questions using the proper function and summation/integration:

1. We first sum:
$$\sum_{0}^{10} p(x) = 1 \implies$$
$$\implies c \cdot 0 + c \cdot 1 + \ldots + c \cdot 10 = 1 \implies$$
$$\implies 55 \cdot c = 1 \implies c = \frac{1}{55} = 0.018.$$
2. We now integrate:

$$\int_0^{10} c \cdot x dx = 1 \implies$$

$$\implies \left. \frac{c \cdot x^2}{2} \right|_0^{10} = 1 \implies$$

$$\implies 50 \cdot c = 1 \implies c = 0.02.$$

Notice the differences in the two questions: X changes from discrete to continuous when allowed to take any value (other than just the integer values 0, 1, ...). Moreover, we have a probability *mass* function in the first question, compared to a probability *density* function in the second question. Finally, we use one of the probability axioms (that the probability of the whole sample space has to equal 1), but we do so through a *summation* in the first question, and an *integration* in the second one.

Exercise 6.3. Valid probability density functions?

Are the following valid probability density functions for a continuous random variable?

- $f(x) = 0.01, 0 \leq x \leq 100$?
- $f(x) = \lambda \cdot e^{-\lambda \cdot x}, x \geq 0$?
- $f(x) = \lambda \cdot e^{-\mu \cdot x}, x \geq 0$, if we are told that $\lambda \neq \mu$?

Commonly used continuous distributions

We now proceed to study some very commonly used and observed continuous distributions. We begin with the *uniform* distribution, where every real value has the same "relative likelihood," before then moving to the *exponential* distribution. The exponential distribution and its related *Erlang/Gamma* distributions are closely related, and are "sibling" distributions to the *Poisson*, so we establish that relationship. We finish this part with one of the most ubiquitous distributions: the famous *normal* distribution.

The uniform distribution

We begin from the simplest continuous distribution, the **uniform** distribution. In essence, the uniform distribution for continuous random variables mimics its discrete counterpart, where everything is equally likely. However, since we are discussing continuous random variables which are allowed to take any value, we no longer require the probability of each outcome to be equal; instead, the fact of the "equal likelihood" here implies that all values of the probability density function $f(x)$ are equal. Formally, we define the **uniform distribution** for continuous random variables as follows.

Definitions and Theorems 6.6. The uniform distribution

The uniform distribution, with bounds a and b, describes a random variable whose value falls within bounds a and b:

it is commonly written as $U(a, b)$. All intervals of the same length within a and b have the same probability.

Example 6.7. Equal probability for intervals in the uniform distribution

Consider a continuous random variable X that is uniformly distributed within bounds $a = 3$ and $b = 7$. Then, the following statements are true:

- $P(X = 5) = 0$: this is true for all continuous random variables, remember!
- $0 < P(x_1 \leq X \leq x_2) \leq 1$ for all $x_1 > 3, x_2 > 3$ and $x_1 < x_2$: the probability of getting a value that is greater than 3 is positive.
- $P(X \leq 5) = P(X \geq 5)$: both intervals ($X \leq 5$ and $X \geq 5$) have the same length, and hence their probabilities are the same.
- $P(3 \leq X \leq 5) \geq P(6 \leq X \leq 7)$: the first interval is larger than the second, so it must be more probable of happening.
    - What is more: the first interval is *twice the length* of the first one, so its probability has to also be *twice the probability* of the second one.

Its probability density function and the cumulative density function are shown next:

$$f(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a \le x \le b \\ 0, & \text{otherwise.} \end{cases}$$

$$F(x) = \int_{-\infty}^{x} f(y)dy = \begin{cases} 0, & \text{if } x < a \\ \frac{x-a}{b-a}, & \text{if } a \le x \le b \\ 1, & \text{if } x > b \end{cases}$$

Example 6.8. Digital and analog dice

Assume two random number generators emulating real-life dice. The first one is a digital dice that produces an **integer number** between 1 and 6; the second is an analog dice that produces a **real number** between 1 and 6. Assume that the first one's value is then a random variable X following a discrete uniform distribution, whereas the second one's value is a random variable Y following a continuous uniform distribution. What is the probability that X is less than or equal to 3? What is the probability that Y is less than or equal to 3?

For both, we can use the cumulative distribution functions for the discrete and continuous uniform distribution versions:

- For the digital dice (discrete):

$$P(X \le 3) = \frac{3-1+1}{6-1+1} = \frac{3}{6} = 0.5.$$

- For the analog dice (continuous):

$$P(X \le 3) = \frac{3-1}{6-1} = \frac{2}{5} = 0.4.$$

We note here that due to the continuous nature of continuous random variables, we do not discriminate whether the bounds are inclusive or exclusive. That is, in the previous example, it wouldn't matter if the analog dice could produce a number as small as 1 or as big as 6, or whether it had to be strictly bigger than (or strictly smaller than) the bounds provided.

Expectedly, the uniform distribution's probability density function is *uninteresting* visually, as it is only a straight line parallel to the axis, with such a "height" (recall that $f(x) = \frac{1}{b-a}$) so as to make the rectangular area beneath it equal to 1. We show this in the next Figure:

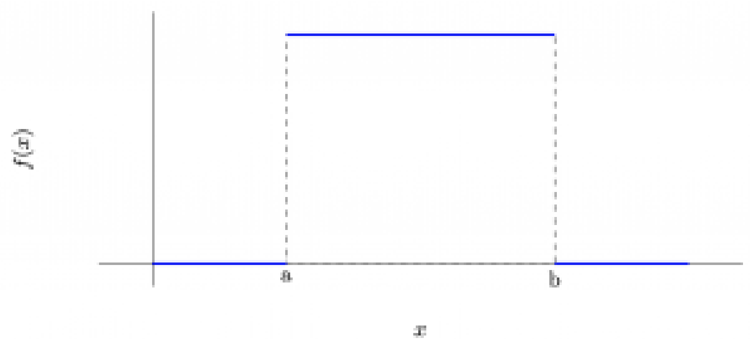

Figure 6.6. A plot of the uniform distribution probability density function. Note that we have [latex]f(x)=\frac{1}{b-a}[/latex] everywhere between [latex]a[/latex] and [latex]b[/latex]. This leads to the rectangular area below the function to be equal to [latex]f(x)\cdot \left(b-a\right)=1[/latex], as required by the axioms of probability. The uniform distribution is zero everywhere else (below [latex]a[/latex] or above [latex]b[/latex]).

Before moving to the next distribution, let us solve one more example using the uniform distribution.

Example 6.9. Totally random buses

Assume you are told that the next bus will arrive at any point in the next 5 to 15 minutes. Hence, in this case, the time until the next bus shows up is uniformly distributed. Then, what is the probability the bus arrives in the next:

- 2 minutes?
- 7 minutes?
- 10 minutes?
- 18 minutes?

This is a uniform distribution with $a = 5, b = 15$. Hence, we can use the cumulative distribution function to answer all questions:

- $P(T \leq 2) = F(2) = 0$, as $2 < a$.
- $P(T \leq 7) = F(7) = \frac{7-5}{15-5} = 0.2$, as $a \leq 7 \leq b$.
- $P(T \leq 10) = F(10) = \frac{10-5}{15-5} = 0.5$, as $a \leq 10 \leq b$.
- $P(T \leq 18) = F(18) = 1$, as $18 > b$.

Or, visually, say for the bus arriving in the next 10 minutes:

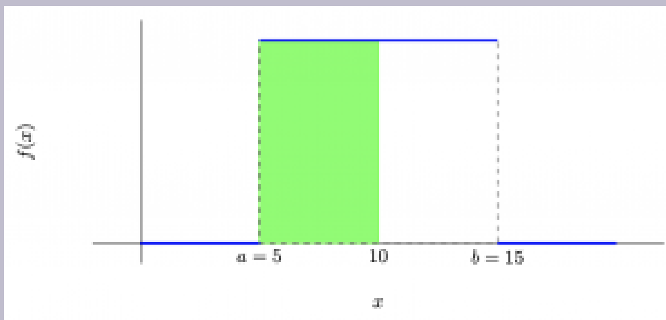

Figure 6.7. The probability of the bus arriving in 10 minutes, marked as the area in green.

Exercise 6.4. Broken receiver

A receiver receives the bit 0 when the current passing through it is smaller than 0.4 units; above 0.4 units, it receives the bit 1. The receiver is broken and receives random values, uniformly distributed between 0.2 and 0.7 units all the time. What is the probability that the receives assumes it has received the bit 0?

The exponential distribution

We now move to one of the most important and consequential probability distributions. The exponential distribution takes its name from the fact that it is based on the exponential function (as can be also seen visually later). Its applications are numerous: from the time until the next patient arrives, to the distance between two random cracks on a highway, and to the game minutes until a soccer player scores their next goal. The exponential distribution is used to model a series of different real-life situations, especially when pertaining to the time until the next event of interest happens.

Definitions and Theorems 6.7. The exponential distribution

A continuous random variable X defined over the interval of $[0, \infty)$ is exponentially distributed if it has probability density function given by $f(x) = \begin{cases} \lambda \cdot e^{-\lambda x}, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0, \end{cases}$ where $\lambda > 0$ is a strictly positive parameter. We sometimes write that $X \sim Exp(\lambda)$ if it follows the exponential distribution with rate λ.

In the definition, we provide the probability density function as $f(x) = \lambda \cdot e^{-\lambda x}$, and we can use that to derive the cumulative distribution function, through proper integration. Specifically, we have:

$$F(x) = \int_0^x f(y)dy = \int_0^x \lambda \cdot e^{-\lambda y} dy = -e^{-\lambda y}\Big|_0^x = 1 - e^{-\lambda x}.$$

Hence, the two functions (probability density and cumulative distribution) for the exponential distribution are:

$$f(x) = \begin{cases} \lambda \cdot e^{-\lambda x}, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0. \end{cases}$$

$$F(x) = \begin{cases} 1 - e^{-\lambda x}, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0. \end{cases}$$

We also present the two functions visually for different values for parameter λ in the next figure. We begin with the probability density function. We notice the "exponential tail" of the exponential distribution in all four of the λ values plotted. We also notice how the probability density function begins with a high value and gets closer and closer to zero as x increases, hence the "tail" in the exponential distribution pdf.

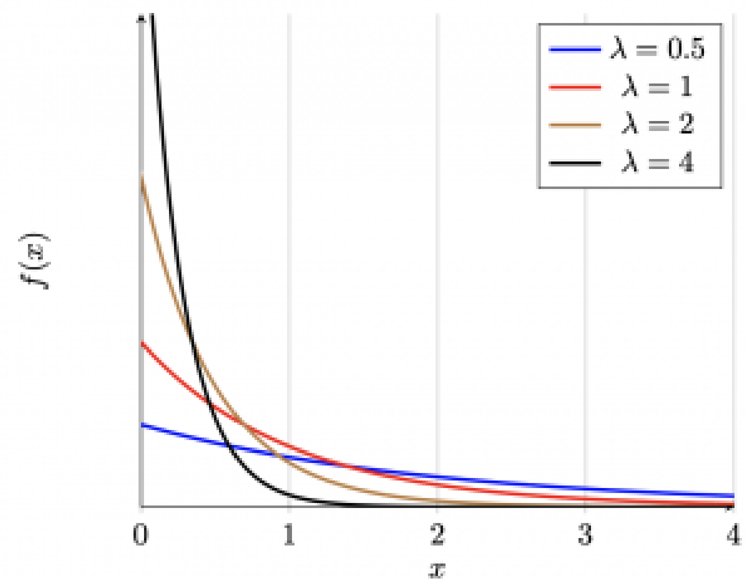

Figure 6.8. The exponential distribution pdf for different rates [latex]\lambda[/latex].

Similarly, we may plot the cumulative distribution function over the same range of λ values.

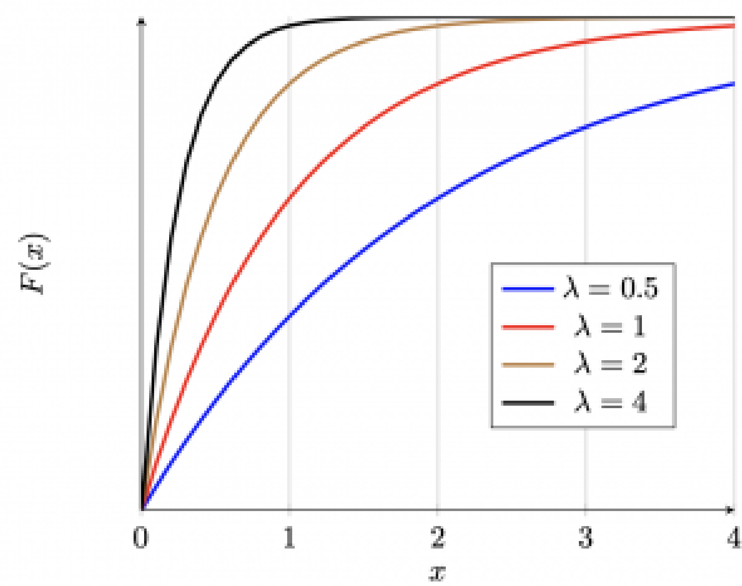

Figure 6.9. The exponential distribution cdf for different rates [latex]\lambda[/latex].

One of the many applications that the exponential distribution sees in practice has to do with quantifying the probability of **the time until the next event** or **the interarrival time between two events**. With the term **interarrival time** we mean the amount of time that passes between two *consecutive* events. When events happen with some rate λ, we can quantify the risk or chance that the next event will happen within some time interval using the exponential distribution. Let us motivate this better with an example.

Example 6.10. Studying an emergency department at a hospital

We are studying an emergency department after hours. Patients arrive requiring medical assistance with a rate of $\lambda = 1$ patient every 12 minutes; the department assumes that the amount of time between two consecutive patient arrivals follows an exponential distribution with the given rate.

- What is the probability that the first patient of the day arrives within the first 5 minutes?
- What is the probability that the first patient of the day arrives after the first 2 but before the first 5 minutes?

These amounts of time (time to first event, time to next event) are exponentially distributed, and hence probability questions relating to how big or small these amounts of time would be can be answered by either properly using the cumulative distribution function or integrating the probability density function (as in any other continuous distribution). Let us answer the questions from the previous example:

Example 6.11. Studying an emergency department at a hospital

Let us define T as the time that elapses until the first patient of the day shows up. We know that T is exponentially distributed with rate $\lambda = 1$ patient every 12 minutes. Hence:

- $P(T \leq 5') = F(5) = 1 - e^{-\lambda 5} = 1 - e^{-\frac{1}{12} \cdot 5} = 0.3408.$
  - We could have similarly answered this question as: $P(T \leq 5') = \int_0^5 f(x)dx = \int_0^5 \frac{1}{12} e^{-\frac{1}{12}x} dx = 0.3408.$
- $$P(2' \leq T \leq 5') = F(5) - F(2) = 1 - e^{-\lambda 5} - \left(1 - e^{-\lambda 2}\right) = e^{-\frac{1}{12} \cdot 2} - e^{-\frac{1}{12} \cdot 5} = 0.1872.$$
  - Or, again, using the pdf: $P(2' \leq T \leq 5') = \int_2^5 f(x)dx = \int_2^5 \frac{1}{12} e^{-\frac{1}{12}x} dx = 0.1872.$

You may have seen signs noting the number of days since the last accident.. Let's use that as an inspiration for our next exercise.

Exercise 6.5. Zero days since last accident

Assume that you work in a facility that has typically a rate of $\lambda = 2$ major accidents per year. Further assume that the time until the next major accident follows an exponential distribution. Answer the following two probability questions.

- What is the probability that the next accident happens in the next year?
- What is the probability that the next accident

happens in the next 3 months?

The last time we solved questions with *rates* and *events happening* we were **counting these events** considering their number a discrete distribution: specifically, the **Poisson** distribution. Let us see if we can formulate some sort of relationship here so as to address our déjà vue feeling we are getting!

The exponential distribution and its relationship to the Poisson distribution

We have already motivated the fact that these two appear to be "sibling" distributions. Let us go back to a question we addressed in our previous chapter's set of activities. Specifically, let us revisit Chapter 5's Problem 25. We repeat this question (and provide its answer) here for convenience.

Example 6.12. Problem 25 Chapter 6: revisited

We saw in class the probability mass function for a Poisson distributed random variable with rate λ: it is $p(x) = P(X = x) = e^{-\lambda} \cdot \frac{\lambda^x}{x!}$. Assume that $\lambda = 3$ per year. What is the probability that there will be no events in the next year? Can you say that this means that the next event will happen more than a year from now?

Let T be the time of the next event: what is $P(T > 1 \text{ year})$?Let X be the number of events during the next year. Then, we have that: $P(T > 1 \text{ year}) = P(X = 0) = e^{-\lambda} \cdot \lambda^0/0! = e^{-3} = 0.05$.

Note that we could also find the probability that the next event does happen during the next year:

$$P(T \leq 1 \text{ year}) = 1 - P(X = 0) = 1 - e^{-\lambda} \cdot \lambda^0/0! = 1 - e^{-3} = 0.95.$$

Let us put the previous result in perspective. **The time to the next event is exponentially distributed if the number of events is distributed as a Poisson random variable! And vice versa!**

Definitions and Theorems 6.8. The relationship between the Poisson and exponential distributions

Let X be a Poisson distributed random variable with rate λ, representing the number of events happening. Then, the distance/time T between events is an exponentially distributed continuous random variable with rate λ.
In table format, we may write that:

Table 6.1. A table representing the relationship (similarities and differences) between the exponential and Poisson distributions.

Exponential distribution	**Poisson distribution**
Continuous	Discrete
Takes values in $[0, +\infty)$.	Takes value in $\{0, 1, \ldots\}$.
Requires parameter/rate λ.	Requires parameter/rate λ.
pdf: $f(x) = \lambda e^{-\lambda x}, x \geq 0.$	pmf: $p(x) = e^{-\lambda}\frac{\lambda^x}{x!}, x = 0, 1, \ldots$
Typically models time/distance to next event/occurrence.	Typically models number of events/occurrences within some time/distance.

Example 6.13. No accident in x days (continued)

How many days until the next accident? In many real-life cases, we assume that the <u>time to the **next** event</u> follows an exponential distribution. For this example, assume that you work in a facility that has typically seen a rate of $\lambda = 2$ major accidents per year.

- What is the probability the next accident happens in the next year?
- What is the probability that the next accident

happens in the next 1 month?

Let X be the time until the next accident. Recall that $\lambda = 2$ per year, or (equivalently) $\lambda = 2$ per 12 months. We then have:

1. $P(X \leq 1 \text{ year}) = F(1) = 1 - e^{-2 \cdot 1} = 0.8647.$
2. $P(X \leq 1 \text{ month}) = F(1) = 1 - e^{-\frac{2}{12} \cdot 1} = 0.1535.$

Note how we used $\lambda = 2/12 = \dfrac{1}{6}$ for the second question's rate.

Exercise 6.6. Overdoses and access to naloxone

Naloxone (and other alternatives available) are a life-saving intervention for people who are overdosing. Let us assume that requests for naloxone at a medical facility follow interarrival times that are exponentially distributed with a rate of 48 requests every 30 days. Assume the medical facility is open for 24 hours every day.

- What is the probability that the next request for naloxone arrives in the next hour?
- What is the probability that there are exactly 5 requests for naloxone during the next day?

Definitions and Theorems 6.9. Memoryless random variables

A random variable X is said to be *memoryless* (i.e., without memory) if
$P(X > s + t | X > s) = P(X > t)$.

In plain terms, the memorylessness property states that information available to us for what has happened so far does not alter our perception for the future. This is pretty uncommon, as we will very soon see. Let us first check what it means with an example.

Example 6.14. Memorylessness and the exponential distribution

A car transmission fails in time that is exponentially distributed with a rate of 1 every 80,000 miles. What is the probability that the transmission does not fail within its first 40,000 miles?

We need $P(T > 40{,}000 \text{ miles})$, when knowing that T is exponentially distributed with rate $\lambda = 1/80000$. We have:

$$\begin{aligned} P(T > 40{,}000 \text{ miles}) &= 1 - P(T \leq 40{,}000 \text{ miles}) = 1 - F(40000) = \\ &= e^{-\frac{1}{80000} \cdot 40000} = e^{-0.5} = 0.6065 = 60.65\%. \end{aligned}$$

Alright, this was expected, and followed exactly the exponential distribution theory. Let us switch things up a little:

Example 6.15. Memorylessness and the exponential distribution (continued)

For the same car, assume we know that its transmission has been working for 80,000 miles already. What is the probability that the transmission does not fail in the next 40,000 miles? Recall that the time until the transmission fails (T) is exponentially distributed.

Can we prove that this is true for exponentially distributed random variables, *in general*? This is left as an activity for you to do in class!

Exercise 6.7. Pokémon Go

In Pokémon Go, there are special Pokémon that you can *catch* which are **shiny variants** of the original ones. We assume that the time it takes for a player to find a shiny variant is exponentially distributed with a rate of 1 every 10 days. What is the probability that a player does not find a shiny Pokémon in the next 10 days, given that they will not find a shiny Pokémon in the first 5 days? That is, letting T

be the number of days, a continuous random variable, until they encounter the next shiny Pokémon, what is $P(T > 10 | T > 5)$?

All in all, though, the argument we can make is that **exponentially distributed random variables are memoryless.** How unique is this memorylessness property? If a random variable was, say, uniformly distributed, would knowing some information about the past affect our probability calculation?

Example 6.16. Memorylessness and the uniform distribution

Assume that the time (in minutes) until the next customer shows up is uniformly distributed in $[0, 60]$. What is the probability the next customer shows up after the first minute? What is the probability the next customer shows up between the 59th and 60th minute, given that no customer has shown up until the 58th minute?

Let T be the time to the next customer arrival. In the first part, we are looking for $P(T > 1)$:

$$P(T > 1) = \frac{59}{60}.$$

Now, note that this answer is *significantly different* than the answer we would get by calculating $P(T > 59 | T > 58)$, which is:

$$P(T > 59|T > 58) = \frac{P(T > 59)}{P(T > 58)} = \frac{1/60}{2/60} = \frac{1}{2}.$$

From the example, we may deduce that the uniform distribution is not memoryless. Memorylessness is a rare property (actually, the only **continuous distribution** to possess the property is the exponential, as we will show in our activities section).
We finish this description with an observation/realization for how **the exponential distribution and the Poisson distribution go hand-in-hand**:

Definitions and Theorems 6.10. Relationship between the Poisson and the exponential distributions

If the time between events is **exponentially distributed** with rate λ then the number of events in some time t is **Poisson distributed** with rate $\lambda \cdot t$, and vice versa.

We will play with this relationship (and introduce Gamma/Erlang distributed random variables, too) even more in the next part of the chapter. But first, let us introduce a small, yet interesting, extension to the (traditional) exponential distribution.

General exponential distribution

The exponential distribution we have discussed so far only requires a single parameter: the rate $\lambda > 0$. In some cases, we may be interested in a "shifted version" of the exponential distribution (as in the following figure).

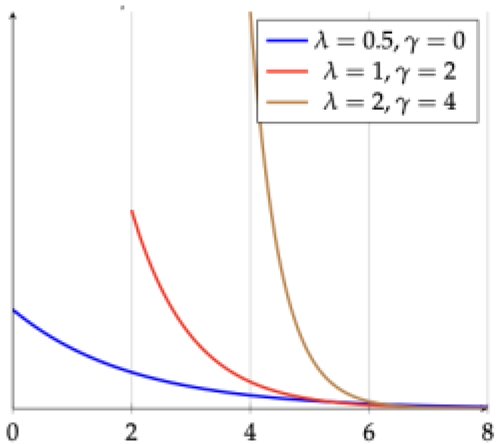

Figure 6.10. The general exponential distribution probability density function visualized for different values of [latex]\lambda[/latex] and [latex]\gamma[/latex].

We call the "shift" the location parameter and we represent it with $\gamma > 0$. Some textbooks employ μ instead of γ. In this class, I will try to reserve μ for something different.
Hence, the general exponential distribution is a two-parameter distribution requiring the presence of a rate $\lambda > 0$ and a location parameter $\gamma > 0$, rendering the pdf and the cdf of the general exponential distribution as follows:
(formula causing problems in exports, insert text of formula)

Example 6.17. Doctor FaceTime

A doctor sees patients in time that is exponentially distributed with rate 1 patient every 40 minutes. However, every patient will spend at least 10 minutes logged in the appointment while they are answering survey questions. In essence, this means that no patient will leave before these 10 minutes are up.

What is the probability the next patient is seen for:

- more than 30 minutes?
- more than 1 hour?
- more than 2 hours?

Let T be the time the next patient will require: T is exponentially distributed with rate $\lambda = 1/40$ minutes and $\gamma = 10$ minutes. Then, we have:

- $$P(T > 30 \text{ minutes}) = 1 - P(T \leq 30 \text{ minutes}) = 1 - F(30) = e^{-\frac{1}{40}\cdot(30-10)} = e^{-0.5} = 0.607$$
- $$P(T > 1 \text{ hour}) = 1 - F(60) = e^{-\frac{1}{40}\cdot(60-10)} = e^{-5/4} = 0.287$$
  .
- $$P(T > 2 \text{ hours}) = 1 - F(120) = e^{-\frac{1}{40}\cdot(120-10)} = e^{-11/4} = 0.064$$
  .

The Gamma and the Erlang distribution

Assume again that you are given a rate λ with which some events happen. So far, we have addressed two related questions:

1. What is the probability that the **next event** happens during some time interval? This is addressed through defining and using an **exponentially distributed random variable** (continuous in nature).
2. What is the probability that we see **a number of events** during some time interval? This is addressed through defining and using a **Poisson distributed random variable** (discrete in nature).

It is time to address a third question: what is the probability that the k-th event happens during some time interval? Like the first question, this is addressed using a continuous random variable; like the second question, we need a number of events to happen first.

Definitions and Theorems 6.11. The Gamma distribution

A continuous random variable X defined over the interval of $[0, \infty)$ is **Gamma distributed** if it has probability density function given by:

$$f(x) = \begin{cases} \frac{\lambda^k \cdot x^{k-1} \cdot e^{-\lambda x}}{\Gamma(k)}, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0, \end{cases}$$

where $\lambda > 0$ and $k > 0$ are given parameters and $\Gamma(k)$ is the Gamma function.

Note: In most examples, we will only deal with integer values of k, and hence $\Gamma(k) = (k-1)!$.

We sometimes write that $X \sim \text{Gamma}(k, \lambda)$ if it follows the Gamma distribution with rate λ and shape parameter k.

When k is a positive integer number, the Gamma distribution is referred to as the **Erlang distribution**. In essence, the definition follows.

Definitions and Theorems 6.12. The Erlang distribution

A continuous random variable X defined over the interval of $[0, \infty)$ is **Erlang distributed** if it can be written as **the summation of exponentially distributed random variables** $X = \sum_{i=1}^{k} X_i$, where X_i is an exponentially distributed random variable.

When X is Erlang distributed it has probability density function given by:

$$f(x) = \begin{cases} \frac{\lambda^k \cdot x^{k-1} \cdot e^{-\lambda x}}{(k-1)!}, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0, \end{cases}$$

where real $\lambda > 0$ and integer $k > 0$ are given parameters.

Erlang distribution PDF curves for various λ and k values
The Erlang distribution probability density function visualized for different values of λ and k. The curves show how the distribution shape changes with different rate (λ) and shape (k) parameters.

Due to the nature of the Γ function (and of $(k-1)!$), it is typically easier to integrate the pdf after we plug in the value for k that we are interested in. For example, consider the problem from our motivation.

Example 6.18. Congratulations, you are our 10th customer

A store, which is open for 8 hours every day, gives a gift card to the (exactly) 10th customer of every day. The store has observed that customers show up at a rate of 1 every 20 minutes (exponentially distributed). What is the probability the 10th customer of the day shows up in the second half of the day?

We are interested in the 10th customer, so we use an Erlang distribution with $k = 10$. The rate is 1 customer per 20 minutes, or $\lambda = 3$ customers per hour.

The second half of the day starts at hour 4, so we need:

$$P(X > 4) = 1 - P(X \leq 4)$$

where $X \sim \text{Erlang}(10, 3)$.

Using the CDF of the Erlang distribution:

$$F(x) = 1 - e^{-\lambda x} \sum_{i=0}^{k-1} \frac{(\lambda x)^i}{i!}$$

For $x = 4$, $k = 10$, $\lambda = 3$:

$$P(X \leq 4) = 1 - e^{-12} \sum_{i=0}^{9} \frac{12^i}{i!}$$

Therefore,

$$P(X > 4) = e^{-12} \sum_{i=0}^{9} \frac{12^i}{i!} \approx 0.7576$$

There is approximately a 75.76% chance that the 10th customer arrives in the second half of the day.

We will check some of the properties of the Gamma/Erlang distribution (such as its mean and variance) during our next chapter; we will also discuss the interplay between the exponential, Erlang, and Poisson distributions during the Activities section. Stay tuned!

The Normal Distribution

The **normal distribution** (also called the **Gaussian distribution**) is perhaps the most important probability distribution in statistics. It appears naturally in many contexts (due to the Central Limit Theorem) and has a characteristic bell-shaped curve. Due to its immense applicability, special tables have been devised to help us calculate probabilities of interest. We often refer to them as z -tables, especially when pertaining to the so-called standard normal distribution.

Definitions and Theorems 6.13. The normal distribution

A continuous random variable X is **normally distributed** with mean μ and variance σ^2 if it has probability density function:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

We write $X \sim N(\mu, \sigma^2)$.

The distribution is defined through two parameters, often written as μ and σ^2. While their formal definition is in the next chapter, we quickly provide an intuitive interpretation. Specifically:

- μ: also referred to as the mean. It is a **location parameter**, showing the center of the distribution
- σ^2: also known as the variance. It is a **scale parameter**, revealing the spread of the distribution
- σ: the standard deviation: it is the square root of the variance.

Definitions and Theorems 6.14. Properties of the Normal distribution

The normal distribution possesses a series of useful properties. Most notably:

1. **Symmetry:** The normal distribution is symmetric about its mean μ.
2. **Bell-shaped:** The curve has a characteristic bell shape.
3. **Asymptotic:** The tails extend to $\pm\infty$ but never quite touch the x-axis.
4. **Empirical Rule (68-95-99.7 rule):**

- Approximately 68% of values fall within $\mu \pm \sigma$
- Approximately 95% of values fall within $\mu \pm 2\sigma$
- Approximately 99.7% of values fall within $\mu \pm 3\sigma$

5. **Mean = Median = Mode:** All three measures of center are equal to μ. Again, all these definitions will be further studied in the next chapters.

The Standard Normal Distribution

The **standard normal distribution** is a special case of the normal distribution with $\mu = 0$ and $\sigma^2 = 1$.

We denote a standard normal random variable as $Z \sim N(0, 1)$. Its probability density function is given by:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

Something particularly useful about the standard normal distribution is that any normally distributed random variable $X \sim N(\mu, \sigma^2)$ can be transformed to the standard normal, through a simple z-transformation:

$$Z = \frac{X - \mu}{\sigma}$$

Example 6.19. Thermometer accuracy

Returning to our motivation problem: A chef is using a thermometer to check if chicken is adequately cooked (internal temperature ≥ 165°F). The thermometer reads 166°F. Assuming the thermometer has normally distributed errors with mean 0 and standard deviation 2°F, what is the probability the chicken is adequately cooked?

Let X be the true temperature. The thermometer reading is 166°F, so we model:

$$\text{Reading} = X + \epsilon$$

where $\epsilon \sim N(0, 4)$ (variance = $2^2 = 4$).

Given the reading is 166, we want $P(X \geq 165)$.

The true temperature X given a reading of 166 is approximately $N(166, 4)$.

We need to find $P(X \geq 165)$:

$$Z = \frac{X - 166}{2}$$

$$P(X \geq 165) = P\left(Z \geq \frac{165 - 166}{2}\right) = P(Z \geq -0.5)$$

From the standard normal table:

$$P(Z \geq -0.5) = P(Z \leq 0.5) \approx 0.6915$$

There is approximately a 69.15% probability that the chicken is adequately cooked.

Using the standard normal distribution table (z-table)

How do we work with the standard normal distribution? The standard normal table (z-table) provides values of the cumulative distribution function: $\Phi(z) = P(Z \leq z)$

Here are the steps for using the table:

1. Standardize your normal random variable by doing the proper transformation: $Z = \dfrac{X - \mu}{\sigma}$.
2. Convert your probability question to one involving Z.
3. Look up the value in the standard normal table.
4. Use properties of probabilities to get your answer, such as:
    - $P(Z > z) = 1 - P(Z \leq z) = 1 - \Phi(z)$
    - $P(Zz)$ (by symmetry)
    - $P(a < Z < b) = \Phi(b) - \Phi(a)$

Example 6.20. Test scores

Test scores are normally distributed with mean 75 and standard deviation 10. What percentage of students score between 70 and 85?

Let $X \sim N(75, 100)$ be the test score.

We want $P(70 < X < 85)$.

1. Standardize:

- $$Z_1 = \frac{70 - 75}{10} = -0.5.$$
- $$Z_2 = \frac{85 - 75}{10} = 1.0.$$

2. Therefore:
   $P(70 < X < 85) = P(-0.5 < Z < 1.0) = \Phi(1.0) - \Phi(-0.5)$
   .
3. From the standard normal table:
   - $\Phi(1.0) \approx 0.8413$
   - $\Phi(-0.5) \approx 0.3085$
4. Hence:
   $P(70 < X < 85) = 0.8413 - 0.3085 = 0.5328$
   .

Approximately 53.28% of students score between 70 and 85.

Relationship between the exponential, Poisson, and Erlang distributions

These three distributions are intimately related through Poisson distributed random variables:

Table 6.2. A table summary of the exponential, Poisson, and Erlang distributions.

Distribution	Type	Question Answered
Exponential λ	Continuous	Time until the *next* event
Poisson λt	Discrete	Number of events in time t
Erlang (k, λ)	Continuous	Time until the *k-th* event

Key relationship: If $X_1, X_2, \ldots, X_k$ are independent exponential random variables with rate λ, then $X_1 + X_2 + \cdots + X_k \sim \text{Erlang}(k, \lambda)$.

Exercise 6.8. Practice exercises

1. Customers arrive at a bank at a rate of 2 per minute. What is the probability that the 5th customer arrives within 3 minutes?
2. The heights of adult males are normally distributed with mean 70 inches and standard deviation 3 inches. What proportion of adult males are taller than 6 feet (72 inches)?
3. A machine produces parts with lengths that are normally distributed with mean 10 cm and standard deviation 0.2 cm. Parts are acceptable if their length is between 9.7 cm and 10.3 cm. What percentage of parts are acceptable?
4. If $X \sim \text{Erlang}(4, 2)$, find $E[X]$ and $\text{Var}(X)$.
5. Given that $Z \sim N(0, 1)$, find:

    a. $P(Z < 1.5)$

b. $P(Z > -1.0)$
c. $P(-1 < Z < 2)$

Activities

Activity 1: Basic continuous probability distribution properties

Let X be a continuous random variable measuring the current (in milliamperes, mA) in a wire with probability density function (pdf) given by $f(x) = 0.05$, for $0 \leq x \leq \alpha$. Answer the following questions.

Problem 1: Valid pdf?

What is α in order for $f(x)$ to be a valid pdf?

Problem 2: Constructing cumulative distribution functions

What is the cumulative distribution function?

Problem 3: Calculating probabilities

The wire is said to be overheating if the current is more than 10mA.

It is also said to be working within the normal range of operations if it is between 5 and 10mA. Answer the following questions:

- What is the probability the wire is overheating?
- What is the probability the wire is within its normal range?

Activity 2: The exponential distribution

The time until the next customer arrives is exponentially distributed with a rate of 1 customer every 10 minutes. Answer the following questions.

Problem 4: Calculating probabilities

Let T be the time until the next customer arrives. What is the probability the next customer shows up in the next:

- 1 minute?
- 5 minutes?
- 10 minutes?
- 20 minutes?

That is, what is

- $P(T \leq 1 \text{ minutes})$?
- $P(T \leq 5 < spanclass = "mathinline" > \text{ minutes } < /span >)$?
- $P(T \leq 10 \text{ minutes})$??
- $P(T \leq 20 \text{ minutes})$?

This is the big difference between using a rate for a Poisson or for an exponential distribution. For the Poisson distribution, we first convert the rate into the time units of the question. For the exponential distribution, we simply multiply the rate by the necessary time interval length!

For example, for a rate $\lambda = 3$ per minute, the probability we see 5 customers in 3 minutes would be written as

$$P(X = x) = e^{-\lambda} \cdot \frac{\lambda^x}{x!} \implies P(X = 10) = e^{-9} \cdot \frac{9^5}{5!},$$

because λ would be set equal to 9 (customers per 3 minutes).

On the other hand, for the same rate, the probability the next customer shows up within the first 3 minutes would be written as

$$P(T \leq t) = 1 - e^{-\lambda t} \implies P(T \leq 3) = 1 - e^{-3 \cdot 3} = 1 - e^{-9}.$$

Problem 5: Inverting the question

The employee of the store wants to take a break, but they do not want to miss the next customer arrival. How long should the break be for in order to have a 50% chance of not missing the next customer?

Problem 6: Taking a break

In the same store, assume that the employee takes a 2 minute break during which no customer arrived. What is the probability the next customer does not arrive in the next 5 minutes now? Contrast it to your answer for $P(T \leq 5)$ in Problem 4.

Based on your answer in Problem 6, we must be deducing (again, see the Theory part for more examples) that the exponential distribution is indeed **memoryless**. In the next activity, we prove that property for all exponentially distributed random variables.As a reminder, we say that a distribution is memoryless if for random variable X following that distribution, we have that

$$P(X > s + t | X > s) = P(X > t).$$

For example, in the case of the time of arrival of the next customer (random variable T) in more than 5 minutes from now, we would write $P(T > 5)$. After we take a break for 2 minutes, then that same probability would be $P(T > 7 | T > 2)$.

Activity 3: Memorylessness

Let's check the proof!

Problem 7: Memorylessness proof

In the first step of the proof, we will calculate the right hand side of what we want to show: $P(X > s + t | X > s)$. Using conditional probabilities , what is this equal to?

$$P(X > s + t | X > s) =$$

Problem 8: Memorylessness proof (cont'd)

In the second step of the proof, we need to make one observation. If $A \subseteq B$ then $A \cap B = A$. Is this the case here for the denominator of your answer in Problem 7? Specifically what can you write about $P(X > s + t \ \cap \ X > s)$?

$$P(X > s + t \ \cap \ X > s) =$$

Problem 9: Memorylessness proof (cont'd)

In the third step of the proof, consider the original identity we are trying to show: $P(X > s + t | X > s) = P(X > t)$. Assume that X is exponentially distributed with rate λ: that is, $P(X > t) = e^{-\lambda t}$. Fill in the blanks below to get the result. Hints are given along the way (look at the final equation!).

$$\begin{aligned} P(X > s + t | X > s) &= \frac{P(X > s + t \ \cap \ X > s)}{P(X > s)} = && \text{(from Prob. 7)} \\ &= \frac{P(X > s + t)}{P(X > s)} = && \text{(from Prob. 8)} \\ &= \frac{}{} = && \\ &= \underline{} = P(X > t). && \end{aligned}$$

The next worksheet problem comes pre-filled. However, going through this will help with understanding why the exponential distribution is unique.

Problem 10: Only the exponential distribution!

So, is the exponential distribution one of many distributions to be memoryless? The answer is a resounding no: it is the **only** continuous distribution to be memoryless. To show that, we need to use all of the tools in our toolbox.

Like we did in Problem 9:

$$\begin{aligned} P(X > s+t | X > s) &= \frac{P(X > s+t \ \cap \ X > s)}{P(X > s)} = & \text{(from Prob. 7)} \\ &= \frac{P(X > s+t)}{P(X > s)}. & \text{(from Prob. 8)} \end{aligned}$$

We now note that if X is memoryless, then $P(X > s+t | X > s) = P(X > t)$:

$$P(X > s+t | X > s) = P(X > t) \implies \frac{P(X > s+t)}{P(X > s)} = P(X > t)$$
$$\implies P(X > s+t) = P(X > s) \cdot P(X > t).$$

For simplicity, define $\overline{F}(x) = P(X > x)$ and hence our previous equality becomes:

$$\overline{F}(s+t) = \overline{F}(s) \cdot \overline{F}(t).$$

Now, take the logarithms on both sides!

$$\ln \overline{F}(s+t) = \ln \overline{F}(s) \cdot \overline{F}(t) = \ln \overline{F}(s) + \ln \overline{F}(t).$$

Once again, for simplicity, define $g(x) = \ln \overline{F}(x)$, rendering our final equality as:

$$g(s+t) = g(s) + g(t).$$

The **only continuous function** that satisfies $g(s+t) = g(s) + g(t)$ is the linear one (see the citation for the proof in the end), hence we deduce that $g(x) = \beta \cdot x$, for some β.

Combine everything:

1. We need $P(X > s+t) = P(X > s) \cdot P(X > t)$ for memorylessness to hold.
2. Equivalently, after some definitions, we saw this is equivalent to $g(s+t) = g(s) + g(t)$, where $g(x) = \ln \overline{F}(x)$.
3. We saw though that this is only true if $g(s)$ is linear, that is $g(x) = \beta \cdot x$.

We replace backwards: $g(s) = \beta \cdot s \implies \ln \overline{F}(s) = \beta \cdot s \implies \overline{F}(s) = e^{\beta \cdot s} \implies F(s) = 1 - e^{-\beta \cdot s}$. Recall that the exponential distribution has $F(x) = 1 - e^{-\lambda \cdot x}$. Pretty close! Letting $\beta = -\lambda$ finishes the proof!

Problem 11: Just checking

Circle the correct choice in each of the three questions.

The exponential distribution is the only *continuous* probability distribution that is memoryless.

True or False

The exponential distribution is the only probability distribution that is memoryless.

True or False

Assume that the time until the next event is exponentially distributed. At 1pm, we calculated the probability that the next event happens before or on 1.30pm is 30%. It is 1.30pm now, and no event has happened yet. What is the probability that the next

event happens before or on 2pm? Pick the correct answer in the multiple choice question.

1. It is higher than 30%: it was 30% at 1pm and it never happened, so now it should happen with higher certainty.
2. It is lower than 30%: it was 30% at 1pm and it never happened, so now we should consider that maybe our probability calculation was off to begin with.
3. It is equal to 30%: it was 30% at 1pm and while it never happened, memorylessness states that the probability is still 30%.

Activity 4: Exponential, Poisson, and Erlang

A manufacturing process requires the completion of 4 small steps: pre-processing, processing, inspection, packaging. Each step of the process requires time that is that is exponentially distributed with a rate of 1 completion every 3 minutes. That is, all steps are exponentially distributed with $\lambda = 1/3$ minutes. The steps have to be performed sequentially. Answer the following questions.

Problem 12

What is the probability that the "pre-processing" step alone (that is, **the first step alone**, from its start to its end) is completed within 4 minutes?

Problem 13

An inspector shows up to watch the operations take place. They only have time to be there for 10 minutes. What is the probability that there are exactly 3 steps that are completed in the 10 minutes after the inspector is there?

Problem 14

What is the probability that a manufacturing order is completed within 10 minutes (all 4 steps, one after the other)?

We can answer Problem 14 using random variable T for the time until 4 steps are completed: T is Erlang with $k = 4$ steps and $\lambda = 1/3$ minutes. We could also calculated this using random variable X for the number of steps that are completed in 10 minutes: X is a Poisson random variable with $\lambda = 10/3$. Then, we can show that:

$$P(T \leq 10 \text{ minutes}) = P(X \geq 4 \text{ steps}).$$

Activity 5: The normal distribution

In this part of the worksheet, we turn our focus to the normal distribution. For this next part, we assume that $\Phi(z)$ is the standard normal distribution cumulative function. We can use the z-table provided in the last page to find $\Phi(z)$!

Problem 15: Converting to z values

Let's practice with converting to the proper z values. Let X be a normally distributed random variable with $\mu = 10, \sigma^2 = 4$.

- $X = 12 \implies z =$
- $X = 8 \implies z =$
- $X = 4 \implies z =$

Problem 16: Simple normal distribution probabilities

For the previous random variable $X \sim \mathcal{N}(10, 4)$, find the probabilities. Use the z values you calculated earlier.

- $P(X \leq 12) =$
- $P(X \geq 4) =$
- $P(4 \leq X \leq 12) =$

Problem 17: Interesting probabilities

As we saw in class, the normal distribution is centered at μ. A follow-up question would be to find the range of values centered at μ that satisfy a certain probability. Let's see an example here: what is the probability that X is within 1 unit from its center (μ), that is what is the probability that X is between 9 and 11? How about 2 units from its mean?

- $P(9 \leq X \leq 11) =$
- $P(8 \leq X \leq 12) =$

Problem 18

Let's now focus on the opposite problem. What should the range be (centered at μ) so that the probability of the range is 50%? In essence, what should a be in order for

$P(\mu - a \leq X \leq \mu + a) = 0.5$? Remember that earlier we calculated two range probabilities:

1. $P(9 \leq X \leq 11) = 0.383$.
2. $P(8 \leq X \leq 12) = 0.6826$.

Based on that, we should anticipate a to fall somewhere above 1 unit but below 2 units. But how big should it be exactly? Recall that we assume that $X \sim \mathcal{N}(10, 4)$.

Problem 19

For $X \sim \mathcal{N}(10, 4)$, we want to see how big the range should be in order to have a probability equal to 95% that X falls in that range. In essence, we are now interested in $P(\mu - a \leq X \leq \mu + a) = 0.95$. Graphically:

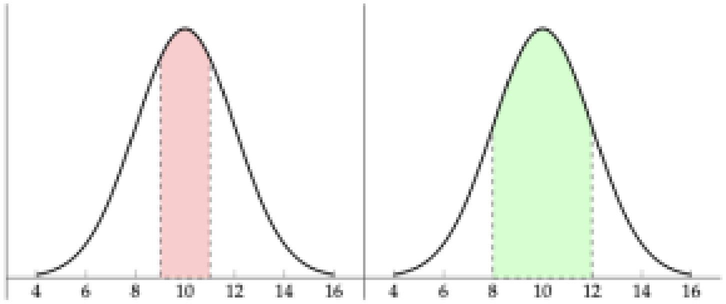

Figure 6.11. What should [latex]a[/latex] be for [latex]P(\mu-a\leq X\leq \mu+a)=0.95[/latex]? Here we show in red the area for [latex]P(\mu-1\leq X\leq \mu+1)=0.383[/latex] and in green the area for [latex]P(\mu-2\leq X\leq \mu+2)=0.6826[/latex]. It should make sense that [latex]P(\mu-a\leq X\leq \mu+a)=0.95[/latex], [latex]\mu-a[/latex] and [latex]\mu+a[/latex] have to be points located even farther from the center [latex]\mu[/latex].

What should a be for $P(\mu - a \leq X \leq \mu + a) = 0.95$? Here we show in red the area for $P(\mu - 1 \leq X \leq \mu + 1) = 0.383$ and in green the area for $P(\mu - 2 \leq X \leq \mu + 2) = 0.6826$. It should make sense that $P(\mu - a \leq X \leq \mu + a) = 0.95$, $\mu - a$ and $\mu + a$ have to be points located even farther from the center μ.

After you have found a, try to draw the resultant area!

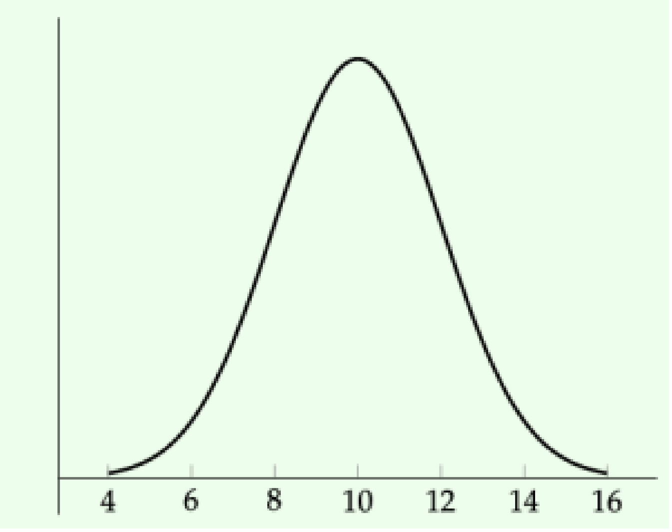

Figure 6.12. An empty normal distribution for you to draw on.

Problem 20

We may have started observing that a does not depend on μ all that much. Instead it depends on σ. For example, seeing as we may write $P(\mu - a \leq X \leq \mu + a) = 0.5$ as a probability of z as follows:

1. $z_1 = \frac{\mu - \alpha - \mu}{\sigma} = -\frac{\alpha}{\sigma}$
2. $z_2 = \frac{\mu + \alpha - \mu}{\sigma} = \frac{\alpha}{\sigma}$.
3. Note that $z_1 = -z_2$.

Based on that: $P(\mu - a \leq X \leq \mu + a) = P(z_1 \leq Z \leq z_2)$. Now recall that $P(z_1 \leq Z \leq z_2) = \Phi(z_2) - \Phi(z_1) = \Phi(z_2) - \Phi(-z_2) = \Phi(z_2) - (1 - \Phi(z_2)) = 2\Phi(z_2) = 2\Phi(\frac{\alpha}{\sigma}) - 1$.

With that in mind, answer the following three questions. Try to answer them generally, not only for $X \sim \mathcal{N}(10, 4)$.

- $P(\mu - \sigma \leq X \leq \mu + \sigma) =$
- $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) =$
- $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) =$

Based on our answers, we have the following realization: σ is very important. Any normally distributed random variable probability can be expressed as a "distance" in terms of "σ". In essence:

See this link for an interesting mnemonic, called the **68-95-99.7 rule**: https://en.wikipedia.org/wiki/68-95-99.7_rule

Activity 6: Contrasting exponentials

Consider two exponentially distributed random variables X_1, X_2 with rates λ_1, λ_2.

Problem 21

What is the probability of $X_1 > X_2$, given that $X_2 = x$?

- $P(X_1 > X_2 | X_2 = x) = P(X_1 > x) =$

Problem 22

What is the probability of $X_1 > X_2$, in general? Recall the total probability law? We can apply this to continuous distributions, too! We cannot sum here, but we may integrate. Let X_1 be random variable distributed with pdf $f(\cdot)$ and X_2 be a random variable distributed with pdf $g(\cdot)$, then:

Use this to answer the following question:

- $P(X_1 > X_2) =$

From this last part, we see that for two exponentially distributed random variables X_1, X_2 with rates λ_1, λ_2, respectively, we have:

Problem 23

Two employees are using a website to place an order with a supplier at exactly the same time. The first person is more tech savvy and completes an order with rate 1 order every 3 minutes. The second person is just starting the job and learning, so they are a little slower and complete an order with rate 1 order every 5 minutes. Both times are exponentially distributed. What is the probability that the second person completes the order faster than or equal to the time the first person takes to complete an order?

A few key takeaways for the exponential, Poisson, and Erlang distributions follow:

Food for thought 6.1.

As a summary of the exponential distribution:

1. If the time between events is exponentially distributed, then the time until the k-th event ($k > 1$) is Erlang distributed, and the number of events within some time is Poisson distributed.

   Customers arrive with time that is exponentially distributed with rate $\lambda = 3$ customers every hour. Then:

   - "What is the probability the next customer shows up in 10 minutes?" is an **exponential distribution** question.
   - "What is the probability there are more than one customer in the next 20 minutes" is a **Poisson distribution** type of question. Recall that we will update λ to be in 20 minute intervals $\implies \lambda = 1$ customer every 20 minutes.
   - "What is the probability the second customer shows up within 20 minutes" is an **Erlang distribution** question.

2. The exponential distribution is **memoryless**! Hence: $P(T > s + t | T > s) = P(T > t).$

If $X_1, X_2, \ldots, X_k$ are independent exponentially distributed random variables with rates $\lambda_1, \lambda_2, \ldots, \lambda_k$ respectively, then the probability X_i is the smallest one (i.e., $P(X_i < X_1, X_i < X_2, \ldots)$) is: $\dfrac{\lambda_i}{\lambda_1 + \lambda_2 + \ldots + \lambda_k}$.

7. Expectations and variances

Theory

Chapter objectives

Learning outcomes

After this chapter, you will be able to:

- Define and explain with examples what expectations and variances are.
- Calculate the expectation and variance of discrete and continuous random variables.
- Calculate the expectation and variance of functions of discrete and continuous random variables.
- Recall and use basic properties of expectations and variances.

Chapter questions

1. What is the meaning of an **expectation** and the meaning of a **variance**?
2. How do we calculate the **expectation** of a:

    - discrete random variable?
    - continuous random variable?

3. How do we calculate the **variance** of a:

    - discrete random variable?
    - continuous random variable?

4. What is the **standard deviation**?
5. What are some useful properties of expectations and variances?

Motivation: Printer lifetime

A printer has a working lifetime that is exponentially distributed with rate $\lambda = 1$ broken printer every 3 years. In English, we typically replace the printer every 3 years. Assume the company decides to change the printer every 2 years (if it hasn't broken down) or when it breaks down (whichever happens first). What is the expected time the company keeps a printer?

Expectation

When describing a random variable and its probability distribution, we sometimes are interested in answering a simple question "what should I expect?" Seeing as a random variable is inherently, well, random, expectations are important and they reveal a "center" of the probability distribution.

Definitions and Theorems 7.1. Expectation

With the term expectation (sometimes we use the term mean or expected value), we imply a measure of the center of the probability distribution. Intuitively, we may think of the expected value of a random variable X as an "average" of the values that X is allowed to take weighted by their respective probabilities.

This definition is very open-ended, so we provide more specific definitions for discrete and continuous random variables in the next subsections.

Discrete random variables

Definitions and Theorems 7.2. Expectation of a discrete random variable

Let X be a numerically-valued discrete random variable with sample space S and probability mass function $p(x)$. Then, the expected value of X is written as $E[X]$ and is calculated as:

$$E[X] = \sum_{x \in S} x \cdot p(x).$$

The expected value is commonly referred to as the *mean* and is also written as μ.

Example 7.1. An unfair die

Consider an “unfair” die with sample space $S = \{1, 2, 3, 4, 5, 6\}$ and $p(1) = 1/3, p(2) = 1/6, p(3) = 1/6, p(4) = 1/6, p(5) = 1/12, p(6) = 1/12$. What is the expected value?

- $E[X] = 1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{12} + 6 \cdot \frac{1}{12} = \frac{33}{12} = 2.75.$

Note how the value we expect can never actually happen, as the die can only take the values of 1, 2, 3, 4, 5, or 6!

Exercise 7.1. Fair die expectation

What about for a fair die, where each side (1, 2, 3, 4, 5, or

6) are equally probable? What is the expectation for this die?

Continuous random variables

Definitions and Theorems 7.3. Expectation of a continuous random variable

Let X be a numerically-valued real random variable defined over $(-\infty, +\infty)$ and probability density function $f(x)$. Then, the expected value of X is written as $E[X]$ and is calculated as:

$$E[X] = \int_{-\infty}^{+\infty} x \cdot f(x)\, dx.$$

The expected value is commonly referred to as the *mean* and is also written as μ.

Example 7.2. A rating system

A company is rating their employees with a system that assigns a score between 1 and 5. We assume the score is continuous (that is, a score of 4, 4.2, and 4.31478 are all valid) and the probability with which score x appears is given by pdf $f(x) = \frac{3}{124} \cdot x^2$, for $1 \leq x \leq 5$.

What is the expected rating score of a random employee?

- $$E[X] = \int_1^5 x \cdot f(x)\, dx = \frac{3}{124} \int_1^5 x^3\, dx = \frac{117}{31} = 3.77.$$

Exercise 7.2. Good employee ratings

Good employees are the ones that receive a rating score of 4 or above. Their scores are distributed with a slightly different distribution: they follow pdf $f(x) = \frac{2}{9} \cdot x$ for $4 \leq x \leq 5$. What is the expected rating score of a good employee?

Properties of the expectation

The expectation satisfies the following properties:

1. Let α be a real number and X be a random variable. Then:

$$E[\alpha \cdot X] = \alpha \cdot E[X].$$

2. Let X, Y be two random variables. Then:

$$E[X + Y] = E[X] + E[Y].$$

    - This generalizes to as many random variables as you would want to. In essence, we have for n random variables $X_1, X_2, \ldots, X_n$:

$$E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i].$$

3. Combining 1 and 2, we have the following. Let α, β be two real numbers and X, Y be two random variables. Then:

$$E[\alpha \cdot X + \beta \cdot Y] = \alpha \cdot E[X] + \beta \cdot E[Y].$$

    - Once again, this can be generalized. For n random variables $X_1, X_2, \ldots, X_n$ and n real numbers $\alpha_1, \alpha_2, \ldots, \alpha_n$:

$$E\left[\sum_{i=1}^{n} \alpha_i \cdot X_i\right] = \sum_{i=1}^{n} \alpha_i \cdot E[X_i].$$

4. Let $g(X)$ be a function of the random variable X. Then, the expectation of $g(X)$ is denoted by $E\left[g(X)\right]$ and is equal to:

- for discrete random variables X with sample space S:
  $E\left[g(X)\right] = \sum_{x \in S} g(x) \cdot p(x).$
- for continuous random variables X (assuming they are defined over $\left(-\infty, +\infty\right)$:
  $E\left[g(X)\right] = \int_{-\infty}^{+\infty} g(x) \cdot f(x).$

Let us see a couple of examples.

Example 7.3. Profit expectation

A company makes \$2,000 if they sell 4 units, \$1,800 if they sell 3 units, \$1,200 if they sell 2 units, lose \$1,000 if they sell 1 unit, and lose \$3,000 if they sell no units. Each event from 0 to 4 customers is equally probable. How much should they expect to make?

Let us use the discrete case expectation of a function:

$$E\left[g(X)\right] = \sum_{x=0}^{4} g(x) \cdot p(x) =$$
$$= 2000 \cdot \frac{1}{5} + 1800 \cdot \frac{1}{5} + 1200 \cdot \frac{1}{5} - 1000 \cdot \frac{1}{5} - 3000 \cdot \frac{1}{5} =$$
$$= \$1000.$$

Example 7.4. Circuit heat

Let X be a continuous random variable measuring the current (in milliamperes, mA) in a wire with pdf $f(x) = 0.05$, for $0 \leq x \leq 20$. The heat produced from the current is given by the function $g(x) = 10 \cdot x$ (with x in milliamperes). What is the mean heat produced by the current?

Here we use the continuous case, as X is a continuous random variable:

$$E\left[g(X)\right] = \int_{x=0}^{20} g(x) \cdot f(x) \cdot dx = \int_{x=0}^{20} g(x) \cdot f(x) dx = \\ = \int_{x=0}^{20} 10 \cdot x \cdot 0.05 \cdot dx = \int_{x=0}^{20} 0.5 \cdot x \cdot dx = 100.$$

Variance

We have established that the expectation describes the center of the distribution. A secondary quantity that we are interested in is how the random variable behaves around its center: Is it close to its mean most of the time? Does it vary a lot? This is captured by the concept of variance.

Definitions and Theorems 7.4. Variance

With the term variance, we refer to the expectation of the squared deviation from the mean. Variance captures how spread out the values of a random variable are.

This definition is very open-ended, so we provide more specific definitions for discrete and continuous random variables in the next subsections.

Discrete random variables

Definitions and Theorems 7.5. Variance of a discrete random variable

Let X be a numerically-valued discrete random variable with sample space S and probability mass function $p(x)$. Then, the variance of X is written as $Var[X]$ and is calculated as:

$$Var[X] = \sum_{x \in S} (x - \mu)^2 \cdot p(x) = E[(X - \mu)^2].$$

The variance is also written as σ^2. The *standard deviation* is defined as $\sigma = \sqrt{Var[X]}$.

Continuous random variables

Definitions and Theorems 7.6. Variance of a continuous random variable

Let X be a numerically-valued real random variable defined over $(-\infty, +\infty)$ and probability density function $f(x)$. Then, the variance of X is written as $Var[X]$ and is calculated as:

$$Var[X] = \int_{-\infty}^{+\infty} (x - \mu)^2 \cdot f(x)\, dx = E[(X - \mu)^2].$$

The variance is also written as σ^2. The *standard deviation* is defined as $\sigma = \sqrt{Var[X]}$.

Properties of the variance

The variance satisfies the following properties:

1. Let α be a real number and X be a random variable. Then:

$$Var[\alpha \cdot X] = \alpha^2 \cdot Var[X].$$

2. The following computational formula is often useful:

$$Var[X] = E[X^2] - (E[X])^2.$$

3. If X, Y are two **independent** random variables, then:

$$Var[X + Y] = Var[X] + Var[Y].$$

   - This generalizes to as many **independent** random variables as you would want to. In essence, we have for n independent random variables $X_1, X_2, \ldots, X_n$:

$$Var\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} Var[X_i].$$

4. If X, Y are two **independent** random variables and α, β are real numbers, then:

$$Var[\alpha \cdot X + \beta \cdot Y] = \alpha^2 \cdot Var[X] + \beta^2 \cdot Var[Y].$$

Expectations and variances of common distributions

In this section, we provide the expectations and variances for the most common probability distributions we have encountered so far.

Bernoulli

Recall that a Bernoulli random variable takes value 1 with probability p and 0 with probability $1 - p$. We have:

- $E[X] = p$.
- $Var[X] = p(1 - p)$.

Binomial

The Binomial distribution counts the number of successes in n independent Bernoulli trials, each with success probability p. We have:

- $E[X] = np$.
- $Var[X] = np(1 - p)$.

Geometric

The Geometric distribution counts the number of trials until the first success, where each trial has success probability p. We have:

- $E[X] = \frac{1}{p}$.

- $Var[X] = \frac{1-p}{p^2}$.

Hypergeometric

The Hypergeometric distribution with parameters N, K, n represents sampling without replacement. We have:

- $E[X] = n\frac{K}{N}$.
- $Var[X] = n \cdot \frac{K}{N} \cdot \frac{N-K}{N} \cdot \frac{N-n}{N-1}$.

Poisson

The Poisson distribution with rate parameter λ models the number of events in a fixed interval. We have:

- $E[X] = \lambda$.
- $Var[X] = \lambda$.

Example 7.5. Chasing cars

A transportation engineer is counting vehicles that are passing through an intersection. They have observed that vehicles pass following a Poisson distribution with rate 1 vehicle every 30 seconds.

- The expected number of vehicles in the next 30 seconds is $\lambda = 1$.
- The expected time until the next vehicle is $\frac{1}{\lambda} = \frac{1}{1/30 \text{ seconds}} = 30 \text{ seconds}$.
- The expected time until the 10th vehicle passes is $\frac{10}{\lambda} = \frac{10}{1/30 \text{ seconds}} = 300 \text{ seconds} = 5 \text{ minutes}$
.

Be careful with the rate you are using! In general, given a rate λ in some time unit, then if we are asked to find an expectation in time t
, we need to replace λ with $\lambda \cdot t$.

Example 7.6. Chasing cars: part 2

A transportation engineer is counting vehicles that are passing through an intersection. They have observed that vehicles pass following a Poisson distribution with rate 1 vehicle every 30 seconds. How many vehicles should they expect to see in 3 hours?

- This is still a Poisson distribution with a rate of 1 every 30 seconds.
- That said, it would be easier to transform the rate into the period asked — 3 hours.
- $3 \text{ hours} = 10800 \text{ seconds} \implies \lambda = 360 \text{ vehicles per } 3 \text{ hours}$

Uniform

Recall that there is a discrete and a continuous uniform distribution. We have:

- **Discrete** between a and b, that is X takes values in $a, a+1, \ldots, b$:
    1. $E[X] = \dfrac{a+b}{2}$.
    2. $Var[X] = \dfrac{(b-a+1)^2 - 1}{12}$.
- **Continuous** in (a, b):
    1. $E[X] = \dfrac{a+b}{2}$.
    2. $Var[X] = \dfrac{(b-a)^2}{12}$.

Exponential

The Exponential distribution with rate parameter λ models the time until an event occurs. We have:

- $E[X] = \dfrac{1}{\lambda}$.

- $Var[X] = \frac{1}{\lambda^2}$.

Gamma and Erlang

The Gamma distribution with parameters λ, k (and the Erlang distribution when k is an integer) models the time until the kth event. We have:

- $E[X] = \frac{k}{\lambda}$.
- $Var[X] = \frac{k}{\lambda^2}$.

Normal

Last, but not least, we see the normal. The good news is that μ and σ^2 are both in the definition of the distribution!

- $E[X] = \mu$.
- $Var[X] = \sigma^2$.

Food for thought

Are two distributions with the same expectation and variance the same distributions? The answer is no: consider for a counterexample an exponential distribution with $\lambda = 0.5$ and a normal distribution $\mathcal{N}(2, 4)$. Their means are both equal to 2 and their variances are both equal to 4, but they are categorically not the same distribution (see the next figure).

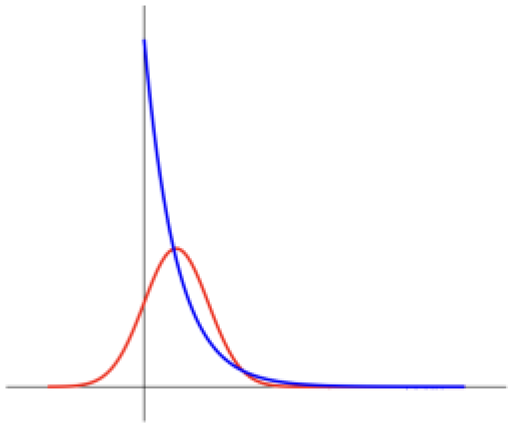

Figure 7.1. An example of how two distributions can have the same mean and variance but yet not be similar at all. The red curve shows a normal distribution $\mathcal{N}(2, 4)$, *while the blue curve shows an exponential distribution with* $\lambda = 0.5$.
Both have mean = 2 and variance = 4.

Review

Table 7.1. Discrete random vari

Name	**Parameters**	**Values**	**pmf**
Bernoulli	$0 < p < 1$	$\{0, 1\}$	$p(0) = 1 - p$ $p(1) = p$
Binomial	$0 \leq p < 1$, $n \geq 0$	$\{0, 1, \ldots, n\}$	$p(x) = \binom{n}{x} p^x \cdot (1$
Geometric	$0 < p < 1$	$\{1, 2, \ldots\}$	$p(x) = (1 - p)^{x-1} \cdot$
Hypergeometric	$N, K, n \geq 0$	$\{1, 2, \ldots\}$	$p(x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$
Poisson	$\lambda > 0$	$\{0, 1, \ldots\}$	$p(x) = e^{-\lambda} \frac{\lambda^x}{x!}$
Uniform	-	$[a, b]$	$p(x) = \frac{1}{b - a + 1}$

Table 7.2. Continuous random variables

Name	Parameters	Values	pdf
Uniform	–	$[a, b]$	$f(x) = \dfrac{1}{b-a}$
Exponential	$\lambda > 0$	$[0, +\infty)$	$f(x) = \lambda \cdot e^{-\lambda x}$
Gamma	$\lambda > 0, k > 0$	$[0, +\infty)$	$f(x) = \dfrac{\lambda^k \cdot x^{k-1} \cdot e^{-\lambda x}}{\Gamma(k)}$
Erlang	$\lambda > 0$, integer $k > 0$	$[0, +\infty)$	$f(x) = \dfrac{\lambda^k \cdot x^{k-1} \cdot e^{-\lambda x}}{(k-1)!}$
Normal	μ, σ^2	$(-\infty, +\infty)$	$f(x) = \dfrac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

Activities

Activity 1: Basic expectation and variance properties

First, let's practice some basic expectation and variance properties. Answer the following questions. If False, correct the statement.

Problem 1: True or False

- $\boldsymbol{E[3X+2] = 3E[X]}$

- True or False

$$\boldsymbol{E\left[3X^2\right] = 3E\left[X^2\right]}$$

- True or False

$$\boldsymbol{Var\left[3X+2\right] = 9Var\left[X\right]+4}$$

- True or False

For the next one, you may assume that X and Y are independent random variables.

- $\boldsymbol{Var\left[3X+Y\right] = 9Var\left[X\right]+Var\left[Y\right]}$
  - True or False

Activity 2: Rating the latest Marvel movie

A movie magazine decides to allow its reviewers to provide a rating ranging from 1 to 4 stars. Here, we are specifically focusing on the latest blockbuster in the Marvel Cinematic Universe. In the next two problems, we will see what the expected stars the movie will get if reviewers are allowed to provide an integer number of stars, or when they are allowed to providen *any real number* of stars.

Problem 2: Integer stars

Assume that the number of stars for any movie is represented as a **discrete random variable** X with pmf equal to $p(x) = \dfrac{x^2}{30}$ for $x = 1, 2, 3, 4$. What is the expected number of stars for any movie (that is, what is the expectation of X)? What is the variance of X?

- $E[X] =$
- $Var[X] =$

Problem 3: Real stars

Now, assume that the number of stars (again, from 1 to 4) a reviewer can give to a movie can be represented as a **continuous random variable** X with pdf equal to $f(x) = \frac{x^2}{21}$ for $1 \leq x \leq 4$. What is the expected value of X in this case? How about the variance of X?

- $E[X] =$
- $Var[X] =$

Activity 3: A printer replacement policy

An office has bought a new printer which is supposed to have lifetime that is **exponentially distributed** with an expected lifetime at 2 years. Answer the following questions.

Problem 4

What is the probability that the printer still works after 2 years?

Problem 5

Based on your answer in Problem 4, if the company buys $n = 30$ printers for their 30 offices, how many of them should we expect to still work after 2 years?

Problem 6

The company is starting a new policy. They will replace the printer either when it breaks down (recall that it breaks down in time that is exponentially distributed with like in Problem 4), or when it becomes 2 years old, whichever comes first. What is the expected lifetime of every printer the company buys?

Note that if we allowed a printer to work until it breaks down, then we would (on average) expect to replace a printer every two years – as this is the given expectation of the exponential distribution. But if we replace any printer the moment they are 2 years old, then the expectation should go down, shouldn't it?

Activity 4: The law of **total expectation**

In this activity, we derive the law of total expectation for both discrete random variables (in the form of a summation) and continuous random variables (in the form of an integral).

Problem 7: A simple case

Let us begin with something simple. An experiment is successful 90% of the time (and failed the remaining time). We perform 10 experiments. How many should we expect to be successful?

Problem 8: External conditions

Let us complicate this slightly. Once again we perform 10 experiments, where an experiment can be successful or failed. However, the success probability depends on some external conditions. If the conditions are good, the probability of success is 95%; in average conditions, the probability becomes 90%; in bad conditions, the probability is lower at 75%. All experiments will take place at the same conditions; so either all experiments will have good conditions, or all experiments will have average conditions, or all experiments will have bad conditions.

Assuming conditions are equally probable (that is, good/average/bad conditions appear $\frac{1}{3}$ of the time), what is the expected number of successful experiments now?

Problem 9: Generalizing the result

Can we generalize the previous result? What if we had m different possible conditions, each appearing with probability $\pi_i, i = 1, \ldots, m$ and each leading to probability of success p_i? How many experiments should we expect to be successful if we perform $n = 10$ experiments?

Based on your answers so far, we observe that if we can partition the space in m mutually exclusive and collectively exhaustive events A_i each with probability of appearing equal to $P(A_i)$, then the expected value of random variable X can be found by:

Definitions and Theorems 7.7. The law total expectation for discrete random variables

$$E[X] = \sum_{i=1}^{m} E[X|A_i] \cdot P(A_i).$$

How do you think this should look like for continuous random variables?

Problem 10

Consider a continuous random variable with pdf $f(x) = \frac{1}{2}(1 + \theta \cdot x)$ for $-1 \leq x \leq 1$, where θ is uniformly distributed between 0 and 1. What is the expected value of X?

The law of total expectation applies to continuous random variables, too. Consider X, Y as continuous random variables, such that we know $E[X|Y = y]$. Also assume that Y has pdf $g(y)$. Then, we have:

Definitions and Theorems 7.8. The law total expectation for continuous random variables

$$E\left[X\right] = \int_{-\infty}^{+\infty} E\left[X|Y=y\right] \cdot g(y)dy.$$

Activity 5: Expectations of functions

This is a slightly bigger question, that I am anticipating you can work on after lecture. Recall that for any function $g(X)$ of a random variable X, we can calculate its expectation as:

$$E\left[g(X)\right] = \begin{cases} \sum_S g(x) \cdot p(x), & \text{if } X \text{ is discrete,} \\ \int_S g(x) \cdot f(x)dx, & \text{if } X \text{ is continuous.} \end{cases}$$

Problem 11

Consider again the printer from Activity 3. The speed with which the printer works (and prints documents) is a function of its age. When the printer is x years old, its speed is given by $g(x) = \frac{\sqrt{x+1}}{x+1}$ velocity units. For example, when it is just bought, and the printer is 0 years old, its speed is equal to 1 velocity unit; on the other hand, after 2 years its speed drops to $\sqrt{3}/3$ velocity units.

What is the expected speed of a printer if the company implements the policy of replacing the printer when it breaks down or when it becomes 2 years old, whichever comes first?

8. The central limit theorem

Theory

Chapter objectives

Learning outcomes

After this chapter, you will be able to:

- Explain why the normal distribution appears often in real life.
- Recall and use the central limit theorem.

Chapter questions

1. If X and Y are normally distributed, then what is $X+Y$ distributed as? How about $X-Y$?
2. Is this true for all linear combinations of normally distributed random variables? Is $Z=\sum_{i=1}^{n} a_i X_i$ normally distributed if all $X_1, X_2, \ldots, X_n$ are normally distributed?
3. Why is the **normal distribution** so prominent around us?

4. What does the **central limit theorem** state?

Motivation

The normal distribution

Why is the normal distribution so ubiquitous? Why is it that in many instances we see normally distributed quantities around us?

Testing a hypothesis

Consider the case of trying to predict the outcome of an election. A good way to do so would be to pick a sample n of potential voters, and ask them what they would be voting for. Say x say they are voting for Candidate 1: this could lead you to deduce that x/n is the proportion of the vote that Candidate 1 would get in the general election. But, what can you say for the distribution of the proportion? How probable is it that your prediction is off?

Introduction

First, we motivate why the central limit theorem applies; later in the chapter, we state the theorem in its entirety. We finish this chapter with an example.

Motivating the central limit theorem

Say we throw a "fair" die (uniform distribution of getting any of the six numbers) 100,000 times and we collect back the appearances of each number. We then plot our results and observe that, as expected, every number appears equally probably. Now, let's consider a game of Monopoly. In this board game, you throw 2 dies and the summation of the two numbers is the number of steps you are expected to take: note that this number goes from 2 (both dies land on the side of 1) to 12 (both dies land on the side of 6). We already saw earlier how the probability of getting a 7 is higher than the rest. There seems to be an interesting pattern emerging... Let's investigate this more!

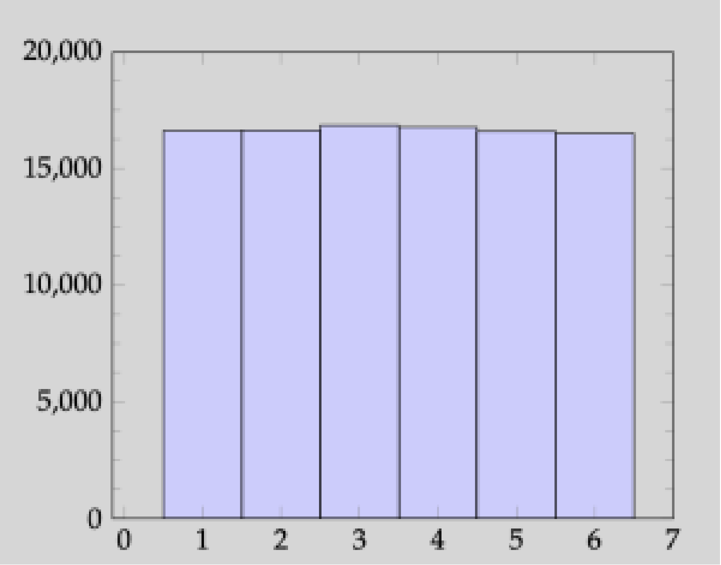

Figure 8.1. The number of occurrences for each number from 1 to 6 for one fair die.

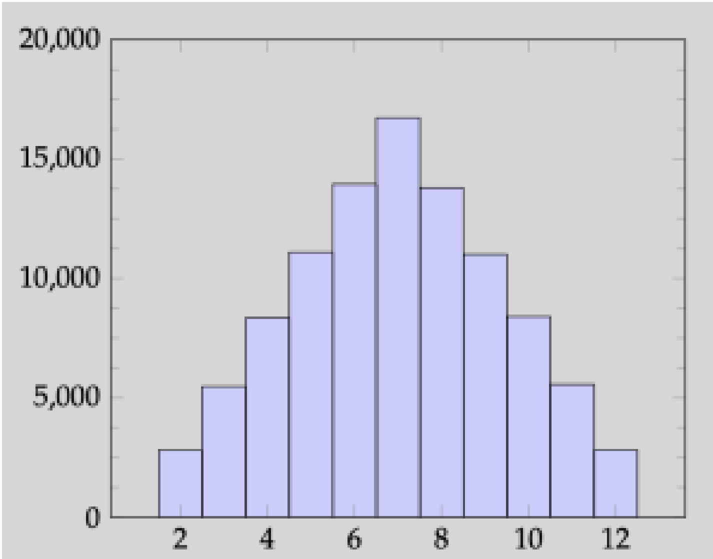

Figure 8.2. The number of occurrences for numbers from 2 to 12 for two fair dies.

We now proceed to show what happens when we toss 5 and 10 dies. The pattern is even clearer now: it seems like **the summation of many random variables with the same distribution follows a normal distribution**! Let's see whether we can formally state what we observe.

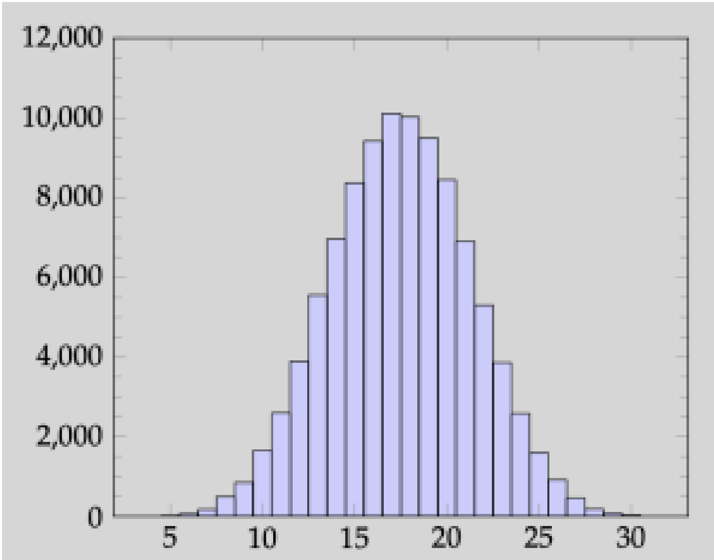

Figure 8.3. The number of occurrences for numbers from 5 to 30 for five fair dies.

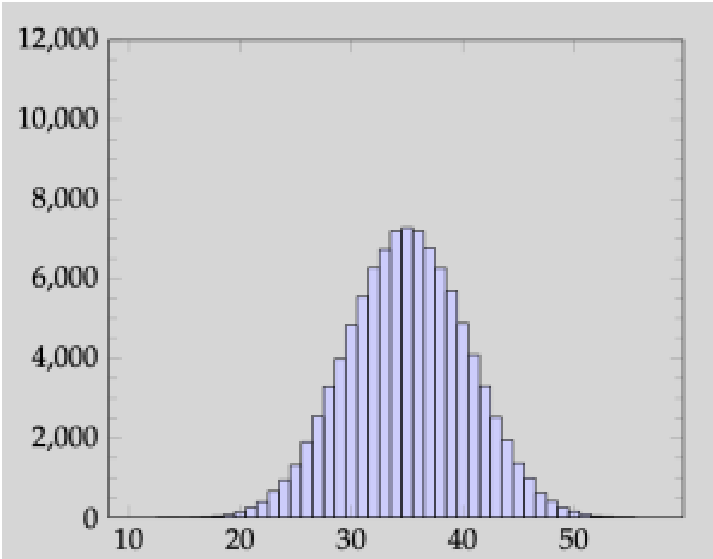

Figure 8.4. The number of occurrences for numbers from 10 to 60 for ten fair dies. The distribution closely follows a normal curve (shown as dashed line).

The central limit theorem

Expectation and variance review

Let us recall a few important properties from calculating expectations and variances. Given a series of independent random variables $X_i, i = 1, \ldots, n$, we have that:

- $E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i].$

- $Var\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} Var[X_i]$.

Moreover, recall that:

- $E[\alpha \cdot X] = \alpha \cdot E[X]$.
- $Var[\alpha \cdot X] = \alpha^2 \cdot Var[X]$.

Combining, we have that for quantities $\frac{\sum_{i=1}^{n} X_i}{n}$, we get:

- $E\left[\frac{\sum_{i=1}^{n} X_i}{n}\right] = \frac{\sum_{i=1}^{n} E[X_i]}{n}$.
- $Var\left[\frac{\sum_{i=1}^{n} X_i}{n}\right] = \frac{\sum_{i=1}^{n} Var[X_i]}{n^2}$.

Assuming that we have $\mu = E[X_1] = E[X_2] = \ldots = E[X_n]$ and $\sigma^2 = Var[X_1] = Var[X_2] = \ldots = Var[X_n]$, we may write that:

$$E\left[\sum_{i=1}^{n} X_i\right] = n \cdot \mu \qquad E\left[\frac{\sum_{i=1}^{n} X_i}{n}\right] = \frac{n \cdot \mu}{n} = \mu$$

$$Var\left[\sum_{i=1}^{n} X_i\right] = n \cdot \sigma^2 \quad Var\left[\frac{\sum_{i=1}^{n} X_i}{n}\right] = \frac{n \cdot \sigma^2}{n^2} = \frac{\sigma^2}{n}$$

The full theorem

We are now ready to state the central limit theorem in its entirety.

Definitions and Theorems 8.1. The central limit theorem

Let $X_i, i = 1, \ldots, n$ be a series of independent, identically distributed random variables with expected value $E[X_i] = \mu$ and variance $Var[X_i] = \sigma^2$. Define $Z = \sum_{i=1}^{n} X_i$ (i.e., as the summation of all random variables X_i) and $Y = \sum_{i=1}^{n} X_i/n$ (i.e., as the average of all X_i).

Then:

- Z follows a normal distribution when n is large enough with parameters

$$\mu_Z = \sum_{i=1}^{n} E[X_i] = n \cdot \mu \text{ and}$$

$$\sigma_Z^2 = \sum_{i=1}^{n} Var[X_i] = n \cdot \sigma^2.$$

$$Z \sim \mathcal{N}(n \cdot \mu, n \cdot \sigma^2)$$

- Y follows a normal distribution when n is large enough with parameters

$$\mu_Y = \frac{1}{n} \sum_{i=1}^{n} E[X_i] = \mu \text{ and}$$

$$\sigma_Y^2 = \frac{1}{n^2} \sum_{i=1}^{n} Var[X_i] = \frac{\sigma^2}{n}.$$

$$Y \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

Example 8.1. Totally random buses

The time you have to wait for a bus every day is uniformly distributed between 0 and 4 minutes. What is the probability you have to wait for more than or equal to 2.2 minutes for the bus on average in the next 300 days?

We may now fully use the central limit theorem.

- As $n = 300$ (big enough), we know that the average time you will wait for the bus in the next 300 days is normally distributed with mean μ and variance $\sigma^2/300$.
- The time you wait for a single bus is uniformly distributed with mean $\mu = \frac{0+4}{2} = 2$ minutes and variance $\sigma^2 = \frac{(4-0)^2}{12} = 4/3$ minutes^2.
- Combining, the average time T you wait for the bus follows $\mathcal{N}(2, \frac{4}{900})$.

Example 8.2. Totally random buses (cont'd)

We are interested in $P(T > 2.2) = 1 - P(T \leq 2.2)$. First let us convert to the proper z value:

$$Z = \frac{X - \mu}{\sigma} = \frac{2.2 - 2}{2/30} = 3.$$

Looking at the z-table:

$$P(T \leq 2.2) = 0.9987 \implies P(T \geq 2.2) = 1 - 0.9987 = 0.0013.$$

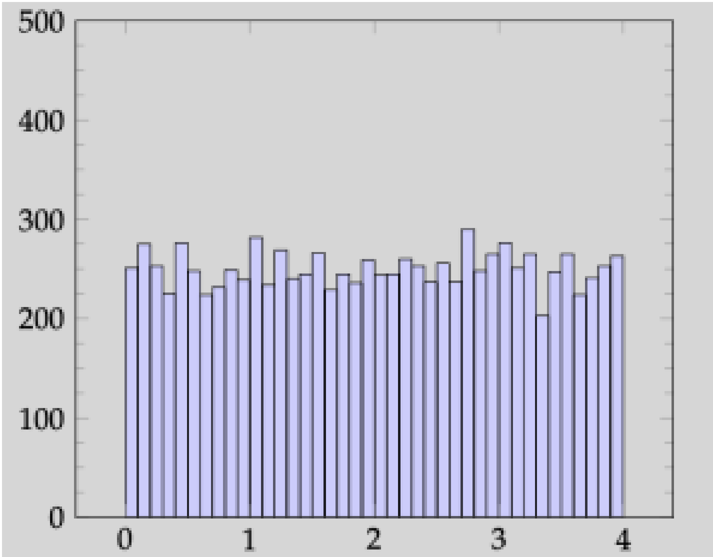

Figure 8.5. The amount of time we wait for one bus.

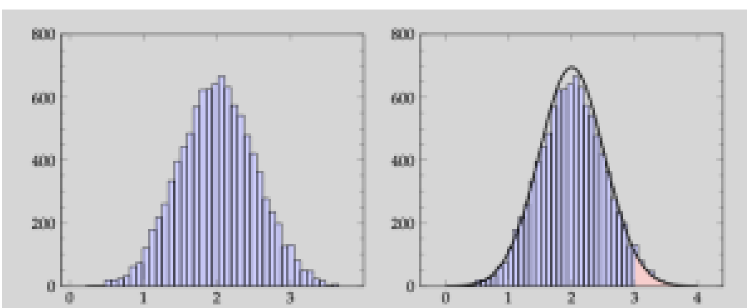

Figure 8.6. The average amount of time we spend waiting for a bus for a total of [latex]n=5[/latex] days (left), followed by a visualization of the "probability" of waiting (on average) for more than 3 minutes in 5 days. It is shown in red on the right side of the figure.

Activities

Activity 1: Normally distributed random variables

We begin this section by mentioning something very important:

Adding two (or more) independent normal distributed random variables together results in a normal distribution.

In mathematical terms: if $X_i, i = 1, \ldots, n$ are independent random variables distributed normally, then $Z = \sum_{i=1}^{n} X_i$ is also normally distributed. The same is true for any linear combination, i.e., $Z = \sum_{i=1}^{n} a_i \cdot X_i$ is normally distributed. Recall that to fully describe a normally distributed random variable, we simply need its expectation and variance.

Problem 1: Expectation and variance of the sum of normally distributed random variables

Consider two normally distributed random variables $X \sim \mathcal{N}\left(\mu_X, \sigma_X^2\right)$ and $Y \sim \mathcal{N}\left(\mu_Y, \sigma_Y^2\right)$, which are independent. What is the expectation and variance of $X + Y$?

In general, $Z = \sum_{i=1}^{n} a_i \cdot X_i$ is normally distributed with

$\mu = E\left[Z\right] = \sum_{i=1}^{n} a_i \cdot E\left[X_i\right]$ and

$$\sigma^2 = Var[Z] = \sum_{i=1}^{n} a_i^2 \cdot Var[X_i].$$

Problem 2: Application

You own a portfolio of stocks that consists of 5 stocks S_1 and 10 stocks S_2. Let X_1 be the price of stock S_1 and X_2 the price of stock S_2 one year from now. X_1 is normally distributed with $\mathcal{N}(50, 100)$ and X_2 is normally distributed with $\mathcal{N}(60, 100)$. Last, assume that the two stocks are totally unrelated (i.e., they are independent of one another).

What is the probability that your portfolio is worth more than $1000 one year from now? What is the probability that your portfolio is worth less than $800 one year from now?

Problem 3: Comparing random variables

Consider two normally distributed random variables X and Y, which are independent. Can you make the claim that $X - Y$ is normally distributed? What is the expectation and variance of $X - Y$?

Problem 4: Comparing stock prices

Again, consider that X_1 is the price of stock S_1 and X_2 is the price of stock S_2 one year from now. Both are normally distributed:

$X_1 \sim \mathcal{N}(50, 100)$ and $X_2 \sim \mathcal{N}(60, 100)$. Finally, assume that X_1, X_2 are independent.

What is the probability that S_1 (a single stock of that) is worth more than S_2 (again, consider only a single stock of that one, too) a year from now? In mathematical terms, what is $P(X_1 > X_2)$?

What about the case where two random variables (distributed normally) are **not independent**? One could make the argument that two stocks affect each other, and knowing that one has gone up may affect our perspective of the other one going up, too. In this case, where independence is hard to assume and use, can we still compare two stocks (or calculate the probability of our portfolio being above a desired value a year from now)?

The answer is yes! However, we need to introduce ideas like **dependence** and measure it with **covariance** and **correlation**. More on that in the next activities.

Activity 2: Using the central limit theorem

In the online course world, certain classes have thousands of students: these classes are sometimes referred to as *Massive Open Online Courses* (MOOC). On average there are 8000 students in a course. For the purposes of this exercise, we assume that a student successfully finishes a MOOC 7.5% of the time and that all students behave independently. Assume that probability is the same regardless of the course. Finally, assume that every class has **exactly** 8000 students.

Answer the following questions.

Problem 5: Setting up the distribution

What is the best distribution to model the number of students to successfully finish a single class? What is the mean and the variance of the number of students who successfully finish a single class?

Problem 6: Setting up the central limit theorem

Now, consider the case of an online course provider, such as Coursera. Assume the total number of classes the provider offers is equal to 1000. For each individual class, the number of students who successfully finish that specific class follows the distribution you found in Problem 5.

What distribution does the average number of students graduating from all class of the provider follow? The average number of students graduating in all 1000 classes can be found by summing the number of graduates in each class and dividing by 1000.

Activity 3: Setting up hypothesis testing

Recall that we have made the hypothesis that a student successfully finishes a class 7.5% of the time. We have also found that the number of students successfully finishing one class follows a binomial distribution (see Problem 5) and the average number of students across 1000 courses follows a normal distribution (based on Problem 6).

The online course provider has decided to check whether this

“7.5%” success rate is true or not. They have decided to survey 15 of the 1000 courses they offer and they found the following:

Table 8.1. Counts of students who successfully “pass” in 15 different courses.

Course 1	588	Course 2	645	Course 3	632
Course 4	623	Course 5	635	Course 6	641
Course 7	644	Course 8	611	Course 9	630
Course 10	628	Course 11	569	Course 12	637
Course 13	635	Course 14	677	Course 15	610

Problem 7: Checking the numbers

What should the average number of graduating students in these 15 classes be distributed as? Once again, to find the average number of graduates, simply add up the numbers for the 15 classes and divide by $n = 15$. What is the mean and the variance of the distribution?

Problem 8: Using the central limit theorem

From the numbers they found (see the tabulated data from Activity 3) that on average a pretty big 627 students in each class successfully finishes it .

Based on the distribution you have identified in Problem 7, what is the probability that there are more than 627 students on average finishing each class?

Problem 9: Huh?

Hopefully, you have gotten a very, very small probability for Problem 8. This implies that the numbers they got appear to be *highly* improbable. While we are at it, let's calculate one more probability. What is the probability that the number of students that (on average) finish successfully each course is more than 610?

Problem 10: Rejecting a hypothesis

Considering the small probability you got for the average to be as high as 610 or more, what can you deduce for the success rate of 7.5%? Should they believe it? Do *you* think it is valid? Or is the true success rate higher/lower?

Congratulations, you just performed a fully-fledged data analysis experiment!

9. Jointly distributed random variables

Theory

Chapter objectives

Learning outcomes

After this chapter, you will be able to:

- Describe and recognize jointly distributed random variables.
- Define joint, marginal, and conditional probability mass functions for discrete random variables.
- Define joint, marginal, and conditional probability distribution functions for continuous random variables.
- Use joint, marginal, and conditional probability mass and distribution functions to calculate probabilities.

Motivation: “Can you hear me now”?

Not all of us pay attention all the time in a Zoom call; sometimes it is our fault (we are distracted or busy), but others it is not (technical difficulties, bad reception). So the question becomes: how many times does something need to be repeated before you hear it? Note that it does not only depend on whether you are paying attention (which is a random variable), but also on whether you are having a clear connection (another random variable).

Jointly distributed random variables

Real life and its outcomes can be viewed as a combination of random events, rather than a single random event. Succeeding in an exam has many factors that do not rely on only your preparation: you need to be healthy and well-rested, you need to be focused during the exam, you need to have luck at your side, you need to have a calculator whose batteries are still working. And even when all of these things align, you also need to be there on time, which means that you need to catch a bus, that there is no construction causing traffic jams, etc. We can go on like this *a lot*.

The truth is that in this class we have focused on single random variables that are distributed their own way. What about the case where two random variables are distributed alongside each other?

But, wait? Did we not discuss the probability of two events happening hand-in-hand during the first chapters of this book? And did we not discuss specifically what happens if those two events are independent or not?

You are correct. We have discussed what happens when two events are happening at the same time. We also discussed what the probability is that an event happens given another event happening. However, there are two caveats in our discussion earlier:

1. We only focused on discrete (countable) events: it is time we see what this implies in the continuous space too.
2. We saw this in terms of events and sets. We are now going to have that discussion in terms of distributions, probability mass/density functions.

Definition

We begin with the definition of jointly distributed random variables.

Definitions and Theorems 9.1. Jointly distributed random variables

Let X and Y be two random variables. The probability distribution that defines their *simultaneous* behavior is referred to as a **joint probability distribution**. The two random variables X and Y are then called **jointly distributed random variables**.

Example 9.1. Jointly distributed random variables

Here are some examples of jointly distributed random variables.

- The times you have to repeat yourself on the phone and your signal reception.

- The grade you receive in an exam and the amount of sleep you've had the night before.
- The performance of two or more stocks in your portfolio.
- The box office of a movie and the critical reception.

We observe here that jointly distributed does not imply immediate effect. For example, a student could get a very high grade in an exam, even if they slept very little the night before; or a movie could make a lot of money in the box office, despite being universally hated by reviewers. However, jointly distributed random variables imply that what we see is a combination of random variables rather than outcome of a single random variable.

Exercise 9.1. Single or jointly distributed?

Are the following better modeled as a single random variable or as jointly distributed random variables?

- Getting a higher grade in an exam than the person sitting next to you?
- Throwing a die?
- Throwing two dies and having the first die land on a higher number than the second one?

Another example

Securing a position after college might require some effort. If the economy is doing well ("is good"), then a student could get more job interview invitations, and consequently there are more chances for a job opportunity. If the economy is average, or if it is outright bad, then a student may struggle to get interviews and/or a job.

Based on our definitions, the state of the economy is a random variable. The same can be said about the number of job interviews that a student gets invited to. In the end of the day, the number of interviews that a student needs to go on before they secure a position after graduation is a **jointly distributed random variable**. Let's assume that the probabilities are as given in the table below.

Table 9.1. Number of job interviews required to get a job depending on the state of the economy.

X = job interviews to get a job	Y = state of the economy		
	Bad	**Average**	**Good**
1	0.01	0.05	0.20
2	0.03	0.05	0.18
3	0.03	0.12	0.08
≥4	0.08	0.12	0.05

Jointly distributed discrete random variables

If X and Y are discrete random variables, then (X, Y) is called a jointly discrete bivariate random variable.

Definitions and Theorems 9.2. Joint probability mass function

The **joint probability mass function** is defined as:

$$f_{XY}(x,y) = P(X = x, Y = y).$$

It follows the next three properties:

1. $f_{XY}(x,y) \geq 0, \forall x, y.$
2. $\sum_x \sum_y f_{XY}(x,y) = 1.$
3. $P((X,Y) \in A) = \sum \sum_{(x,y)\in A} f_{XY}(x,y)$

.

A couple of quick notes about the notation here. You will observe that the joint probability mass function is given by $f_{XY}(x,y)$. We had previously reserved $f(\cdot)$ for continuous random variables, keeping $p(\cdot)$ for discrete ones. For convenience, we only use $f(\cdot)$ for joint probability distributions.

Additionally, we notice that there is a subscript in the function. The subscript is supposed to reveal which random variables the function is including. For example $f_{XY}(3,4)$ would imply that the function is considering random variables X and Y and is asking for them to be equal to 3 and 4, respectively.

Marginal probability mass function

The marginal probability mass function addresses questions involving only one of the two (or more) jointly distributed random variables. In essence, given two (or more) jointly distributed random variables, what is the probability that one of them satisfies a certain property? That is, the marginal distribution of a random variable answers the question: “what is the probability that X takes a certain value, regardless of Y?”

Definitions and Theorems 9.3. Marginal probability mass function

The **marginal probability mass function** (marginal pmf) of a discrete random variable is computed by summing over all possible values of the other random variable. For two random variables, X and Y:

1. The marginal distribution of X: $$f_X(x) = P(X = x) = \sum_y f_{XY}(x, y).$$
2. The marginal distribution of Y: $$f_Y(y) = P(Y = y) = \sum_x f_{XY}(x, y).$$

Example 9.2. Getting a job after college

Using the table from earlier, what is the probability that:

1. a student gets a job in 1 interview and that the economy is good? The probability is 20%.
2. a student gets a job in less than or equal to 3

interviews and the economy is average? We sum: 0.05 + 0.05 + 0.12 = 22%.

3. a student gets a job in more than 3 interviews? We are after the probability of $P(X>3)=P(X\geq 4)$. Based on the definition of marginal distributions: $P(X\geq 4)=0.08+0.12+0.05=0.25.$ The probability is 25%.
4. the economy is good?We are after $P(Y=\text{Good})$: $P(Y=\text{Good})=0.20+0.18+0.08+0.05=0.51$. The probability is 51%.

For calculating marginal distributions in discrete random events given in tabular format, we may also add up the probabilities in the columns and rows:

Table 9.2. Number of job interviews required to get a job depending on the state of the economy. We now supplement the table with the marginal probabilities $f_X(x)$ and $f_Y(y)$.

X = job interviews	Y = state of economy			$f_X(x)$
	Bad	**Average**	**Good**	
1	0.01	0.05	0.20	**0.26**
2	0.03	0.05	0.18	**0.26**
3	0.03	0.12	0.08	**0.23**
≥4	0.08	0.12	0.05	**0.25**
$f_Y(y)$	**0.15**	**0.34**	**0.51**	**1**

Conditional probability mass function

On the other hand, the conditional probability mass function addresses questions involving one of the variables, given information about one or more of the other jointly distributed random variables. That is, the conditional distribution of a random variable answers the question: "what is the probability that X takes a certain value, given that Y has taken another value already?"

Definitions and Theorems 9.4. Conditional probability mass function

The **conditional probability mass function** (conditional pmf) of a discrete random variable given values for the other ones is computed by dividing the joint pmf of all over the marginal pmf of the others. For two random variables, X and Y:

1. The conditional distribution of X given $Y = y$:
$$f_{X|Y=y}(x) = f_{X|y} = P(X = x|Y = y) = \frac{f_{XY}(x,y)}{f_Y(y)}.$$
2. The conditional distribution of Y given $X = x$:
$$f_{Y|X=x}(y) = f_{Y|x} = P(Y = y|X = x) = \frac{f_{XY}(x,y)}{f_X(x)}.$$

Of course, for the conditional pmf to make sense, we need that $f_X(x) > 0$ and $f_Y(y) > 0$.

Example 9.3. Conditional probability

What is the probability that a student gets a job in 1 interview if we know that the economy is good?

This is the definition of a conditional probability. Specifically, we want to calculate $P(X = 1|Y = \text{Good})$.

$$P(X = 1|Y = \text{Good}) = \frac{f_{XY}(1, \text{Good})}{f_Y(\text{Good})} = \frac{0.2}{0.51} = 0.3922.$$

One full example

Before finishing the discussion on discrete jointly distributed random variables, let us solve a bigger, full example where we need to derive all of the special probability functions:

Example 9.4. Jointly distributed discrete random variables

Two discrete random variables X and Y have a joint distribution of $f_{XY}(x, y) = \dfrac{x+y+1}{c}$, for x and y equal to 0, 1, or 2.

1. What should c be?
2. What is $P(X \leq 1, Y = 1)$?
3. What is $P(Y = 1)$?
4. What is $P(X \leq 1|Y = 1)$?

Solution:

1. From the second property of joint pmfs:

$$\sum_{x=0}^{2}\sum_{y=0}^{2}\frac{x+y+1}{c}=1 \implies \frac{1}{c}(1+2+3+2+3+4+3+4+5)=1 \implies c=27.$$

2. With the full joint pmf:

$$P(X\leq 1, Y=1)=\sum_{x=0}^{1} f_{XY}(x,1)=\frac{2}{27}+\frac{3}{27}=\frac{5}{27}.$$

3. Computing the marginal pmf:

$$f_Y(y)=\sum_{x=0}^{2} f_{XY}(x,y)=\frac{3y+6}{27}.$$

$$P(Y=1)=f_Y(1)=\frac{9}{27}=\frac{1}{3}$$

4. Computing the conditional pmf:

$$f_{X|Y=1}(x)=\frac{f_{XY}(x,1)}{f_Y(1)}=\frac{x+2}{9}$$

$$P(X\leq 1|Y=1)=f_{X|Y=1}(0)+f_{X|Y=1}(1)=\frac{2}{9}+\frac{3}{9}=\frac{5}{9}.$$

Jointly distributed continuous random variables

If X and Y are continuous random variables, then (X, Y) is called a jointly continuous bivariate random variable.

Definitions and Theorems 9.5. Joint probability density function

The **joint probability density function** is defined as $f_{XY}(x, y)$. Like for simple continuous random variables, f_{XY} reveals a relative likelihood rather than a probability value. It also follows three properties:

1. $f_{XY}(x, y) \geq 0, \forall x, y$.
2. $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{XY}(x, y)\, dx\, dy = 1$.
3. $$P((X, Y) \subset R) = \iint_R f_{XY}(x, y)\, dx\, dy$$

.

Example 9.5. A chemical mixture

A product is a mixture of two materials: let the volume of material 1 used be represented as X, and the volume of material 2 used be represented as Y. The joint probability density function of the two random variables is:

$$f_{XY}(x, y) = c(2x + 3y), \quad 0 \leq x \leq 1, 0 \leq y \leq 1$$

1. What is c?

From the second property:

$$\int_0^1 \int_0^1 c(2x + 3y)\, dx\, dy = 1 \implies c \int_0^1 (3y + 1)\, dy = 1 \implies c\frac{5}{2} = 1 \implies c = \frac{2}{5}.$$

2. What is the probability the first material has volume less than or equal to 0.5, and the second material has volume between 0.25 and 0.5?

$$P(X \leq 0.5, 0.25 \leq Y \leq 0.5) = \int_{0}^{0.5} \int_{0.25}^{0.5} \frac{2}{5}(2x + 3y)\, dy\, dx = \frac{13}{160}.$$

Marginal probability density function

The marginal probability density function plays the same roles as its discrete counterpart. Recall though that the continuous nature of the random variable(s) forces us to use the density function as a proxy of probability through proper integration!

Definitions and Theorems 9.6. Marginal probability distribution function

The **marginal probability density function** (marginal pdf) of a continuous random variable is computed by integrating over all possible values of the other random variable. For two random variables, X and Y:

1. The marginal distribution of X:

$$f_X(x) = \int_{-\infty}^{+\infty} f_{XY}(x, y)\, dy.$$

2. The marginal distribution of Y:

$$f_Y(y) = \int_{-\infty}^{+\infty} f_{XY}(x, y)\, dx.$$

Example 9.6. Computing marginal pdfs

For the chemical mixture with $f_{XY}(x, y) = \frac{2}{5}(2x + 3y)$:

Marginal pdf of X:

$$f_X(x) = \int_0^1 \frac{2}{5}(2x + 3y)\, dy = \frac{1}{5}(4x + 3).$$

Marginal pdf of Y:

$$f_Y(y) = \int_0^1 \frac{2}{5}(2x + 3y)\, dx = \frac{1}{5}(6y + 2).$$

Probability calculations:

$$P(X \leq 0.5) = \int_0^{0.5} \frac{1}{5}(4x + 3)\, dx = 0.4.$$

$$P(0.25 \leq Y \leq 0.5) = \int_{0.25}^{0.5} \frac{1}{5}(6y+2)\,dy = \frac{17}{80}.$$

Conditional probability density function

Like in the discrete case, the conditional probability density function helps us address probability issues for one of the jointly distributed random variables given information about the other(s).

Definitions and Theorems 9.7. Conditional probability distribution function

The **conditional probability distribution function** (conditional pdf) of a continuous random variable given values for the other ones is computed by dividing the joint pdf of all over the marginal pdf of the others. For two random variables, X and Y:

1. The conditional distribution of X given $Y = y$:
$$f_{X|Y=y}(x) = f_{X|y} = \frac{f_{XY}(x,y)}{f_Y(y)}.$$
2. The conditional distribution of Y given $X = x$:
$$f_{Y|X=x}(y) = f_{Y|x} = \frac{f_{XY}(x,y)}{f_X(x)}.$$

Once again, the conditional pdf is only defined when $f_X(x) > 0$ and $f_Y(y) > 0$.

Example 9.7. Conditional probability

What is the probability the first material has a proportion less than or equal to 50%, given that the second material has a proportion equal to 30%?

First, we calculate $f_{X|Y=y}(x)$:

$$f_{X|Y=y}(x) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{\frac{2}{5}(2x+3y)}{\frac{1}{5}(6y+2)} = \frac{4x+6y}{6y+2}.$$

Replacing $Y = 0.3$:

$$f_{X|Y=0.3}(x) = \frac{4x+1.8}{3.8}.$$

Finally:

$$P(X \leq 0.5|Y = 0.3) = \int_0^{0.5} \frac{4x+1.8}{3.8}\, dx = 0.3684.$$

Review

A very brief summary of this chapter follows. For two random variables X and Y that are *jointly* distributed, we have the following:

Joint pmf/pdf

- **TL;DR:** How are both variables distributed *simultaneously*?
- Denoted by $f_{XY}(x,y)$.
- For **discrete**: what is $P(X=x \text{ and } Y=y)$?
- For **continuous**: what is the relative likelihood of X having the value of x and Y getting the value of y?

Marginal pmf/pdf

- **TL;DR:** I am only interested in one of the random variables.
- Denoted by $f_X(x)$ or $f_Y(y)$.
- For **discrete**: Let's lose Y. What is $P(X=x)$?
- For **continuous**: Let's lose Y. What is the relative likelihood of X getting the value of x?

Conditional pmf/pdf

- **TL;DR:** I am given information on one of the random variables.
- Denoted by $f_{X|y}(x)$ or $f_{Y|x}(y)$.
- For **discrete**: I know what Y is! It is equal to y. What is $P(X=x|Y=y)$?
- For **continuous**: I know what Y is! It is equal to y. What is the relative likelihood of X taking value x now?

Recall that all definitions shown here for both discrete and continuous random variables may be extended to more than 2 random variables $X_1, X_2, \ldots$

Activities

Before we get started, please remember (it is crucial to differentiate these two notions): we refer to X and Y (upper case, by convention) as two random variables, whereas x and y (lower case) are values they take.

Activity 1: Jointly distributed discrete random variables

In the next four questions, we focus on a pair of *discrete* random variables X and Y, distributed with joint probability mass function

$$f_{XY}(x,y) = c \cdot \frac{x+1}{y}.$$

At this point, it is useful to remind ourselves that the probability mass function values *are actual probabilities*, since we are discussing about discrete random variables.

Problem 1: Joint probability mass functions

Assume that $X = \{1,2,3\}$ and $Y = \{10,20\}$, that is X is allowed to take values 1, 2, or 3, and Y is allowed to be equal to either 10 or 20. What should c be in order for this to be a valid joint pmf?

Problem 2: Table form

We may construct a table! Based on your answer in Problem 1, we may collect the different probability mass function values in tabular form. Fill the table below with the actual probabilities for each pair of values.

Table 9.3. Empty table to fill in your probabilities from Problem 2.

		Y	
3-4		10	20
	1		
X	2		
	3		

Verify once again that indeed the summation all of them is equal to 1, just to be sure.

Problem 3: Marginal probabilities

Where may we find the *marginal probabilities* for X or Y alone? In the table form we saw earlier, they are found by summing over a row or a column, depending on which one we are looking for! In this case, what is:

(2) $P(X=1)$? $P(Y=20)$?

- $P(X = 1) =$
- $P(Y = 20) =$

Problem 4: Conditional probabilities

Where may we find the *conditional probabilities* for X given $Y = y$ or for Y given $X = x$? In the table form we saw earlier, they are found by focusing on one column or row, and then dividing the probability we are looking for over the summation of all elements in that column or row. In our example, what is:

- $P(X = 1|Y = 20)$?
- $P(Y = 20|X = 1)$?

- $P(X = 1|Y = 20) =$
- $P(Y = 20|X = 1) =$

Of course, this table form is valuable; but it is also limited to smaller sample spaces. What happens when we are dealing with a huge number of cases? In that case, we need to resort to the actual formulations for each and every one of our probability calculations. We'll see how that algebraic way of dealing with probabilities works in Activity 3 (in Page 6).

First, though, let's take a walk in the realm of continuous random variables in the next activity.

Activity 2: Jointly distributed continuous random variables

Let X and Y be two continuous random variables that are allowed to take any value between 0 and 1: that is, $0 \leq X \leq 1$ and $0 \leq Y \leq 1$. We further assume that they are jointly distributed with probability density function:

$$f_{XY}(x,y) = \frac{12}{11}\left(x^2 + y^2 + xy\right).$$

Problem 5: Calculating a probability

Recall that with continuous random variables, we need to integrate properly to calculate a probability. With that in mind, what is the probability that $0.3 \leq X \leq 0.7$ and $Y \geq 0.75$?

Problem 6: Deriving a marginal distribution

Again, like you did earlier, derive the two marginal distributions. However, remember, that we are no longer summing! In the continuous space, we integrate. With that in mind, what is the marginal distribution of X? What is the marginal distribution of Y ?

Problem 7: Deriving the conditional distribution

What is $P(X \leq 0.5|Y = 1)$?

Think about your approach above. Could you have calculated the probability $P(X \leq 0.5|Y = y)$ for any value y?

Problem 8: Final touch

This is a little tougher. What is $P(X \leq Y)$? To help you get started, we have begun the solution approach, but do take a look at the hint for another explanation..

From the law of total probability for continuous random variables, we have:

$$P(X \leq Y) = \int_0^1 P(X \leq Y|Y = y) f_Y(y) dy = \int_0^1 P(X \leq y) f_Y(y) dy = \dots$$

Activity 3: Discrete, but infinite

Consider two discrete random variables X and Y that take on integer values $X \geq 0$ and $1 \leq Y \leq 5$. That is, X could be

3, 107, or 0, and Y could be equal to 1, 2, 3, 4, or 5. Their joint probability mass function is given by:

$$f_{XY}(x, y) = e^{-y} \cdot \frac{y^x}{5 \cdot x!}.$$

As a side note, remember that

Problem 9: Using the joint pmf

Let's start easy. What is the probability that both X and Y are equal to 1? That is, what is $P(X = 1 \cap Y = 1)$?

Problem 10: Deriving a marginal distribution

What is the probability that $Y = 1$, regardless of what X is? That is, what is $P(Y = 1)$?

After answering this, what is the probability that $Y = y$, for any value y? Is it always $\frac{1}{5} = 20\%$?

Problem 11: Deriving a conditional distribution

What is the probability that $X \geq 1$ given that $Y = 1$? That is, what is $P(X \geq 1 | Y = 1)$?

10. Jointly distributed random variables: extensions and common distributions

Theory

Chapter objectives

Learning outcomes

After this chapter, you will be able to:

- Calculate and use the expectation and variance of jointly distributed random variables.
- Define, calculate, and use the conditional expectation.
- Recognize independence in joint distributions.
- Use independence to calculate probabilities.
- Quantify the level of dependence using covariance and correlation.
- Calculate the expectation and variance of multiple random variables, independent or not.
- Find the pmf/pdf of a function of a random variable.

- Recognize multinomial distributions.
- Calculate probabilities (including marginal and conditional ones) for multinomially distributed random variables.
- Recognize bivariate normal distributions and explain them and their correlations.
- Calculate probabilities (including marginal and conditional ones) for bivariate normally distributed random variables.

Chapter questions

1. How do we calculate the **expectation and variance of a single random variable** in jointly distributed random variables?

   You need the **marginal distribution**.
2. How do we calculate the ***conditional*** **expectation and variance** of a single random variable given the other in jointly distributed random variables?

   You need the **conditional distribution**.
3. How do we calculate the **expectation of a function of both random variables** at the same time?

   You need the **joint distribution**.
4. How do we check if two random variables that are jointly distributed are **independent**?
5. What is the **covariance and correlation** of two random variables? How is it calculated?
6. When do we use the **multinomial** distribution?

- What is its joint pmf?
- What is its marginal pmf?
- What is its conditional pmf?

7. When do we use the **bivariate normal** distribution?

    - What is its joint pdf?
    - What is its marginal pdf?
    - What is its conditional pdf?

8. How do we calculate the **pmf/pdf of a function of a random variable h(X)**? Letting $u(y) = h^{-1}(y)$, we have:

    - How do we find the pmf of $Y = h(X)$ if X is discrete?
    - How do we find the pdf of $Y = h(X)$ if X is continuous?

Motivation

What should I expect?

Consider two jointly distributed random variables (X, Y). What should I expect X to be? What should I expect Y to be? What should I expect X to be if I already know what Y is? Does that change or does it stay the same?

Dependence

Knowing whether the value of X affects Y or not is important. Consider, for example, a movie studio planning a series of superhero movies. If the first movie is unsuccessful, and is panned by critics and the audience, then the studio may want to rethink the sequel and subsequent movies. We would want to know the level of dependence between two random variables to help us make decisions better.

Success or failure? More like full success, or somewhat success, or ...

In Chapter 5, we introduced a lot of discrete distributions. One of the most fundamental ones is the binomial distribution. Its premise is simple: perform an experiment n times and count the number of successes, assuming the remainders are failures. This works pretty well when we have two outcomes: for example, a patient may have an infection or not, a student may pass a class or not, etc.

What happens when the number of outcomes is higher than 2? What if a patient may have a severe infection, or a moderate infection, or no infection? What if a student can get an A, a B, a C, a D, or fail a class?

Normally distributed random variables with correlation

We sometimes are aware that a specific random variable is normally distributed. However, its exact parameters may depend (and in turn may also affect) another normally distributed random variable. For example, consider a Sunday night at HBO. A TV series starting at 10pm may expect a normally distributed share of viewers with a known mean and standard deviation. However, if the TV series showing at 9pm has its grand finale, we may anticipate a higher viewership for the 10pm show, too! This relationship needs to be modeled somehow…

Expectations and variances

Once again, we have discussed expectations and variances before; however, those were in the setting of single random variables. Here, we generalize to two or more random variables. As a motivating example, consider a student taking two classes: the student may be interested in the expected grade in one of the two classes alone.

This may ring a bell. Recall that during our last chapter, we discussed the **marginal pmf/pdf** of jointly distributed random variables (X, Y). We specifically said that they come into play when we want to answer the question "what is the probability that X takes a certain value, regardless of Y?" Well, we will use this to calculate expectations and variances. Specifically, we want to answer the questions:

1. "what is the expected value that X takes, regardless of Y?"
2. "what is the variance of X, regardless of Y?"

Of course, both are easily adapted for Y (regardless of X).

Definitions and theorems 10.1. Expectations and variances from marginals

Let (X, Y) be two jointly distributed random variables with marginal pmf/pdf $f_X(x)$ and $f_Y(y)$. Then:

Table 10.1. [Author adds caption]

Quantity	Discrete	Continuous
$E[X] = \mu_X$	$\sum_x x f_X(x)$	$\int_{-\infty}^{+\infty} x f_X(x) dx$
$\text{Var}[X] = \sigma_X^2$	$\sum_x x^2 f_X(x) - \mu_X^2$	$\int_{-\infty}^{+\infty} x^2 f_X(x) dx -$
$E[Y] = \mu_Y$	$\sum_y y f_Y(y)$	$\int_{-\infty}^{+\infty} y f_Y(y) dy$
$\text{Var}[Y] = \sigma_Y^2$	$\sum_y y^2 f_Y(y) - \mu_Y^2$	$\int_{-\infty}^{+\infty} y^2 f_Y(y) dy - \mu$

Example 10.1. Applying the formulae – A chemical mixture

Last time we saw a chemical mixture problem with the volumes of two materials. Here, X and Y are continuous random variables between 0 and 1 representing the material volumes with joint pdf $f_{XY}(x, y) = \frac{2}{5}(2x + 3y)$. Recall that last time we calculated the marginal pdf for X and Y as $f_X(x) = \frac{1}{5}(4x + 3)$ and $f_Y(y) = \frac{6y + 2}{5}$.

1. What is the expectation and the variance of the volume of the first material?
2. What is the expectation and the variance of the volume of the second material?

For the first material, recall that $f_X(x) = \frac{1}{5}(4x+3)$:

$$E[X] = \int_0^1 x \cdot \frac{1}{5}(4x+3)dx = \frac{17}{30} = 0.566\ldots$$

and

$$\mathrm{Var}[X] = \int_0^1 x^2 \cdot \frac{1}{5}(4x+3)dx - (E[X])^2 = \frac{71}{900} = 0.0788\ldots$$

For the second material, recall that $f_Y(y) = \frac{6y+2}{5}$:

$$E[Y] = \int_0^1 y \cdot \frac{6y+2}{5} dy = 0.6$$ and

$$\mathrm{Var}[Y] = \int_0^1 y^2 \cdot \frac{6y+2}{5} dy - (E[Y])^2 = \frac{11}{150} = 0.073\ldots$$

Exercise 10.1. Practice exercises

Practice with the following joint distributions:

- **Discrete:** $f_{XY}(x, y) = \dfrac{x \cdot y}{18}$, $x = 1, 2$, $y = 1, 2, 3$.
- **Continuous:** $f_{XY}(x, y) = x + y$, $0 \leq x \leq 1, 0 \leq y \leq 1$.

Conditional expectations and variances

Conditional expectations and variances can also be defined in the case of two jointly distributed random variables X, Y. They would answer the question: "what should I expect X to be if I know that Y is equal to y?" Clearly, the same question can be asked for Y.

Definitions and theorems 10.2. Conditional expectations and variances

The conditional mean and variance of random variable X given a value for random variable $Y = y$ and of random variable Y given $X = x$ are:

Table 10.2. [Author adds caption]

Quantity	Discrete	Continuous
$E[X\|y] = \mu_{X\|y}$	$\sum_x x f_{X\|y}(x)$	$\int_{-\infty}^{+\infty} x f_{X\|y}(x)$
$\text{Var}[X\|y] = \sigma^2_{X\|y}$	$\sum_x x^2 f_{X\|y}(x) - \mu^2_{X\|y}$	$\int_{-\infty}^{+\infty} x^2 f_{X\|y}(x$
$E[Y\|x] = \mu_{Y\|x}$	$\sum_y y f_{Y\|x}(y)$	$\int_{-\infty}^{+\infty} y f_{Y\|x}(y)$
$\text{Var}[Y\|x] = \sigma^2_{Y\|x}$	$\sum_y y^2 f_{Y\|x}(y) - \mu^2_{Y\|x}$	$\int_{-\infty}^{+\infty} y^2 f_{Y\|x}(y$

Example 10.2. Back to the chemical mixture

Recall that we had calculated during the previous chapter that $f_{X|y}(x) = \dfrac{4x+6y}{6y+2}$. What is the expectation and the variance of the volume of the first material, given that the second material's volume is equal to 0.6?

Applying the traditional expectation and variance formulae, we have:

$$E[X|y] = \int_0^1 x\frac{4x+6\cdot 0.6}{6\cdot 0.6+2}dx = 0.5595$$ and

$< spanstyle = "text - align : initial; font - size : 1em;" > \mathrm{Var}[X|y] = \int_0^1 x^2\frac{4x+6\cdot 0.6}{6\cdot 0.6+2}dx - 0.5595^2 = 0.0799.$

Exercise 10.2. Practice exercises

Find the conditional distributions of the two joint distributions below:

- **Discrete:** $f_{XY}(x,y) = \frac{x\cdot y}{18}$, $x = 1, 2$, $y = 1, 2, 3$.
- **Continuous:** $f_{XY}(x,y) = x + y$, $0 \leq x \leq 1, 0 \leq y \leq 1$.

Then, find:

- For the first one (the discrete distribution): $E[X|Y=2]$, $\mathrm{Var}[X|Y=2]$.
- For the second one (the continuous distribution): $E[Y|X=0.5]$, $\mathrm{Var}[Y|X=0.5]$.

Expectations of functions

Finally, as far as expectations are concerned, we take a look at the expectation of a function. Let $h(X,Y)$ be a function of two jointly distributed random variables X, Y. Very similarly to what we did for expectations of functions of single random variables, we get:

Definitions and theorems 10.3. Expectation of functions of jointly distributed variables

For a function $h(X, Y)$ of jointly distributed random variables:

Discrete:

$$E[h(X,Y)] = \sum_x \sum_y h(x,y) f_{XY}(x,y).$$

Continuous:

$$E[h(X,Y)] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h(x,y) f_{XY}(x,y) dxdy.$$

Recall that for random variable X and function $g(X)$, we have the expectation as:

- **Discrete:** $E[g(X)] = \sum_x g(x)p(x)$
- **Continuous:** $$E[g(X)] = \int_{-\infty}^{+\infty} g(x) f(x) dx$$

Example 10.3. Again with the chemical mixture

The chemical mixture's quality is evaluated by function $h(X, Y) = 3X + 7Y$. We note here that material 2 is preferable to material 1, and hence the quality is severely favored when material 2 has higher volume. The maximum possible quality is equal to 10 (when both material volumes are equal to 1): in general, the quality ranges from 0 to 10. **What is the expected mixture quality?**

We can calculate this as:

$$E[h(X,Y)] = \int_0^1 \int_0^1 (3x + 7y)\frac{2}{5}(2x + 3y)dxdy = \int_0^1 \frac{1}{5}(42y^2 + 23y + 4)dy = 5.9.$$

Independence

Independence is a fundamental property of events and random variables. In the past we have discussed independence for events: events A and B are independent if $P(A|B) = P(A)$ or $P(A \cap B) = P(A) \cdot P(B)$. This property proved very useful for calculating basic probabilities. Now, we extend its definition to jointly distributed random variables.

Definitions and theorems 10.4. Independence for random variables

Two random variables X, Y are **independent** if any of the following statements hold:

1. $f_{XY}(x,y) = f_X(x)f_Y(y), \forall x, y$
2. $f_{X|y}(x) = f_X(x), \forall x, y$
3. $f_{Y|x}(y) = f_Y(y), \forall x, y$
4. $P(X \in A, Y \in B) = P(X \in A) \cdot P(Y \in B), \forall A, B$

In plain terms, the four statements say the same thing; but they do it in different ways. The first one claims that two independent random variables will see their joint pmf/pdf be equal to the product of the individual marginal pmfs/pdfs. **This is typically the easiest way to show (or not) independence of two random variables.** If we find the marginal pmf/pdf of X and Y and their product does not always equate to their joint pmf/pdf, then the two variables are not independent.

The second and the third statements are similar to the first definition of independence of events. In essence, they claim that the conditional pmf/pdf of one random variable given the other is equal to the marginal pmf/pdf.

The last statement is interesting, but it needs to be shown for any two sets of values A and B. It states that the probability of both X belonging to set A and Y belonging to set B can be found through the product of the individual probabilities.

Example 10.4. Discrete random variable independence

Consider two discrete random variables with joint pmf

$f_{XY}(x,y) = e^{-3}\dfrac{2^x}{x! \cdot y!}$ for $x, y \geq 0$. Are random variables X and Y independent?

This looks very intimidating, but we can use some well-known facts from calculus to obtain the answer. Recall that $\sum_{i=0}^{\infty} \dfrac{\alpha^i}{i!} = e^{\alpha}$. This will be useful.

Now, to find the marginal pmfs:

1. $f_X(x) = \sum_{y=0}^{\infty} e^{-3}\dfrac{2^x}{x! \cdot y!} = e^{-3}\dfrac{2^x}{x!} \cdot e = e^{-2}\dfrac{2^x}{x!}$
2. $f_Y(y) = \sum_{x=0}^{\infty} e^{-3}\dfrac{2^x}{x! \cdot y!} = e^{-3}\dfrac{1}{y!} \cdot \sum_{x=0}^{\infty} \dfrac{2^x}{x!} = e^{-3}\dfrac{1}{y!} \cdot e^2 = e^{-1}\dfrac{1}{y!}$

We then observe that $f_{XY}(x,y) = f_X(x) \cdot f_Y(y)$, **showing independence**.

We could have used statements 2 and 3 to show the same thing!

Example 10.5. Discrete random variable independence (alternative approach)

Consider two discrete random variables with joint pmf $f_{XY}(x,y) = e^{-3}\dfrac{2^x}{x! \cdot y!}$ for $x, y \geq 0$. Are random variables X and Y independent?

We could first find the conditional pmfs:

1. $$f_{X|y}(x) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{e^{-3}\frac{2^x}{x!\cdot y!}}{e^{-1}\frac{1}{y!}} = e^{-2}\frac{2^x}{x!} = f_X(x)$$
2. $$f_{Y|x}(y) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{e^{-3}\frac{2^x}{x!\cdot y!}}{e^{-2}\frac{2^x}{x!}} = e^{-1}\frac{1}{y!} = f_Y(y)$$

We then observe that $f_{X|y}(x) = f_X(x)$ and $f_{Y|x}(y) = f_Y(y)$, **showing independence again**.

The last statement would prove a little tougher, as we would need to show that it is true for any pair of sets of values. Let's see an example where independence does not hold.

Example 10.6. Continuous random variable independence

Consider two continuous random variables with joint pdf $f_{XY}(x,y) = x + y, 0 \leq x \leq 1, 0 \leq y \leq 1$. Are the random variables independent?

Similarly to the previous example, let's calculate the marginal pdfs:

1. $$f_X(x) = \int_0^1 (x+y)dy = xy + \left.\frac{y^2}{2}\right|_0^1 = x + \frac{1}{2}$$
2. Similarly, $f_Y(y) = y + \frac{1}{2}$

We now note that

$f_X(x) \cdot f_Y(y) = xy + \frac{1}{2}x + \frac{1}{2}y + \frac{1}{4}$, which does not reveal independence, as the product is not always equal to $x + y$. At this point we may claim that the two events are **not independent**. The same can be said using statements 2 and 3:

1. $f_{X|y}(x) = \dfrac{x+y}{y+\frac{1}{2}}$
2. Similarly, $f_{Y|x}(y) = \dfrac{x+y}{x+\frac{1}{2}}$

Again, these two are not necessarily equal to $f_X(x)$ or $f_Y(y)$, respectively.

Last, let us consider statement 4: if we are able to find at least one pair of values A, B such that $P(X \in A, Y \in B) \neq P(X \in A) \cdot P(Y \in B)$ should be enough to disprove independence.

Example 10.7. Continuous random variable independence (statement 4)

Consider two continuous random variables with joint pdf $f_{XY}(x, y) = x + y, 0 \leq x \leq 1, 0 \leq y \leq 1$. Are the random variables independent?

Consider $A = [0, 0.25]$ and $B = [0.5, 1]$. We have:

$$P(X \in A, Y \in B) = \int_0^{0.25} \int_{0.5}^{1} (x+y) dy dx = \int_0^{0.25} \left(0.5x + \frac{3}{8}\right) dx = 0.109375.$$

On the other hand:

1. $$P(X \in A) = \int_0^{0.25} f_X(x) dx = \int_0^{0.25} (x + \frac{1}{2}) dx = 0.15625$$
2. $$P(Y \in B) = \int_{0.5}^{1} f_Y(y) dy = \int_{0.5}^{1} (y + \frac{1}{2}) dy = 0.625$$

We observe that

$P(X \in A) \cdot P(Y \in B) = 0.15625 \cdot 0.625 = 0.09765625$ which is not equal to $P(X \in A, Y \in B) = 0.109375$. Hence, the two random variables are **not independent**.

Alright, so we are able to tell if two random variables are independent or not. Another useful metric though, would be to be able to tell **how dependent** two random variables are.

Covariance

What if two variables are not independent? Then, in that case, the two random variables may affect one another. In plain terms, knowing information about one may affect the probabilities of the other, either positively or negatively! To help us quantify this dependence we introduce covariance and correlation.

Definitions and theorems 10.5. Covariance

Covariance is a measure of the *association* between two random variables. For two random variables X and Y, we define *covariance* as:
$\sigma_{XY} = \mathrm{Cov}[X,Y] = E[(X - E[X]) \cdot (Y - E[Y])] = E[XY] - E[X] \cdot E[Y]$.

Remember the definition of variance for a single random variable? It was: $\sigma_X^2 = \mathrm{Var}[X] = E[(X - E[X])^2] = E[X^2] - (E[X])^2$. This is very similar to the definition of covariance, extended to include jointly distributed random variables.

Key observations:

1. If $X \geq E[X]$ whenever $Y \geq E[Y]$ and if $X \leq E[X]$ whenever $Y \leq E[Y]$, then the covariance will be **positive**.
2. If $X \geq E[X]$ whenever $Y \leq E[Y]$ and if $X \leq E[X]$ whenever $Y \geq E[Y]$, then the covariance will be **negative**.
3. Finally, and very importantly: **two independent random variables X, Y will have $\sigma_{XY} = \mathrm{Cov}[X,Y] = 0$. The inverse is not necessarily true.**

Example 10.8. A small Florida example

In Gainesville, FL, summer days are classified as either sunny or rainy. Whenever it is sunny, Floridians go to watch a local baseball team; whenever it is rainy, they tend to forget their umbrellas and they need to buy one. If it is rainy, profits skyrocket for an umbrella selling grocery store and they make $4,500; at the same time, the local team only makes $1,000. If it is sunny, the grocery store only makes $500; the team though makes $2,500 from tickets. Summers are sunny in Florida 65% of the time, and rainy the remaining 35%. What is the covariance of the two company profits?

Let U be the umbrella profits, and T the team profits. To help us collect all data we may construct a small table as follows:

Table 10.3. [Author adds caption]

	Sunny	**Rainy**
Probability	0.65	0.35
Team (T)	$2500	$1000
Umbrellas (U)	$500	$4500

- First, calculate the expected profits:
  $$E[U] = 0.65 \cdot 500 + 0.35 \cdot 4500 = \$1900$$
  $$E[T] = 0.65 \cdot 2500 + 0.35 \cdot 1000 = \$1975$$
- Now, on to calculating $(X - E[X]) \cdot (Y - E[Y])$:
  - when it is sunny:
    $$(U - E[U]) \cdot (T - E[T]) = (500 - 1900) \cdot (2500 - 1975) = -805000.$$
  - when it is rainy:
    $$(U - E[U]) \cdot (T - E[T]) = (4500 - 1900) \cdot (1000 - 1975) = -2340000.$$
- Last, calculate the covariance:
  $$\begin{aligned}\mathrm{Cov}(U, T) &= E[(U - E[U]) \cdot (T - E[T])] = \\ &= 0.65 \cdot (-805000) + 0.35 \cdot (-2340000) = \\ &= -1342250.\end{aligned}$$

Covariance is easier to calculate as $E[X \cdot Y] - E[X] \cdot E[Y]$ when the probabilities are given in joint pmf/pdf format.

Example 10.9. Covariance for continuous random variables

Earlier, we saw an example of two jointly distributed continuous random variables X and Y with pdf $f_{XY}(x,y) = x + y, 0 \leq x \leq 1, 0 \leq y \leq 1$. We actually calculated their marginal pdfs:

- $f_X(x) = x + \frac{1}{2}$
- $f_Y(y) = y + \frac{1}{2}$

We also found that these two are not independent. What is their covariance?

Recall that we can calculate the covariance of X and Y as $E[XY] - E[X] \cdot E[Y]$.

- $E[X] = \int_0^1 x(x + \frac{1}{2})dx = \frac{7}{12}$
- $E[Y] = \int_0^1 y(y + \frac{1}{2})dy = \frac{7}{12}$
- Also:
  $$E[XY] = \int_0^1 \int_0^1 xy(x+y)dydx = \frac{1}{3}$$

Finally: $\sigma_{XY} = \frac{1}{3} - \frac{7}{12} \cdot \frac{7}{12} = -\frac{1}{144}$.

Correlation

The problem with covariance is that it is not normalized. That is, the covariance is "unit-dependent." If we want to contrast the relationship between two pairs of random variables X_1, Y_1 and X_2, Y_2, then the covariances may not reveal which pair showcases more dependence than the other. A very big covariance or a very small covariance do not necessarily imply the actual level of dependence. This is why we introduce **correlation**, a measure that directly relates its value to the magnitude of dependence. It makes sense to introduce a "unit-agnostic" measure of dependence, in the form of the **correlation**.

Definitions and theorems 10.6. Correlation

Correlation is a measure of the linear relationship between two random variables X and Y. It is calculated as: $\rho_{XY} = \frac{\text{Cov}[X, Y]}{\sqrt{\text{Var}[X]}\sqrt{\text{Var}[Y]}} = \frac{\sigma_{XY}}{\sigma_X \cdot \sigma_Y}$.

By definition, the correlation will be at most equal to 1 (fully positively correlated), and at least equal to -1 (fully negatively correlated). Positive value of correlation imply a positive dependence (that is, higher values of X are typically associated with higher values of random variable Y and vice-versa), while negative values for ρ imply a negative dependence. Notice also that the numerator of correlation is the covariance itself, normalized by the product of the individual standard deviations (square roots of the variances). A few observations we can make based on this definition:

1. $-1 \leq \rho \leq 1$.
2. When X and Y are independent, then $\sigma_{XY} = \rho_{XY} = 0$.
3. $\rho_{XY} = 1$ implies that X and Y are fully positively correlated.
4. $\rho_{XY} = -1$ implies that X and Y are fully negatively correlated.

Example 10.10. Back to Florida

What is the correlation in the Florida example?

First, we calculate individual variances:

- $\text{Var}[U] = 0.65 \cdot (500 - 1900)^2 + 0.35 \cdot (4500 - 1900)^2 = 3640000$
- $\text{Var}[T] = 0.65 \cdot (2500 - 1975)^2 + 0.35 \cdot (1000 - 1975)^2 = 508471.25$

Since we already calculated $\text{Cov}(U, T) = -1342250$, we can compute the correlation as:

$$\rho_{UT} = \frac{-1342250}{\sqrt{3640000} \cdot \sqrt{508471.25}} = -0.9866.$$

Hence, as expected, the two profits are almost totally negatively correlated!

Example 10.11. Correlation for continuous random variables

Another example we saw earlier: $f_{XY}(x,y) = x + y$, $0 \le x, y \le 1$ with:

- $f_X(x) = x + \frac{1}{2}$
- $f_Y(y) = y + \frac{1}{2}$
- $E[X] = \frac{7}{12}$
- $E[Y] = \frac{7}{12}$
- $\sigma_{XY} = -\frac{1}{144}$

What is the correlation?

First, we find the variances:

$$\begin{aligned}\mathrm{Var}[X] &= \int_0^1 x^2 f_X(x)dx - (E[X])^2 = \\ &= \int_0^1 x^2(x + \frac{1}{2})dx - (\frac{7}{12})^2 = \\ &= \frac{5}{12} - (\frac{7}{12})^2 = \frac{11}{144}.\end{aligned}$$

From symmetry we may deduce that $\mathrm{Var}[Y] = \frac{11}{144}$, too. Finally, we combine to get:

$$\rho_{XY} = \frac{-\frac{1}{144}}{\sqrt{\frac{11}{144}} \cdot \sqrt{\frac{11}{144}}} = -\frac{1}{11}.$$

When x and y restrict each other

This part of the chapter serves as more of a reminder from calculus. When taking the integral of more than one variable at the same time, we need to be very careful with the bounds we are using.

Let us see this with an example. Assume (X, Y) are two jointly distributed continuous random variables with joint pdf equal to $f_{XY}(x, y) = 3 \cdot (x + y)$. Moreover, assume that $X, Y \geq 0$ and (and this is important!) $X + Y \leq 1$.

Note how the value of random variable X affects the range of values that Y is allowed to take; and vice versa. We need to be very careful with how we proceed in this case. There are three things we need to be able to do:

1. Calculate probability for **both random variables at the same time** and expectations for a function of both random variables.
2. Calculate marginal distributions **for one random variable at a time**.
3. Calculate probabilities, expectations, and variances **for one random variable (forgetting the other exists)** and calculate conditional probabilities, expectations, and variances **for one random variable setting the other equal to a value**.

Example 10.12. "Both at the same time"

Let us specifically focus on these three items for the pdf provided: $f_{XY}(x, y) = 3 \cdot (x + y)$ for $x, y \geq 0$, such that $x + y \leq 1$. Typical questions that you will encounter when dealing with joint distributions include:

- Verify that $f_{XY}(x,y)$ is a valid pdf.
- What is the probability that $X \leq 0.3$ and $Y > 0.5$?

To answer these questions, we need to allow x and y to consider all values they can get. For example, if x is allowed to go from 0 to 1, then y is allowed to go from 0 to $1-x$. On the other hand, if y is allowed to go from 0 to 1, then x is only allowed to go from 0 to $1-y$.

What we could do wrong: we could possibly allow both x and y to go from 0 to 1, allowing $x+y$ to potentially go higher than 1, breaking the requirement.

Finally:

Verify that $f_{XY}(x,y)$ is a valid pdf.
$\int_0^1\int_0^{1-x} 3(x+y)dydx = \int_0^1 3(xy+\frac{y^2}{2})\Big|_0^{1-x} dx = \int_0^1 3(\frac{1}{2}-\frac{x^2}{2})dx = (\frac{3}{2}x-\frac{1}{2}x^3)\Big|_0^1 = 1$
or
$\int_0^1\int_0^{1-y} 3(x+y)dxdy = \int_0^1 3(\frac{x^2}{2}+yx)\Big|_0^{1-y} dy = \int_0^1 3(-\frac{y^2}{2}+\frac{1}{2})dy = (-\frac{1}{2}y^3+\frac{3}{2}y)\Big|_0^1 = 1$
.

What is the probability that $X \leq 0.3$ and $Y > 0.5$?

$$\begin{aligned}\int_0^{0.3}\int_{0.5}^{1-x} 3(x+y)\,dy\,dx &= \int_0^{0.3} 3\left(xy+\frac{y^2}{2}\right)\Bigg|_{0.5}^{1-x} dx \\ &= \int_0^{0.3}(1.125-1.5x-1.5x^2)\,dx \\ &= \left(\frac{9x}{8}-\frac{3x^2}{4}-\frac{x^3}{2}\right)\Bigg|_0^{0.3} \\ &= 0.2565\end{aligned}$$

Let us now check the second case: one at a time.

Example 10.13. “One at a time”

Typical questions in this category would include to:

- Find the marginal distribution of X.
- Find the marginal distribution of Y.

To answer the questions, we need to either express y as a function of x or express x as a function of y.

What we could do wrong: we could possibly allow both x and y to go from 0 to 1, allowing $x + y$ to potentially go higher than 1, breaking the requirement.

Let’s see how we could go about solving this:

- Find the marginal distribution of X.
$$f_X(x) = \int_0^{1-x} 3(x+y)dy = \frac{3}{2} - \frac{3x^2}{2}.$$
- Find the marginal distribution of Y.
$$f_Y(y) = \int_0^{1-y} 3(x+y)dx = \frac{3}{2} - \frac{3y^2}{2}.$$

The third case involves forgetting that one of the variables exists.

Example 10.14. “One alone”

Typical questions in this category:

- Find the probability that $X \leq 0.3$.
- What is the expectation of Y?

To answer the questions, we simply use the bounds as given!

What we could do wrong: we could possibly try to restrict x or y as a function of the other, when the other no longer exists — as we do not care about it in the setup.

Solutions:

- Find the probability that $X \leq 0.3$.
$$\int_0^{0.3} f_X(x)dx = \int_0^{0.3} (\frac{3}{2} - \frac{3x^2}{2})dx = 0.436$$
.
- What is the expectation of Y?
$$\int_0^1 yf_Y(y)dy = \int_0^1 y(\frac{3}{2} - \frac{3y^2}{2})dy = 0.375$$
.

Extension to more than 2 random variables

Everything we have discussed today can be generalized to more than 2 random variables. More specifically, for a multivariate jointly distributed random variable $(X_1, X_2, \ldots, X_n)$ (hence n random variables), with joint pmf/pdf $f(x_1, x_2, \ldots, x_n)$, we have expectations and variances:

Definitions and theorems 10.7. Expectation and variances for multiple joint random variables

Discrete: $E[X_i] = \sum_x x_i f_{X_i}(x_i) = \mu_{X_i}$ and

$$\mathrm{Var}[X_i] = \sum_x (x_i - \mu_{X_i})^2 f_{X_i}(x_i) = \sigma^2_{X_i}.$$

Continuous:

$$E[X_i] = \int_{-\infty}^{+\infty} x_i f_{X_i}(x_i) dx_i = \mu_{X_i}$$ and

$$\mathrm{Var}[X_i] = \int_{-\infty}^{+\infty} (x_i - \mu_{X_i})^2 f_{X_i}(x_i) dx_i = \sigma^2_{X_i}.$$

Similarly, for independence:

Definition 10.8 Independence for multiple random variables

Random variables $X_1, X_2, \ldots, X_n$ are **independent** if and only if for all values of $x_1, x_2, \ldots, x_n$, we have:
$f_{X_1 X_2 \ldots X_n}(x_1, x_2, \ldots, x_n) = f_{X_1}(x_1) \cdot f_{X_2}(x_2) \cdots f_{X_n}(x_n)$
.

Example 10.15 A 4-component machine

Recall the machine (from the previous chapter) that consists of four components, whose lifetimes (in years) are jointly distributed with the following pdf: $f_{X_1X_2X_3X_4}(x_1, x_2, x_3, x_4) = 3 \cdot e^{-2x_1}e^{-x_2}e^{-3x_3}e^{-0.5x_4}$

Are the random variables from the pdf in this example independent?

First, we calculate all 4 marginal pdfs:

1. $f_{X_1}(x_1) = 2e^{-2x_1}$
2. $f_{X_2}(x_2) = e^{-x_2}$
3. $f_{X_3}(x_3) = 3e^{-3x_3}$
4. $f_{X_4}(x_4) = 0.5e^{-0.5x_4}$

Finally, we have that: $f_{X_1X_2X_3X_4}(x_1, x_2, x_3, x_4) = f_{X_1}(x_1)f_{X_2}(x_2)f_{X_3}(x_3)f_{X_4}(x_4)$, so they are **independent**.

We finish this part of the chapter with the following very important result:

Definitions and theorems 10.9. Linear combinations of random variables

For a linear combination of random variables, we have:

Expectation (always holds):

$$E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i].$$

Variance (general case):

$$\begin{aligned}\mathrm{Var}\left[\sum_{i=1}^{n} a_i X_i\right] &= \sum_{i=1}^{n} a_i^2 \mathrm{Var}[X_i] + \sum_{i=1}^{n} \sum_{j=1: i\neq j}^{n} a_i a_j \cdot \mathrm{Cov}[X_i, X_j] \\ &= \sum_{i=1}^{n} a_i^2 \mathrm{Var}[X_i] + \sum_{i=1}^{n} \sum_{i<j} 2 a_i a_j \cdot \mathrm{Cov}[X_i, X_j]\end{aligned}$$

.

When we have independence, $\mathrm{Cov}[X_i, X_j] = 0$ for all X_i, X_j, and hence, for multiple **independent** random variables, we have: $E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i]$ and $\mathrm{Var}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i^2 \mathrm{Var}[X_i]$.

Distribution of a function

We have already discussed what we expect will happen for a function of a random variable. We repeat the definitions here for convenience:

Definitions and theorems 10.10. Expectation of a function of a random variable

Definition: Let $g(X)$ be a function of a random variable X. Then, the expectation of $g(X)$ is denoted by $E\left[g(X)\right]$ and is equal to:

- for discrete random variable X with sample space S: $E\left[g(X)\right] = \sum_{x \in S} g(x) \cdot p(x)$.
- for continuous random variable X: $$E\left[g(X)\right] = \int_{-\infty}^{+\infty} g(x) \cdot f(x) dx.$$

It is time we discuss how the function is **distributed**: rather than addressing questions of expectation ("what should I expect the function value to be?"), we will be addressing questions of probability ("what is the probability the function value is…").

Some examples of why this would be useful:

- What is the probability my profits are higher than $2000 today?
  - My profits depend on the number of customers, discrete random variable X.
- What is the probability the circuit overheats?
  - The heat of the circuit is a function of its current, continuous random variable X.
- What is the probability the crop has high yield?

- The yield of a crop is a function of the location temperature, continuous random variable X.

Formally, let $Y = h(X)$ be a **one-to-one** transformation of a random variable X to a random variable Y. The one-to-one transformation is important: it implies that solving $y = h(x)$ provides us with a unique solution. Assume that the solution is $x = h^{-1}(y) = u(y)$.

Example 10.16. Inverse functions

- $Y = X^2 \implies x = u(y) = \sqrt{y}$.
- $Y = 2\ln x \implies x = e^{y/2}$.

Definitions and theorems 10.11. Distribution of a function

Definition: Let $Y = h(X)$ be a one-to-one function of random variable X to Y. X is distributed with pmf/pdf $f_X(x)$. Then, the pmf/pdf of random variable $Y = h(X)$ can be found using the **chain rule**:

1. Discrete X: $f_Y(y) = f_X(u(y))$.
2. Continuous X: $f_Y(y) = f_X(u(y)) \cdot |u'(y)|$, where $u'(y)$ is the derivative of function $u(y)$.

Why is that true? Let's take a look at the derivation for both discrete and continuous random variables, starting with the discrete case.

Function of a discrete random variable.

Consider a discrete random variable X and a function $Y = h(X)$. Let's consider the probability mass function for Y: it would be $f_Y(y) = P(Y = y)$. More specifically, we have: $f_Y(y) = P(Y = y) = P\left(h(X) = y\right) = P\left(X = u(y)\right)$, where $u(y) = h^{-1}(y)$ is the inverse function. Now, that last term $\left(P\left(X = u(y)\right)\right)$ is easy to calculate! $P\left(X = u(y)\right) = f_X\left(u\left(y\right)\right)$.

Function of a continuous random variable.

Now, consider a continuous random variable X and a function $Y = h(X)$. We can't immediately deal with $P(Y = y)$ like earlier, because Y is a continuous random variable! Instead, consider the cumulative distribution function for Y: it would be $F_Y(y) = P(Y \leq y)$. Like earlier, we do our transformation: $F_Y(y) = P(Y \leq y) = P\left(h(X) \leq y\right) = P\left(X \leq u(y)\right)$, where $u(y) = h^{-1}(y)$ is the inverse function. Now, that last term $\left(P\left(X \leq u(y)\right)\right)$ is again straightforward! $P\left(X \leq u(y)\right) = \int_{-\infty}^{u(y)} f_X\left(u\left(y\right)\right)$. Recall that our goal is to derive the pdf of Y, that is $f_Y(y)$. We also remember how for continuous random variables we have that the pdf is the derivative of the cdf, or in mathematical terms, $f_Y(y) = \left(F(y)\right)'$. Combining everything:

$$f_Y(y) = (F(y))' = \left(\int_{-\infty}^{u(y)} f_X(u(y)) \right)' = f_X(u(y)) \cdot u'(y)$$

. Almost there: this works like a charm, **if** $\boldsymbol{u'(y) \geq 0}$ (equivalently, if $u(y)$ is non-decreasing). When $u'(y) < 0$ (i.e., when $u(y)$ is decreasing), then we have to take the absolute value, as our pdf can never be negative. This sums it up: $f_Y(y) = f_X(u(y)) \cdot |u'(y)|$.

Example 10.17. Printer speed

A printer has speed that is equal to $h(x) = \dfrac{\sqrt{x+1}}{x+1}$, where x is the condition of the printer. The condition of the printer is a continuous random variable distributed exponentially with rate $\lambda = 1$. What is the probability the printer is faster than 0.5?

First of all, let's see what we have:

- X is the condition of the printer, a random variable with $f_X(x) = \lambda \cdot e^{-\lambda x}$ and $x \geq 0$.
- Y is the speed of the printer, a random variable which is a function of X and has $Y = h(x) = \dfrac{\sqrt{x+1}}{x+1}$. By definition $0 \leq y \leq 1$.
- We may solve for $u(y)$:
$$y = \frac{\sqrt{x+1}}{x+1} \implies x = 1 - \frac{1}{y^2} \implies u(y) = \frac{y^2-1}{y^2}$$

.

Based on the chain rule: $f_Y(y) = e^{-\frac{y^2-1}{y^2}} \cdot \frac{2}{y^3}$.

Finally, we have:

$$P(Y > 0.5) = \int_{0.5}^{1} e^{-\frac{y^2-1}{y^2}} \cdot \frac{2}{y^3} dy = 0.9502.$$

The multinomial distribution

Flashback! Let's review together the **binomial distribution**, one of the first discrete probability distributions we studied together. We had the following setup.

What if we perform n independent trials of the same experiment? Each trial may result in a success (with probability p) or a failure (with probability $q = 1 - p$). Let X be the number of successes we observe: then X is said to be binomially distributed with parameters n and p. Some interesting things about the binomial distribution:

- pmf: $P(X = x) = p(x) = \binom{n}{x} p^x (1 - p)^{n-x}$, for $0 \leq x \leq n$.
- expectation: $E[X] = np$.
- variance: $Var[X] = np(1 - p)$.

What if we tried to generalize this? We will still perform n independent trials; however now each trial will result in one of k

outcomes (instead of just two). Each outcome appears with each own probability p_i. Clearly we need $\sum_{i=1}^{k} p_i = 1$. This is called the **multinomial distribution**. Formally:

Definitions and theorems 10.12. The multinomial distribution

Definition: Let X_i be the number of times that outcome i appears in n independent trials. Each outcome i appears with probability p_i such that $\sum_{i=1}^{k} p_i = 1$. Then, $(X_1, X_2, \ldots, X_k)$ is distributed following a multinomial distribution with parameters n and $p_i, i = 1, \ldots, k$. The joint probability mass function is given by:

$$P(X_1 = x_1, X_2 = x_2, X_3 = x_3, \ldots, X_k = x_k) = \frac{n!}{x_1! x_2! \ldots x_k!} p_1^{x_1} p_2^{x_2} \ldots p_k^{x_k}$$

. Note that $\sum_{i=1}^{k} x_i = n$.

Example 10.18. Grades

According to the grade disparity website at UIUC, a student registering for CS 101 gets an A 73% of the time, a B 17% of the time, a C 6% of the time, a D 2% of the time, and an F the remaining 2% of the time. In a section of the class, there are 20 students. What is the probability that:

1. 10 students get an A and 10 students get a B?
2. Everyone gets an A?
3. 12 students get an A, 5 students get a B, 2 students get a C, and 1 student gets a D?

This is a multinomial distribution with $n = 20$, $p_1 = 0.73$, $p_2 = 0.17$, $p_3 = 0.06$, $p_4 = 0.02$, $p_5 = 0.02$ for A, B, C, D, and F, respectively. Let X_1, X_2, X_3, X_4, X_5 be the number of students getting an A, B, C, D, F. Then, we have:

1. 10 students get an A and 10 students get a B?
$$P(X_1 = 10, X_2 = 10, X_3 = 0, X_4 = 0, X_5 = 0) =$$
$$= \frac{20!}{10!10!0!0!0!} 0.73^{10} 0.17^{10} 0.06^{0} 0.02^{0} 0.02^{0} =$$
$$= 0.00016.$$
2. Everyone gets an A?
$$P(X_1 = 20, X_2 = 0, X_3 = 0, X_4 = 0, X_5 = 0) =$$
$$= \frac{20!}{20!0!0!0!0!} 0.73^{20} 0.17^{0} 0.06^{0} 0.02^{0} 0.02^{0} =$$
$$= 0.00185.$$
3. 12 students get an A, 5 students get a B, 2 students get a C, and 1 student gets a D?
$$P(X_1 = 12, X_2 = 5, X_3 = 2, X_4 = 1, X_5 = 0) =$$
$$= \frac{20!}{12!5!2!1!0!} 0.73^{12} 0.17^{5} 0.06^{2} 0.02^{1} 0.02^{0} =$$
$$= 0.00495.$$

The marginal distribution

Let us derive the marginal distribution of $f_{X_1 X_2 \cdots X_k}(x_1, x_2, \ldots, x_k) = P(X_1 = x_1, X_2 = x_2, X_3 = x_3, \ldots, X_k = x_k)$ for the multinomial distribution.

First of all, a little notation. Like we did earlier, the marginal distribution of, say, X_i can be written as $f_{X_i}(x_i)$. Remember that X_i is the random variable capturing the number of outcomes i we see in n tries.

Now, let outcome i be a "success"; otherwise, a "failure". Does this ring a bell? Also, remember that outcome i happens with probability p_i; everything else with probability $1 - p_i$. Hence, X_i is the number of successes we see in n independent tries. What is X_i distributed like when we view it like this?

Food for thought 10.1.

The marginal distribution of X_i is the binomial distribution: i.e., every single one of the X_i is **binomially distributed** with parameters n, p_i.

Example 10.19. Grades

For the example discussed earlier, what is the probability that:

1. 10 students get an A?

2. at most 1 students fails?

Also, how many students are expected to get each grade?

The first one is binomially distributed with $n = 20, p = 0.73$. The second one is binomially distributed with $n = 20, p = 0.02$. Overall, we have:

1. For the first probability calculation:
$$P(X_1 = 10) = \binom{20}{10} 0.73^{10} 0.27^{10} =$$
$$= 0.01635.$$
2. For the second probability calculation:
$$P(X_5 \leq 1) = = \binom{20}{0} 0.02^0 0.98^{20} + \binom{20}{1} 0.02^1 0.98^{19} =$$
$$= 0.6676 + 0.2725 = 0.9401.$$

To answer the expectation question, if each outcome is binomially distributed with $n = 20$ and p_i, the expectations are:

- A: 14.6
- B: 3.4
- C: 1.2
- D: 0.4
- F: 0.4

The conditional distribution

Now, let us consider the conditional distribution of the multinomial distribution. Say that among all outcomes we already know that X_j

has happened x_j times. This implies that there is no uncertainty about x_j of the n tries. Let's keep that in mind.

Furthermore, if we were to remove these x_j outcomes, what we are left with is $n - x_j$ tries; however these tries do not have all k outcomes happening, but instead only $k-1$.

Let us consider this with an example: if we have 20 students taking a class, and we know that 15 students ended up with an A, then the stochastic nature of this distribution only affects the remaining 5 students — after we remove the students whose grade is known to be an A.

Finally, in the remaining outcomes, we know that x_j is missing. However, we also know that if we sum the probabilities of all outcomes we should be getting 1. In the case of the grades from before, after we remove the As, we get a summation of probabilities equal to $p_2 + p_3 + p_4 + p_5 = 0.27 \neq 1$. To fix this issue, we renormalize the remainder of the probabilities. Instead of p_i, we now use $q_i = \dfrac{p_i}{\sum_{\ell \neq j} p_\ell}$. Summing up:

Food for thought 10.2.

The conditional distribution of $X_1, X_2, \ldots, X_{j-1}, X_{j+1}, \ldots, X_k$ given $X_j = x_j$ is the multinomial distribution again but with parameters $n - x_j$, $q_i = \dfrac{p_i}{\sum_{\ell \neq j} p_\ell}$.

Example 10.20. Grades

Back to the example we have been using. We have just been informed that 3 students failed. What is the probability that:

1. 10 students get an A and 7 students get a B?
2. 10 students get an A, 5 students get a B, and 2 students get a C?

Both are multinomial distributions with parameters $n = 20 - 3 = 17$ and

$$q_1 = \frac{p_1}{p_1 + p_2 + p_3 + p_4} = \frac{0.73}{0.98} = 0.7449;$$

$$q_2 = \frac{p_2}{p_1 + p_2 + p_3 + p_4} = \frac{0.17}{0.98} = 0.1735;$$

$q_3 = 0.0612$; and $q_4 = 0.0204$. Then, we have:

$$P(X_1 = 10, X_2 = 7, X_3 = 0, X_4 = 0) =$$
$$= \frac{17!}{10!7!0!0!} 0.7449^{10} 0.1735^{7} 0.0612^{0} 0.0204^{0} =$$
$$= 0.0048$$

$$P(X_1 = 10, X_2 = 5, X_3 = 2, X_4 = 0) =$$
$$= \frac{17!}{10!5!2!0!} 0.7449^{10} 0.1735^{5} 0.0612^{2} 0.0204^{0} =$$
$$= 0.01265.$$

The bivariate normal distribution

Similarly to what we did for the binomial and its extension to the multinomial, we will also extend the normal distribution. What if, we have two jointly distributed variables that are *individually* normally distributed with their own means and variances? In essence, what if we have the three-dimensional pdf portrayed in the following figure.

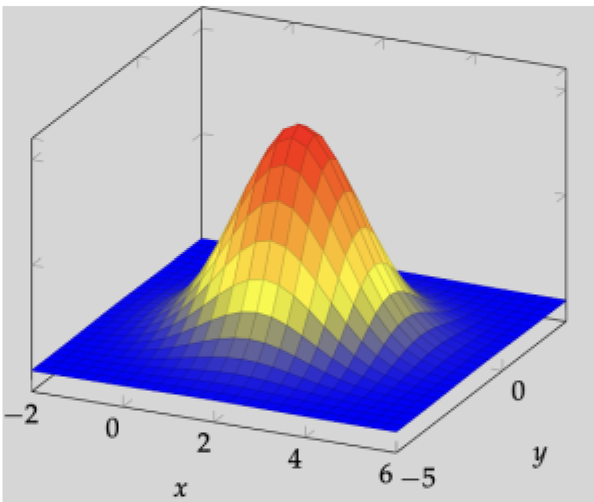

Figure 10.1. The bivariate normal distribution joint probability density function.

Formally, we provide the definition that follows:

Definition 10.13. The bivariate normal distribution

Consider two normally distributed random variables X, Y with means μ_X, μ_Y and variances σ_X^2, σ_Y^2. That is, $X \sim \mathcal{N}\left(\mu_X, \sigma_X^2\right)$ and $Y \sim \mathcal{N}\left(\mu_Y, \sigma_Y^2\right)$. We also assume that the two random variables are correlated with correlation ρ_{XY}.

Then, two random variables X and Y with the above parameters are **jointly distributed with a bivariate random distribution** if:

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} \cdot e^{\frac{-z}{2\left(1-\rho_{XY}^2\right)}},$$

where

$$z = \frac{\left(x-\mu_X\right)^2}{\sigma_X^2} - \frac{2\rho_{XY}\left(x-\mu_X\right)\left(y-\mu_Y\right)}{\sigma_X\sigma_Y} + \frac{\left(y-\mu_Y\right)^2}{\sigma_Y^2}$$

.

We observe that ρ_{XY} plays an important role. When $\rho_{XY} = 0$, we have the simplified version of the bivariate random distribution as:

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y} \cdot e^{-\left(\frac{\left(x-\mu_X\right)^2}{2\sigma_X^2} + \frac{\left(y-\mu_Y\right)^2}{\sigma_Y^2}\right)}.$$

However, we know that $\rho_{XY} = 0$ implies that X and Y are independent. And, for two independent random variables, we know that their joint pdf is equal to the product of the individual pdfs. Let's see if that is the case here:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\cdot\sigma_X} e^{-\frac{\left(x-\mu_X\right)^2}{2\sigma_X^2}}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\cdot\sigma_Y} e^{-\frac{\left(y-\mu_Y\right)^2}{2\sigma_Y^2}}$$

$$f_X(x) \cdot f_Y(y) = \frac{1}{\sqrt{2\pi} \cdot \sigma_X} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}} \cdot \frac{1}{\sqrt{2\pi} \cdot \sigma_Y} e^{-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}} = \frac{1}{2\pi\sigma_X\sigma_Y} \cdot e^{-\left(\frac{(x-\mu_X)^2}{2\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right)} = f_{XY}(x, y)$$

.

This independence is shown in the next figure. What happens when $\rho > 0$ or $\rho < 0$?

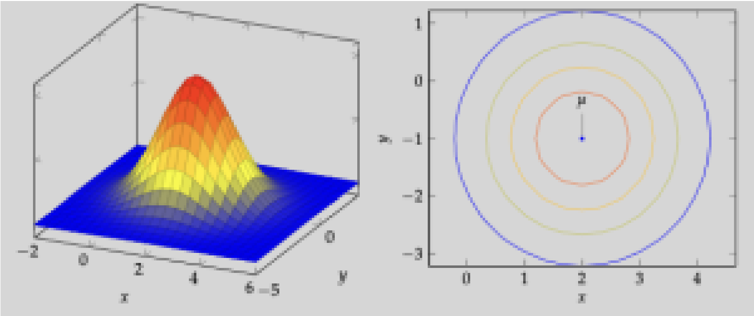

Figure 10.2. The bivariate normal distribution and its contour plot. Here, we have that [latex]\rho_{XY}=0[/latex].

But, what happens when $\rho = 1$ or $\rho = -1$? And how is it represented visually?

When we have positive correlation, this implies that higher/lower values of X will imply higher/lower values of Y and vice-versa (for Y and X). This is showcased with the pdf and the contour in the following figure, which appears to be "positively" skewed.

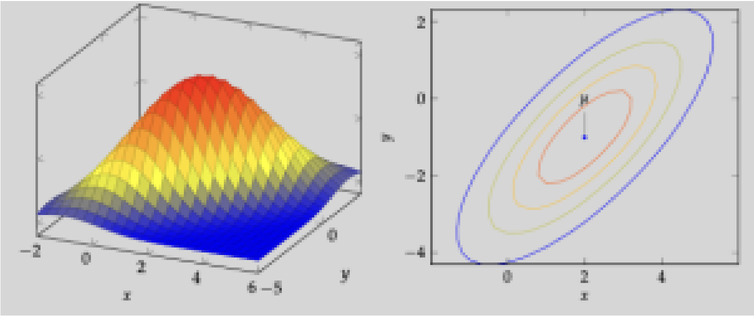

Figure 10.3. The bivariate normal distribution and its contour plot. Here, we have that [latex]\rho_{XY}>0[/latex].

On the other hand, if $\rho_{XY} < 0$, this means that higher/lower values of X will lead to lower/higher values of Y and vice-versa (for Y and X). This is exactly the opposite. This is again shown visually with the pdf and the contour in the following figure, which appears to be "negatively" skewed.

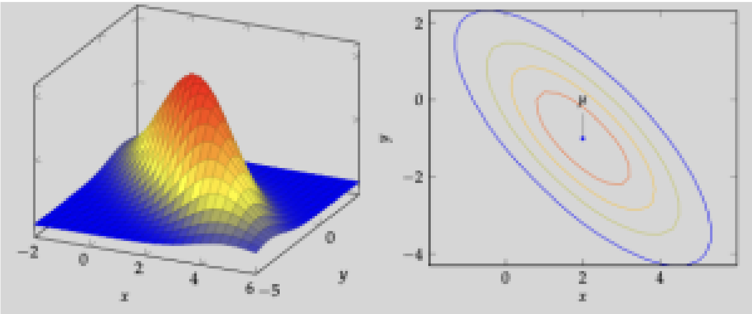

Figure 10.4. The bivariate normal distribution and its contour plot. Here, we have that [latex]\rho_{XY}<0[/latex].

Food for thought 10.3.

What happens when $\rho = 1$ or $\rho = -1$? Let's leave this as food for thought.

The marginal and the conditional distribution

Much like what we did for the multinomial distribution, we may also derive the marginal and conditional distributions for the bivariate normal distribution. More specifically, both the *marginal* and the *conditional* distributions for the bivariate normal distribution are normal distributions themselves!

Marginal pdf:

$X \sim \mathcal{N}\left(\mu_X, \sigma_X^2\right)$

$Y \sim \mathcal{N}\left(\mu_Y, \sigma_Y^2\right)$

Conditional pdf:

$X|Y = y \sim \mathcal{N}\left(\mu_{X|Y=y}, \sigma^2_{X|Y=y}\right)$

$\mu_{X|Y=y} = \mu_X + \rho_{XY}\left(\frac{\sigma_X}{\sigma_Y}\right)(y - \mu_Y)$

$\sigma^2_{X|Y=y} = \sigma^2_X\left(1 - \rho^2_{XY}\right)$

$Y|X = x \sim \mathcal{N}\left(\mu_{Y|X=x}, \sigma^2_{Y|X=x}\right)$

$\mu_{Y|X=x} = \mu_Y + \rho_{XY}\left(\frac{\sigma_Y}{\sigma_X}\right)(x - \mu_X)$

$\sigma^2_{Y|X=x} = \sigma^2_Y\left(1 - \rho^2_{XY}\right)$

For the conditional pdf, the question from earlier comes back. What if X and Y are independent? What if they are perfectly correlated ($\rho_{XY} = 1$ or $\rho_{XY} = -1$)?

We finish this chapter with a big, comprehensive example that combines information from this chapter. Pay close attention to the derivations and calculations that follow!

Example 10.21. The bivariate normal distribution

A class has two exams, both of which have grades that are normally distributed with $\mu_1 = 80, \mu_2 = 82.5$ and $\sigma_1^2 = 100, \sigma_2^2 = 225$. Finally the two exams are positively correlated with $\rho = 0.6$. What is the probability that:

1. a random student scores over 75 in Exam 2?
2. a random student scores over 75 in Exam 2 given that they scored an 85 in the first exam?
3. the sum of the two exams of a random student is less than or equal to 175?
4. a random student did better on the second exam than the first exam?

Let's get to it. Let X_1 be the grade of the first exam, and X_2 the grade of the second exam. Then:

We know that $X_2 \sim \mathcal{N}\left(82.5, 225\right)$. Hence:

$$z = \frac{75 - 82.5}{15} = -0.5.$$

$$P(X_2 > 75) = 1 - P(X_2 \leq 75) = 1 - \Phi(z) = \Phi(-z) = \Phi(0.5) = 0.6915$$

.

We also know that $X_2 | X_1 \sim \mathcal{N}\left(\mu_{X_2|X_1}, \sigma^2_{X_2|X_1}\right)$. We calculate:

$$\mu_{X_2|X_1=85} = \mu_{X_2} + \rho_{X_1X_2}\left(\frac{\sigma_{X_2}}{\sigma_{X_1}}\right)(85 - \mu_{X_1}) = 82.5 + 0.6 \cdot \frac{15}{10} \cdot 5 = 87$$

.

$$\sigma^2_{X_2|X_1=85} = \sigma^2_{X_2}\left(1 - \rho^2_{X_1X_2}\right) = 225 \cdot 0.64 = 144$$

.

$$z = \frac{75 - 87}{12} = -1.$$

$$\begin{aligned} P(X_2 > 75 | X_1 = 85) &= 1 - P(X_2 \leq 75 | X_1 = 85) \\ &= 1 - \Phi(z) = \Phi(-z) = \Phi(1) = 0.8413 \end{aligned}$$

.

Hence, knowing that the student did better than average

in the first exam changes our perspective for their probability to do well in the second exam, too.
The sum of two normally distributed random variables is also normally distributed! Additionally, we have:

$$E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E\left[X_i\right]$$

[The following equation is causing problems]

$$Var\left[\sum_{i=1}^{n} a_i X_i\right]$$

$$= \sum_{i=1}^{n} a_i^2 Var\left[X_i\right] + \sum_{i=1}^{n} \sum_{j=1:i\neq j}^{n} a_i a_j Cov\left[X_i, X_j\right]$$

$$= \sum_{i=1}^{n} a_i^2 Var\left[X_i\right] + 2a_i a_j \cdot \sum_{i=1}^{n} \sum_{i<j} Cov\left[X_i, X_j\right]$$

[The above equation is causing problems]

In our case, we have two random variables, so:

$$E\left[X_1 + X_2\right] = E\left[X_1\right] + E\left[X_2\right] = 162.5$$

$$Var\left[X_1 + X_2\right] = Var\left[X_1\right] + Var\left[X_2\right] + 2Cov\left[X_1, X_2\right] = 325 + 2\sigma^2_{X_1 X_2}$$

.
To calculate $\sigma^2_{X_1 X_2}$ we use the definition of correlation (see Lecture 12):

$$\rho_{X_1 X_2} = \frac{\sigma^2_{X_1 X_2}}{\sigma_{X_1}\sigma_{X_2}} \implies 0.6 = \frac{\sigma^2_{X_1 X_2}}{10 \cdot 15} = \sigma^2_{X_1 X_2} = 90$$

.
This leads to a final variance of $Var\left[X_1 + X_2\right] = 505$. Finally:

$$X_1 + X_2 \sim \mathcal{N}\left(162.5, 505\right).$$

$$z = \frac{175 - 162.5}{\sqrt{505}} = 0.56.$$

$$P(X_1 + X_2 \leq 175) = \Phi(z) = \Phi(0.56) = 0.7123$$

.

For this question, we want $X_2 > X_1 \implies X_2 - X_1 > 0$. The difference of two normally distributed random variables is—again—normally distributed! Its details:

$$E\left[X_2 - X_1\right] = E\left[X_2\right] - E\left[X_1\right] = 2.5$$

$$Var\left[X_2 - X_1\right] = Var\left[X_1\right] + Var\left[X_2\right] - 2Cov\left[X_1, X_2\right] = 125$$

$X_2 + X_1 \sim \mathcal{N}\left(2.5, 125\right)$.

$$z = \frac{0 - 2.5}{\sqrt{125}} = -0.22.$$

$$\begin{aligned} P(X_2 > X_1) = P(X_2 - X_1 > 0) &= 1 - P(X_2 - X_1 \leq 0) \\ &= 1 - \Phi(z) = \Phi(-z) = \Phi(0.22) = 0.5871 \end{aligned}$$

.

Activities

Activity 1: Jointly distributed discrete random variables

During our last worksheet, we saw two discrete random variables $X = \{1, 2, 3\}$ and $Y = \{10, 20\}$ that were jointly distributed with probability mass function:

$$f_{XY}(x,y) = \frac{20}{27} \cdot \frac{x+1}{y}.$$

Additionally, we can calculate the marginal distribution of X and Y
:

$$f_X(x) = \sum_y f_{XY}(x,y) = \frac{20}{27}\frac{x+1}{10} + \frac{20}{27}\frac{x+1}{20} = \frac{x+1}{9},$$
$$f_Y(y) = \sum_x f_{XY}(x,y) = \frac{20}{27}\frac{2}{y} + \frac{20}{27}\frac{3}{y} + \frac{20}{27}\frac{4}{y} = \frac{180}{27y} = \frac{60}{9y}.$$

Finally, recall that the conditional distribution for X given $Y = y$ can be calculated as:

$$f_{X|Y=y}(x) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{\frac{20}{27} \cdot \frac{x+1}{y}}{\frac{60}{9y}} = \frac{x+1}{9}.$$

It seems that the conditional and the marginal distribution are the same – this is not always the case! This only happens when X and Y are independent. More on that later today.

Problem 1: Expectations and variances

What is the expectation of X and what is the variance of X?

Problem 2: Conditional expectations and variances

What is the expectation of X and what is the variance of X *given that* $Y = 10$?

Problem 3: Independent? Covariance? Correlation?

As hinted at when we calculated the marginal and conditional distributions, it appears that $f_X(x) = f_{X|Y=y}(x)$. Equivalently, we could make the observation that $f_{XY}(x, y) = f_X(x) \cdot f_Y(y)$. Hence, the two random variables X and Y are independent!

Based on your this observation of independence, what is the covariance? What is the correlation?

- $\sigma_{XY} = Cov\left[X, Y\right]$?
- $\rho_{XY} = Corr\left[X, Y\right]$?

Activity 2: Jointly distributed continuous random variables

Consider two jointly distributed *continuous* random variables X, Y with domains $0 \leq X \leq 2$ and $0 \leq Y \leq 1$ and with joint probability density function equal to:

$$f_{XY}(x, y) = \frac{3}{4} x^3 y^2.$$

Problem 4: Warm-up with marginal distributions

Let's repeat what we had done during the previous chapter. What are the marginal distributions of X and Y?

- $f_X(x) = \int\limits_0^1 f_{XY}(x,y)dy =$
- $f_Y(y) = \int\limits_0^2 f_{XY}(x,y)dx =$

Problem 5: Getting the expectations

What are the expectations of X and Y? Don't forget that they are defined over *different domains*!

Problem 6: Independent?

Are X and Y independent? Why/Why not?

Activity 3: When X and Y restrict each other

Assume that random variables X and Y are jointly distributed with probability density function $f_{XY}(x, y) = \frac{1}{4}(x + y)$ defined over $0 \leq X \leq Y \leq 2$. Note how random variable X always takes values that are at most as big as the value of random variable Y. X and Y are not independent; this is clear from their definition, as knowing the one restricts the values the other one may take. What is the covariance of X and Y then? Well, to answer that we will need a lot of things. Namely, we need:

1. the marginal distributions $f_X(x), f_Y(y)$;
2. the expectation $E[X], E[Y]$;
3. the expectation of function $X \cdot Y$: $E[X \cdot Y]$;
4. finally, you'll get $Cov[X, Y] = E[X \cdot Y] - E[X] \cdot E[Y]$.

Let's get to it!

Problem 7: Marginal distributions

What are the marginal distributions of X and Y? Verify that X and Y are not independent by checking whether $f_X(x) \cdot f_Y(y)$ is different than $f_{XY}(x, y)$. Here, we help you get $f_X(x)$ started as a hint.

$$f_X(x) = \int_x^2 f_{XY}(x, y)dy =$$

Problem 8: Expectation of X and Y

Using the marginal distributions from earlier, what is the expectation of X and what is the expectation of Y?

Problem 9: Expectation of $X \cdot Y$

And.. what is the expectation of $X \cdot Y$?

Problem 10: Covariance and correlation

Almost there. So, what is the covariance and what is the correlation of X and Y? For the correlation calculations as well as for any other remaining parts in this specific activity, use that $Var\left[X\right] = \dfrac{43}{180}$ and $Var\left[Y\right] = \dfrac{3}{20}$.

Problem 11: Applying the variance identity

In the Theory part of the chapter, we discuss how the variance identity becomes :

$$Var\left[aX + bY\right] = a^2 Var\left[X\right] + b^2 Var\left[Y\right] + 2abCov\left[X, Y\right].$$

So, assume we are interested in the variance of random variable $Z = 3 \cdot X + 2 \cdot Y$: what can it be evaluated as?

Activity 4: The multinomial distribution

A product is manufactured in one of three factories located in Fargo, ND, Atlanta, GA, and Portland, OR. The product can be of high or low quality (assume these are the only two possibilities). The three factories have slightly different quality outputs, that are provided in Table 1.

Table 10.3. The high and low quality output rates of each factory. For example, Fargo, ND will have 95% of their products be of high quality, whereas Atlanta, GA will only have 92% of their products be of high quality.

	Fargo, ND	Atlanta, GA	Portland, OR
High quality	95%	92%	97.5%
Low quality	5%	8%	2.5%

Last, assume that Fargo, ND is responsible for 40% of the production in the United States, with Atlanta, GA and Portland, OR sharing the remaining production.

Problem 12: Law of total probability

Remember the law of total probability? Let's apply it here (in its discrete form). What is the probability a random product you pick up is of low quality?

Problem 13: An application for the multinomial

You pick 10 products at random. What is the probability that exactly 4 of them were produced in Fargo, ND, exactly 3 of them were produced in Atlanta, GA, and the other 3 of them were produced in Portland, OR?

Problem 14: Marginal and conditional distributions

You pick 10 products at random again. Answer the two probability questions here:

1. What is the probability that at most one of them was produced in Atlanta, GA?
2. What is the probability 4 of them were produced in Fargo and the other 4 of them were produced in Portland given that 2 of them were manufactured in Atlanta?

A takeaway message from Activity 4 is the following:

Food for thought 10.4.

In general:

1. The marginal distribution of a multinomial distribution is the binomial.
2. The conditional distribution of a multinomial distribution is another multinomial.

Activity 5: The bivariate normal distribution

We have asked a set of people about their views on climate change and associated sustainability policies proposed to curb its effects. Specifically, we want to measure the net favorability of each question (measured in $\%$) defined as the $\%$ of respondents agreeing minus the $\%$ of respondents disagreeing. The survey in several different states has shown that the net favorability in the climate change question is $X \sim \mathcal{N}(4, 4)$, while the net favorability in the policies question is $Y \sim \mathcal{N}(2, 9)$. In English, we assume the net favorability in agreeing that climate change is a real threat is (continuous) random variable X normally distributed with mean $\mu_X = 4$ and variance $\sigma_X^2 = 4$; while the net favorability in agreeing with policies proposed is another normally distributed random variable called Y with mean $\mu_Y = 2$ and $\sigma_Y^2 = 9$.

Problem 15: Simple normal

Let's jog our memories with the normal distribution. Answer the two next probability questions.

- What is the probability that there is more than a 6% net favorability difference for climate change? That is, what is $P(X > 6)$?
- What is the probability that there is less than or equal to a 1% net favorability difference adopting sustainability policies? That is, what is $P(Y \leq 1)$?

Problem 16: Independent?

Let's now look at the problem from a "data analyst" perspective. Do you think it is fair to assume that these two answers are independent? Or would they be correlated? And is that correlation positive or negative? Simply a sentence of explaining why yes/no would suffice here.

[Add title]

Hopefully you have agreed that the two are indeed correlated. From now on, please assume that we have a positive correlation equal to $\rho_{XY} = 0.5$.

Problem 17: The case of NC

We are analyzing the results from North Carolina. It appears that in NC respondents agree that climate changes is an important threat. Specifically, we observe that the net favorability of this question is $X = 3\%$. What is the probability that respondents disagree with policies to promote sustainability given that $X = 3\%$? That is, what is $P(Y \leq 0|X = 3)$? Recall that we have $\rho_{XY} = 0.5$.

Problem 18: Overwhelming support for one; so-and-so on the other

What is the probability of getting a state that both has a higher than 6% net favorability for the first question but is very close on the second question? Let's define "very close" as being between -1% and +1% of net favorability. In mathematical terms, what is $P\left((X > 6) \cap -1 \leq (Y \leq 1)\right)$?

Activity 6: The velocity of a particle

A particle's velocity (measured in m/s) in a gas is a continuous random variable (let it be V) with pdf $f_V(v) = av^2e^{-bv}, v > 0$, where b is a constant that depends on the temperature of the gas and the mass of the particle. While we could make the case that the speed of anything can be no larger than the speed of light, we do allow speed to go up to infinity here, as it makes the math a little easier � so please consider that $v \in (0, +\infty)$.

Problem 19: Back to basics

Once again, what should a be in order for $f_V(v)$ to be a valid pdf? Note that your result will be a function fo b.

Problem 20: The expectation of a function

Back to expectation properties now. The kinetic energy of a particle is $W = \frac{1}{2}mV^2$, where V is the velocity (the random variable from earlier). What is the expected kinetic energy of the particle? Of course, as in Problem 1, your final answer will be a function of b.

Problem 21: The probability density function of a function

Assuming again that the kinetic energy of a particle is $W = \frac{1}{2}mV^2$, where V is the velocity random variable: what is the pdf of W?

The key point here to remember is that this is a very useful derivation:

> *Food for thought 10.5.*
>
> Let X be a random variable distributed with pdf $f_X(X)$. Also consider Y a random variable that is a function of X, as in $Y = g(X)$. Let $u(y)$ be the inverse function, i.e., $u(y) = g^{-1}(y)$. Then, Y is distributed with pdf: $f_Y(y) = f_X(u(y)) \cdot |u'(y)|$
> .

11. Estimators

Theory

Chapter objectives

Learning outcomes

After this chapter, you will be able to:

- Describe point estimation.
- Explain the difference between bias and variance of a point estimator.
- Evaluate the bias, variance, mean square error of a point estimator.
- Compare two point estimators and pick the better one.

Motivation: Inferring parameters

In most applications, we have a good enough idea on the distribution that we need to follow. When a company makes vehicles, we could pick some of them to check for their quality and count how many are of high quality: this can be modeled as a binomial or a hypergeometric distribution. When a student takes an exam, they will expect to do similarly to their previous exams plus or minus some points if they prepare in a similar manner: their score can be modeled as a normal distribution.

However, one of the questions we need to answer is: what are the parameters of the distributions? What is the mean and variance of that normal distribution? What is p in a binomial distribution? So far, we have been given this as part of our data. What happens when we are given data and need to infer their values, based on real-life observations?

Chapter questions

1. What is the **bias** of an estimator $\hat{\Theta}$ for parameter θ?
   $\text{bias}[\hat{\Theta}] = E[\hat{\Theta}] - \theta$
2. How do we calculate the **expected value** of an estimator?

   - $E[aX + bY + c] = aE[X] + bE[Y] + c$
   - $E[\Sigma X_i] = \Sigma E[X_i]$
   - If all X_i are distributed the same way (say from some population X), then $E[\Sigma X_i] = \Sigma E[X_i] = nE[X]$

3. How do we calculate the **variance** of an estimator $\hat{\Theta}$ for parameter θ?

   - $Var[aX + bY + c] = a^2Var[X] + b^2Var[Y]$, **if X and Y are independent**
   - $Var[\Sigma X_i] = \Sigma Var[X_i]$, **if X and Y are independent**
   - If all X_i are distributed the same way (say from some population X) **and are independent**, then $Var[\Sigma X_i] = \Sigma Var[X_i] = nVar[X]$

4. How do we calculate the **mean square error** of an estimator $\hat{\Theta}$?

◦ MSEs are useful for comparing two estimators $\hat{\Theta}$�, $\hat{\Theta}$�.

Motivation: Predicting an election

Before an election takes place, we see many polls. Some of them appear to better resemble the final result (after the election); others fail to capture reality. Given this data based on a *sample* of the whole *population*, what can we say about the election? What can we say about the probabilities of one candidate versus another?

Statistical inference

Statistical inference takes us from the sample to the whole. For example, consider any of the next scenarios:

- We interviewed 50 people about the next election. What do the results imply for the general election?
- We picked a sample of 10 cars and performed a crash test. What do the observations imply for the whole production line?
- We collected exit interview data from 100 alumni. What do their answers imply for the starting salary of our alumni?

Right away, we can make some intuitive observations:

1. Checking a sample, rather than the whole, saves us time and effort.
2. Checking a sample, rather than the whole, comes with a loss of information.
3. Checking a sample, rather than the whole, we want to recreate the whole.

Statistical inference theme

Flow diagram showing the statistical inference process. On the left is an oval labeled 'Population X ~ f(x,θ)' in red. A blue arrow points from it to a circle labeled 'Sample X1,X2,...,Xn' in yellow. Another blue arrow points from the sample to a rectangle labeled 'Statistics' in blue. A dashed line with a question mark goes from Statistics back to Population, illustrating the inference process from sample statistics back to population parameters.
Figure 1: The theme of statistical inference. We collect a **sample** of the whole **population** that we analyze to get some **statistic**, which we then use to infer information about the population.

Example 11.1. Sample averages and variances

Assume a large population X: you decide to collect only 5 random variables X_1, X_2, X_3, X_4, X_5. Then, we may calculate the sample average $\dfrac{X_1 + X_2 + X_3 + X_4 + X_5}{5}$ and the sample variance and use those in lieu of the population mean (unknown) and the population variance (unknown).

For example, say I want to figure out the average height of every Chicago resident, I could (i) travel to Chicago, (ii) ask 10 people about their height, and (iii) calculate the average of these 10 people. Is this the true average?

Statistics

We proceed with some preliminaries that will be used throughout the next few classes.

Definition and theorems 11.1. Statistic

A statistic is *any* value obtained by random data.

Well, this is not very useful. The only recurring theme here is *random* data. In essence, the definition claims that any value that is different for differently obtained data can be considered a statistic!

Example 11.2. Height statistics

Assume that the heights of the 5 people in the leadership team of a student chapter are: 60, 67, 72, 63, 60. Then, the height of the second person picked (67) is a statistic. The average height is another statistic. Finally, getting the height of the first person and multiplying it by 3 and adding to it the height of the last person is *also* a statistic.

Clearly, some statistics are more useful than others. For example, in our earlier discussion the average is more useful than the last statistic.

Exercise: 11.1. Are these statistics?

True or False:

- The sample variance. **True**
- The value of the first element of a sample. **True**
- The value of the first element of a sample minus 3. **True**

Some basic properties of statistics:

1. **Statistics depend on the sample selected:** the value a statistic gets will be different depending on the sample selected. If I select 5 students in the class and report their average exam score (a statistic), it will be different depending on the 5 students selected.
2. **Statistics are functions of the sample selected:** we can use the same "formula" or follow the same "approach" to estimate the value of a statistic given a sample, no matter the sample.
3. **Statistics are random variables:** statistics are distributed as random variables. Certain values may appear more often than others, we can define expectations for statistics, etc.

Sampling distribution

So, if a statistic is a random variable, what is its distribution? The distribution of a statistic, called the **sampling distribution** depends on three things:

1. The distribution of the whole population. Of course we should expect that the distribution of the population will be reflected when looking at a sample!
2. The size of the sample. Once again, it should make sense that the bigger the sample we pick the more accurately we will reflect the population distribution.
3. The way the sample was selected. We will not devote a lot of time in this: but, picking a sample in a non-random way will affect the distribution we see.

Example 11.3. Back to the normal distribution

Assume you have a population where each individual is distributed following a normal distribution with mean μ and variance σ^2. You pick, at random, a set of n individuals X_i. What is the average distributed as?

We have seen that the average $Y = \frac{\sum X_i}{n}$ is also normally distributed with the same mean μ and variance $\frac{\sigma^2}{n}$. This normal distribution $\mathcal{N}(\mu, \sigma^2/n)$ is the sampling distribution.

Point estimators

Let's put everything formally. Let X be a population distributed with some pdf $f(x, \theta)$, where θ is some unknown parameter. By the way, you may treat θ as a vector of multiple parameters.

Furthermore, let $X_1, X_2, \ldots, X_n$ be a series of random elements picked from the population. They form the sample of size n that was selected. By definition, seeing as $X_1, X_2, \ldots, X_n$ all come from the same place, they are identically distributed and independent random variables.

Definition and theorems 11.2. Point estimators

We define point estimator(s) $\hat{\Theta}$ as a statistic that is used to approximate the unknown parameter(s) θ.

By definition, $\hat{\Theta}$ is a function of the sample selected ($X_1, X_2, \ldots, X_n$), hence we may say that $\hat{\Theta}$ is a random variable depending on the sample ($\hat{\Theta} = h(X_1, X_2, \ldots, X_n)$, where $h(\cdot)$ is some function).

Of course, once we have picked a sample then $\hat{\Theta}$ can be calculated and assigned a value. This value is called the **point estimate** $\hat{\theta}$. To summarize, $\hat{\Theta}$ is the general statistic used (e.g., the point estimator can be found if we take the average and add 2) whereas $\hat{\theta}$ is the value the estimator receives for a specific sample (e.g., for our sample, the average is 7 so the point estimate is 9). To summarize, $\hat{\Theta}$ is typically a "formula" or an "expression", whereas $\hat{\theta}$ is an actual number.

Common point estimators

Table 11.1. Single Population

Parameters	**Point estimators**
Population mean μ	Sample average $\hat{\Theta} = \overline{x}$
Population variance σ^2	Sample variance $\hat{\Theta} = s^2$
Population proportion p	Sample proportion $\frac{\hat{n}}{n}$

Table 11.2. Two populations

Parameters	**Point estimators**
Difference in population means $\mu_1 - \mu_2$	Difference in sample averages $\hat{\Theta} = \overline{x}_1 - \overline{x}_2$
Ratio in population variances $\frac{\sigma_1^2}{\sigma_2^2}$	Ratio in sample variance $\frac{s_1^2}{s_2^2}$
Difference in population proportions $p_1 - p_2$	Difference in sample proportions $\hat{\Theta} = \frac{\hat{n}_1}{n_1} - \frac{\hat{n}_2}{n_2}$

What makes a good estimator?

Every estimator has two main items we want to evaluate it by: **accuracy** and **precision**. In statistics terms, we refer to them as **bias** and **standard error** or **variance**. We present the effect that each of the two would have to our estimation process: decreasing the bias would lead to better results *on average*; decreasing the standard error/variance would lead to smaller dispersion.

Bias

Definition and theorems 11.3. Bias

The **bias** of a point estimator $\hat{\Theta}$ for parameter θ is defined as:

$$Bias(\hat{\Theta}) = E[\hat{\Theta}] - \theta$$

An estimator is **unbiased** if $Bias(\hat{\Theta}) = 0$, i.e., $E[\hat{\Theta}] = \theta$.

Variance and Standard Error

Definition and theorems 11.4. Variance and Standard Error

The **variance** of a point estimator $\hat{\Theta}$ is:

$$Var(\hat{\Theta}) = E[(\hat{\Theta} - E[\hat{\Theta}])^2]$$

The **standard error** is the square root of the variance:

$$SE(\hat{\Theta}) = \sqrt{Var(\hat{\Theta})}$$

Mean Square Error

Definition and theorems 11.5. Mean Square Error

The **mean square error** (MSE) of an estimator $\hat{\Theta}$ is:

$$MSE(\hat{\Theta}) = E[(\hat{\Theta} - \theta)^2] = Var(\hat{\Theta}) + [Bias(\hat{\Theta})]^2$$

The MSE combines both bias and variance into a single measure of estimator quality.

Example 11.4. Evaluating estimators

Consider two estimators for the population mean μ:

- Estimator 1: $\hat{\Theta}_1 = \overline{X}$ (sample mean)

- Estimator 2: $\hat{\Theta}_2 = X_1$ (first observation)

Both are unbiased: $E[\hat{\Theta}_1] = \mu$ and $E[\hat{\Theta}_2] = \mu$.

However, their variances differ:

- $Var(\hat{\Theta}_1) = \dfrac{\sigma^2}{n}$
- $Var(\hat{\Theta}_2) = \sigma^2$

Since $\dfrac{\sigma^2}{n} 1$, the sample mean is a better estimator!

Activities

Activity 1: Anchoring activity

Let us begin with a small game activity. Consider a (discrete) **uniformly distributed population** X that can take any integer value between 1 and α, where $\alpha > 1$ is sadly unknown. To estimate it, we will try three point estimators. Given a sample of $n = 3$ numbers (call them X_1, X_2, X_3) from the population, we will estimate α as:

(2) $$\hat{\Theta}_1 = \frac{X_1 + X_2 + X_3}{3}. \quad \hat{\Theta}_2 = \frac{X_1 + 2X_2 + X_3}{3}.$$
$$\hat{\Theta}_3 = \frac{2}{3}(X_1 + X_2 + X_3) - 1.$$

Problem 1: Using estimators

First, navigate yourself to the website

http://vogiatzis.web.illinois.edu/random.html

and generate three numbers. Call them X_1, X_2, X_3. What is the point estimate you'd get for α with each of the three estimators?

$$X_1 = \qquad, \quad X_2 = \qquad, \quad X_3 =$$

$$\hat{\theta}_1 = \qquad \hat{\theta}_2 = \qquad \hat{\theta}_3 =$$

Problem 2: Estimator bias

If we are trying to estimate α, recall that the bias of some estimator $\hat{\Theta}$ can be calculated as

$$bias\left[\hat{\Theta}\right] = E\left[\hat{\Theta}\right] - \alpha.$$

Moreover, recall that because X_1, X_2, X_3 were obtained from the same population X we know that the following are true:

$$E\left[X_1\right] = E\left[X_2\right] = E\left[X_3\right] = E\left[X\right] = \mu$$
$$Var\left[X_1\right] = Var\left[X_2\right] = Var\left[X\right] = \sigma^2.$$

You do not need the variance in this question, but you may need it later today!

Finally, remember that the mean μ depends on the distribution we have, so in our case it would be $\mu = \dfrac{1+\alpha}{2}$ (seeing as the numbers obtained come from a discrete uniform distribution between 1 and α).

Putting all of the above together, what is the bias of each of the three estimators? Your bias could possibly be a function of α!

$$bias\left[\hat{\Theta}_1\right] =$$
$$bias\left[\hat{\Theta}_2\right] =$$
$$bias\left[\hat{\Theta}_3\right] =$$

You must have gotten that the bias of estimator $\hat{\Theta}_3$ is equal to zero! This is great news. Let us check what its variance is.

Problem 3: Estimator variance

What is the variance of $\hat{\Theta}_3$? Recall that the variance too can be a function of the unknown parameter α. The variance of a uniformly distributed discrete random variable between a and b is $\dfrac{(b-a+1)^2-1}{12}$. In our case, our population X is between 1 and α so the formula gives us a variance of

$$Var\left[X\right] = \frac{\alpha^2-1}{12}.$$

- $Var\left[\hat{\Theta}_3\right] =$

Problem 4: Estimator variance for different sample sizes

How would the variance change if we picked a sample of $n = 5$ observations $(X_1, X_2, X_3, X_4, X_5)$ and then calculated the estimator as $\hat{\Theta} = \frac{2}{5}(X_1 + X_2 + X_3 + X_4 + X_5) - 1$? Would it increase, decrease, or stay the same? What happens as n increases more and more?

- $Var\left[\hat{\Theta}\right] =$

Takeaway from Activity 1

This brings us to our first realization. With an unbiased estimator, larger samples will lead to smaller variances!

Let's make it interesting. Use this estimator to produce some values from the website and note here your best estimate for what α is. What is the biggest number you can possibly obtain from this website? The biggest number that can be produced from the website is…

My best estimate is…

Activity 2: Weird point estimators

Assume that a population is distributed with pdf $f(x) = c\,(1 + \theta x), -1 \leq x \leq 1$, where θ is an unknown parameter, and c a constant.

Problem 5: Back to basics

Let's return to the basics for a second! What should c be equal to in order for $f(x)$ to be a valid continuous pdf?

Takeaway from Problem 5

If we are doing things right, θ should disappear after taking the integral. This implies that θ can take any from a series of values, hence being a **parameter** rather than a **constant**. A parameter can be any thing; a constant has to take on a specific value.

Problem 6: Where did you come up with this?

Assume you obtain a sample of n observations. Consider the sample average $\overline{X} = (X_1 + X_2 + \ldots + X_n)/n$. Show that $\hat{\Theta} = 3\overline{X}$ is an **unbiased estimator** for θ.

Problem 7: Variance and standard error

What is the standard error of the point estimator $\hat{\Theta} = 3\overline{X}$? The standard error can be calculated as

$$SE\left[\hat{\Theta}\right] = \sqrt{Var\left[\hat{\Theta}\right]}.$$

Takeaway from Problem 7

This is quite common: when dealing with an unknown parameter (in our case, θ), the bias and the variance can depend on the value that the parameter actually has. So, how do we compare estimators for unknown parameters?

Activity 3: Comparing point estimators

Assume we have collected a sample of $n = 3$ observations X_1, X_2, X_3 coming from a population X distributed with *some pdf* with unknown μ and known $\sigma^2 = 16$. We have devised three point estimators for the unknown population mean:

- Get the average from the first two observations omitting the third, i.e., $\hat{\Theta}_1 = \dfrac{X_1 + X_2}{2}$.

Add the "odd" observations once and the "even" observations doubled and divide everything by 4, i.e., $\hat{\Theta}_2 = \dfrac{X_1 + 2X_2 + X_3}{4}$. Once again omit the third observation and simply add the first two and divide by 4, i.e., $\hat{\Theta}_3 = \dfrac{X_1 + X_2}{4}$.

Now, for one more time in this set of activities, we go ahead and calculate the bias and variance of each of the estimators.

- We have for the first estimator, $\hat{\boldsymbol{\Theta}}_{\mathbf{1}}$:

$$
\begin{aligned}
bias\left[\hat{\Theta}_1\right] &= E\left[\hat{\Theta}_1\right] - \mu = E\left[\frac{X_1 + X_2}{2}\right] - \mu = \frac{1}{2}E\left[X_1\right] + \frac{1}{2}E\left[X_2\right] - \mu = \\
&= \frac{1}{2}E\left[X\right] + \frac{1}{2}E\left[X\right] - \mu = E\left[X\right] - \mu = \mu - \mu = 0. \\
Var\left[\hat{\Theta}_1\right] &= Var\left[\frac{X_1 + X_2}{2}\right] = \frac{1}{4}Var\left[X_1 + X_2\right] = Var\left[X_1\right] + \frac{1}{4}Var\left[X_2\right] = \\
&= \frac{1}{4}Var\left[X\right] + \frac{1}{4}Var\left[X\right] = \frac{1}{2}Var\left[X\right] = \frac{1}{2}\sigma^2 = 8.
\end{aligned}
$$

- Similarly, for the second estimator, $\hat{\boldsymbol{\Theta}}_{\mathbf{2}}$, we would end up with:

$$
\begin{aligned}
bias\left[\hat{\Theta}_2\right] &= 0. \\
Var\left[\hat{\Theta}_2\right] &= 6.
\end{aligned}
$$

- Finally, for the third estimator, $\hat{\boldsymbol{\Theta}}_{\mathbf{3}}$:

$$bias\left[\hat{\Theta}_3\right] = -\frac{\mu}{2}.$$
$$Var\left[\hat{\Theta}_3\right] = 2.$$

Or, in tabular form:

Table 11.3. The bias and variance for the three estimations $\hat{\Theta}_1, \hat{\Theta}_2, \hat{\Theta}_3$.

	$\hat{\Theta}_1$	$\hat{\Theta}_2$	$\hat{\Theta}_3$
bias	0	0	$-\mu/2$
variance	8	6	2

Problem 8: Comparison I

What is the MSE of each of the estimators? Which estimator is the best according to its MSE, if we have been told that $\mu > 4$?

Takeaway from Problem 8

Note how the result would have changed if $\mu < 4$. On the other hand, there is no way that the first estimator has the smallest variance among the three.

Problem 9: Observation

Earlier in this activity, we observe that the first two estimators are unbiased (i.e., zero bias). In general, assume you are collecting a sample of n observations $(X_1, X_2, \ldots, X_n)$ and are using $\hat{\Theta} = a_1 X_1 + a_2 X_2 + \ldots + a_n X_n$ to estimate the unknown mean. What condition should $a_1 + a_2 + \ldots + a_n$ satisfy in order for $\hat{\Theta}$ to have bias equal to zero?

Activity 4: Uniform distribution

Last time, we worked with estimating the upper bound of a discrete uniformly distributed population X; today, we begin with the continuous version! Assume we are trying to estimate the upper bound of a continuous uniformly distributed population X. In English: if we have a random number generator producing (real) numbers between 0 and α, how can we estimate what α is after collecting some data from the generator?

Problem 10: Intuition

For example, say we have obtained $X_1 = 7.7, X_2 = 15.21, X_3 = 9.1$, then what is a "good" estimate for the true value of α? Is saying that all numbers that are produced by this generator are between 0 and 10 true? How about between 0 and 20?

A good estimate we can come up with for the upper bound of the generator is that it is ...

Problem 11: Bias for $n = 3$

Assume that we use the following estimator:

$$\hat{\Theta} = \max\{X_1, X_2, \ldots, X_n\},$$

where n is the number of data points we have collected. For example, if $X_1 = 7.7, X_2 = 15.21, X_3 = 9.1$ are the data obtained then $n = 3$, and $\hat{\theta}$ (the estimate obtained) is 15.21. **What is the bias of the estimator?** Assume that we have already obtained that

$$E\left[\hat{\Theta}\right] = \frac{n}{n+1} \cdot \alpha.$$

$$bias\left[\hat{\Theta}\right] =$$

Problem 12: Bias for $n \to \infty$

What happens to the bias as we increase the number of observations from the generator? Specifically, what is $\lim_{n\to\infty} bias\left[\hat{\Theta}\right]$?

Takeaway from Problem 12

We see that the bias goes to 0 as we obtain more and more observations. This is great news! We define **a consistent estimator** (sometimes also called an *asymptotically* consistent estimator) as an estimator that has bias converging to 0 as the number of data points used in its calculation increases. In practice, this means that the estimates obtained get closer and closer to the true value of the parameter we are estimating as we use more and more data.

Problem 13: Variance for $n = 3$

Let's go back to our estimator $\hat{\Theta} = \max\{X_1, X_2, \ldots, X_n\}$ for $n = 3$ (i.e., only 3 data points from the generator). What is the variance of the estimator? Assume we have already calculated that

$$E\left[\hat{\Theta}^2\right] = \frac{n}{n+2} \cdot \alpha^2.$$

$$Var\left[\hat{\Theta}\right] =$$

Problem 14: The mean square error

What is the MSE for the estimator $\hat{\Theta}$ for $n = 3$? How about for $n = 10$? When $n \to \infty$?

- For $n = 3$: $MSE = bias^2\left[\hat{\Theta}\right] + Var\left[\hat{\Theta}\right] =$
- For $n = 10$:
  $MSE =$

As $n \to \infty$:
$\lim_{n \to \infty} MSE =$

Activity 5: Comparing point estimators

Assume that a population is distributed with pdf $f(x) = \theta\left(x - \frac{1}{2}\right) + 1,\ 0 \leq x \leq 1$, where θ is an unknown parameter. We have also been able to collect a sample of n observations and have calculated their sample average as $\overline{X}$. We have been trying estimators for θ and we want to compare three of them and pick the best:

1. $\hat{\Theta}_1 = 2\overline{X} - 1$
2. $\hat{\Theta}_2 = 12\overline{X} - 6$

Problem 15: MSE

Which one of the two has the smallest mean square error?

Problem 16: Application

For the previous population from Problem 6, we have collected a sample of 6 items and found them equal to $X_1 = 0.8, X_2 = 0.83, X_3 = 0.95, X_4 = 0.72, X_5 = 0.85, X_6 = 0.65$. What is a good estimate for θ? Use the point estimator that provided the best MSE from Problem 15.

Activity 6: Point estimators for the exponential distribution

Assume that a population X is exponentially distributed; however, we have no idea what λ is. You recalled one thing though from IE 300: **the expected time between events is equal to** $\frac{1}{\lambda}$. This gives you an idea. You will wait to observe the times between some events and use them to estimate λ. More specifically, you will estimate λ as 1 over the average time between events.

For example, if the sampled times between two consecutive events are equal to 3 minutes, 2 minutes, 5 minutes, 4 minutes, 2 minutes, 2 minutes, you will estimate λ as 1 over 3 minutes!

Problem 17: Biased or not?

Is this estimate for λ biased or not? In mathematical terms, is $\hat{\lambda} = \frac{1}{\overline{X}}$, where $\overline{X}$ is the average of n observations, biased?

Problem 18: MSE

What is the mean square error for the estimator $\hat{\lambda} = \dfrac{1}{\overline{X}}$ for the unknown rate of an exponential distribution after observing n data points?

12. Methods of point estimation

Theory

Chapter objectives

Learning outcomes

After this chapter, you will be able to:

- Find point estimators for unknown parameters.
- Use the method of moments to find point estimators for unknown parameters.
- Use maximum likelihood estimation to find point estimators for unknown parameters.
- Use Bayesian estimation to find point estimators for unknown parameters.
- Compare and identify when it is easiest to use likelihood and when log-likelihood.
- Propose new point estimators for unknown parameters based on these methods.
- Propose new point estimators for unknown parameters based on prior information.
- Estimate the unknown rate of an exponential

distribution, or the unknown success probability of a Bernoulli distribution using these methods.

Chapter questions

1. What does the method of moments state?
2. What is the k-th moment of a population X?
3. What is the k-th moment of a sample $X_1, X_2, \ldots, X_n$?
4. How do I obtain estimates for the unknown parameters solving a system of equations?

Motivation

"I guess it is exponentially distributed. But what is λ?"

Often in practice, we can determine the type of distribution from the context (exponential for waiting times, normal for measurements, Poisson for counts), but we still need to estimate the specific parameters. How do we find these parameters from our data? Say we have observed some arrivals for an exponentially distributed random variables, and we have captured these times: how do we use that sample to estimate the true rate of arrivals λ?

Estimating the mortality risk

We call mortality risk of a hospital the probability of death occurring for any patient admitted to the hospital. The question we need to answer is "what is the mortality risk" of a given hospital? It depends on many factors, such as the type of conditions the patients admitted in this hospital have; the equipment of the hospital; the personnel of the hospital; among many, many others. Our intuition says the following, though: could we not *observe* the hospital for a period of time and then deduce what the risk is based on the obtained data? Is this fair/unfair/correct/misleading? What is the number of deaths in a hospital truly distributed as?

Motivation: Heads or Tails?

We flip a coin 10 times and we get 6 Heads and 4 Tails. Do you believe it is a fair coin? What does the method of moments and the maximum likelihood estimation method say about this situation?

Both methods would estimate $p = \dfrac{6}{10} = 0.6$. But intuitively, we might think "it's probably close to fair, just random variation." How can we incorporate our prior belief that coins are usually fair?

Estimation

In the previous chapter, we saw what makes a good point estimator $\hat{\Theta}$. We would like for it to have:

- small **bias** (zero, if possible). $bias = E\left[\hat{\Theta}\right] - \theta$.
- small **variance** (minimum among all estimators). $Var\left[\hat{\Theta}\right]$.
- small **mean square error**. $MSE = bias^2 + Var\left[\hat{\Theta}\right]$.
- We also defined the **relative efficiency** of two estimators

$$\hat{\Theta}_1, \hat{\Theta}_2 \text{ as } \frac{MSE\left(\hat{\Theta}_1\right)}{MSE\left(\hat{\Theta}_2\right)}.$$

So, *given* two or more estimators, you may calculate these items and infer which one to use/which one is better. However, where do these estimators come from? When faced with the problem of recognizing a parameter based on data, what can we do? In the rest of the chapter, we will work on deriving, using, and comparing three methods of point estimation:

1. **Method of moments** estimators.
2. **Maximum likelihood** estimators.
3. **Bayesian** estimators.

Before we get to their details, we provide a definition and a motivating example.

Definitions and theorems 12.1. Method of estimation

Assume we are provided a population X distributed with unknown parameter(s) θ. We want to estimate θ. Given a series of observations (sample) $X_1, X_,2 \ldots, X$, how to come up with a "good" point estimator $\hat{\Theta}$?

Example 12.1. Mortality risk

Let us go back to our original motivating example with calculating/estimating the mortality risk of a hospital based on observations. Say, we have been observing the hospital over the last 2 months, and we have observed 18 deaths in the first 150 patient admissions. What would we estimate the mortality rate as?

Some more examples we may consider follow:

- How to estimate the rate of an exponential distribution?
- How to estimate the probability of success in a Bernoulli distribution?
- How to estimate the mean of a normal distribution?

All of these are "point estimation problems." This is because their goal is to identify a good (single or "point") value for the missing parameter.

Method of Moments

The method of moments is a leading methodology for finding good estimators for unknown parameters. As the name reveals, it focuses on estimation through the use of **moments**.

Idea behind the method of moments

Assume we have a population X distributed with some unknown parameter(s) θ. We can obtain a sample of n observations $X_1, X_2, \ldots, X_n$. The method of moments says to:

1. Calculate all necessary population moments: $E\left[X^k\right]$ for $k = 1, 2, 3, \ldots$ (as many as needed).
2. Calculate the corresponding sample moments: $\frac{1}{n}\sum_{i=1}^{n} X_i^k$ for $k = 1, 2, 3, \ldots$.
3. Equate the two and solve for θ.

Definitions and theorems 12.2. Population and sample moments

The k-th **population moment** of a random variable X is defined as: $E\left[X^k\right]$.

The k-th **sample moment** of a sample $X_1, X_2, \ldots, X_n$ is defined as: $\frac{1}{n}\sum_{i=1}^{n} X_i^k$.

Note that the first sample moment is nothing more than the sample average $\overline{X}$! A common mistake for students using moments for the first time is that they assume that the second sample moment corresponds to the sample variance, but this is incorrect. In reality, the second sample moment is the sum of squares of the individual samples divided by the sample size: $\frac{1}{n}\sum_{i=1}^{n} X_i^2$.

Example 12.2. Our first population moments

Assume $f(x) = \frac{1}{2}(1 - \alpha \cdot x)$ defined for $-1 \leq x \leq +1$, where α is some parameter. What are the first three population moments?

- first population moment:
$$E[X] = \int_{-1}^{+1} x \cdot f(x)\,dx = \int_{-1}^{+1} x \cdot \frac{1}{2}(1 - \alpha \cdot x)\,dx = -\frac{\alpha}{3}.$$
- second population moment:
$$E\left[X^2\right] = \int_{-1}^{+1} x^2 \cdot f(x)\,dx = \int_{-1}^{+1} x^2 \cdot \frac{1}{2}(1 - \alpha \cdot x)\,dx = \frac{1}{3}$$
- third population moment:
$$E\left[X^3\right] = \int_{-1}^{+1} x^3 \cdot f(x)\,dx = \int_{-1}^{+1} x^3 \cdot \frac{1}{2}(1 - \alpha \cdot x)\,dx = -\frac{\alpha}{5}$$

We observe that most moments are a function of the unknown parameter, but not always! Some (like the second population moment in Example 12.2) can be a constant number (even possibly zero).

Before we proceed to apply the method of moments, broadly defined as equating the corresponding population and sample moments, let us turn to one more example for the sample moments now.

Example 12.3. Our first sample moments

Assume $f(x) = \frac{1}{2}(1 - \alpha \cdot x)$ defined for $-1 \leq x \leq +1$, where α is some parameter. We have been observing this random variable, and we have collected a sample of $n = 6$ observations: $X_1 = 0.1, X_2 = 0.3, X_3 = -0.05, X_4 = 0.33, X_5 = 0.12, X_6 = 0.4$. What are the first three sample moments?

- first sample moment: $\frac{1}{n}\sum_{i=1}^{6} X_i = 0.2$.
- second sample moment: $\frac{1}{n}\sum_{i=1}^{6} X_i^2 = 0.0643$

.

- third sample moment: $\frac{1}{n}\sum_{i=1}^{6} X_i^3 = 0.02159$.

We are ready to formally define the process behind the method of moments.

Definitions and theorems 12.3. The method of moments

The **method of moments** estimates parameters by equating population moments with sample moments. Formally, its steps are as follows:

1. Express the population moments ($E[X], E[X^2]$, …) in terms of the unknown parameters.
2. Calculate the corresponding sample moments from your data.
3. Set population moments equal to sample moments and solve for the parameters.

For k unknown parameters, we *typically* use the first k moments:

- First moment: $E[X] = \frac{1}{n}\sum_{i=1}^{n} X_i$
- Second moment: $E[X^2] = \frac{1}{n}\sum_{i=1}^{n} X_i^2$
- …

An *implementation* note: as we saw earlier some population moments may be constants (not a function of the parameters) or may even disappear. If that is the case, we skip that equality and proceed to the next one. We stop when we have enough equations to solve the system. As a general rule, if you have k unknown parameters you are trying to estimate you will need k equations.

Example 12.4. Our first method of moments estimator

Assume we have a continuous random variable X over $[0,1]$ and distributed with pdf $f(x) = \frac{1}{2}(1 - \alpha \cdot x)$, where α is some unknown parameter.

We have collected a sample of $n = 5$ observations: $X_1 = 0.7, X_2 = 0.77, X_3 = 0.65, X_4 = 0.5, X_5 = 0.83$. Based on this, what is the method of moments estimator for α?

We only have one unknown parameter, so we will need only one equation: the first moment. Let us calculate both moments and then equate them:

Population 1st moment:

$$E[X] = \int_0^1 x \cdot \frac{1}{2}(1 - \alpha \cdot x)dx = \frac{1}{2}\int_0^1 (x - \alpha x^2)dx = \frac{1}{2}\left[\frac{x^2}{2} - \frac{\alpha x^3}{3}\right]_0^1 = \frac{1}{2}\left(\frac{1}{2} - \frac{\alpha}{3}\right) = \frac{1}{4} - \frac{\alpha}{6}$$

.

Sample 1st moment:

$$\frac{1}{5}\sum_{i=1}^{5} X_i = \frac{1}{5}(0.7 + 0.77 + 0.65 + 0.5 + 0.83) = 0.69$$

.

Equating the two:

$$\frac{1}{4} - \frac{\alpha}{6} = 0.69 \implies \frac{\alpha}{6} = 0.25 - 0.69 = -0.44 \implies \alpha = -2.64$$

.

The method of moments estimator for an exponential distribution

Assume we have obtained a sample of n observations $X_1, X_2, \ldots, X_n$ with average

$\overline{X} = \dfrac{X_1 + X_2 + \ldots + X_n}{n}$. We also assume that the population is exponentially distributed with rate λ. What is the method of moments estimator for λ?

First, recognize that for an exponential distribution with rate λ we have $E[X] = \dfrac{1}{\lambda}$. Therefore, the first population moment is $E[X] = \dfrac{1}{\lambda}$.

The first sample moment is the sample average $\overline{X}$.

Equating the two: $\dfrac{1}{\lambda} = \overline{X} \implies \lambda = \dfrac{1}{\overline{X}}$.

Hence, the method of moments estimator for λ is $\hat{\lambda} = \dfrac{1}{\overline{X}}$.

The method of moments estimator for a Bernoulli distribution

Once again, consider that we have a population producing random variables distributed as Bernoulli with probability of success p. We have obtained a sample of $n = 10$ observations with 7 successes (let them be $X_i = 1$) and 3 failures ($X_i = 0$). What is the method of moments estimator for the unknown p?

Recall that $E[X] = p$ for a Bernoulli random variable. Hence:

Population moment: $E[X] = p$.

Sample moment:

$$\overline{X} = \frac{1}{10}\sum_{i=1}^{10} X_i = \frac{1}{10}(1+1+1+1+1+1+1+0+0+0) = \frac{7}{10} = 0.7$$

.

Equate: $p = \overline{X} = 0.7$.

Hence, the method of moments estimator for the unknown p is $\hat{p} = 0.7$.

Maximum Likelihood Estimation (MLE)

Another very popular estimation technique is the maximum likelihood estimation, more commonly known as MLE. It relies on the use of likelihood functions that are then optimized to find the most likely set of parameters that led to the sample being obtained. Let us study it together next.

Idea behind maximum likelihood estimation

In the maximum likelihood estimation, the idea is simple: pick that value of parameter θ for which the obtained data $X_1, X_2, \ldots, X_n$ is **most likely to have been observed**.

Definitions and theorems 12.4. Likelihood function

The **likelihood function** of a sample of n observations $X_1, X_2, \ldots, X_n$ is defined as $L(\theta) = f(X_1, \theta) \cdot f(X_2, \theta) \cdot \ldots \cdot f(X_n, \theta) = \prod_{i=1}^{n} f(X_i, \theta)$. Observe how the likelihood function is only a function of θ as $X_i, i = 1, \ldots, n$ are known quantities.

So, the maximum likelihood estimator is the value $\hat{\theta}$ that maximizes the likelihood function $L(\theta)$.

Definitions and Theorems 12.5. Maximum Likelihood Estimation

The **maximum likelihood estimator** (MLE) is the value of the parameter that makes the observed data most likely, in the sense that it **maximizes the likelihood function**.

Its main steps are as follows:

1. Derive the **likelihood function**

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta)$$

   - Sometimes it may be easier: derive the **log-likelihood**

$$\ell(\theta) = \ln L(\theta) = \sum_{i=1}^{n} \ln f(x_i; \theta)$$

2. Take the derivative with respect to θ.
3. Set the derivative equal to zero and solve for θ.
4. Verify it's a maximum (second derivative test or check endpoints).

To find that, we can take the derivative of the likelihood function with respect to θ, equate it to 0, and solve for θ, when possible. That is: $\dfrac{\partial L(\theta)}{\partial \theta} = 0$. If easier, we may opt to plot the likelihood function and "see" where it is maximized. More on these approaches next, as well as in the Activities section later in the chapter.

Example 12.5. Our first MLE estimator

We have:

- pdf $f(x) = \frac{1}{2}(1 - \alpha \cdot x)$;
- sample $X_1 = 0.7, X_2 = 0.77, X_3 = 0.65, X_4 = 0.5, X_5 = 0.83$.

We build the likelihood function as:

$$L(\alpha) = f(X_1) \cdot f(X_2) \cdot f(X_3) \cdot f(X_4) \cdot f(X_5)$$
$$= \frac{1}{2}(1 - 0.7\alpha) \cdot \frac{1}{2}(1 - 0.77\alpha) \cdot \frac{1}{2}(1 - 0.65\alpha) \cdot \frac{1}{2}(1 - 0.5\alpha) \cdot \frac{1}{2}(1 - 0.83\alpha)$$
$$= \frac{1}{32}(1 - 0.7\alpha)(1 - 0.77\alpha)(1 - 0.65\alpha)(1 - 0.5\alpha)(1 - 0.83\alpha)$$

Taking the derivative and equating to 0 leads to $\alpha = 1.88$. This can also be seen from a plot of the likelihood function.

Finally, observe how we got a different estimator here compared to the method of moments! Is this always the case?

Extension to log-likelihood

Since the likelihood involves a product of n pdf values, it comes as no surprise that our end result may be a little difficult to control and use. This is why we may also introduce the **log-likelihood**:

Definitions and theorems 12.6. Log-likelihood function

The **log-likelihood function** of a sample of n observations $X_1, X_2, \ldots, X_n$ is defined as

$$\ln L(\theta) = \ln f(X_1, \theta) + \ln f(X_2, \theta) + \ldots + \ln f(X_n, \theta) = \sum_{i=1}^{n} \ln f(X_i, \theta)$$

.

Observe how also the log-likelihood function is only a function of θ as $X_i, i = 1, \ldots, n$ are known quantities. Contrary to the simple likelihood function, the log-likelihood is a summation which makes it easier to differentiate. In summary, why do we prefer to use the log-likelihood?

- Products become sums (often easier to differentiate).
- Exponentials simplify.
- Same maximizer as likelihood, because the logarithm is monotonic.

Example 12.6. Our first MLE estimator using log-likelihood

We have:

- pdf $f(x) = \frac{1}{2}(1 - \alpha \cdot x)$;
- sample

$$X_1 = 0.7, X_2 = 0.77, X_3 = 0.65, X_4 = 0.5, X_5 = 0.83$$

.

We build the log-likelihood function as:

$$\ln L(\alpha) = \ln f(X_1) + \ln f(X_2) + \ln f(X_3) + \ln f(X_4) + \ln f(X_5) =$$
$$= \ln \frac{1}{2}(1 - 0.7\alpha) + \ln \frac{1}{2}(1 - 0.77\alpha) +$$
$$+ \ln \frac{1}{2}(1 - 0.65\alpha) + \ln \frac{1}{2}(1 - 0.5\alpha) + \ln \frac{1}{2}(1 - 0.83\alpha).$$

Here, we note that $\left(\frac{1}{2}(1 - X_i\alpha)\right)' = \frac{X_i}{\alpha X_i - 1}$. Hence, in our case we have:

$$\frac{\partial \ln L}{\partial \alpha} = 0 \implies$$
$$\implies \frac{0.7}{1 - 0.7\alpha} + \frac{0.77}{1 - 0.77\alpha} + \frac{0.65}{1 - 0.65\alpha} + \frac{0.5}{1 - 0.5\alpha} + \frac{0.83}{1 - 0.83\alpha} = 0 \implies$$
$$\implies \alpha = 1.88.$$

The key takeaway from this is that the result obtained using the likelihood or the log-likelihood function will be the same.

The MLE estimator for an exponential distribution

Assume we have obtained a sample of n observations $X_1, X_2, \ldots, X_n$ with average $\overline{X} = \frac{X_1 + X_2 + \ldots + X_n}{n}$. We also assume that the population is exponentially distributed with rate λ. What is the MLE estimator for λ?

First, we build the log-likelihood function as:

$$\ln L(\lambda) = \ln \lambda e^{-\lambda X_1} + \ln \lambda e^{-\lambda X_2} + \ldots + \ln \lambda e^{-\lambda X_n}$$
$$= \ln \lambda - \lambda X_1 + \ln \lambda - \lambda X_2 + \ldots + \ln \lambda - \lambda X_n = n \ln \lambda - \lambda (X_1 + X_2 + \ldots + X_n)$$

.

Again, find the maximizer:

$$\frac{\partial \ln L(\lambda)}{\partial \lambda} = 0 \implies (n \ln \lambda - \lambda (X_1 + X_2 + \ldots + X_n))' = 0 \implies \frac{n}{\lambda} - (X_1 + X_2 + \ldots + X_n) = 0 \implies n - \lambda \sum_{i=1}^{n} X_i = 0 \implies \lambda = \frac{n}{\sum_{i=1}^{n} X_i} = \frac{1}{\overline{X}}$$

.

12.1. Food for thought

Observe how we have reached the same result as when using the method of moments. Recall that this is **not necessarily** always the case.

Say we had not wanted to use the log-likelihood and instead used the simple likelihood function $L(\lambda)$: $L(\lambda) = \lambda e^{-\lambda X_1} \cdot \lambda e^{-\lambda X_2} \cdot \ldots \cdot \lambda e^{-\lambda X_n} = \lambda^n \cdot e^{-\lambda (X_1 + X_2 + \ldots + X_n)} = \lambda^n \cdot e^{-\lambda \sum_{i=1}^{n} X_i}$. Take the derivative: $\frac{\partial L(\lambda)}{\partial \lambda} = 0 \implies \left(\lambda^n \cdot e^{-\lambda \sum_{i=1}^{n} X_i}\right)' = 0 \implies n \cdot \lambda^{n-1} \cdot e^{-\lambda \sum_{i=1}^{n} X_i} - \lambda^n \cdot \sum_{i=1}^{n} X_i \cdot e^{-\lambda \sum_{i=1}^{n} X_i} = 0$. Observe how we can simplify quite a bit: we may divide by λ^{n-1} (because we know that $\lambda > 0$). This gives us: $n \cdot e^{-\lambda \sum_{i=1}^{n} X_i} - \lambda \cdot \sum_{i=1}^{n} X_i \cdot e^{-\lambda \sum_{i=1}^{n} X_i} = 0$. We may also divide by $e^{-\lambda \cdot \sum_{i=1}^{n} X_i}$ because it is also definitely positive. This leads to the much more manageable: $n - \lambda \cdot \sum_{i=1}^{n} X_i = 0 \implies \lambda = \frac{n}{\sum_{i=1}^{n} X_i} = \frac{1}{\overline{X}}$.

We now present a very important note for the likelihood and the log-likelihood functions:

> *12.2. Food for thought*
>
> Note how getting the same result with the log-likelihood was significantly easier due to the nature of this probability density function.

The MLE estimator for a Bernoulli distribution

Once again, consider that we have a population producing random variables distributed as Bernoulli with probability of success p. We have obtained a sample of $n = 10$ observations with 7 successes (let them be $X_i = 1$) and 3 failures ($X_i = 0$). What is the MLE estimator for the unknown p?

Based on the MLE method we first need to calculate the likelihood function. Recall that for a Bernoulli random variable its probability mass function (as it is a discrete random variable) is $P(0) = 1 - p$ and $P(1) = p$. Without loss of generality, assume we arrange the observations with the successes first (the first, say, 7 observations) and the failures next (the remaining $10 - 7 = 3$ observations).

We are now ready to build the likelihood function: $L(p) = \left(\prod_{i=1}^{7} p\right) \cdot \left(\prod_{i=8}^{10} (1-p)\right) = p^7 \cdot (1-p)^{10-7} = p^7 \cdot (1-p)^3$. The derivative of the likelihood function can be found as: $\frac{\partial L}{\partial p} = \left(p^7 \cdot (1-p)^3\right)' = 7p^6(1-p)^3 - 3(1-p)^2 p^7$.

Now, equate this to 0 to get the maximizer: $7p^6(1-p)^3 - 3(1-p)^2p^7 = 0 \implies 7\,(1-p) - 3p = 0 \implies \hat{p} = 0.7$.

In the above, we make the assumption that $p \in (0,1)$: that is, it cannot be 0 or 1. If we allowed this to be the case, then solving would give three solutions $p=0, p=1, p=0.7$. However, the first two solutions are *minima* rather than *maxima*, and we would still pick $\hat{p}=0.7$ as our estimator.

More examples

Example 12.7. A delivery problem

We believe the times it takes to deliver a package are identically distributed with the same unknown mean μ and variance σ^2. We have collected information on 10 packages and the time to delivery (in hours) are: 49.1, 47.9, 48.6, 50.4, 49.5, 49.8, 48.2, 50.3, 45.2, 46.2. What are good mean and variance estimators for the normal distribution using the method of moments?

We have two unknown parameters (mean and variance), so we will need at least two population and sample moments. Let us take the first two:

Population 1st moment: $E\left[X^1\right]=E[X]=\mu$.

Population 2nd moment:

$$E\left[X^2\right]=Var[X]+(E[X])^2=\sigma^2+\mu^2.$$

Sample 1st moment: $\frac{1}{10}\sum_{i=1}^{10} X_i^1=48.52$.

Sample 2nd moment: $\dfrac{1}{10}\sum_{i=1}^{10} X_i^2 = 2356.844$.

Equating the two and solving the system of equations, we get $\hat{\mu} = 48.52$ and $\hat{\sigma}^2 = 2.6536$.

Example 12.8. A discrete distribution

Assume we have a discrete random variable X defined over $0, 1, 2, 3, 4$ and distributed with probabilities $p(0) = \dfrac{\theta_1}{3}, p(1) = \dfrac{\theta_1}{6}, p(2) = \dfrac{\theta_1}{6}, p(3) = \dfrac{\theta_2}{2}, p(4) = \dfrac{\theta_2}{2}$.

Now, assume we have collected a sample of $n = 10$ observations: $0, 1, 1, 3, 4, 2, 2, 3, 4, 1$. Based on this, what is the method of moments estimators for θ_1 and for θ_2?

Now, let's see. At first glance we have two estimators.. But we know better than that. We probably remember that $\sum_{x=0}^{4} p(x) = 1$, which implies that:

$$\sum_{x=0}^{4} p(x) = 1 \implies \frac{\theta_1}{3} + \frac{\theta_1}{6} + \frac{\theta_1}{6} + \frac{\theta_2}{2} + \frac{\theta_2}{2} = 1 \implies \frac{2\theta_1}{3} + \theta_2 = 1$$

. Based on that, if we knew, say θ_1 we could obtain θ_2 right away. Let us get the first moments and equate them:

$$E[X] = 0 \cdot \frac{\theta_1}{3} + 1 \cdot \frac{\theta_1}{6} + 2 \cdot \frac{\theta_1}{6} + 3 \cdot \frac{\theta_2}{2} + 4 \cdot \frac{\theta_2}{2} = \frac{\theta_1 + 7\theta_2}{2}$$

.

$$\frac{1}{n} \sum_{i=1}^{10} X_i = \frac{1}{10}(0+1+1+3+4+2+2+3+4+1) = 2.1$$

.

Finally, we have a system of equations at our hands!

System:

$$\frac{2\theta_1}{3} + \theta_2 = 1$$

$$\frac{\theta_1 + 7\theta_2}{2} = 2.1$$

Solution:

$$\hat{\theta}_1 = \frac{42}{55}$$

$$\hat{\theta}_2 = \frac{27}{55}$$

Comparing methods

Let us do a quick comparison of the two methods we have seen so far: we also present some of the nice properties the two methods have.

The **method of moments** is simple, easy to calculate, and will always work. **Maximum likelihood estimation** is generally more efficient, and uses the data we have (the sample we have collected) optimally.

That said, both methods will provide us with good estimators: additionally, often the two estimation methods will provide us with different parameter estimates.

Bayesian Estimation

During this chapter, we have discussed two methods to identify "good" estimators $\hat{\Theta}$ for a series of unknown parameters:

- **the method of moments**.

    1. Compute the moments of the population, calculated as $E\left[X^k\right]$.
    2. Compute the moments of the sample (empirical moments), calculated as $\frac{1}{n}\sum_{i=1}^{n} X_i^k$.
    3. Equate the two and solve a system of equations for the unknown parameters.

- **maximum likelihood estimation**.

    1. Calculate the likelihood function as $L\left(\theta\right) = f(X_1, \theta) \cdot f(X_2, \theta) \cdot \ldots \cdot f(X_n, \theta)$.
    2. Or the log-likelihood function as $\ln\left(L\left(\theta\right)\right) = \ln\left(f(X_1, \theta)\right) + \ln\left(f(X_2, \theta)\right) + \ldots + \ln\left(f(X_n, \theta)\right)$.
    3. Find the maximizer (usually found by setting the derivative per each unknown parameter equal to 0).

Both these methods have one thing in common: they require no prior information to work, but instead they base all of their observations on the obtained sample. What if I already know some more information about what is going on?

Bayesian estimation through an example

We begin in a slightly different way than usually. We begin with an example to help us build intuition! Assume I carry 3 coins with me:

1. One with both sides showing Heads.
2. One with both sides showing Tails.
3. One that is fair and has a side of Heads and a side of Tails.

Assume I randomly pick one coin and start flipping it. I report to you the number of tries (n) and the number of Heads (x). For example, I may tell you $n = 8, x = 5$ or $n = 2, x = 0$, and so on.

Example 12.9. Flipping the coin: first take

I let you know that I flipped the coin three times and got Heads both times: $n = 3, x = 2$. What are the method of moments and the maximum likelihood estimators for p?

We will have $E[X] = p$ and $\frac{1}{3} \cdot (1 + 1 + 0) = \frac{2}{3}$, and equating will give $\hat{p} = \frac{2}{3}$. The likelihood function is $L(p) = p^2 \cdot (1 - p)$, and maximizing will also give $\hat{p} = \frac{2}{3}$.

But… I carry three coins with me. Shouldn't I use this information somehow?

1. Can it be my "2-Heads" coin?
2. Can it be my "2-Tails" coin?
3. Does it have to be my "50-50" coin?

This is the key to realizing what Bayesian estimation brings to the table: extra information in the form of prior probabilities for the parameters that are unknown.

Bayesian estimation

We separate the discussion between discrete sets for the values the parameter can take (like in the previous example where I carried 3 distinct coins with me) and between continuous sets, where the parameter can be any real number in a range of values.

For discrete parameter values

Before describing the method, we provide some notation:

- **prior probabilities** ("priors"): the probability of seeing a certain parameter $P\left(\theta\right)$.
- **likelihood probabilities** ("likelihoods"): the likelihood of seeing an outcome *given* a certain parameter $P\left(X|\theta\right)$.
- **posterior probabilities** ("posteriors"): the multiplication of the two $P(\theta) \cdot P\left(X|\theta\right)$.

12.3. Food for thought

A quick note about the likelihoods: those are calculated in identical manner as the likelihood function in the maximum likelihood estimation method!

The Bayesian estimation method then states that: “the higher the posterior probability, the better the chance of having that parameter!”

This is it! This is the whole method!

Definitions and theorems 12.7. Bayesian estimation

In Bayesian estimation, we combine two sources of information:

1. **Prior distribution** $\pi(\theta)$: Our belief about θ before seeing data
2. **Likelihood** $L(\theta|x_1, \ldots, x_n)$: Information from the observed data

We use Bayes' theorem to get the **posterior distribution** $\pi(\theta|x_1, \ldots, x_n)$:

$$\pi(\theta|x_1, \ldots, x_n) = \frac{L(\theta|x_1, \ldots, x_n) \cdot \pi(\theta)}{\int L(\theta|x_1, \ldots, x_n) \cdot \pi(\theta) d\theta}$$

In other words: **Posterior** $\propto$ **Likelihood** × **Prior**

Example 12.10. Flipping the coin: second take

Let us go back to the example where I carried three coins ("2-Heads"; "2-Tails"; and "50-50") and I picked one at random. After 3 tries, we got 2 Heads: $n = 3, x = 2$. Let's see what we have for these three distinct cases of $p = 1, p = 0, p = 0.5$:

- priors $P(p)$: probability of picking a certain coin, that is $P(p = 1) = P(p = 0) = P(p = 0.5) = \frac{1}{3}$.
- likelihoods $P(X = 2|p)$: likelihood function of seeing two Heads for each coin. For example, the likelihood function for the $p = 0.5$ coin with $x = 2$ Heads in $n = 3$ tries would be: $p^2 \cdot (1 - p) = 0.5^2 \cdot 0.5 = 0.125$.
- posteriors $P(p) \cdot P(X = 2|p)$: we will need to calculate this for each coin.

Let us put this in table format.

Table 12.1. The possible parameters values ($p = 0, 0.5, 1$), their prior probabilities, their likelihood, and their poster probabilities.

parameter	prior	likelihood	posterior
p	$P(p)$	$P(X=2\|p)$	$P(p) \cdot P(X=2\|p)$
0	$\frac{1}{3}$	$0^2 \cdot 1^1 = 0$	$\frac{1}{3} \cdot 0 = 0$
1	$\frac{1}{3}$	$1^2 \cdot 0^1 = 0$	$\frac{1}{3} \cdot 0 = 0$
0.5	$\frac{1}{3}$	$0.5^2 \cdot 0.5^1 = 0.125$	$\frac{1}{3} \cdot 0.125 = 0.0416\overline{6}$

The maximum value (and only non-zero probability!) is achieved for the “50-50” coin so it must be this!

See? It is pretty intuitive. Of course, we may complicate things by making the probability of picking a coin a little more general.

Example 12.11. Flipping the coin: third take

I still have three types of coins on me. But, given that I am an adult that carries money wherever I go, I carry more actual coins (“50-50”) than novelty coins (“2-Heads”, “2-Tails”). More specifically, I carry 6 real coins and 1 of each novelty coin. I take a coin out and toss it twice and get two Heads! Which coin is it?

Let us try the table format again:

Table 12.2. The possible parameters values ($p = 0, 0.5, 1$), their prior probabilities, their likelihood, and their poster probabilities.

parameter	prior	likelihood	posterior
p	$P(p)$	$P(X=2\|p)$	$P(p) \cdot P(X=2\|p)$
0	$\frac{1}{8}$	$0^2 = 0$	$\frac{1}{8} \cdot 0 = 0$
1	$\frac{1}{8}$	$1^2 = 1$	$\frac{1}{8} \cdot 1 = 0.125$
0.5	$\frac{3}{4}$	$0.5^2 = 0.25$	$\frac{3}{4} \cdot 0.25 = 0.1875$

The maximum value is still achieved for a “50-50” coin, so we are inclined to think we picked one. Note how much closer the posteriors are, though..

12.4. Food for thought

It would actually take one more Heads to change our parameter estimation towards the “2-Heads” novelty coin! Why is that?

Normalizing may also be useful. Instead of looking at the posterior values as they are in the end, we may turn them into actual probability values to help compare them. To normalize simply take each posterior and divide it by the summation of all posterior probabilities. For example, in our third take (see above) we would end up with probabilities:

- $P(p=0) = \dfrac{0}{0+0.125+0.1875} = 0$.
- $P(p=1) = \dfrac{0.125}{0+0.125+0.1875} = 0.4$.
- $P(p=0.5) = \dfrac{0.1875}{0+0.125+0.1875} = 0.6$.

This helps us quantify our parameter estimation even more. There is a 40% chance we picked the “2-Heads” coin and a 60% chance we picked one of the “50-50” coins.

Let’s work one more example before we move to the continuous case.

Example 12.12. A computer vision system

A machine learning algorithm for computer vision is trained to observe the first vehicle that passes from an intersection at or after 8am every day. Then, it reports the time from that vehicle to the next one again and again. We assume this time is exponentially distributed but with unknown λ.

- If the first vehicle of the day was a personal car, then $\lambda_1 = 1$ per minute.
- If the first vehicle of the day was a motorcycle, then $\lambda_2 = 1$ per 5 minutes.
- If the first vehicle was a truck, then $\lambda_3 = 1$ per 10 minutes.
- If the first vehicle was a bike, then $\lambda_4 = 1$ per 12 minutes.

We observe a sample of 5 times: $X_1 = 9$ minutes, $X_2 = 8.5$ minutes, $X_3 = 8$ minutes, $X_4 = 10.5$ minutes. What is the probability of each parameter λ?

First, we need to calculate the prior probabilities $P(\lambda)$. The first vehicle of the day is:

- a personal car with probability $\dfrac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4} = 0.732$ (why?),
- a motorcycle with probability $\dfrac{\lambda_2}{\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4} = 0.146$,

- a truck with probability $\dfrac{\lambda_3}{\lambda_1+\lambda_2+\lambda_3+\lambda_4} = 0.073$,
- or a bike with probability $\dfrac{\lambda_4}{\lambda_1+\lambda_2+\lambda_3+\lambda_4} = 0.049$.

With these in hand, we calculate the likelihood functions as being $\lambda \cdot e^{-\lambda \cdot X_1} \cdot \lambda \cdot e^{-\lambda \cdot X_2} \cdot \ldots \cdot \lambda \cdot e^{-\lambda \cdot X_n}$ since we have an exponentially distributed random variable. For example, if $\lambda = \lambda_1 = 1$, then we would have $1 \cdot e^{-1 \cdot 9} \cdot 1 \cdot e^{-1 \cdot 8.5} \cdot 1 \cdot e^{-1 \cdot 8} \cdot 1 \cdot e^{-1 \cdot 10.5} = e^{-36} = 2.32 \cdot 10^{-16}$. Finally:

Table 12.3. The possible parameters values ($\lambda = 0.06, 0.1, 0.2, 1$), their prior probabilities, their likelihood, and their poster probabilities.

parameter	prior	likelihood	posterior
λ	$P(\lambda)$	$P(X_1, X_2, X_3, X_4 \vert \lambda)$	$P(\lambda) \cdot P(X_1, X_2, X_3, X_4 \vert \lambda)$
$\lambda_1 = 1$	0.732	$2.32 \cdot 10^{-16}$	$1.70 \cdot 10^{-16}$
$\lambda_2 = 0.2$	0.146	$1.19 \cdot 10^{-6}$	$1.74 \cdot 10^{-7}$
$\lambda_3 = 0.1$	0.073	$2.73 \cdot 10^{-6}$	$1.99 \cdot 10^{-7}$
$\lambda_4 = 0.06\overline{6}$	0.049	$1.79 \cdot 10^{-6}$	$8.77 \cdot 10^{-8}$

From the results it seems that the vehicle that first passed today is more likely a truck!

If we wanted to assign probability values to each type of vehicle, we would report:

- personal car: $\frac{6.24 \cdot 10^{-17}}{6.24 \cdot 10^{-17} + 1.43 \cdot 10^{-7} + 1.81 \cdot 10^{-7} + 8.18 \cdot 10^{-8}} = 1.54 \cdot 10^{-10} \approx 0$.
- motorcycle: 0.3527.
- truck: 0.4458.
- bike: 0.2015.

We can now move to the continuous case.

For continuous parameter values

Let us again begin with an example. The method is largely still the same; but the definitions of some of the items change slightly to accommodate the continuous nature of the unknown parameter(s).

Say we have a coin that is made with the goal of being fair; that is, "50-50". But, materials fail and get deposited more on one side than the other resulting in different compositions for the probability of Heads and Tails. Say, in the end, the probability of Heads is normally distributed with $\mathcal{N}\left(0.5, 0.01\right)$, that is a mean of $\mu = 0.5$ and a variance $\sigma^2 = 0.01 \implies \sigma = 0.1$. Visually, we would get the distribution of the next figure.

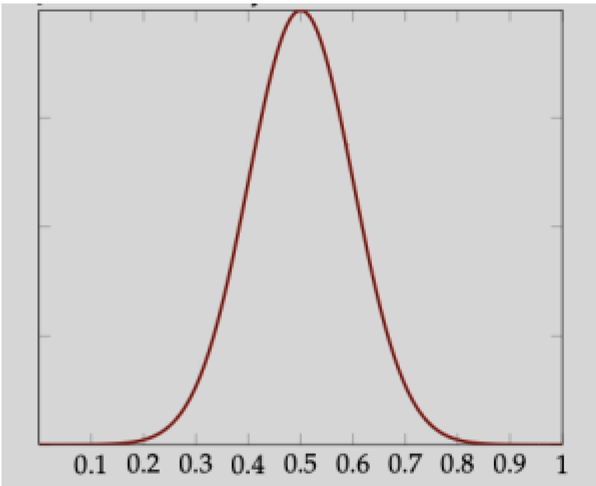

Figure 12.1. The distribution of the probability of getting Heads in the continuous version of the problem. We see how [latex]p=0.5[/latex] is more likely, but we can get values as low as 0.1, 0.2, *or as high as* 0.8, 0.9, *albeit with very small likelihood.*

Now that we know this, say we tossed a coin 10 times, and got 10 straight times Heads! Recall that both the method of moments and the maximum likelihood estimation method would simply assume that the coin has $p = 1$ and proceed.

Getting 10 Heads in 10 tosses would be highly improbable for a coin that is "50-50", but it could mean that I have a biased coin towards Heads. So, what should our estimate be?

First, calculate the likelihood function, the way we did during the maximum likelihood estimation calculations. In this case, it would be $L(p) = p^{10}$. Let's plot that (see next figure).

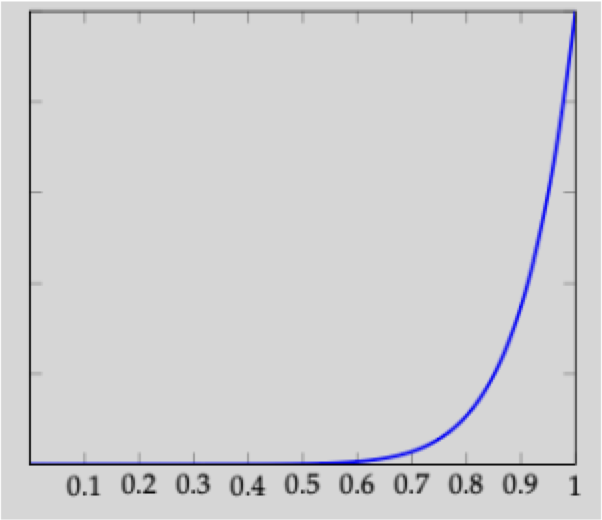

Figure 12.2. The likelihood function of getting 10 Heads after tossing a coin 10 times. It is maximized at [latex]p=1[/latex], which would then be our maximum likelihood estimator.

In Bayesian estimation for discrete-valued parameters earlier, we calculated $P(\theta)$ (priors) with $P(X|\theta)$ (likelihoods) to obtain a series of posteriors that we would compare. In the continuous version, we calculate $f(\theta)$ (prior distribution) with $L(\theta)$ (likelihood function) to obtain a posterior distribution that we would then find the maximizer at! Confused? Let's look at this visually again in the next figure.

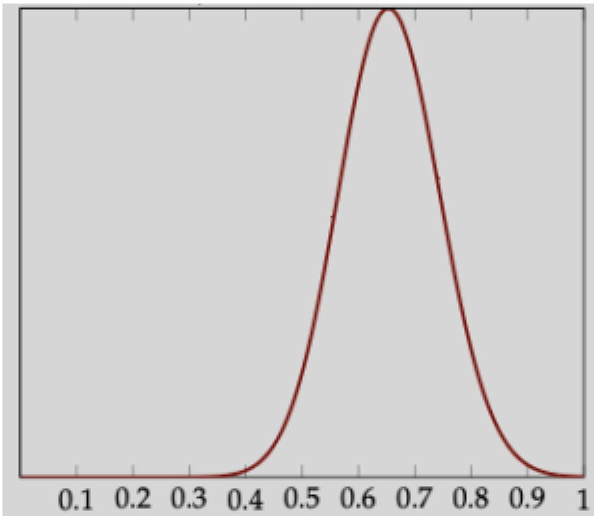

Figure 12.3. The posterior distribution, found by multiplying [latex]f(\theta)[/latex] (the pdf of the normal distribution [latex]\mathcal{N}\left(0.5, 0.01\right)[/latex]) with the likelihood function [latex]L(\theta)[/latex]. The maximizer here is the Bayesian estimator and is found at [latex]\hat{p}=0.6531[/latex].

Let us define the notation for the method then:

- **prior distribution**: the distribution of the real-valued and continuous parameter θ, $f(\theta)$.
- **likelihood function**: the likelihood function, built just as in the maximum likelihood estimation metho, $L(\theta)$.
- **posterior distribution**: the multiplication of the two functions $f(\theta) \cdot L(\theta)$.

The Bayesian estimation method for continuous parameters states that: "the estimate is found by maximizing the posterior distribution."

And, yes! This again sums it up. Let us view one example from beginning to end using the method.

Example 12.13 Mortality risk

We call mortality risk of a hospital the probability of death occurring for any patient admitted to the hospital. The mortality risk in US hospitals is in general **exponentially distributed** with a mean at 1.5% (that is, $\lambda = \frac{1}{1.5\%}$). You have been observing a hospital and have seen 25 deaths in the first 150 patient admissions. What is the Bayesian estimator for the true mortality rate of the hospital?

Right away, distinguish between two items:

1. the prior distribution that we believe the mortality risk to be distributed as (exponential)
2. the mortality rate itself is a Bernoulli random variable (p and $1 - p$); in our case, we have a sample we have collected (25 deaths in 150 admissions) to help us estimate p.

So, let us start collecting what we need one-by-one.

Prior distribution: $f(p) = \frac{1}{1.5} e^{-\frac{1}{1.5}p}$.

Likelihood function: $L(p) = p^{25} \cdot (1-p)^{125}$.

Posterior distribution:

$$f(p) \cdot L(p) = \frac{1}{1.5} e^{-\frac{1}{1.5}p} \cdot p^{25} \cdot (1-p)^{125}.$$

To maximize, get the derivative and equate to 0 to get:

$$\frac{\partial f(p) \cdot L(p)}{\partial p} = \frac{4}{9} e^{-\frac{2}{3}p} (p-1)^{124} p^{24} \left((p-226)p + 37.5\right) = 0 \implies \left((p-226)p+37.5\right) = 0 \implies p = 0.16605$$

.

The maximizer is, then, at $\hat{p} = 0.16605$.

If we want to, we can see the same result visually. First, plot our prior beliefs/distribution:

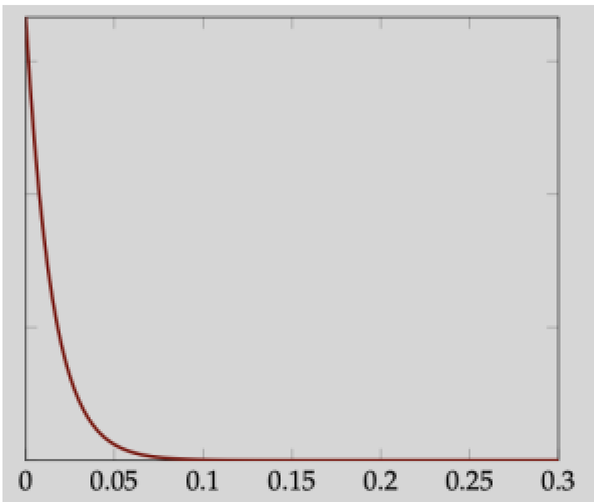

Figure 12.4. Our prior belief distribution.

Then, plot our likelihood function based on the sample collected:

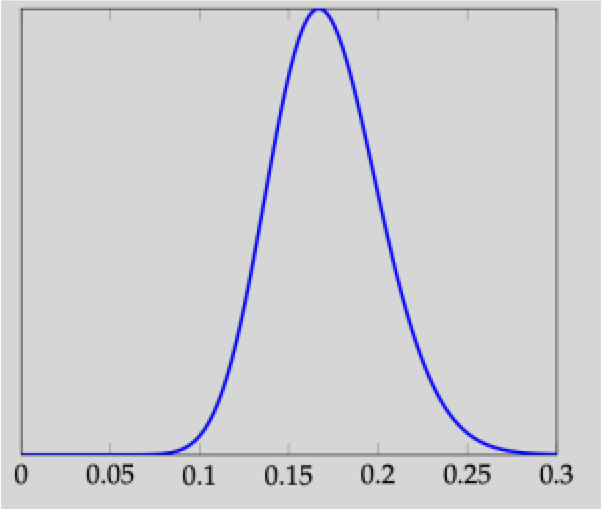

Figure 12.5. Our likelihood function.

And finally plot the posterior distribution, and check that the maximizer is indeed at $\hat{p} = 0.16605$:

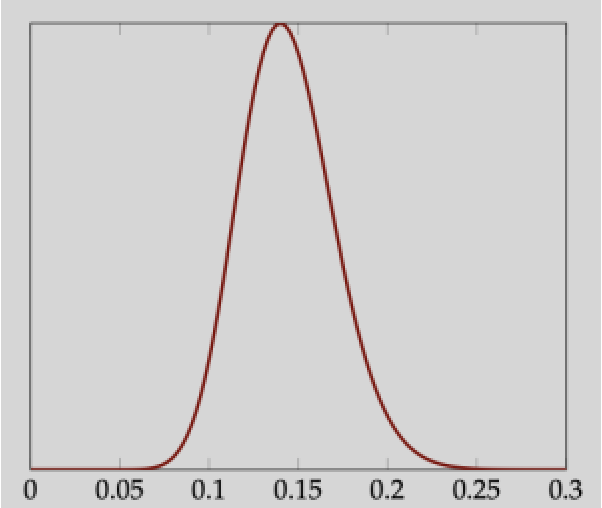

Figure 12.6. Our posterior distribution.

One last example

Let us work on one more example for continuously distributed parameters. Assume we have a population distribution with pdf $f(x) = (\theta + 1)x^{\theta}$, for $0 \leq x \leq 1$. Moreover, assume that θ is not totally random, but is instead distributed with pdf $f(\theta) = \frac{1}{12}(3 - \theta)$, defined over $-2 \leq \theta \leq 2$. Assume we have collected a sample of $X_1 = 0.9, X_2 = 0.89, X_3 = 0.76, X_4 = 0.96$. What is the Bayesian estimator for θ?

You may inspect the solution visually as a homework assignment. Algebraically, though, we would multiply the prior distribution ($f(\theta)$) with the likelihood function ($L(\theta)$) to obtain the posterior distribution. In mathematical terms: $f(\theta) = \frac{1}{12}(3 - \theta)$.

$L(\theta) = (\theta+1)X_1^{\theta} \cdot (\theta+1)X_2^{\theta} \cdot (\theta+1)X_3^{\theta} \cdot (\theta+1)X_4^{\theta} = (\theta+1)^4(X_1 \cdot X_2 \cdot X_3 \cdot X_4)^{\theta} = (\theta+1)^4 0.5844096^{\theta}$

. $f(\theta) \cdot L(\theta) = \frac{1}{12}(3 - \theta) \cdot (\theta + 1)^4 0.5844096^{\theta}$.

Getting the derivative of the posterior, and equating it to 0, we get:

$\frac{\partial f(\theta)L(\theta)}{\partial \theta} = 0 \implies 0.0447628 \cdot 0.58441^{\theta}(1+\theta)^3(17.4783 + \theta(-11.3083 + \theta)) = 0$

.

We get three possible solution: $\theta = -1$, $\theta = 1.85$, or $\theta = 9.46$. We note that the last one cannot happen as θ is between -2 and 2. Between the two remaining possible solutions, we compare their posterior distribution values:

- $f(-1) \cdot L(-1) = \frac{1}{12}(3 - (-1)) \cdot ((-1) + 1)^4 0.5844096^{-1} = 0$
.
- $f(1.85) \cdot L(1.85) = \frac{1}{12}(3 - 1.85) \cdot (1.85 + 1)^4 0.5844096^{1.85} = 2.34$
.

Hence, $\hat{\theta} = 1.85$ is the maximizer and the Bayesian estimator.

12.5. Food for thought

I lied.. Here is the visual version of the posterior also. It is clear that 1.85 is indeed the maximizer!

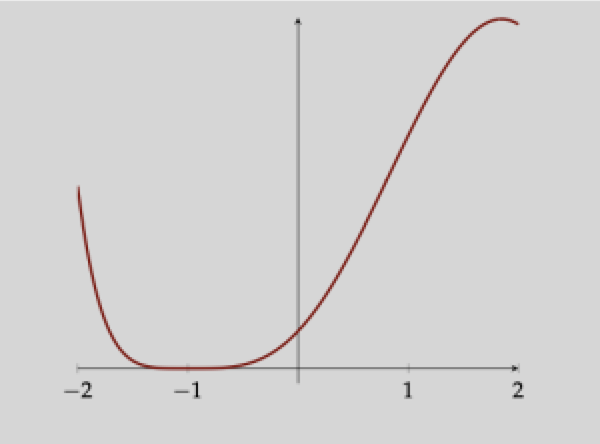

Figure 12.7. The posterior function represented visually, showing a maximum achieved at 1.85.

Examples

Example 12.14. Bernoulli with Beta prior

Suppose we observe n Bernoulli trials with k successes. We use a Beta(α, β) prior for p.

Prior: $\pi(p) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1}(1-p)^{\beta-1}$

Likelihood: $L(p|k) = p^k(1-p)^{n-k}$

Posterior: Beta$(\alpha + k, \beta + n - k)$

Bayesian estimate (posterior mean):

$$\hat{p}_{Bayes} = \frac{\alpha + k}{\alpha + \beta + n}$$

Interpretation:

- This is a weighted average of the prior mean $\frac{\alpha}{\alpha+\beta}$ and the MLE $\frac{k}{n}$
- As $n \to \infty$, the estimate approaches the MLE (data dominates prior)
- With small n, the prior has more influence

Example 12.15. Coin flipping with prior belief

We flip a coin 10 times and get 6 heads. We use a

Beta(5,5) prior (centered at 0.5, moderately strong belief in fairness).

Prior parameters: $\alpha = 5, \beta = 5$

Data: $n = 10, k = 6$

Posterior: Beta$(5 + 6, 5 + 4)$ = Beta(11, 9)

Bayesian estimate:

$$\hat{p}_{Bayes} = \frac{11}{11 + 9} = \frac{11}{20} = 0.55$$

Compare to MLE: $\hat{p}_{MLE} = \frac{6}{10} = 0.6$

The Bayesian estimate is "pulled" toward 0.5 by our prior belief!

Example 12.16. Poisson with Gamma prior

Suppose we observe Poisson data with total count $\sum x_i = s$ over n observations. We use a Gamma (a, b) prior for λ.

Prior: Gamma(a, b)

Likelihood: $L(\lambda) = \frac{\lambda^s e^{-n\lambda}}{\prod x_i!}$

Posterior: Gamma$(a + s, b + n)$

Bayesian estimate (posterior mean):

$$\hat{\lambda}_{Bayes} = \frac{a + s}{b + n}$$

Compare to MLE: $\hat{\lambda}_{MLE} = \dfrac{s}{n}$

Advantages and Disadvantages

As an outro for the chapter, we compare Bayesian estimation to Method of moments and MLE (frequentist approaches to estimation).

Advantages of Bayesian estimation:

- Incorporates prior knowledge naturally
- Provides full distribution (posterior), not just point estimate
- Natural framework for updating beliefs as new data arrives
- Works well with small sample sizes
- Provides probability statements about parameters

Disadvantages:

- Requires choosing a prior (subjective choice)
- Can be computationally intensive for complex models
- Results depend on prior choice (though less so with more data)
- May be more difficult to communicate to non-statisticians

Summary

We have now seen three major approaches to point estimation:

Table 12.4. A table formatting for the key idea and advantages of each of the three methods of estimation that we discussed in this chapter.

Method	Key Idea	Advantages
Method of Moments	Match sample and population moments	Simple, always works
MLE	Maximize probability of observed data	Optimal properties, uses all data
Bayesian	Update prior with data	Incorporates prior knowledge, gives full distribution

Exercise 12.1

1. Suppose you observe 3 successes in 5 Bernoulli trials. Calculate the Bayesian estimate for p using:
   - Beta(1,1) prior (uniform)
   - Beta(2,2) prior
   - Beta(10,10) prior (strong belief in fairness)

   Compare how the prior strength affects the estimate.
2. Show that with a uniform prior Beta(1,1), the Bayesian estimate equals the MLE for Bernoulli data.
3. For Poisson data with $\sum x_i = 20$ from $n = 10$ observations, calculate the Bayesian estimate using Gamma(2,1) prior.

Activities

Activity 1: Streaming services and their data

A TV streaming service has collected a lot of information from several customers and is interested in analyzing it for patterns about their streaming habits. More specifically, they have collected data on two items: how much time they have spent using the service (from login to closing the tab/exiting the app) and how many episodes they have watched. Part of the data follows.

You will use the time data (first row, provided in hours) in Problems 1 and 2; you will also need the count data (second row, provided in number of episodes) in Problem 3.

Let us help them find good estimators using the method of moments! As a reminder, for the method of moments, we:

1. Calculate (generally, there are exceptions) as many population and sample moments as the unknown parameters.
2. Equate each population moment with the corresponding sample moment. Typically the population moment(s) will be a function of the unknown parameter(s), whereas the sample moment(s) will be a value based on the observations.
3. Solve a system of equations for the unknown parameters.

Problem 1: The method of moments for a uniform distribution

The streaming service assumes that people spend time that is **uniformly distributed** in $[\theta, 2\theta]$ using their service. Use the method of moments to provide an estimator for θ, $\hat{\Theta}$. What is the point estimate $\hat{\theta}$ you get when using the data of the previous table?

You'll only need the first row of the previous table.

Problem 2: The general uniform distribution

What if the streaming time is not distributed in $[\theta, 2\theta]$ but is instead uniformly distributed in $[a, b]$? How would we go about estimating a and b? What are the point estimates $\hat{a}$ and $\hat{b}$ when you use the data from the earlier table?

Again, you will simply need the first row from the earlier table:

You *may* also need that $Var[X] = \dfrac{(b-a)^2}{12}$ for a uniformly distributed random variable X.

Problem 3: Number of episodes as a Poisson distributed random variable

Let us turn our focus to the number of episodes users watch. We assume the number of episodes watched during a session is a Poisson distributed random variable with rate λ; alas, λ is not known.

Using the method of moments, provide an estimator for λ. Use that estimator on the data from the table to obtain a point estimate $\hat{\lambda}$. As a reminder, here are the data on the number of episodes per session:

Activity 2: Coming up with an estimator

The outcome of an experiment is a number between 0 and 1 (where 0 is considered an utter failure and 1 is considered a big success). The outcome is distributed with pdf

$$\boldsymbol{f(x) = \theta\left(x - \frac{1}{2}\right) + 1},$$ defined over $0 \leq x \leq 1$, where θ is an unknown parameter.

Problem 4: Applying the method of moments

Using the method of moments, obtain an estimator for θ. You may assume that you have been given a sample $X_1, X_2, \ldots, X_n$ and have obtained the sample average (sample first moment) as $\overline{X}$.

12.6. *Food for thought*

Hopefully, you have gotten that $\hat{\Theta} = 12\overline{X} - 6$, where $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ is the average of n observations.

If you did not get that result, go back, find the mistake, and correct it. Did you take the first population and the first sample moments and equate them? Did you remember to equate the first sample moment (the sample average $\overline{X}$) to the first population moment (expected value $E\left[X\right] = \int_0^1 x f(x) dx = \frac{\theta + 6}{12}$)?

Problem 5: MSE

What is the MSE of your estimator? If you get stuck remember you will need that $E\left[\overline{X}\right] = E\left[X\right]$ and $Var\left[\overline{X}\right] = \frac{1}{n} Var\left[X\right]$. Finally, keep in mind that both the bias and the variance may end up being a function of the unknown parameter or of the sample size.

Problem 6: Application

For this experiment, we have collected a sample of 6 items and found them equal to $X_1 = 0.8, X_2 = 0.83, X_3 = 0.95, X_4 = 0.72, X_5 = 0.85, X_6 = 0.65$. What is a good point estimate for θ? Use the point estimator you came up with in Problem 5. Using that estimate, what is the probability the experiment is at least a moderate success? Assume that moderate success is any value above 0.75.

Activity 3: The Pareto distribution

Problem 7: A 2×2 system

The Pareto distribution (named after Italian mathematician and engineer, Vlifredo Pareto) is described by $f(x) = \frac{\alpha \beta^{\alpha}}{x^{\alpha+1}}$ for $x \geq \beta$. Originally it was designed to apply to the distribution of wealth where a small portion of the population accounts for a large part of the wealth; but it has been successfully applied in other settings, too. In this exercise, you are asked to estimate α and β using the method of moments. You may assume that $\alpha > 2$ and $\beta \geq 1$.

No need to actually solve the system of equations! Simply set it up as if you had a sample of n observations $X_1, X_2, \ldots, X_n$ and leave it as a system of equations.

Activity 4: Special case

In this last activity, we have you run into a weird case. What if the first moment does not work? Then, we need to take the second moments and equate them. Here is such a case.

Problem 8: A weird situation

Let X be a continuous population distributed with probability density function $f(x) = \frac{1}{2\beta}e^{-\frac{|x|}{\beta}}, \quad x \in (-\infty, +\infty)$. Using the method of moments provide an estimator for β.

12.7. Food for thought

So, in this case, we observe that we do not always take the first k moments when there are k parameters that we are estimating. Instead, we need to only consider moments that are a function of the parameters. In the above case, the first moment of X ended up being a constant (not a function of β) so we had to take the second moments and equate them.

Activity 5: Our first MLE

As a reminder, to get the maximum likelihood estimators, we:

1. build the likelihood function $L(\theta)$ by multiplying the probability mass function (for discrete) or the probability density function (for continuous) for each of the sample values.
2. either plot $L(\theta)$, or (for larger sample sizes) get the derivative(s) $\frac{\partial L(\theta)}{\partial \theta}$ for the unknown parameter(s) and equate to zero.
3. if plotting, check to find the maxima it attains; if using the derivative(s), solve the system and obtain the maxima.
    - When we have multiple maxima, then pick the absolute biggest amongst them.

Maximum likelihood estimators can get pretty tough to calculate as we have samples of larger and larger size, so let us start easy.

Problem 9: Building a likelihood function

Assume we have been told that a continuous population X is distributed with pdf $f(x) = \frac{1}{2}(1+\theta x), -1 \leq x \leq 1$, where θ is a parameter with $-1 \leq \theta \leq 1$. We have collected a sample of size $n = 3$: $X_1 = 0.25, X_2 = 0.85, X_3 = -0.50$. What is the likelihood function $L\left(\theta\right)$?

Problem 10: MLE for a sample of $n = 3$ observations

Based on your likelihood function from Problem 1, what is the maximum likelihood estimator for θ? You can solve this problem in two ways:

1. Plot the (degree 3) polynomial you obtained for $L\left(\theta\right)$. Then find the value for θ that leads to the maximum $L\left(\theta\right)$. If there are multiple maxima, pick the "highest" one.
2. Get the derivative $\frac{\partial L\left(\theta\right)}{\partial \theta}$ and set it equal to 0. Then solve the equation to obtain the value for θ. If there are multiple solutions, pick the one that leads to maximum $L\left(\theta\right)$.

Problem 11: When the derivatives (and the plotting!) lie

We cannot *always* visualize and pick the maximum or get the derivatives and trust that the maximum is the estimator we are looking for. Let's see a case where this "issue" happens.

Recall again how continuous population X is distributed with pdf $f(x) = \frac{1}{2}(1+\theta x), -1 \leq x \leq 1$. Also recall that we have

$-1 \leq \theta \leq 1$. We again pick a random sample of size $n = 3$. This time, though, it is: $X_1 = 0.75, X_2 = 0.65, X_3 = 0.80$. What is the maximum likelihood estimator now?

So, this was weird, especially if we decided to take derivatives. Let us recap:

12.8. Food for thought

1. **If we plot the likelihood function** $L(\theta) = f(X_1) \cdot f(X_2) \cdot f(X_3)$, or $L(\theta) = \left(\frac{1}{2}\right)^3 \cdot (1 + 0.75 \cdot \theta) \cdot (1 + 0.65 \cdot \theta) \cdot (1 + 0.80 \cdot \theta)$,

we obtain:

It seems like the maximum is achieved for very very large values of θ.

Recall though that $-1 \leq \theta \leq +1$. Then, we have:

The maximum is clearly attained at

If we take the derivative and set to 0, we get:

$$\frac{\partial L(\theta)}{\partial \theta} = 0 \implies 0.275 + 0.401875 \cdot \theta + 0.14625 \cdot \theta^2 = 0 \implies \theta = \begin{cases} -1.45964 \\ -1.28822 \end{cases}.$$

Neither is between -1 and $+1$, hence, before deciding the maximum, we'd need to also check the values at -1 and $+1$. Comparing, we get:

$$L(-1) = \left(\frac{1}{2}\right)^3 \cdot (1 + 0.75 \cdot (-1)) \cdot (1 + 0.65 \cdot (-1)) \cdot (1 + 0.80 \cdot (-1)) = 0.0021875.$$

$$L(+1) = \left(\frac{1}{2}\right)^3 \cdot (1 + 0.75 \cdot 1) \cdot (1 + 0.65 \cdot 1) \cdot (1 + 0.80 \cdot 1) = 0.6496875.$$

Contrasting, we would pick [Is something missing here?]

As a parenthesis, here are the two "zero derivative" points: they are not both maxima! And, as discussed earlier, even the local maximum here is not as big as the "global" maximum achieved at $\hat{\theta} = 1$.

Activity 6: Streaming services and their data (reloaded)

Let us revisit the example from last time. Remember that TV streaming giant with the data they had collected? Well, they are back and they'd like to see what else they can do to come up with estimators.As a reminder, they have provided you with data on the number of episodes people watch during a session (an integer number).

Table 12.5. The data for Activity 6. Here we show 10 sessions of a streaming service, and the number of episodes watched in each session.

	Session #									
	1	2	3	4	5	6	7	8	9	10
# of episodes	1	3	3	3	2	5	1	5	3	1

Let us help them find a good estimator using the maximum likelihood method this time around.

Problem 12: Poisson rates in the general case

You have been provided with the actual observations in the sample. What if we had not collected a sample yet, though? Can you still calculate the maximum likelihood estimator as a function of the sample, whatever it may be?

Let's put this to practice for the rate of a Poisson distributed population. What is λ as a function of the sample collected $X_1, X_2, \ldots, X_n$? Don't use the provided data just yet. See if you can derive the MLE as a general function of $X_1, X_2, \ldots, X_n$.

Problem 13: Poisson rates in the general case (reloaded)

While the previous result is not terribly difficult to derive, it is still confusing sometimes to take the derivative of a product of functions, as is the case with general likelihood.

Following our logic from earlier, the likelihood function for a general sample of size n (let it be $X_1, X_2, \ldots, X_n$) would be:

$$L(\lambda) = e^{-\lambda} \cdot \frac{\lambda^{X_1}}{X_1!} \cdot e^{-\lambda} \cdot \frac{\lambda^{X_2}}{X_2!} \cdot \ldots e^{-\lambda} \cdot \frac{\lambda^{X_n}}{X_n!}.$$

We would need the derivative of this , and this is not always easy.
In the notes, we mentioned something called the log-likelihood. For the log-likelihood, you should still calculate the likelihood function but then take its logarithm to obtain $\ln\left(L(\theta)\right)$. Then, you can take its derivative and equate it to 0. Using the logarithm properties we may calculate

$$\ln\left(L(\lambda)\right) = \ln\left(e^{-\lambda} \cdot \frac{\lambda^{X_1}}{X_1!}\right) + \ln\left(e^{-\lambda} \cdot \frac{\lambda^{X_2}}{X_2!}\right) + \ldots + \ln\left(e^{-\lambda} \cdot \frac{\lambda^{X_n}}{X_n!}\right),$$

which is **so much easier to differentiate**! What would be the maximum likelihood estimator?

12.9. *Food for thought*

The takeaway message is that the estimator is of course the same either way! Using the likelihood or the log-likelihood to differentiate and equate to 0 will always give the same result.

Problem 14: Solving the example

Based on your answer in Problem 12 or 13, use the data provided for the number of episodes watched per session and calculate the maximum likelihood estimator for the rate λ. Is it the same as the rate you got when you used the method of moments?

12.10. *Food for thought*

This rings a bell. This is the same as the estimator we obtained last time using the method of moments! Are these two methods (MLE and method of moments) always giving us the same estimators?

Activity 7: Extra details

In this activity, we see a couple of special cases and answer the following questions:

1. Are the method of moments and MLE always giving us the same estimators?See Problem 6 (where they are the same)and Problem 7 (where they are not).
2. Is there a time when the actual estimator is different than the one obtained when setting the derivative(s) equal to 0? Or maybe where the parameter can be calculated without derivatives?See Problem 3 (earlier) and Problems 8-9.
3. How can we estimate two or more parameters using MLE?See Problem 10 (comes pre-filled).

Problem 15: Not always the same

Consider the following probability density function $f(x) = \theta x^{\theta-1}$ defined over $0 \leq x \leq 1$. Assume we have been provided a sample of size n: $X_1, X_2, \ldots, X_n$. We can apply the method of moments to get:

$$E\left[X\right] = \frac{1}{n}\sum_{i=1}^{n} X_i \implies \int_0^1 xf(x)dx = \overline{X} \implies \int_0^1 \theta x^\theta dx = \overline{X} \implies$$

$$\implies \theta \frac{x^{\theta+1}}{\theta+1}\bigg|_0^1 = \overline{X} \implies \frac{\theta}{\theta+1} = \overline{X} \implies \hat{\theta} = \frac{\overline{X}}{1-\overline{X}}.$$

How about the maximum likelihood estimator? What would it be?

12.11. Food for thought

So, as we just saw the method of moments and the maximum likelihood estimation method match often; but not always.

Problem 16: When the derivatives "lie" (revisited)

We go back to the streaming service. For convenience, here is some data they obtained for the streaming time (continuous random variable) for 10 sessions.

Let us assume that they believe people to be watching episodes in time that is uniformly distributed and continuous in $[\theta, 2\theta]$. What is the likelihood function based on the sample they have obtained?

Problem 17: When the derivatives "lie" (revisited)

Hopefully, we have gotten that $L(\theta) = \left(\frac{1}{\theta}\right)^n$, where $n = 10$ in our case. This is an always decreasing function! Because of that, for $L(\theta)$ to be maximum, we want θ to be smallest..

What is the smallest value that θ is allowed to take? Recall that we need $\theta \leq X_i$ and $2\theta \geq X_i$ for **all our observations** X_i ! This means that $\theta \leq 0.75$ and $2\theta \geq 0.75$, $\theta \leq 1.20$ and $2\theta \geq 1.20$, and so on. With that in mind, what is the maximum likelihood estimator for θ?

> *12.12. Food for thought*
>
> To easily solve this? Only pick the smallest and the largest observation and then only write these two constraints: $\theta \leq 0.75$ and $2\theta \geq 1.43$, which implies that $\theta \leq 0.75$ and $\theta \geq 0.715$, which finally gives that $\hat{\theta} = 0.715$. Cool, no? No derivatives at all!

This last problem comes pre-solved. **Please study the derivation** for an example of using MLE for two parameters!

Problem 18: The normal distribution

Assume that you have n observations from a normally distributed population with unknown mean and variance. What are the maximum likelihood estimators for μ and σ^2?

Let's build the likelihood function:

$$L(\mu,\sigma) = f(X_1)\cdot f(X_2)\cdot f(X_3)\cdot f(X_4)\cdot f(X_5) = \\ = \frac{1}{\sqrt{2\pi}\cdot\sigma} e^{-\frac{(X_1-\mu)^2}{2\sigma^2}} \cdot \frac{1}{\sqrt{2\pi}\cdot\sigma} e^{-\frac{(X_2-\mu)^2}{2\sigma^2}} \cdots \frac{1}{\sqrt{2\pi}\cdot\sigma} e^{-\frac{(X_5-\mu)^2}{2\sigma^2}}.$$

Focus on the exponent calculation for a second. First of all, we know that $e^a \cdot e^b = e^{a+b}$. Hence, we need to sum the exponents. We would have:

$$-\frac{(X_1-\mu)^2}{2\sigma^2} - \frac{(X_2-\mu)^2}{2\sigma^2} - \frac{(X_3-\mu)^2}{2\sigma^2} - \frac{(X_4-\mu)^2}{2\sigma^2} - \frac{(X_5-\mu)^2}{2\sigma^2} = \\ = \frac{-X_1^2 + 2\mu X_1 - \mu^2 - X_2^2 + 2\mu X_2 - \mu^2 - \ldots - X_5^2 + 2\mu X_5 - \mu^2}{2\sigma^2} = \\ = -\frac{\sum X_i^2 - 2\mu \sum X_i + n\mu^2}{2\sigma^2}.$$

Finally, this gives us:

$$L(\mu,\sigma) = \left(\frac{1}{\sqrt{2\pi}\cdot\sigma}\right)^n \cdot e^{-\frac{\sum X_i^2 - 2\mu\sum X_i + n\mu^2}{2\sigma^2}}.$$

Now, on to the derivatives... Why plural? Well, we have two unknown parameters! First, for μ:

$$\frac{\partial L(\mu,\sigma)}{\partial \mu} = 0 \implies \frac{\partial\left[\left(\frac{1}{\sqrt{2\pi}\cdot\sigma}\right)^n \cdot e^{-\frac{\sum X_i^2 - 2\mu\sum X_i + n\mu^2}{2\sigma^2}}\right]}{\partial \mu} = 0.$$

Let's use some derivative properties. First of all, the left part of the function is a constant as far as μ is concerned. Hence, we can just take it out:

$$\frac{\partial\left[\left(\frac{1}{\sqrt{2\pi}\cdot\sigma}\right)^n\cdot e^{-\frac{\sum X_i^2-2\mu\sum X_i+n\mu^2}{2\sigma^2}}\right]}{\partial\mu}=\left(\frac{1}{\sqrt{2\pi}\cdot\sigma}\right)^n\cdot\frac{\partial\left[e^{-\frac{\sum X_i^2-2\mu\sum X_i+n\mu^2}{2\sigma^2}}\right]}{\partial\mu}.$$

Then, for the right part of the function, we have that

$$\left(e^{-g(x)}\right)'=-(g(x))'e^{-g(x)}.$$

Specifically, in this case we have

$$g(\mu)=\frac{\sum X_i^2-2\mu\sum X_i+n\mu^2}{2\sigma^2}\text{ and }(g(\mu))'=2\mu n-2\sum X_i.$$

Combining:

$$\begin{aligned}\frac{\partial L(\mu,\sigma)}{\partial\mu}=0\implies&\frac{\partial\left[\left(\frac{1}{\sqrt{2\pi}\cdot\sigma}\right)^n\cdot e^{-\frac{\sum X_i^2-2\mu\sum X_i+n\mu^2}{2\sigma^2}}\right]}{\partial\mu}=0\implies\\&\implies\left(\frac{1}{\sqrt{2\pi}\cdot\sigma}\right)^n\cdot\left(2\sum X_i-2\mu n\right)e^{-\frac{\sum X_i^2-2\mu\sum X_i+n\mu^2}{2\sigma^2}}=0\\&\implies\left(2\sum X_i-2\mu n\right)=0\implies\hat{\mu}=\frac{\sum X_i}{n}=\overline{X}.\end{aligned}$$

This is true as both $\left(\frac{1}{\sqrt{2\pi}\cdot\sigma}\right)^n>0$ and $e^{-\frac{\sum X_i^2-2\mu\sum X_i+n\mu^2}{2\sigma^2}}>0$. Now, for the derivative as far as σ is concerned:

$$\frac{\partial L(\mu,\sigma)}{\partial \sigma} = 0 \implies \frac{\partial\left[\left(\frac{1}{\sqrt{2\pi}\cdot\sigma}\right)^n \cdot e^{-\frac{\sum X_i^2 - 2\mu\sum X_i + n\mu^2}{2\sigma^2}}\right]}{\partial \sigma} = 0.$$

Recall that $(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$. Using for simplicity that $g(\mu) = \sum X_i^2 - 2\mu\sum X_i + n\mu^2$, leads to:

$$\begin{aligned}
\frac{\partial L(\mu,\sigma)}{\partial \sigma} = 0 &\implies \frac{\partial\left[\left(\frac{1}{\sqrt{2\pi}\cdot\sigma}\right)^n\right]}{\partial \sigma}\cdot e^{-\frac{g(\mu)}{2\sigma^2}} + \left(\frac{1}{\sqrt{2\pi}\cdot\sigma}\right)^n \cdot \frac{\partial\left[e^{-\frac{g(\mu)}{2\sigma^2}}\right]}{\partial \sigma} = 0 \\
&\implies -\left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \frac{n}{\sigma}\cdot e^{-\frac{g(\mu)}{2\sigma^2}} + \left(\frac{1}{\sqrt{2\pi}\cdot\sigma}\right)^n \cdot e^{-\frac{g(\mu)}{2\sigma^2}} \cdot \frac{g(\mu)\mathrm{e}^{-\frac{g(\mu)}{2\sigma^2}}}{\sigma^3} = 0 \\
&\implies \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \mathrm{e}^{-\frac{g(\mu)}{2\sigma^2}}\left(-\frac{n}{\sigma} + \frac{g(\mu)}{\sigma^3}\right) = 0 \implies \\
&\implies -\frac{n}{\sigma} + \frac{g(\mu)}{\sigma^3} = 0 \implies -n\sigma^2 + g(\mu) = 0 \implies \\
&\implies \hat{\sigma}^2 = \frac{g(\mu)}{n} = \frac{\sum X_i^2 - 2\mu\sum X_i + n\mu^2}{n}
\end{aligned}$$

Replacing that $\mu = \overline{X}$, we get:

$$\hat{\sigma}^2 = \frac{\sum X_i^2}{n} - \overline{X}^2.$$

Activity 8: A simple Bayesian estimation

As a reminder, to get the Bayesian estimators, we:

1. identify our prior probabilities/distribution: this quantifies what we think the unknown parameters are *before* observing

any sample.

2. build the likelihood function $L(\theta)$ by multiplying the probability mass function (for discrete) or the probability density function (for continuous) for each of the sample values (**exactly what we did last time for MLE**).
3. compute the posterior probabilities/distribution by multiplying the priors with the likelihood.
4. find the maximum posterior probability or find the point where the posterior distribution attains its maximum.

    - This latter part can be done by either comparing all probabilities and picking the biggest one or
    - by getting the derivative(s) of the posterior distribution for each of the unknown parameter(s) and setting it to 0

Problem 19: Creating a table

For smaller problems, it may be useful to create a table. In the table, we store the prior probabilities (based on past information or prior beliefs), the likelihood functions (based on the sample), and computer the posterior probabilities. Let us see this in an example.

A school can be in one of three categories: $R1, R2, R3$. $R1$ schools place 90% of their graduates in good jobs, $R2$ schools place 75% of their graduates in good jobs, and $R3$ schools place 50% of their graduates in good jobs. You may assume that each school has a $\frac{1}{3}$ chance of appearing.

You have been observing a school and have found that 82 of their last 100 graduates have been placed in good jobs. Is the school an $R1$, $R2$, or $R3$ school? Equivalently, if p is the probability of placing a graduate in a good position, what is $\hat{p}$ for the specific school begin observed?

Table 12.6. The empty table for Problem 19. You are expected to fill in the priors, likelihoods, and posteriors for the different parameter values.

	priors	likelihood	posterior
	$\pi(p)$	$L(X\|p)$	$\pi(p) \cdot L(X\|p)$
$p = 0.9$			
$p = 0.75$			
$p = 0.5$			

Problem 20: Normalizing the result

Based on your calculations, you probably noticed that the posterior probabilities are very small. Can you normalize them so as to answer the question: "how certain are you that the school is an R1/R2/R3 institution?"

Activity 9: Streaming services and their data (Bayesian remix)

We turn one last time to the data from the TV streaming giant. As a reminder, they have given you data on the time people spend during a session (continuous); as well as the number of episodes people watch during a session (discrete number). In this activity, we focus solely on the number of episodes streamers watch per session.

Table 12.7. The data for Activity 9. Here we show 10 sessions of a streaming service, and the number of episodes watched in each session. Note that these are exactly the same numbers as in Table 12.5 of Activity 6.

	Session #									
	1	2	3	4	5	6	7	8	9	10
# of episodes	1	3	3	3	2	5	1	5	3	1

Problem 21: Poisson with prior beliefs

The streaming giant believes that there are three types of customers. For all three types the number of episodes they watch is Poisson distributed; they do have different rates λ, though. The three types of customers are:

- 75% of their clientele watches a number of episodes that is Poisson distributed with $\lambda = 1$.
- 10% of their clientele watches a number of episodes that is Poisson distributed with $\lambda = 3$.
- 15% of their clientele watches a number of episodes that is Poisson distributed with $\lambda = 4$.

Using this new information, and assuming that all data has been collected from one type of customers alone, what is the Bayesian estimator for λ?

Problem 22: Poisson with more general prior beliefs

Here we go in the general, continuous case. What if the streaming giant was wrong and customers are not discrete (that is $\lambda = 1, 3,$ or 4), but are instead continuous (λ is *anything* in the 1 to 4 range)?

For this problem, you may assume that λ is uniformly distributed between 1 to 4. What is the Bayesian estimator for λ in this case?

12.13. Food for thought

We may summarize this result as follows. When presented with discrete cases, we will calculate posterior *probabilities* (normalize them, if we prefer) and report the maximum among them. In the continuous case, we will calculate the posterior *distribution* and find the maximizer (potentially by plotting it and observing it, or by setting the derivative equal to 0, when possible).

Activity 10: Coins

Problem 23: Step-by-step

Assume you carry with you three coins with probability of Heads $p = 0.25, 0.5, 0.75$. You pick a coin and you flip it. Which coin do you (believe you) have picked to flip if:

- the first coin comes up Heads?
- the second coin comes up Heads?
- the third coin comes up Tails?
- the fourth coin comes up Tails?
- the fifth coin comes up Tails?

Problem 24: Coins again

You still have an unfair coin at your disposal that brings Heads with unknown probability p. This time though, your coin is not one of three like earlier. Instead, the probability p is uniformly distributed between 0.4 and 0.6.

You decide to run an experiment to estimate p. You will toss the coin as many times as necessary to get Heads and record the number of times until Heads are observed. You have obtained $N_1 = 5, N_2 = 4, N_3 = 5$.

With this in mind, what is the Bayesian estimator for p?

> *12.14. Food for thought*
>
> This is interesting! If our maximizer is above the largest allowable value, or below the smallest allowable value, then we need to pick the largest/smallest one! You may also show this graphically, if you prefer, to see what happens.

Activity 11: Continuous case only

What if both the distribution we are estimating is continuous **and** the distribution of the parameter is continuous at the same time?

Problem 25: Normal and exponential

The time between two consecutive vehicles passing through an intersection is exponentially distributed with unknown rate λ; however, we do know that λ is normally distributed with $\mu = 2$ per minute and variance $\sigma^2 = 0.25$. In essence, this states that typically 2 vehicles would pass every minute, but this can go lower to 1 every 2 minutes (when it is less busy) and higher to 7 every 2 minutes (when it is much busier).

A new day has just begun and we would like to see how many vehicles to expect in the next 2 hours. During the first 5 minutes, we saw the first vehicle appear in $T_1 = 0.5$ minutes, $T_2 = 1$ minutes, $T_3 = 2$ minutes, $T_4 = 1.5$ minutes. Recall that these are times between two consecutive vehicles.

With this in mind, what is the Bayesian estimator for λ?

13. From basic probability theory to machine learning

Introduction

Throughout these last few chapters, we have been building a vocabulary and a set of notations to use in order to talk about probability and probability models. We have learned to describe the world around us through *events* and *sample spaces*, to *count* the number of ways events can result in, and to *update* probabilities when new information comes along.

We also established the basic vocabulary in the form of *probability distributions*, discrete and continuous, so that we can represent real-life problems as mathematical models. We saw the *ubiquity of the normal distribution* as a consequence of the central limit theorem, and we used that ubiquity in our attempts to *estimate* missing parameters for the probability models we encounter.

At each and every step of the way, the underlying question has always been the same: *given uncertainty*, *given variability*, *how do we reason and make decisions*? That question turns out to be at the heart of machine learning and artificial intelligence, and more so in their intersection with operations research and industrial engineering.

Machine learning, the branch of mathematics, computer science, statistics, and engineering that is responsible for modern medical diagnoses, smart spam filters, and recommendation systems for your next album to listen to, is at its very core an **application of probability theory**. This chapter aims to make this connection clear and help guide you in your next ventures in probability. We will not be introducing any new notation or technical derivations; instead, we will show how every major idea from this textbook reappears in machine learning and artificial intelligence.

Random experiments and our world

The world is filled with random experiments! Recall our definition from the first chapter: a random experiment is any activity with outcomes that differ even when executed under the same conditions. In that chapter, we discussed coin flips, dice rolls, and card games; but consider also:

- A radiologist examining an X-ray and having to decide: is there a tumor present?
- An autonomous vehicle observing the road ahead and having to decide: is that a pedestrian or a mailbox?

In the technical sense, both are random experiments: here are some observations (evidence) and we need to act on them under uncertainty. The outcome is not known in advance nor immediately observable: what is observable are brightness pattern changes in an X-ray, distances and shapes from moving pictures, word frequencies in an email, etc. The challenge we face is to reason from those observable features towards a decision in an uncertain environment. This is exactly the problem that machine learning is designed to solve. And the tools it reaches for are precisely the ones we have developed in this textbook.

Data as evidence

When we first discussed probability, we described what we referred to as the *frequentist* view: the probability of an event is the relative frequency with which it occurs across many repetitions of the same experiment. We then showed how to estimate it by simply counting. This is also the core of how a machine learning system learns: it is shown many examples–millions of X-rays, perhaps–and it starts counting: how often does a pattern appear when a tumor is present? These counts are estimates of probability: the machine learning system is, in essence, building an empirical picture of reality through the samples it studies. The more data we provide it, the better the system will *learn*.

This points to one of the most important practical lessons in machine learning: **data is the currency of reasoning**. More so in the presence of uncertainty: in probabilistic reasoning, more data will always mean better decisions. Why is that? Based on our tools from this chapter, more data means clearer probability distribution patterns, more better means better estimators for unknown parameters, which leads to better decisions. Of course, this leaves the question of how much data do we really need? And what kind of data?

Probability distributions and what the world looks like

A major step between the basic probability theory presented in this textbook and its use in machine learning is the concept of a *probability distribution*: we saw many distributions in the textbook (both discrete and continuous), and each of them had one job: to provide us with a complete, concise description of how likely each outcome is across the sample space.

In practice, the quantities that machine learning systems care about (such as the weight of a manufactured component, the time until a machine fails, the demand for a product on a given day) tend to follow recognizable distributional patterns. Many continuous measurements cluster around a central value and taper off symmetrically in both directions, following a shape known as the *normal* distribution. Counts of events (defects per batch, arrivals per hour) often follow a *Poisson* distribution. Times between failures or between customer arrivals frequently follow an *exponential* distribution.

Why is this important and interesting? Well, if we can identify *which distribution* governs a quantity, then we immediately know a lot about it. We know where to *expect* most outcomes to fall; we know how surprised to be by an extreme value and how *varied* and "all-over-the-place" the quantity values can get; we, perhaps most importantly, know how to compute the probability of any event or combination of events! The entire machinery and tools of this textbook apply directly once we know the distribution that best models our world.

Machine learning models are **assumptions about distributions**. When an engineer builds a model to predict the remaining useful life of a bearing, they are making assumptions of the sort: "I believe the distribution of failure times has roughly this shape," and then they proceed to estimate: "I am now going to use data to estimate the exact parameters of said distribution." The model is the probability distribution that the engineer chose. Then, the data and their observations helped them narrow it down precisely.

Conditional probability is the heart of prediction

Of all the ideas in this textbook, the one that most directly powers modern machine learning is conditional probability. We remind you again of the definition: $P\left(E_1 \middle| E_2\right)$ is the probability of event E_1 occurring *given that we already know* E_2 *has occurred*. In the flight example we gave earlier, once we knew that the first leg was delayed, we had to quite dramatically update our estimates about whether we'd make it on time for our next connection: in other terms, new information reshaped our probability calculations.

Machine learning is, at its core, the systematic computation of conditional probabilities and their use to reshape the probability landscape. When a spam filter classifies an email as spam, it is computing a conditional probability of the form: "*What is the probability this email is spam, given its observed features*?" When a predictive maintenance system flags a machine as likely to fail soon, it is computing: "*What is the probability of failure within the next 24 hours, given the current sensor readings*?" And when a medical imaging algorithm highlights a suspicious region in an X-ray, it is computing: "*What is the probability this tissue is malignant, given these pixel intensities and textures*?"

In each case, the structure of the question is the same: given some observed evidence, estimate the probability of a desired event. The entirety of supervised machine learning is about estimating conditional probability from data.

Bayes' theorem: evidence turned belief

In the textbook, we spend significant time establishing, deriving, and using the Bayes' theorem, arguably the most celebrated result in probability theory. It is derived immediately from the definition of conditional probability, and it deserves special attention here because it underlies some of the most powerful ideas in machine learning and AI. As a reminder, Bayes' theorem states that $P\left(A|B\right) = \dfrac{P\left(B|A\right) \cdot P(A)}{P(B)}$.

As a formula and conceptually, Bayes' theorem is remarkable for its ability to "reverse" the conditional information. If we know how likely the evidence B is when A is true ($P\left(B|A\right)$), and we know our prior belief about how likely A is before seeing any evidence ($P(A)$), then we can compute our updated belief after seeing B ($P(A|B)$).

In the language of machine learning, $P(A)$ is the *prior* *probability* (what we believe to be true before any observations) and $P(A|B)$ is the *posterior probability* (our updated belief after we observe data). Bayes' theorem is the mechanism for updating our beliefs in the face of new evidence.

From an industrial engineering perspective, if a quality inspector has historic data about the percentage of defects, then this is their prior probability. A newly installed sensor flagging parts as defective with known accuracy are new evidence that can come in: that is, what is the probability of a defect given the sensor flag? Bayes' theorem helps us address the question and often leads to surprising results: even a highly accurate sensor, applied to a process with a very low defect rate, can still produce more false alarms than true detections.

An entire body of machine learning, called Bayesian machine learning, is built on this powerful mechanism. Rather than treating model parameters as fixed unknown constants waiting to be estimated, Bayesian methods treat them as random variables with their own distributions. Prior beliefs about reasonable parameter values are updated with data to produce posterior distributions. Finally, the fact that predictions are uncertain themselves means that our possible answer would be a range of plausible values, with associated probabilities.

From probability theory to decision-making

Let us bring this full circle. We began this textbook with a Monopoly player needing to roll a 6 or 7, and a card game player wondering whether to play. In both cases, the goal was not probability simply as a calculation, but as a tool for making a decision. Should I take the risk? Should I play? Machine learning systems face the same challenge. A classifier that outputs $P(\text{spam}|\text{specific email}) = 0.73$ has done useful calculations for us, but eventually someone must decide to either block the email or let it through? A machine learning model that "alerts" $P(\text{failure within 24 hours}) = 0.41$ is informative, but a maintenance crew needs to decide whether shutting down the machine now before a failure is prudent or not.

The bridge between probability theory and decision theory is through **hypothesis testing**. In hypothesis testing, we formally test (and try to disprove) a statement about the "status quo." Do we believe a machine is operating well within parameters or not? Is the percentage of voters agreeing with a policy greater than 50%? In these types of questions, we have two types of errors: Type I and Type II or α and β.

In an ideal world, we'd make decisions with both errors being zero: alas, this is almost never the case and a decision-maker needs to consider the cost of making a mistake. If the cost of missing a failure is catastrophic and a planned shutdown is cheap, even a 20% failure probability might justify intervening early and often. If the cost of unnecessary downtime is enormous, you might wait until the probability gets closer to, say 70%. Probability theory tells us *what the world looks like*; decision theory tells us *what to do about it*.

This is where industrial engineering and operations research meet machine learning. Industrial engineering has always been (among many, many other things!) about optimizing systems under uncertainty (e.g., scheduling production, managing inventory, routing vehicles, allocating resources). The probabilistic models developed in this textbook, extended by distributions, conditional probabilities, Bayes' theorem, are the exact tools that modern industrial engineers use to make those decisions intelligently and automatically.

In lieu of an adieu

From the outside, machine learning and artificial intelligence may appear foreign. An imposing collection of black-box algorithms with deep mathematical connections. And, it is a matter of fact that the true mathematical depth of many of these models extends beyond the scope of the textbook. That said, even those begin from the basic building blocks presented in this textbook: underneath all of it, the fundamental reasoning is basic probability. A model can be described as a distribution. Training is akin to estimation. Prediction is an involved and adaptable conditional probability. Evidence updates belief. Decisions are made by weighing probabilities against consequences in the form of hypotheses to be tested and rejected.

As a student, you have learned to think in exactly this way through the chapters of the textbook. You can now describe a random experiment, enumerate its outcomes, count favorable cases, compute probabilities, condition on new information, and test whether events are independent. You can suspect the distribution your data follows, and calculate important information about where you expect the outcomes to be, and how variable they are. You have learned to estimate missing parameters. These are not preliminary steps before the "real modeling" and the "real work" begins. These steps are the real work, and this is the foundation of all data-driven intelligence.

The next time you use a navigation app that reroutes because of a closed street, or you observe a medical device (such as a fitness ring or watch) that flags an anomaly, or when you encounter a production system that schedules its own maintenance based on its data collected, you will know something about how and why it works the way it does. How and why it is reasoning about uncertainty, in the language of probability, in the same way you have learned to do.

Glossary

Bayes' theorem

One of the most fundamental result in probability theory that allows us to invert a conditional probability.

Given n mutually exclusive and collectively exhaustive events $S_1, S_2, \ldots, S_n$ (called states), with known prior probabilities $P(S_i)$
and m outcomes $O_1, O_2, \ldots, O_m$ with known likelihood probabilities $P(O_j|S_i)$,
then the posterior probability $P(S_i|O_j)$ can be calculated as

$$P(S_i|O_j) = \frac{P(S_i) \cdot P(O_j|S_i)}{\sum_{i=1}^{n} P(S_i) \cdot P(O_j|S_i)}.$$

Bernoulli distribution

The Bernoulli distribution is the probability distribution of a discrete random variable which is allowed to take the value of 1 with some positive probability p and the value of 0 with probability $1 - p$.

The Bernoulli distribution is common when modeling the outcome of a single experiment that can have a binary success-failure, yes-no, true-false outcome.

Possible outcomes: $\{0, 1\}$

Probability mass function:

$$p(x) = \begin{cases} P(X=0) = q = 1-p, \\ P(X=1) = p \end{cases}$$

Cumulative distribution function:

$$F(x) = \begin{cases} 0, & \text{if } x < 0 \\ 1-p, & \text{if } 0 \leq x < 1 \\ 1, & \text{if } x \geq 1 \end{cases}$$

Expectation: $E[X] = p$
Variance: $Var[X] = pq = p\left(1 - p\right)$

binomial distribution

The binomial distribution is the probability distribution of a discrete random variable consisting of the number of successes observed when performing n independent experiments, whose outcomes each correspond to a Bernoulli random variable with probability of success equal to p.

The binomial distribution is common when modeling the outcome of multiple, independent experiments, each of which can have a binary success-failure, yes-no, true-false outcome.
Possible outcomes: $\{0, 1, \ldots, n\}$
Probability mass function: $p(x) = \binom{n}{x} p^x (1 - p)^{n-x}$
Cumulative distribution function:
$F(x) = \sum_{y=0}^{\lfloor x \rfloor} \binom{n}{y} p^y (1 - p)^{n-y}$
Expectation: $E[X] = np$
Variance: $Var[X] = np\left(1 - p\right)$

cardinality

The cardinality of an event E is a measure of its size and it corresponds to the number of elements/outcomes that it contains. It is denoted by $|E|$.

collectively exhaustive

Two or more events are collectively exhaustive when their union make up the whole sample space. Mathematically, $E_1, E_2, \ldots, E_m$ are collectively exhaustive if

$E_1 \cup E_2 \cup \ldots \cup E_m = S$, or equivalently $\bigcup_{i=1}^{m} E_i = S$.

In plain terms, at least one of the events must always happen if the events are collectively exhaustive.

combination

An unordered subset of $r < n$ outcomes selected from a sample space with $n$outcomes.

conditional probability

The probability $P(E_1|E_2)$ that an event E_1 happens given that event E_2 has already happened.

It is defined mathematically as the fraction $P(E_1|E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)}$, and is only defined when $P(E_2) > 0$ (that is, the condition has a positive probability of happening).

cumulative distribution function

The cumulative distribution function (*cdf*) $F(x)$ of a discrete random variable X is the probability that the random variable takes a value that is smaller than or equal to a specific value x. Mathematically, we have $F(x) = P\left(X \le x\right) = \sum_{y:y \le x} P\left(X = y\right) = \sum_{y:y \le x} p(y)$.

For a continuous random variable, the calculation changes to reflect the fact that continuous random variables are allowed to take any value within a range. Hence, for a continuous random variable X with probability density function $f(x)$, we would have $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(y)dy.$

For example, let $p(x) = \frac{x}{6}$ for a discrete random variable X taking on integer values $1, 2, 3$. Then, the probability that X is less than or equal to 2 can be calculated as $F(2) = p(1) + p(2) = \frac{1}{6} + \frac{2}{6} = 0.5.$. On the other hand, for a continuous random variable X taking on any value in $[0, 1]$ with probability density function $f(x) = 2x$, the probability it is less than or equal to 0.5 will be: $F(0.5) = \int_{0}^{0.5} 2xdx = 0.25.$

Discrete random variables

Discrete random variables are random variables that are allowed to take countable, discrete values.

Examples include the number of earthquakes in a year (0, 1, 2, ...), or the letter grade a student receives in a class (A, B, C).

distinguishable permutations

Distinguishable permutations account for sets with multiple same elements. For example, while there are 6 permutations using letters {'a', 'b', 'c'} ("abc", "acb", "bac", "bca", "cab", "cba"), there are only 3 distinguishable permutations using letters {'a', 'a', 'b'} ("aab", "aba", "baa").

equally probable

Events that all have the same probability of happening.

Assuming a sample space S, then any event happening has probability $\frac{1}{|S|}$.

factorial

The factorial of an integer number n is calculated as $n! = n \cdot (n-1) \cdot (n-2) \ldots 1$. By definition, we assume that $0! = 1$.

geometric distribution

The geometric distribution is the probability distribution of a discrete random variable consisting of the number of trials required until we observe the first success, when each success outcome appears independently with probability p.

The geometric distribution is common when modeling the number of experiments until an outcome appears, when each experiment can have a binary success-failure, yes-no, true-false outcome with the same probability p.

Possible outcomes: $\{1, \ldots\}$

Probability mass function: $p(x) = (1-p)^{x-1} \cdot p$

Cumulative distribution function: $F(x) = \sum_{y=0}^{\lfloor x \rfloor} (1-p)^{y-1} \cdot p$

Expectation: $E[X] = \frac{1}{p}$

Variance: $Var[X] = \frac{1-p}{p^2}$

We note here that the geometric distribution is memoryless.

hypergeometric distribution

The hypergeometric distribution is the probability distribution of a discrete random variable consisting of the number of successes observed when performing n experiments, whose success/failure outcomes each correspond to a random choice of a success or a failure from a larger population containing initially K successes and $N - K$ failures.

The hypergeometric distribution is common when modeling the outcome of multiple experiments, each of which can have a binary success-failure, yes-no, true-false outcome, but that all come from the same larger population.

Possible outcomes: $\{0, 1, \ldots, n\}$

Probability mass function: $p(x) = \binom{n}{x} p^x (1 - p)^{n-x}$

Cumulative distribution function: $F(x) = \sum_{y=0}^{\lfloor x \rfloor} \binom{n}{y} p^y (1 - p)^{n-y}$

Expectation: $E[X] = n\frac{K}{N}$

Variance: $Var[X] = n\frac{K}{N}\frac{N-K}{N}\frac{N-n}{N-1}$

Assume we have a large population of N items, K of which are successes. We pick n items from the population. The number of successes X we observe is: 1) distributed as a hypergeometric distribution, if we do not replace the items as we pick them (without replacement), 2) distributed as a binomial distribution, if we replace the items as we pick them (with replacement).

independent event

Two events E_1, E_2 are independent if we have that: $P(E_2|E_1) = P(E_2) \text{ and } P(E_1|E_2) = P(E_1)$. Equivalently, we may write that two events E_1, E_2 are independent if $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$.

law of total probability

Theorem that allows us to calculate the probability of any event B. Let A_i for $i = 1, \ldots, m$ be some collectively exhaustive and mutually exclusive events. Then, we can calculate $P(B)$ as a "weighted average" of all conditional probabilities $P(B|A_i)$. That is, we calculate

$$\begin{aligned} P(B) &= \sum_{i=1}^{m} \underbrace{P(A_i \cap B)}_{P(A_i)\cdot P(B|A_i)} \\ &= \sum_{i=1}^{m} P(A_i) \cdot P(B|A_i) \end{aligned}$$

memoryless

A random variable X is memoryless if $P(X > t + s|X > t) = P(X > s)$.

The geometric distribution and the exponential distribution are both memoryless distributions. What is more: the geometric distribution is the **only discrete distribution that is memoryless**; the exponential distribution is the **only continuous distribution that is memoryless**.

permutation

An ordered sequence of elements.

The number of permutations of n different elements is $P_n = n!$.

The number of permutations of a selection of r elements selected at random among n different elements is

$$P_{n,r} = \frac{n!}{r! \cdot (n-r)!}.$$

Poisson distribution

A discrete random variable X taking nonnegative integer values $0, 1, 2, \ldots$ is distributed as a Poisson random variable with parameter (rate) $\lambda > 0$ if it follows a probability mass function $p(x) = P\left(X = x\right) = e^{-\lambda} \dfrac{\lambda^x}{x!}$.

The Poisson distribution is very common when modeling the number of events that happen during some period (or the number of occurrences in some space) when they happen with rate λ. For example, number of customers arriving at a store in the next hour, assuming they arrive at a rate of 2.2 per 30 minutes; or the number of typos in a book of 200 pages, assuming they happen at a rate of 1.35 typos per page.

Possible outcomes: $\{0, 1, \ldots\}$

Probability mass function: $p(x) = e^{-\lambda} \dfrac{\lambda^x}{x!}$

Cumulative distribution function:

$$F(x) = \sum_{y=0}^{\lfloor x \rfloor} p(y) = \sum_{y=0}^{\lfloor x \rfloor} e^{-\lambda} \frac{\lambda^y}{y!}$$

Expectation: $E[X] = \lambda$

Variance: $Var[X] = \lambda$

probability

A real number between 0 and 1 representing the likelihood of an event happening.

Probabilities satisfy three main rules (the Kolmogorov axioms of probability):

1. $P(E) \geq 0$, for any event E.
2. $P(S) = 1$, that is if an event comprises the whole sample space then its probability is 1.
3. If $E_1, E_2, \ldots, E_m$ are m **mutually exclusive events**, then

$$P\left(E_1 \cup E_2 \cup \ldots \cup E_m\right) = P\left(E_1\right) + P\left(E_2\right) + \ldots + P\left(E_m\right),$$

or even more concisely: $P\left(\bigcup_{i=1}^{m} E_i\right) = \sum_{i=1}^{m} P(E_i).$

probability density function

We define the probability density function (pdf) $f(x)$ of a continuous random variable X as the function whose curve presents the relative likelihood that random variable X takes a value around that location. Mathematically, the area under the curve of the function reveals the probability that random variable X falls within a range of values. A probability density function over sample space S satisfies three rules:

$$f(x) \geq 0, \text{for all } x \in S.$$

$$\int_S f(x)dx = 1.$$

$$P(X \in A) = \int_A f(x)dx.$$

random variable

A random variable (see also random quantity or stochastic variable) is a real-valued function that assigns a number with each element in the sample space.

Examples include the number of Heads outcomes obtained in 10 coin tosses, or the time until the next patient arrives at an emergency department.

Subsets and supersets

A set S_1 is a subset of set S_2 if all elements in S_1 also appear in S_2. We also say that set S_2 is a superset of set S_1.

uniform distribution

Discrete

The uniform distribution is the probability distribution of a discrete random variable with n different outcomes $x_i, i = 1, \ldots, n$ when each outcome is equally probable and has probability of happening $\frac{1}{n}$.

That probability can be zero when $n \to \infty$.

In a special case, a uniformly distributed random variable takes on integer values in $[a, b]$. In that case:

Possible outcomes: $\{a, a+1, \ldots, b\}$

Probability mass function: $p(x) = \frac{1}{b-a+1}, \; x = a, a+1, \ldots, b$

Cumulative distribution function: $F(x) = \frac{\lfloor x \rfloor - a + 1}{b - a + 1}$

Expectation: $E[X] = \frac{a+b}{2}$

Variance: $Var[X] = \frac{(b-a+1)^2 - 1}{12}$

Continuous

The uniform distribution is the probability distribution of a continuous random variable taking on values in a range between a lower bound a and an upper bound b. Every range with the same length has the same probability of happening.

Possible outcomes: $[a, b]$ (or exclusive, as the probability of getting exactly any value x is zero for a continuous random variable).

Probability density function: $f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise.} \end{cases}$

Cumulative distribution function:

$$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x > b \end{cases}$$

Expectation: $E[X] = \dfrac{a+b}{2}$

Variance: $Var[X] = \dfrac{(b-a)^2}{12}$